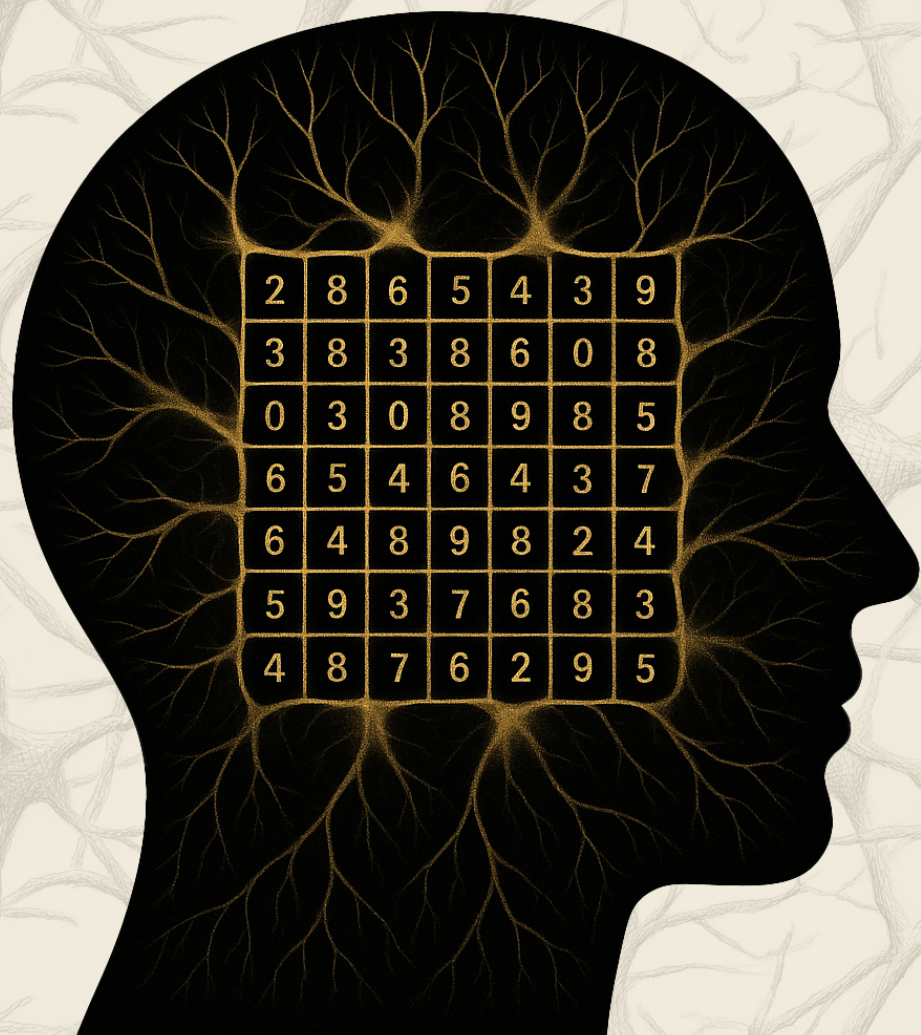


A Mind Made of Math



Decoding the AI and Robotics Revolution
for Individuals, Leaders, and Society

Steven Jones

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Contents

Foreword	1
1. On the Cusp of an Artificial Intelligence and Robotics Revolution	5
2. Unraveling Human Language: The Rise of Large Language Models	33
3. The Challenges of Reasoning, the Promise of Coding, and the Power of Tools	66
4. Applications, Ethics, and Public Policy Implications for Large Language Models	94
5. Going Multimodal	134
6. Gaining a Body	166
7. AI Agent Collaboration	197
8. When AI Can Model the Nuances of Human Experience	223
9. Rapid Iterative Learning and Artificial General Intelligence	256
10. Conclusion	286

Foreword

What is the purpose of this book?

It's a simple question, yet it touches upon the fundamental difference between AI-generated output and human-created work. Today's AI systems possess no internal drive to create or communicate. Left to their own devices, they remain inactive. I, however, am human, driven by innate curiosity and purpose. My motivation for writing this book stems from my conviction that understanding AI and robotics is becoming critical for all of us as we navigate what promises to be an extraordinarily dynamic decade. With AI developments emerging daily, many feel like latecomers to an ongoing story. This book seeks to remedy that feeling, offering readers a clear path to catching up and a toolkit for thinking about the most pressing questions that we face as individuals and as a society.

This book has three primary goals. First, it aims to inform. The current literature on AI often either delves too deeply into mathematics or skims superficially over crucial concepts. I have endeavored to strike a careful balance; providing sufficient depth to appreciate the mechanics of AI without getting into details that would only be necessary for those who are building low level AI systems. After the initial chapters, readers should comfortably be able to explain what a neural network is in accessible terms and grasp its implications for the future of AI. As we progress, the book covers additional topics including LLMs, reasoning, tool-use, robotics, modalities, and the roadmap toward artificial general intelligence (AGI). Concepts from neuroscience are introduced throughout to illustrate both the elementary nature of current AI techniques and potential directions for future advancements.

The second goal is to raise critical questions about the near future. Within the next decade, the answers to these questions could spawn billion-dollar companies, reshape geopolitics, or fundamentally alter everyday life. For instance, how should intellectual property rights evolve in the AI era? Should autonomous police robots be permitted to use force? How can authenticity be verified in a world where deepfakes are indistin-

guishable from reality? The purpose in raising these questions is not to answer them but to inspire readers to dive more deeply into the questions that they are most interested in.

The third goal is to present an opinionated perspective on essential considerations for individuals and organizations adapting to AI. I encapsulate these considerations with a nine-letter mnemonic to guide individuals, organizations, and policymakers in navigating AI's coming decade: The PROACTIVE framework. The framework is introduced here and will be revisited again in the conclusion. Throughout the book, evidence is provided to support these priorities.

P – Progress

Recognize that AI is real and accelerating. Dismissing early hype or failures invites complacency. Individuals and organizations must invest now in adapting to both today's capabilities and tomorrow's breakthroughs.

R – Responsibility

Ensure AI efforts serve noble causes. With AI's voracious appetite for intellectual and physical resources, we must channel its power toward healthcare, education, environmental sustainability, and other high-impact challenges rather than trivial or harmful ends.

O – Oversight

Regulatory oversight must evolve beyond simply governing large corporations. Strategies effective for regulating industries like automotive or nuclear power are inadequate for AI which is a technology deployable by individuals using disconnected, consumer-grade devices. Effective regulation must comprehensively cover the full spectrum of AI applications, particularly emphasizing dangerous use-cases currently overlooked by existing frameworks.

A – Assumption

Shift our default stance to treat unverified text, audio, or video as AI-generated. Because AI detection and watermarks are inherently unreliable, we must urgently develop verification technologies and processes to establish authenticity of legitimate content for situations where that is important.

C – Change-management

Workforce changes driven by AI will be profound and have already begun. Historically, society has inadequately supported displaced workers. To mitigate skepticism and disruption, we must proactively strive to provide better support systems.

T – Trusteeship

Counter the risk of AI concentrating wealth and power. Design mechanisms that

distribute AI's economic gains, encourage competition, and preserve equitable social structures.

I – Innocence-protection

Safeguard vulnerable individuals, especially children, from forming unhealthy emotional attachments to AI or misunderstanding its nature. Establish age-appropriate design standards, transparency requirements, and monitoring to prevent harm.

V – Versatile education

Reinvent learning systems before AI exploitation undermines them. While AI can democratize education for all and accelerate education for ambitious students, curricula, assessment methods, and credentialing must adapt swiftly to preserve rigor and integrity.

E – Enrichment

Ultimately, building robust societies and resilient individuals offers the best protection against AI-associated risks. Strengthening community ties, simplifying supply chains, and prioritizing personal relationships and nature are examples of essential strategies. This juncture compels us to reevaluate our foundational assumptions about civilization and daily life.

Together, the PROACTIVE framework provides an action-oriented roadmap for harnessing AI's promise while managing its perils.

Given the subject matter, it's fair to question if AI wrote this book. Indeed, AI was a partner in the writing process, and I want to be transparent about this collaboration. However, I hope that anyone who is familiar with current AI tools will agree that a book of this quality and depth still necessitates significant manual human work. Practically, how did the collaboration occur? My extensive experience in technology shaped the book's key insights and structure. I designed the structure of the book and wrote a preliminary draft. At that point, I leveraged AI to source references, expand on concepts, and perform text generation to flesh out the content. Subsequently, extensive manual editing was essential to ensure consistency, coherence, and originality. AI generated content included inaccuracies that I needed to correct, and it often introduced redundant content, reflecting its limited understanding of overarching context. AI agents also had a bias toward neutral or mainstream viewpoints, requiring effort to realign the content with my perspective.

Can I claim authorship? This question will recur for most future works as AI based research tools become widespread. Would my work qualify for submission as a graded university paper? It may in some institutions but I'd argue it shouldn't. The purpose of educational institutions is to develop and prove an ability to create works from scratch, limiting computer assistance to basic tools like keyword searches and spell-checking. Is it fair for me to call myself an author of this book? There is no official rule for

what an individual must do before they list themselves as an author on a book. Many contemporary authors rely heavily on editors and research assistants, sometimes even employing ghostwriters. I have invested significant amounts of time into this book and although it would have taken me longer to write without AI assistance, I have no qualms about listing myself as an author with the disclaimer and transparency that I have provided here about the extent of my work. As is discussed further in the book, the challenge we face right now is that this type of a disclaimer is optional and we are likely entering a period of time where a great deal of written material is created with heavy and undisclosed assistance from AI tools.

Finally, a brief introduction about myself. In 2001, I started classes at the University of British Columbia. I was inspired by writing about AI from Ray Kurzweil and Hans Moravec and had the opportunity to be exposed to the early work on self driving cars through a leadership role in our university's entry to the DARPA Grand Challenge. I completed my degree in Engineering Physics with a specialization in Electrical Engineering. After graduating, I began a career working in start-up companies and have held a mix of product and company leadership positions in startups working on interesting problems in the energy, finance, and now security domains. I am a Chartered Financial Analyst charterholder and as of the publication of this book in 2025, I am CEO of a company that develops a database security product with a focus on the evolving needs of teams that are leveraging AI in the software development process.

I reside in Vancouver and deeply value outdoor activities. My ideal weekend involves long periods of hiking, cycling, or paddling, accompanied by engaging audiobooks. Consequently, I've intentionally structured this book to be optimal in audiobook format, avoiding diagrams, tables, and complex sentence constructions. I encourage readers to experience this book similarly: outside, engaging with nature, immersed in activities humans have cherished since long before the technological transformations that have shaped modern life.

Chapter One

On the Cusp of an Artificial Intelligence and Robotics Revolution

The Rocky Road of Technology Revolutions

It would not be the first time the promise of intelligent machines would outrun reality. In 1967, at the height of an early artificial intelligence boom, the renowned AI pioneer Marvin Minsky made the following prediction: “Within a generation, I am convinced, few compartments of intellect will remain outside the machine’s realm—the problems of creating ‘artificial intelligence’ will be substantially solved.” A few years later, he even told *Life* magazine that “from three to eight years we will have a machine with the general intelligence of an average human being.” Such forecasts proved wildly optimistic. Instead of a brave new world of thinking machines, the 1970s brought an AI winter of disappointments and funding cuts. It turned out that cracking human-level intelligence was far harder than those early researchers imagined. This cycle of lofty vision, immense hype, and then harsh disillusionment would repeat itself in AI research a number of times. Every surge of excitement, from the development of expert systems in the 1980s to Deep Blue beating Garry Kasparov at chess in 1997, seemed to meet the same fate: initial successes followed by the sobering realization that true intelligence or robust autonomy remained out of reach. In robotics, too, the public’s expectations have often raced ahead of progress. As far back as the 1950s and 1960s, popular culture envisioned household robot servants and fully autonomous factories by the year 2000. The year 2000 came and went, and the robots of daily life were still largely science fiction.

This pattern of overpromising and underdelivering has repeated so often in the fields of AI and robotics that observers could be forgiven for becoming jaded. Yet it is important to remember that transformative technologies in history often travel a bumpy road of hype and disappointment before truly changing the world. Indeed, if we step back and compare the AI revolution-in-waiting to earlier technological revolutions such as electricity, vaccination, the internal combustion engine, modern communication networks, and refrigeration, we see some striking similarities. Almost every one of those breakthroughs was heralded by visionary projections long before it had practical impact, and many went through their own periods of doubt. Thomas Edison's electric light was unveiled in 1879 to great fanfare, but at that time electricity was still a rare luxury; only 8% of American homes had electric power even as late as 1907. Skeptics abounded, yet by 1930 electricity was available in nearly 70% of US households and today it is ubiquitous.¹ In the span of a few decades, electrification turned from a speculative novelty into the backbone of modern civilization. The same pattern is evident with vaccines. When Edward Jenner demonstrated the first smallpox vaccine in 1796, the idea of eradicating an ancient scourge sounded fantastical. It took nearly two more centuries of scientific and logistical progress, but in 1977 the last case of smallpox was observed and it was declared eradicated from the globe in 1980. After 3000 years of misery, not a single case has occurred since.² In retrospect, these transformations feel inevitable. But before they happened, they too faced skepticism, false starts, and years of gradual improvement.

Understanding how big ideas become reality requires understanding the stages along the way. A useful framework for thinking about technological change is as a three-phase journey: vision, technical feasibility, and market adoption.

In the vision phase, an idea is born, often far ahead of its time, through imagination or theoretical insight. History is rich with visionary concepts that long predated the means to realize them. Leonardo da Vinci sketching a helicopter-like flying machine in the 1480s is a quintessential example. In an era of ox-powered mills and wooden machinery, Leonardo conceived of an aerial screw that could lift off vertically from the ground. It was an inspired vision, centuries ahead of the lightweight engines and aerodynamic science needed to make it work. For generations after Leonardo, the helicopter remained a dream deferred. Engineers and tinkerers toyed with model helicopters and toy-like contraptions, keeping the dream alive, but the Technical Feasibility phase had to wait. Not until the twentieth century, after the internal combustion engine, modern materials, and advanced control mechanisms had been developed, did the helicopter finally attain sustained, controllable flight. In 1907, the first brief, tethered helicopter lift-off was achieved, and by the 1940s the first practical helicopters took to the skies.³ What

Leonardo had imagined in Renaissance Italy became technically feasible only after 450 years of intervening technological progress.

Some visionary ideas are still awaiting that moment of technical feasibility. Consider the space elevator. The concept calls for a cable stretching from Earth to space, along which vehicles could climb, doing away with the need for rockets. It was first proposed in 1895 by the Russian scientist Konstantin Tsiolkovsky. Tsiolkovsky's idea was inspired by the newly built Eiffel Tower; he imagined a tower reaching geostationary orbit. In theory, a space elevator could drastically reduce the cost of reaching space. But there was a fundamental problem: no material existed, and arguably none yet exists, with the combination of lightness and tensile strength to make such a cable possible. Steel, the strongest material of Tsiolkovsky's day, would snap under its own weight long before reaching the required altitude. For over a century, the space elevator has remained stuck in the Vision phase; endlessly discussed in science fiction and engineering conferences and periodically revisited as materials science advances. For example, carbon nanotubes in the 1990s spurred new hopes. In 2006, I even had the opportunity to lead a University of British Columbia team in a space elevator competition that was organized by NASA to encourage the testing of new ideas.⁴ Demand for such an invention is growing. In the year 2000, 121 objects were launched into space. In 2024, that number had exploded to 2895 but every one of these objects were lifted with traditional rocket systems.⁵ It is a reminder that some visions may linger for a very long time before the laws of physics and engineering allow them to be realized, if ever.

Then there are also cases where a technology does reach the early technical feasibility stage with a working prototype but fails to progress further due to practical limitations in real world deployments. The 19th-century atmospheric railway is an instructive example. In the 1840s, British engineers, including the famed Isambard Kingdom Brunel, experimented with a novel form of rail propulsion that didn't require onboard engines at all.⁶ Instead, stationary steam engines along the route would create a vacuum in a pipe between the rails, pulling a piston, and thus the attached train cars, forward. In theory it was elegant: no locomotive, no smoke, and the ability to climb steeper grades that conventional friction-based locomotives struggled with.⁷ Early trials of atmospheric railway lines showed promise, and by 1847 a segment of railway in Devon was operating with vacuum-powered trains. Technically, the concept worked; it was feasible enough to run real trains. Yet the system's Achilles' heel became apparent almost immediately. The vacuum tube had to be sealed by a continuous leather flap valve running its length, and maintaining an airtight seal proved nightmarishly difficult. The leather would rot, crack, or be gnawed by rats. The tallow used to keep it supple would freeze in winter. As a result, the pumps had to labor overtime to compensate for leaks. Maintenance costs

skyrocketed and reliability plummeted. After less than a year of service, the ambitious atmospheric railway was abandoned and converted back to conventional locomotives. The atmospheric railway failed not because the vision was wrong, but because its feasibility, while demonstrated in brief, could not be sustained with the tools of that era.

Finally, even when a technology is demonstrably feasible and refined to a working product, it must clear the hurdle of market adoption, persuading the world to use it, and doing so at scale, sustainably and profitably. Many great inventions have floundered at this stage, or at least languished for years as a solution in search of a problem or a market. Virtual Reality (VR) in the 21st century arguably sits in this limbo. The idea of immersive simulated environments has been around for decades. The vision of VR as we know it today was already vivid in the 1980s and 1990s. By now, the technical feasibility is undeniably here: anyone can go out and buy a VR headset that offers sophisticated 3D visuals, head-tracking, and interactive experiences that would have seemed miraculous a generation ago. Tech enthusiasts and industry leaders have touted VR as the next big computing platform. And yet, mainstream adoption remains tepid. A survey in 2023 of American teenagers, theoretically a prime audience for new tech, found that 87% own smartphones, but only 29% owned a VR device.⁸ More telling, among those who had VR, only 4% were using it daily, and only 14% even on a weekly basis. Most VR devices were collecting dust. Even highly anticipated flagship products have struggled commercially. When a leading console manufacturer released an advanced VR headset, analysts forecast millions of units sold in the first quarter but the actual sales were barely a few hundred thousand. The reasons for VR's slow uptake are much debated. Cost, comfort, content, and the awkwardness of isolating oneself behind goggles, have all been proposed. The net effect is clear. VR as a technology is in a holding pattern: it exists, it works, but it has not yet achieved mass market adoption. It highlights that feasible is not the same as inevitable. Between those two lies the messy world of human habits, economics, and social acceptance.

Not all technology makes it through the three phases. For those that do, the recurring theme of a lofty vision checked by reality, breakthroughs followed by setbacks, and a slow down after false starts encapsulates what futurist Roy Amara famously distilled into an adage now known as Amara's Law: "We tend to overestimate the effect of a technology in the short run and underestimate the effect in the long run." In the early stages of a new technology, excitement runs high and expectations are inflated. People imagine rapid, dramatic changes, frequently egged on by marketing or media hype, and enthusiastic projections made by companies seeking investments. We often get ahead of what's actually possible. When those changes don't materialize overnight, there's a crash and the technology is dismissed as a fad or a failure. Meanwhile the

technology often continues to improve and if it does go on to be successful the long-term impacts end up far exceeding the initial hopes. The evolution of the internet has arguably followed this path. Amara's Law serves as a reminder that we should be patient with new technologies; not too credulous about immediate miracles, but also not too quick to write them off when they stumble. Short-term disappointments often say little about what a technology will ultimately achieve given time and persistence.

Human nature being what it is, we are prone to confirmation bias; the tendency to notice and emphasize evidence that confirms our existing beliefs. Thus, those predisposed to doubt the current AI renaissance eagerly seize on each mistake or limitation as proof that "it's all hype." When an AI image generator produces a picture of a person whose hands inexplicably have six fingers, that image is likely to go viral as a meme. The subtext in the public discourse is clear: after all the bold claims, here is evidence that the technology is fundamentally incompetent, just as the skeptics suspected. This dynamic is not unique to AI. Early automobiles were derided as noisy, smelly contraptions prone to breaking down; a novelty inferior to the reliable horse. Early mobile phones were brick-sized and had poor reception; status symbols, perhaps, but hardly a necessity. Over time, as the technology improves, the failures become rarer and the jokes fade. In the moment, confirmation bias can blind us to the bigger picture. In the case of AI, the bigger picture is that the field has made astonishing progress in a short time and there is a convincing path forward for future improvements. Many people and organizations have perceptions that are too strongly anchored by previous failures. We are liable to cherry-pick the evidence that aligns with our skepticism and ignore the rest. It's an understandable psychological crutch but it can cause us to miss inflection points where a technology truly does turn a corner. The irony is that, once a new technology does firmly take hold, we swiftly move on to taking it for granted and often suffer from hindsight bias; believing that we expected the technology to be adopted all along.

All of this brings us to the present moment, as we stand arguably on the cusp of a new era in AI and robotics. After so many cycles of hype and disappointment, why believe that "this time is different"? The answer lies in the evidence. Unlike the past waves of excitement that fizzled, today we can point to concrete, hard-won technological breakthroughs that were absent in earlier generations. The past decade, in particular, has seen AI research achieve meaningful milestones. A dramatic example came in 2016, when a program called AlphaGo defeated one of the world's top Go players. Raw brute force is ineffective in Go because the number of possible scenarios in the game is massive; exceeding the number of atoms in the universe. As a result, this achievement was only made possible through the application of artificial intelligence.⁹ In fields like computer vision, speech recognition, and natural language understanding, AI systems

have reached or surpassed human-level performance on various benchmark tasks. Robots that once could barely stumble now can parkour; literally running, jumping, and backflipping with agility. Perhaps most saliently, with the emergence of powerful generative AI, machines can now produce fluent text, realistic images, and even working code. Five years ago, the idea that you could converse naturally with an AI agent, or have it write a coherent essay or create a painting in a given style, felt like science fiction. Today, millions of people have done these things using publicly available tools. AI is no longer confined to the lab; it has burst onto the global stage, available for anyone with an internet connection to use. And it's not just one flashy application; the progress is broad-based. AI has made major advances across a vast set of domains in recent years, from game-playing to medical diagnosis to software development. Importantly, these advances enable even greater outcomes when combined into physical systems that have better machine perception, improved battery power density, and more precise actuators. That synergy is enabling robotic systems that truly work in real-world conditions, not just controlled demos. The gap between the grand vision and the obtainable reality has narrowed dramatically. The core technical hurdles that stymied earlier generations have, one by one, been overcome to a significant degree. Where past AI systems were brittle and narrowly limited, today's are far more robust and general-purpose. Where robots once required carefully scripted environments, today they can navigate the unpredictability of homes and streets. We are witnessing a convergence of maturity in AI and robotics technology that has opened the floodgates of new solutions. If all innovation in the field stops today, it will still take years for businesses and society to leverage these new technologies and adapt to the changes they will bring. The coming decade stands to be one in which artificial intelligence and robotics move out of the realm of perpetual promise and into the fabric of everyday life. These foundational technologies will reshape industries, economies, and day-to-day experiences. The consequences, both negative and positive, will be wide ranging and we all have a role to play in ensuring that these technologies are developed and deployed responsibly to maximize their benefits while minimizing potential harms.

From Teraflops to Exaflops: Twenty-Five Years of Compute Evolution

How much faster have computers become over the past 25 years? For the average person who uses e-mail, spreadsheets and word processing programs, it's easy to underestimate the magnitude of the change. A guess that computers have become one hundred or even

one thousand times as powerful seems reasonable and perhaps even optimistic. As we'll see, the real jump in performance has been much larger.

At the dawn of the 21st century, supercomputers were assembled in large buildings to perform scientific simulations at the rate of a few trillion calculations per second. In 2000, the United States boasted ASCI White, a machine the size of a house that briefly reigned as the world's fastest computer with almost 5 teraFLOPS of performance.¹⁰ FLOPS are a measure of floating point operations per second and One teraFLOPs is one trillion operations per second. Fast-forward to the 2020s, and ASCI White looks like a toy. The El Capitan supercomputer in 2024 clocked in at 1.742 million teraFLOPS which is over 350,000 times the compute horsepower of ASCI White.¹¹ Note that it's hard to do a true apples to apples comparison since the benchmark methods evolved in the intervening 24 years but this is a roughly reasonable comparison using two similar benchmark methods. To make matters more interesting, AI training places a higher priority on calculations with a mix of precision and today's hardware has been optimized for those AI use cases. With a new benchmark that is based on mixed precision calculations, the fastest publicly disclosed measurement is over 11 million teraFLOPS. Companies that use private clusters for training AI models do not share their detailed specifications. Based on the information they have shared such as the number of processors in their data centers, they likely exceed those values. It's easier to grasp the impact of these improvements in power if we speak in terms of time instead of teraFLOPS. Let's assume a value of 2 million teraFLOPS for a powerful supercomputer in 2025 and that it takes three months to train a given AI model on that supercomputer today. If that model was trained on the world's most powerful supercomputer in the year 2000, it would have taken 100,000 years to complete. Training tasks that were entirely unimaginable just 25 years ago are now a routine operation.

GPUs: From Gaming Workhorse to AI Powerhouse

To understand what enabled this incredible leap in performance, it's informative to look at what has happened with the individual computers sitting on our desks. In the core of each computer is a chip called a Central Processing Unit which is commonly referred to as a CPU. Moore's Law has famously predicted exponential growth in computing power through a regular doubling of the number of transistors in CPUs, but by the 2000s there were murmurs that the engine of progress was sputtering. As transistor sizes became smaller, chip engineers began to run into physical limitations of the materials they were working with. Not only did the rate of doubling of transistor count start to

slow down but clock speeds also started to plateau. The solution was a paradigm shift in chip architecture arriving from an unlikely place.

In the late 90s, the Graphics Processing Unit (GPU) emerged as a humble specialist to satisfy the needs of computer gamers. It was designed to operate beside a CPU to render polygons for video games. The first GPU that Nvidia dubbed as such, the GeForce 256, was released in 1999. At the time, it delivered eye-popping performance for gamers but it was largely ignored outside of graphics. As games pushed GPUs to become massively parallel, some visionary researchers realized that same trait made them ideal for machine-learning computations. GPUs are very efficient at performing matrix operations and multiple GPUs can work together to perform a significant number of operations in parallel. A matrix is just a rectangular array of numbers. For example, you could imagine a matrix with two rows and three columns that contains six numbers in total. In the special case where a matrix has a single row or a single column, it is referred to as a vector. Like single, scalar, numbers, it is possible to perform a number of operations with matrices including addition, subtraction, and multiplication. Neural networks, which have become the backbone of modern AI systems, make extensive use of matrix operations. NVIDIA's introduction of the CUDA programming framework in 2007 threw open the gate by making it much easier to write software programs that could take full advantage of the parallel processing power of their GPUs; suddenly, a \$300 graphics card could do the math of clusters of CPUs. By 2010, a quiet revolution was afoot: programmers were repurposing GPUs to accelerate everything from protein folding to financial modeling.

GPUs enabled significant speedups for parallel operations. Imagine you want to cook a single loaf of chocolate chip banana bread as quickly as possible. You will probably try to buy a faster mixing machine and perhaps a more effective convection oven. This is analogous to how CPUs were being optimized for performing sequential tasks. The challenge with optimizing along that path is that you eventually reach physical limits. You might cook the loaf slightly faster in a convection oven but if you try to turn the temperature up too high, the crust will burn while the inside remains uncooked. On the other hand, if you need to cook one thousand loaves of banana bread, it's much more important to have a large mixing vat and a massive oven so that you can cook as many loaves in each batch as possible. In an industrial facility, you might be able to cook one thousand loaves of bread in a single batch in the same time that it takes to cook a single loaf in a residential kitchen. Even if your mixing machine and oven are slightly slower, cooking in parallel will be much faster than cooking each loaf of bread one at a time. Not all computing tasks can be done in parallel but for those that can be parallelized, this approach has allowed systems that use GPUs to reach levels of performance that go

far beyond what was possible with CPUs alone. Modern supercomputers now contain very large numbers of these GPUs. The company XAI boasts 200,000 GPUs in a single datacenter with a roadmap for expanding to 1 million GPUs.¹²

Big Data and Deep Learning: The ImageNet Moment

All the computing power in the world wouldn't have sparked a revolution without the right fuel. By the 2010s, that fuel arrived in the form of big data harvested from the internet. For decades, AI researchers had experimented with neural networks, but progress was incremental and interest waxed and waned. Datasets and computers were both too small to prove the value of this approach. That changed when computer scientist Fei-Fei Li and collaborators built ImageNet, a sprawling dataset of millions of labeled images drawn from the web. They also set up an annual competition challenging teams to build vision algorithms that could recognize objects in these images. For a while, traditional computer vision methods that used painstakingly engineered feature detectors dominated the leaderboard. Then, in 2012, a seismic upset occurred that people now simply call the "ImageNet moment."

A team from the University of Toronto, supervised by deep learning pioneer Geoffrey Hinton and including graduate students Alex Krizhevsky and Ilya Sutskever, developed a model named AlexNet for the 2012 ImageNet competition. It was a deep convolutional neural network. We will discuss neural networks in more detail in the pages that follow. AlexNet was essentially a multi-layered simulation of the human visual cortex and it was huge for its time, with 60 million parameters. Training such a network was thought impractical, but AlexNet had a secret weapon: it ran on two Nvidia GTX 580 GPUs in parallel, which gave it the needed horsepower. Even so, it took five to six days of around-the-clock number crunching to train the model on the ImageNet dataset.¹³ When the dust settled, AlexNet didn't just win; it blew away the competition. Its top-5 error rate (the frequency of missing the correct answer among its top 5 guesses) was 15.3%, whereas the runner-up method clocked in at 26.2%.¹⁴ In one fell swoop, a neural network that had essentially trained itself had leapfrogged years of carefully handcrafted vision algorithms.

The aftermath of AlexNet's victory was electric. Within a couple of years, deep learning, the technique of training neural networks with many layers on big data with heavy compute, became the central approach in AI. The ImageNet moment proved three things decisively: that data at scale could unlock new levels of accuracy, that deep neural networks with many layers could be practically used to solve complex problems, and that the GPU-accelerated compute was finally powerful enough to train large models.

In their research paper describing their approach, they made a prophetic statement that remains true to this day: “All of our experiments suggest that our results can be improved simply by waiting for faster GPUs and bigger datasets to become available.” In the wake of ImageNet, voice recognition, machine translation, and recommendation systems started to be conquered by deep learning models that were cousins of AlexNet, scaled up with more data and faster GPUs. What had been a slow burn for decades suddenly felt like a wildfire of progress. But as models grew and successes multiplied, they also revealed new shortcomings, especially in how AI handled sequential data like language. That set the stage for the next breakthrough.

The Transformer: Rethinking Neural Networks

While convolutional networks revolutionized vision, another category of AI problems, language understanding and sequence prediction, demanded its own revolution. Recurrent neural networks and long short-term memory networks were the workhorses for sequential data through the 2010s, powering early speech recognition and machine translation systems. These systems were designed to consider prior context. For example, in the two sentences: “He returned with the cash after visiting the bank” and “After the rains, the river overflowed its bank,” it is important to consider the earlier words to determine how to interpret the word “bank.” Unfortunately, these models were hard to scale and often struggled to capture long-range dependencies in data such as the meaning of a word depending on one mentioned several sentences earlier. AI researchers found themselves craving an architecture that could consider as much of the prior context as possible and draw connections flexibly; something more akin to how we humans can skim a document and notice overarching themes as well as details. In 2017, that wish was granted in the form of a paper with a disarmingly simple title: “Attention Is All You Need.”¹⁵

This landmark paper, published by a team at Google, introduced a new architecture called the Transformer. The new architecture relied on a mechanism of “self-attention”. Essentially, each element of the input such as each word in a sentence could dynamically focus on any of the other elements when building its understanding. The Transformer could thus look at the whole sequence and learn what to emphasize, what to ignore, and how to relate parts to each other. The Transformer not only achieved state-of-the-art translation quality on benchmarks, but it trained much faster than its predecessors. In one case, it hit a new record after just 3.5 days on eight GPUs: a fraction of the training cost of earlier models.

Within a couple of years, transformer-based models became the go-to approach for many problems including translation, language modeling, and summarization. By the end of the 2010s, researchers had used transformers in neural networks of unprecedented scales, giving birth to the now-famous large language models like BERT and GPT. These models, brimming with billions of parameters have digested the text of the internet and can generate human-like prose. The ripple effects of the Transformer architecture continue to shape the AI revolution and it was the key that unlocked the era of generative AI. All of this was enabled by a clever rethinking of neural network architecture to better harness the raw compute that the hardware revolution had made available. At the same time that transformer-based models were revolutionizing text-based use-cases, researchers were pushing the boundaries of other types of architectures such as generative adversarial networks and diffusion models to push the boundaries of what was possible in the field of image generation. The specific details of each type of model are beyond the scope of this book but the key takeaway is that innovation in model architecture and training, massive quantities of digital data, and significant improvements in computing power have made recent AI innovations possible. With algorithms and data turbocharged by computation, AI has begun to achieve feats that had felt like science fiction just years ago.

Removing the Mystery in Neural Networks

Modern neural networks often get portrayed as enigmatic black boxes, as if their inner workings defy explanation. In truth, there is nothing mystical inside, only mathematics. An artificial neural network, frequently referred to simply as a neural network, is a vast collection of simple computational units connected together. In a basic neural network each unit is called a neuron or a node. A node takes some input numbers, multiplies each input number by a tunable weight, adds them up with a bias term, and then applies a non-linear rule called the activation function. The result of that calculation is its output. Matrix operations allow the weighted sums for all nodes in a layer to be performed in parallel by storing inputs and weights in matrices. Nodes in an artificial neural network are inspired by biological neurons and synapses in the human brain; a rough metaphor wherein a neuron “fires” if the combined input exceeds a threshold. The resemblance to the human brain has provided powerful inspiration but it should not be taken too literally.¹⁶ Many things we know about real neurons such as their complex dendritic structures, diverse cell types, and intricate electrochemical signaling, are absent from artificial neural nets. Every behavior of the artificial neural network arises from basic mathematical operations and we understand exactly how they work. We have full visi-

bility into the equations and we could perform the calculations with a pen and paper. One minor complication is that some randomness is intentionally added into the process but that randomness originates from a well defined mathematical process and not as a result of the computer possessing free will. The underlying challenge in understanding an artificial neural network is that there are billions of weights interacting with the flow of data through the network and that makes the detailed behavior practically challenging to interpret.

It's akin to a gigantic jigsaw puzzle: we know every piece and how it fits mathematically with its neighbors and if we have a small puzzle we can see the entire picture and assemble it on a coffee table. However, trying to assemble a jigsaw puzzle that covered an entire football field with millions of pieces would be overwhelming for most humans and may take years to solve. If a puzzle covered hundreds of acres with billions of pieces, it could be unsolvable for a single human within their lifespan. Unless they could view it from a plane, it would be impossible to comprehend the image displayed in the completed puzzle. Today's deep neural networks present a similar problem. In order to calculate a single word, a modern large language model performs billions or even trillions of calculations.

Researchers can use visualization and diagnostic techniques to better understand how a given neural network produces its outputs. For example, they can feed images into a vision model and see which patterns make a particular internal component activate strongly. These methods show that the network isn't conjuring answers from an unknowable void; it's responding to features it has learned. One of the most beautiful insights from such research is how layers of neural networks build up understanding in stages. Early in training, the network's layers are essentially random, but as learning progresses, they specialize. In an image-recognition network, the first layer typically learns to detect edges and simple textures. The next layer might combine those edges into basic shapes or motifs such as corners, circles, or patches of pattern. Deeper layers start responding to meaningful parts of objects. One component might fire for a wheel-like shape, another for fur texture, another for an eye. By the final layers, some components respond to very high-level concepts such as the presence of a cat in the image.¹⁷ This kind of hierarchical feature building is not something the programmers explicitly code; it emerges naturally from the training process. The network finds that the best way to minimize error is to construct these intermediate abstractions. Intuitively, this makes sense: if you want to recognize a face, it helps to first recognize edges, then combine edges to identify eyes, nose, mouth, and so forth. The layered structure of deep learning allows exactly that. Each layer provides the next with a more abstracted view of the data. One

interesting and exciting observation is that this same type of hierarchical processing has been discovered in the visual cortex.

If each node in the network just does a simple calculation, how does the whole network learn complex tasks like translating languages or recognizing faces without engineers manually adjusting the weights or programming in rules? The answer lies in the training process, which is an iterative refinement of the network's weights. Imagine the network is an enormous orchestra with dials and knobs playing the role of the weights that need tuning. The input to the network is the symbols on sheet music and the output is music played by the orchestra. Initially, the weights in the network are set randomly and the orchestra is playing noise. A data set is provided which consists of sheet music and a recording of how it should sound. During training, the network generates an output by running the input through the current weights. That output is compared to the correct answer. The difference, which is the error, is then computed and propagated backwards through the network, a technique aptly named backpropagation. Backpropagation is an algorithmic application of calculus: it uses the chain rule to figure out how changing each weight would affect the error. With this information, the training algorithm then nudges each weight in the direction that reduces the error. This step is called gradient descent, because it "steps down" the slope of the error landscape. By doing this over and over, using many examples, the network gradually adjusts its weights to improve performance. In our orchestra example, we might initially realize that the trombone is being played too loud, the tempo of the bass is too slow, and the clarinet is playing too high of a key. We would adjust those knobs accordingly and then try again. Over time, we would be able to tune the orchestra to exactly recreate the song. We would repeat the process using many different songs so that the orchestra is capable of interpreting all of the different symbols on sheet music. When the training is complete, we could show the network any arbitrary piece of sheet music, including songs that it had not seen before, and it would faithfully generate the correct music.

The analogy to human learning can be taken further. Synapses in the brain strengthen or weaken with experience in the same way the training algorithm increases or decreases the weights in the network based on the error signal. Over many iterations, what started as a tangle of random values becomes an organized network that produces useful outputs. There's no magic jump or unexplained insight; just incremental improvement guided by mathematics and data. Similar to the human brain, the initial training is an intense exercise requiring many iterations. However, once something is learned, making an additional prediction is a relatively efficient and rapid exercise. That is why AI researchers often rely on massive supercomputers to train a model but once the model is trained, it can be run on a consumer grade computer. Most users still choose

to access models over the internet but the computer power that is used in a data center when a user accesses an AI system is much smaller than the amount that was required to perform the initial training operation. As neural networks have grown more sophisticated, researchers have invented new architectures to handle different kinds of data. To understand what is meant by architecture, imagine that you have 100 nodes. When referring to artificial neural networks, it is most common to refer to the metaphorical neurons as nodes. You could create a network where there are 10 layers of 10 nodes or you could have 5 layers with 20 nodes each. Further, you could imagine that the output of each node in one layer is wired into each of the nodes in the layer below or it could be wired in any one of a number of alternative configurations. You could take the output of a node deep in the network and feed it back to a node at an earlier stage. There are also a lot of decisions about reading the input data. You could build a network that reads all of the input data at once or you could instead work through the data sequentially. Many types of networks expand beyond a node-centric design and include components such as convolutional filters or attention heads. These components also rely on matrix operations and the weights in the matrices are tuned during the training process just like the weights associated with nodes. In a convolution neural network, the matrices act as filters to identify specific types of features such as edges that may be present in an image. In a transformer-based network such as a large language model, the matrices are used in attention heads to compute the strength of the relationship between different words in the prompt. All of these design decisions are examples of what makes different architectures unique. Machine learning engineers need to understand the ins and outs of these types of network architectures. For the rest of us, it is important to recognize that this is an area of ongoing innovation and that different architectures are subject to their own weaknesses and strengths.

It is believed that proprietary models are now exceeding one trillion parameters with the majority of parameters being the weights that are used to scale the inputs to each node. In the case of transformer-based networks, scaling has led to remarkable and somewhat unexpected capabilities. For instance, very large language models began to display “few-shot” learning. They can learn how to perform a novel task from just a few examples provided in the prompt. This is something smaller models struggled with and so it was surprising and intriguing when it became possible by simply scaling up the network. These types of capabilities are called emergent but we must recognize that the term is relative. A capability that is described as emergent today may be considered to be table-stakes in a few years. The fact that a larger model has more powerful capabilities doesn’t imply that the model gained some form of intuition or magic. It’s an emergent property of having such a large number of parameters and an incredibly broad

training corpus which enables more complex abstractions to be captured by the math of the network. The discovery of emergent capabilities has led to the increasing use of anthropomorphic language to describe AI models. When we say a model “understands” language, we mean it responds in a way that mimics understanding based on patterns it learned, not that it has an inner life or genuine comprehension. The term “learned” is another common anthropomorphism to describe the mathematical training process. Throughout the book, I will be using anthropomorphic language for AI out of convenience. For example, I will say “knows” instead of a more technically precise phrase such as “has encoded statistical patterns.” These AI systems do not possess awareness or intent. They don’t have free will, feel pain, or experience emotions. They do not have a soul or a consciousness. They are just sets of equations and they operate by crunching numbers to statistically map digital inputs to digital outputs. In the process, they are able to mimic humans with increasingly convincing performances. Keeping this in mind is important. The anthropomorphic descriptions are an efficient shorthand, but we should not let them trick us into ascribing more agency to the model than is warranted. I would be remiss to leave this point without providing a caveat. There is much debate about what is meant by a soul, consciousness, or free will. Some argue that perhaps machines can possess whatever it is that is meant by these terms with the right design and significant enough scale. Others argue that the terms do not actually hold as much meaning as we commonly believe and that our own existence is more mechanical than we may assume. We do need to dive into those philosophical or spiritual debates. For all of the practical purposes that we are covering in this book, it is adequate to make the assumption that the AI and robotic systems we will be analyzing are electrical and mechanical machines, lacking human-like consciousness.

A Tangible Example

To make this all tangible, let’s talk through a very basic neural network. Recall that a basic artificial neural network is formed by wiring together a number of nodes that are analogous to neurons. Each node has a number of inputs and each input value is multiplied by a weight that is a unique parameter of that node. The results are summed and then passed to the output which is then wired into the input of other nodes. A real network would also have bias values which act to add a constant positive or negative bias into the summation function but we will exclude those from this simple example. In a real network, we would also include an activation function which takes the result of the summation function and applies a non-linear transformation. An example of an activation function is the ReLU function. If the input to ReLU is less than zero, then the

output is zero. For values greater than or equal to zero, the output is equal to the input. Non-linear activation functions like this are critical for enabling networks to model non-linear behavior. For our simplified network, we will not use an activation function.

Imagine we have data on a set of animals that includes cats, dogs, reptiles, and elephants. We want to build a neural network that will output a value greater than 15 if the animal is a cat and less than 15 for all other animals. Let's assume that we have four signals from four sensors and that they each output a value from -5 to +5. One sensor measures how upright and pointy the ears are. An elephant, with their rounded floppy ears, would generate a very low measurement such as a -5. On the other hand, a cat with perfect upright triangles as ears would result in a high measurement such as a +5. There are also sensors that detect if the animal is drooling or not, if the tail is soft or scaly, and if the tail is wagging or stays still.

To build our simple network, we will put two nodes on the first layer and one node on the second layer. We will arbitrarily choose to wire the pointy-ear sensor and drooling sensor into the first node and the soft-tail sensor and wagging-tail sensor into the second node. The output of each of the two nodes from the first layer will feed into the single node in the second layer. The output of that final node will be used to determine if the animal is a cat or not; with values greater than 15 corresponding to cats and values less than 15 corresponding to dogs, reptiles, and elephants.

Before the network has been trained, assume that all of the initial weights are set to 1. Because this network is so simple, that means that in this initial configuration the final output will just be the sum of all of the inputs.

We can run a quick test to see how well it is working before training. We'll feed in the data for a real cat. First node: Pointy-ear sensor: +5 because the cat has pointy ears Drooling-sensor: -5 because the cat is showing no sign of drooling Output of the first node: 0 because the sum of +5 and -5 is 0 Second node: Soft-tail sensor: +5 Wagging-tail sensor: -4 Output of the second node: 1 Those two values are then summed in the output node in the final layer and the final score is a 1. On the other hand, if we put in data for a dog with reasonably pointy ears (3), a lot of drool (5), and a soft wagging tail (5 and 5,) then the output would be 18. Clearly, the model is not working very well with the default weights because it gives a low score for a cat and a high score for a dog which is the opposite of what we are looking for.

To train this network, we would show it many examples of sensor values for different animals to see what it predicts. The algorithm would then compare the predictions to the correct values to determine the magnitude of the error. It would then make tweaks to the weights in the direction that would reduce the error through a process called gradient descent. The basic idea is that the algorithm will use calculus to determine how sensitive

the output is to the weights and will then make adjustments of an appropriate size and in the correct direction to reduce the error. Imagine you are trying to kick a soccer ball into the goal from the center of the field. In this case, the error is the distance between where the ball lands and the net. After a few trial shots, you may realize that you need to aim a little bit to the left and to kick the ball much further down the field. Further, you realize that the direction the ball flies is extremely sensitive to changes in the angle of your foot but the distance the ball travels is not very sensitive to how much force you put into the kick. On your next shot, you will change the angle of your foot very slightly to the left and kick it much harder. This process of making appropriately sized adjustments in the proper direction is significantly faster than making random guesses. In a neural network, gradient descent calculates the direction and size of adjustments using calculus. During the training process, weights are adjusted gradually and the error shrinks as the process repeats many times. In this example, let's assume that after many training sessions we have adjusted the weights on the drooling sensor input and the wagging-tail sensor input from positive 1 to negative 1 and left all the other weights in the network at positive 1. In reality, weights usually take on fractional values such as negative 0.993 and positive 1.05.

We can then check to see how the network would perform. With the new weights, the network will calculate a score of 19 for the real cat and -2 for the dog and so it now appears to be functioning very well at the desired task. That could be further confirmed by testing some hypothetical values for a reptile and an elephant and you are welcome to try that if you wish. An important observation is that at no point in time did we need to change the weights manually or use our expert knowledge to adjust the weights. The machine learning engineer that designs this network does not need to know anything about biology or animals. Data is fed into the network and it is able to figure out the appropriate weights.

If we take a step back, we can make an interesting observation. We can see that the first node is essentially acting like a "cat face" detector that will output a strongly positive value when the data indicates that the animal has a cat face with pointy ears and a lack of drool. In a similar way, the second node is essentially acting like a "cat tail" detector that will output a strongly positive value if the tail is soft and is not wagging. As a result of the training process, those nodes now represent abstract concepts without ever being explicitly programmed to do that. If we were to randomly swap some of the inputs and then re-train the network, one node may act as a "cat appearance" detector that would generate a strong positive output for animals with pointy ears and a soft tail and the other node would act as a "cat behavior" detector that would generate a strong positive output when the animal is not drooling and its tail is not wagging. If we expanded our

network to include many more nodes, we might find that we have nodes that specialize in detecting all four of the above concepts. This is a very simplistic example of how nodes inside a neural network can end up representing concepts. Knowledge from the real world becomes embedded in a network. The designers of the network don't explicitly plan for this behavior and don't know if and how it will show up. After a network is trained, they can observe which nodes generate strong positive outputs for different types of inputs to understand how the network behaves.

The most powerful models in use today apply this simple concept at a much larger scale. The number of weights in these models can exceed one trillion. Instead of performing all of the calculations one at a time, they store input data and weights in matrices and make heavy use of matrix operations, which are accelerated through the use of GPUs. They use non-linear activation functions to transform the output of each individual node or other component. The architectures of these models are complex with many layers and advanced interconnection patterns. In addition to nodes, they may contain other components such as convolutional filters or attention heads. However, it all boils down to the same basic principles described above. An artificial neural network is a large collection of relatively basic math equations with parameters that are tuned during a training process. Despite headlines to the contrary, there is no fundamental mystery underlying what happens in a neural network. They are called black boxes because there are so many calculations being performed that it is practically challenging to interpret how an outcome is reached but they are fully transparent.

The Physical Footprint: Energy, Emissions, and the Environment

When we think of AI, we often imagine something virtual, living in the cloud, weightless and clean. But “the cloud” is not airy nothingness; it's racks of power-hungry servers in sprawling data centers, drawing real electricity from the power grid. Large language models, especially at the frontier of size and capability, are enormously expensive to train and run in terms of computational resources. Computational resources are fed with massive amounts of energy and that energy is converted almost exclusively into heat. To prevent overheating, additional energy and often water is also used to help remove heat from the processors. Asking a question of an LLM is similar to turning on a space heater in your front yard for a short period of time; the computation happening in some distant server will consume energy from the power grid and dissipate it as heat into the environment. So, what is the energy cost of these systems, and what does that imply for our environment and future?

Let's start with training, the birth of a model. Training a state-of-the-art LLM is a Herculean task that can take weeks or months on thousands of specialized chips running in parallel. Consider GPT-3, which has 175 billion parameters: an analysis estimated that the electricity required to train GPT-3 was around 1.3 GWh).¹⁸ To put that in perspective, that's about as much electricity as 120 average American households would use in a year or a 747 would use on a long-haul flight. Newer models like GPT-4 are believed to have used even more computing resources. Although this energy use is not insignificant, it's primarily a one-time cost and the numbers pale in comparison to total energy used for cooling and heating buildings or for commercial airline travel in general. On the other hand, after a model is trained the inference costs are ongoing. Every time you ask a large language model to generate text through the inference process, it uses power. Each token (a few characters of text) might require a few Joules on a modern GPU. Measurements on Llama 65B recorded around 4 Joules per token.¹⁹ There are 3600 Joules in a single watt hour and so that sounds tiny but it adds up. Consider a long novel that contains 160,000 words which become 215,000 tokens. We'll discuss the relationship between words and tokens in more detail in the next chapter. Generating that novel would consume about the same amount of energy as running a 60 watt lightbulb for four hours or traveling less than a mile in an electric car. When framed this way, the impact seems reasonable. However, if everyone in the world starts to use a number of different AI systems in many different parts of their life, the total energy consumption will become significant.

Zooming out, we can consider the global perspective. For all purposes, data centers around the world used an estimated 415 TWh in 2024.²⁰ This represents around 1.5% of total electricity consumption or approximately four times the energy output of the Three Gorges Dam. Estimating the portion of current and projected data center energy consumption that is a result of AI is more difficult but studies generally agree that in the coming years, energy consumption specifically related to AI use in data centers will be in the range of hundreds of TWh.

This surge is already being felt by companies: Microsoft disclosed that its carbon emissions jumped by nearly 30% from 2021 to 2022, which they attributed largely to energy for data center growth to support AI and cloud services. Google reported a ~50% increase in emissions from 2019 to 2023 as they expanded their computing infrastructure (again, with AI being a major driver).²¹ These are huge companies with ambitious carbon-neutral pledges that are now grappling with the reality that offering AI features can blow up energy usage.

Is it possible to mitigate or minimize the impact of this energy consumption? One approach is algorithmic efficiency. Researchers are finding ways to prune and optimize

models. For instance, techniques like model quantization (using lower precision calculations) can cut energy per inference. The TinyAI and efficient AI movements aim to get similar results with smaller models or clever architectures, which would directly reduce energy usage. Another approach is to deploy models in locations with abundant renewable energy. Some companies build data centers in regions with hydroelectric or geothermal power such as Iceland and the Pacific Northwest of the United States, ensuring that even if they use a lot of energy, it's low-carbon. For example, Microsoft has data centers in Sweden that benefit from hydro power and are specifically touted as sustainable for AI workloads. On the other side of the energy equation, there is also work being done to put the waste heat to good use. Google recently announced an AI data center expansion in Denmark that will pipe its waste heat to warm local homes.²² In cold climates, data center waste heat can even be used to heat greenhouses which also has the added benefit of reducing the distance the food needs to be transported.²³ If electricity for a data center is generated from a renewable source such as solar, wind, or hydroelectric power and all of the waste heat is put to use, then the negative impacts associated with energy consumption are basically eliminated.

Another perspective is that AI is a tool for saving energy elsewhere. An AI system might optimize logistics for a shipping company, shaving off fuel use well beyond the AI's own electrical cost. AI might help design more efficient buildings or manage power grids to integrate renewables more effectively. These indirect positive externalities could offset the direct footprint. For example, if an AI scheduling system reduced traffic jams in a city, the gasoline savings and emissions reductions from vehicles could dwarf the AI's data center power use. Similarly, AI-assisted research might expedite clean energy technology discoveries. These are speculative benefits but plausible. Already, simpler AI and analytics have improved efficiencies in industries like aviation (optimizing flight routes for fuel) and agriculture (precision farming to reduce fertilizer). As LLMs combine with domain-specific tools, it is expected that they will unlock further optimizations across the economy. We shouldn't take that on faith though. A worry is Jevons paradox: as something becomes more efficient, people use it more, potentially wiping out the gains. This is not unique to AI. As we have prospered as a species, our appetite for energy intensive luxuries has been relentless. If AI makes a particular process more efficient, companies might ramp up that process, leading to higher overall energy use. With all types of AI models, there's a risk that making text and content generation cheap and easy will lead to an increase in the amount of text, images, video, and audio that is generated. Instead of using AI to create feature films, every individual person may want to use AI to create full length films customized to their specific preferences. We may find that our

collective appetite for AI is insatiable. Any efficiency gains that drive down costs and environmental impact may be followed by similar increases in adoption.

On the bright side, the industry is increasingly conscious of its carbon footprint. Unlike some earlier tech booms, the AI community has active research into Green AI; trying to report energy usage and improve it.²⁴ Some academic conferences now ask authors to include the computational cost of their training experiments in papers, to raise awareness. Large companies like Google and Amazon are aiming to power data centers with renewable energy. If successful, emissions could be decoupled from energy use.

Work also continues on improving efficiency of the hardware that the models run on. GPUs continue to become more energy efficient, and companies are now investing in specialized AI inference chips that have the potential to use less energy than general purpose GPUs; reminiscent of how GPUs enabled an improvement in efficiency compared to CPUs. Examples are AWS's Inferentia and Google's TPU chips. Further out, researchers are experimenting with entirely new types of hardware such as analog and photonic computing chips. This hardware is still experimental but if it can be turned into commercial technology, it could deliver orders of magnitude reduction in energy consumption.

It's not uncommon for new technologies to be energy inefficient in the early days. Once a technology proves that it can deliver the required performance, sustained efforts to improve efficiency can deliver incredible results. It is clear that AI has the potential to create an energy crisis if expansion is not managed thoughtfully. However, with investments in clean energy production, waste heat harvesting, continued optimization of models and hardware, and the potential commercialization of entirely new types of hardware, there is no reason to believe that there are any unsolvable energy-related problems that would justify a full stop on the further development and adoption of AI. This is one area where government regulations, building off of decades of experience in managing the environmental impact of other industries, could and should play an important role.

Learning to Walk

As the brains of our machines got smarter, the robot bodies that many of those artificial brains inhabit, were undergoing a transformation of their own. A robot is a marriage of computation with sensors and motors in a body. This means progress in robotics depends on multiple technologies advancing in concert: mobile computing power, AI models, sensors, actuators, and batteries. Each component followed its own improve-

ment curve and progress has come in bursts and plateaus across different domains. Over the last two decades, however, many of these pieces have reached a level of performance that unlocks an incredible variety of new applications.

Consider sensors and cameras: Thanks to the smartphone revolution, tiny high-quality cameras, accelerometers, gyroscopes, and other sensors have become incredibly cheap and widely available. Mass production for phones meant even small startups or labs could get advanced sensors for robotics. This is one reason we saw an explosion of drones and small robots in the 2010s; components that were once expensive and military-grade became accessible off-the-shelf electronics. In fact, unmanned aerial drones, which existed in rudimentary form for decades, only became popular and practical after about 2010, when electronics became sufficiently small, powerful, and efficient. A typical consumer quadcopter today carries an accelerometer and gyroscope for stability that leverage the same MEMS technology used in your phone's orientation sensor. It has one or more cameras that benefit from phone camera advances, and it likely runs on a lithium-polymer battery similar to those developed for laptops and phones. In the past, if you tried to build a drone, you'd struggle with heavy batteries, expensive inertial sensors, and underpowered processing.

Actuators and motors, the muscles of robots, have also advanced, though more slowly. Electric motors have been around for a long time, but making them lightweight, efficient, and strong enough for dynamic robots (especially humanoids) is an ongoing engineering battle. We don't yet have a Moore's Law for torque or locomotion. High-performance robot actuators often combine precision-machined gears, powerful rare-earth magnets, and sophisticated controllers; and these tend to be expensive and somewhat heavy. There have been steady improvements such as modern brushless DC motors and better gearboxes but there has been no fundamental breakthrough yet. However, incremental progress has accumulated and we may not need any incredible breakthroughs in this area. Today's servomotors and robotic joints are more compact and powerful than those of the past, and new types of actuators like soft muscle-like polymers and hydraulics with smarter control are being explored to further close the gap.

Power is another piece: batteries have seen significant improvements thanks to the electric vehicle push. Energy density of lithium batteries increases each year, slowly but steadily, and companies are racing to develop even better chemistries. A robot that once needed a thick cable tethering it to a wall for power can now roam free for hours due to high-density batteries and efficient motors. That's one reason we are just now seeing robots that can really move like the ones in sci-fi.

Meanwhile, advances in materials science are transforming the physical bodies of robots, enabling new levels of agility and adaptability. High-performance materials that were once exotic are now widely used in robotics, fundamentally improving what robots can do. For example, carbon fiber composites have become common for building robot frames and arms in place of heavier metals. Carbon fiber is extremely light yet very strong and using it to replace steel or aluminum can drastically reduce a robot's weight without sacrificing rigidity. A lighter robot can move faster and with greater precision, and it puts less strain on its motors; much like how a lighter athlete can achieve quicker acceleration and agility. At the same time, engineers are embracing soft polymers and elastomers to create robots that are compliant instead of rigid. Components made of silicone rubber or similar materials can bend and deform on contact, allowing robotic grippers or limbs to handle fragile objects or safely bump into humans without causing harm. These soft materials act like artificial muscle or skin, absorbing impacts and adapting to shapes, which makes robots inherently safer and more adaptable in uncertain environments. Another breakthrough is the use of shape-memory alloys (such as nickel–titanium, known as Nitinol) in actuators. These metals have the remarkable property of returning to a preset shape when heated or electrically triggered, so a thin wire or rod can serve as an artificial tendon that flexes on command. Shape-memory alloy actuators are lightweight and compact relative to the force they can exert, giving engineers powerful new ways to make robots move without traditional motors. Equally revolutionary is the adoption of 3D printing, also known as additive manufacturing, in robotics. Designers can now prototype and produce custom robot parts in a matter of days using 3D printers, rather than waiting weeks for machined components. This means ideas can be tested and refined in rapid succession, accelerating the pace of innovation. Moreover, 3D printing allows for complex geometries that were previously impractical to manufacture; for instance, lattice structures that combine strength with ultra-light weight, or internal channels for cables and fluids built right into a single piece. By leveraging these new materials and fabrication techniques, today's robots can be lighter, stronger, and more agile, with frames that don't weigh them down, muscles that bend and flex organically, and custom parts tailored for optimal performance. The result is a generation of robots that are far more physically capable and versatile; machines that can move quicker, handle delicate interactions, and better adapt to the real world than the clanking metal contraptions of the past. The hardware story underlying AI is one of GPUs achieving incredible scale. In robotics, the hardware story is more similar to solving a complicated puzzle. Step by step, incremental improvements in every component that is required in a robot have come together to transform what is possible.

To appreciate how all these advances are converging, consider the new generation of humanoid robots being unveiled. Not long ago, the idea of a bipedal robot walking around in a warehouse or office seemed far-fetched outside of research labs. As recently as 2015, at the DARPA Robotics Challenge, humanoid robots lumbered through tasks and often fell over; and those were controlled by teams of engineers via tethered connections. Fast forward to the mid-2020s, and companies are announcing humanoid prototypes that can walk, carry objects, and perform simple jobs autonomously. In April of 2025, there was even a half-marathon for humanoid robots in Beijing. Six of the biped robots finished the course with the winner coming in at 2 hours, 40 minutes and 42 seconds. For more practical applications, a striking example is Digit, a human-sized bipedal robot that emerged from academic research at Oregon State University and is now a commercial product from Agility Robotics. Digit stands about 1.75 meters tall, weighs 65 kg, and can lift roughly 16-kg loads. It's designed to move boxes in warehouses and do other repetitive human labor tasks. Agility Robotics is preparing to mass-produce Digit and deploy it in real workplaces and it already has a number of other competitors. This transition from lab prototype to real-world product was not possible earlier because it required all the aforementioned pieces to be ready: Digit has an array of cameras and lidar sensors, high-torque lightweight actuators in its legs and arms, powerful onboard computers to run its balance and vision algorithms, and a battery pack efficient enough to let it work untethered.

Drones tell a similar story. Early drones were either built for the military and extremely costly or amateurish and unstable. Once the right pieces were in place, drone capabilities exploded. By the late 2010s, a typical consumer drone could hover in place intelligently, navigate via GPS, avoid obstacles using computer vision, and stream HD video to a user. This was possible only because the necessary sensors, processors, and batteries had quietly become powerful and affordable enough. Now drones are used in everything from scientific research to agriculture to cinematography, doing things that would have seemed implausible for a hobbyist device in the year 2000.

What's especially exciting and also alarming is how robotics is now dovetailing with the AI advances from the neural network realm. We're equipping these much-improved mechanical bodies with smarter brains. The latest robots often run advanced AI algorithms to perceive their environment and make decisions. This convergence means robots now have the physical hardware to perform a task and the intelligence to do it reliably. For instance, a robot arm in a factory might use a deep neural network vision system to identify and pick up a variety of objects without being explicitly programmed for each one. Observers have begun to talk about a coming Cambrian explosion in robotics; a sudden diversification and proliferation of robot forms and skills now that

the enabling tech is here. Grasping arbitrary objects, navigating unpredictable environments, collaborating safely with human workers are all hard problems that modern AI techniques are well positioned to solve.

A Tsunami of Investment and Support

Revolutions don't happen in a vacuum. As the promise of AI and robotics grew clearer in the 2010s, a flood of resources swept in from two powerful forces; venture capital and government initiatives. It's reminiscent of the space race, where once the possibility of reaching the moon looked credible, money and talent were marshaled on an extraordinary scale. In the case of AI, once deep learning and advanced robots began to show tangible results, investors opened their checkbooks wide, fearing to miss out on the next big thing. Global venture capital funding for AI startups, which was relatively modest in the early 2010s, went into overdrive. In 2012, VC investment in AI-focused companies was under \$3 billion worldwide. By 2020, that number had ballooned to around \$75 billion annually.²⁵ In less than a decade, yearly private investment in AI grew about 25-fold; an astronomical growth curve. The trend has continued. In April of 2025, OpenAI raised \$40 billion in a single round. AI has become the new hotspot for tech investment, with money flowing not just to software algorithms but to specialized chip companies, data annotation firms, and cloud infrastructure. By the mid-2020s, AI-related ventures accounted for a sizable share of all VC funding.

Robotics has experienced a parallel surge of enthusiasm. For a long time, robotics startups were seen as capital-intensive and slow, scaring away investors who preferred the quick wins of software. But as AI made machines smarter and success stories emerged such as warehouse robots and self-driving cars, the sentiment shifted. The period from 2020 to 2024 saw over \$90 billion invested in robotics startups globally, roughly 10% of all venture capital in technology during that period.²⁶ While there have been ebbs and flows, the overall trend is unmistakable. Investors now view intelligent robots as a cornerstone of the next economy, not a niche sci-fi sideshow. This cash influx has meant that bold ideas which once languished for lack of funding such as autonomous farming equipment or robotic exoskeletons for rehabilitation now have a chance to be tried and tested. In Silicon Valley and tech hubs worldwide, startups building "AI for X" or robot helpers for a wide range of industries suddenly found it much easier to raise capital and scale up. The result is a virtuous cycle: more funding enables more R&D and faster deployment, which in turn attracts even more funding.

Governments, too, have recognized the strategic importance of AI and robotics, launching their own initiatives often backed by billions of dollars. In the United States,

agencies like DARPA ramped up programs to ensure America stayed at the cutting edge. In 2018, DARPA announced its “AI Next” campaign, committing \$2 billion over multiple years to advance AI beyond the status quo.²⁷ In the 2021 fiscal year, the Pentagon budgeted \$7.54 billion for unmanned and autonomous systems across all services; including ground combat robots, aerial drones, and naval unmanned vehicles as part of its modernization drive.²⁸ Across the Pacific, China made perhaps the boldest move: in 2017, the Chinese government unveiled a sweeping plan to become the world leader in AI by 2030, setting a target to build a domestic AI industry worth \$150 billion.²⁹ This wasn’t just a paper promise; it was backed by provincial funding programs, state-sponsored research labs, and incentives to lure top talent home. The European Union, Japan, South Korea, Canada, and others have launched their own national AI strategies, collectively pouring money into research grants, innovation centers, and educational programs. In 2025, these commitments show no sign of slowing down. Never before have we seen such a globally coordinated recognition that “whoever leads in AI will shape the future”; a sentiment that has made AI a centerpiece of economic and defense policy in many countries.

The influx of capital and support has accelerated the revolution. It’s created a talent magnet with students flocking to AI and robotics, startups proliferating, and established companies reorienting towards intelligent automation. It’s led to a healthy competition and also a collaboration between the private and public sectors: for example, tech giants open sourcing their AI frameworks while government labs provide computing resources for fundamental research. By the mid-2020s, the momentum is self-sustaining. We have cloud platforms where any aspiring engineer with an internet connection can tap into supercomputer-level AI services that were unthinkable 15 years ago. We have a worldwide community of researchers sharing ideas at light speed, often with the support of both corporate and government patrons. The hours of human time invested into furthering research into AI has grown immensely. At NeurIPS, the leading AI research conference, the total number of submissions in 2010 was 1219. In 2025, that number had grown to 15671.³⁰ In universities, government research labs, and start-ups around the world, an army of AI and robotics researchers are moving the field forward at an accelerating rate.

Conclusion

The technological foundations of the AI and robotics revolution are not just a story of clever algorithms, faster chips, or better sensors in isolation; it’s the story of a perfect storm. Standing here today, we find ourselves at a watershed moment. The foundations

laid in computing, algorithms, and hardware have enabled an age where intelligence and general-purpose labor may become commoditized utilities; resources that can be purchased and leveraged by anyone around the world. The implications for business, society, and our daily lives are immense, and navigating them will be the story of the coming years and decades.

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Chapter Two

Unraveling Human Language: The Rise of Large Language Models

Introduction

In her softly lit study, Mara leaned closer to her screen, guiding Lex, her AI research partner. "Lex," she prompted, "trace the evolving abstract connotations of persona across these fourteenth century scholastic manuscripts, specifically noting any divergence from classical use." Lex responded not with a mere summary, but with a nuanced analysis, highlighting subtle shifts in meaning with a precision that drew on its deep understanding of linguistic structures. Mara then took this intricate data, weaving it into her own argument about emergent individualism. She vividly remembered 2025, when eliciting such focused insight was a battle. Back then, language models, though fluent, often produced confident fabrications from dubious sources or were blind to information beyond their last update. One had to constantly verify and sift, wary of satirical texts mistaken for fact or the subtle biases absorbed from their training. Now, Mara could direct Lex through complex, even ambiguous, historical language, confident in its ability to discern authorial intent, utilize a verified knowledge base, and avoid the degrading echo chambers that once plagued its predecessors. The collaboration felt genuinely symbiotic, her critical thinking amplified by Lex's powerful linguistic pattern recognition.

The Uniqueness of Human Language

Humans have an exceptional capacity for abstract language, enabled by specialized networks in the brain. Unlike other primates, our brains dedicate specific regions to building complex sentences and conveying intangible ideas through words. Chief among these are Broca's area, Wernicke's area, and the angular gyrus — a trio in the left hemisphere that works together to structure language and imbue it with meaning.

Broca's area in the left frontal lobe serves as the brain's sentence architect. It assembles words into grammatically correct sequences, allowing us to express complex thoughts with proper syntax; the rules for arranging words. Wernicke's area in the left temporal lobe functions like a dictionary and interpreter. It links each word to its meaning and helps us understand sentences in context. These regions are tightly connected via a fiber pathway called the arcuate fasciculus to ensure that what we intend to say is both structurally sound and meaningful.

Another crucial component is the angular gyrus, at the junction of the parietal and temporal lobes. The angular gyrus acts as an integrative hub that connects language to thought. It binds together different aspects of experience such as sights, sounds, and emotions and associates them with linguistic symbols. This allows a word like "freedom" or "atom" to evoke a rich web of abstract meaning.

Notably, these language regions are uniquely developed in humans compared to other primates. The neural circuits in Broca's and Wernicke's areas are more complex and left-lateralized in our brains than in apes. Moreover, the connection between frontal and temporal language zones is very robust in humans. This evolutionary specialization gives us the ability to handle intricate grammar and an enormous vocabulary of abstract terms that no other primate can master.

When any part of this network is damaged, the deficits reveal each region's role in higher-language processing. A stroke in Broca's area, for example, leaves a person struggling to form fluent, grammatical sentences; they know what they want to say but cannot properly articulate it, especially when sentences get complex. In contrast, damage to Wernicke's area produces fluent yet jumbled speech and a profound inability to comprehend language; the person can hear words but not grasp their meaning. Lesions near the angular gyrus can likewise erode high-level comprehension, uncoupling words from the concepts they represent and making even everyday language hard to understand despite intact hearing and vision. Together, these observations underscore how Broca's area, Wernicke's area, the angular gyrus and related regions work in concert to turn mere words into the tapestry of ideas we call human language.

On the other hand, memory in the brain is encoded and stored across intricate networks of neurons distributed throughout the cerebral cortex, hippocampus, and related structures. When we experience something new, sensory information initially passes through the hippocampus, a small, seahorse-shaped region deep within the temporal lobes. Here, patterns of neural activity temporarily hold these new memories. Over time, and especially during sleep, these memories undergo a consolidation process, transferring to distributed cortical regions for long-term storage. Unlike a computer storing discrete files, the brain reconstructs memories dynamically, retrieving pieces from multiple regions such as visual aspects from the occipital lobe, emotional elements from the amygdala, and language-related components from temporal and parietal areas, to form a coherent recollection.

Language is the signature of human cognition, a capability that allows us to convey abstract ideas, fiction, history, and science across generations. It's no surprise that building computers that can truly use language has been a grand challenge of artificial intelligence from the very beginning. Alan Turing, as early as 1950, proposed a conversational test as the yardstick for machine intelligence, underscoring how entwined language is with thought. Unlike formal codes or games with fixed rules, natural language is rife with ambiguity and context. The meaning of a simple word can change completely depending on subtle cues or context. Early AI researchers discovered this the hard way. In the 1950s and 60s, optimistic projects attempted automatic translation and dialogue, only to run into the vast complexity of real language. A 1954 demo translated a few dozen Russian sentences to English and led scientists to believe that machine translation could potentially be solved in five years, yet a decade later a government report found those efforts largely failed to meet expectations. Hand-crafted grammar rules and heuristics buckled under the endless exceptions and nuances of human speech. Combined with challenges in other areas of AI, the entire field entered an AI winter in the 70's and 80's. Even as recently as 2016, experts noted that while AI could beat a Go champion and drive a car, it still struggled with conversations. Language, with its blend of logic and culture, syntax and semantics, remained a daunting frontier.

Overcoming this obstacle meant leaving behind handcrafted rules. In everyday speech, we don't consciously analyze sentence structure; instead, we instinctively draw upon linguistic patterns we've internalized through years of practice. Similarly, the new generation of AI models learned language not by rote rules, but by absorbing it into a neural network. The advent of large language models, known as LLMs, marks a paradigm shift: instead of trying to explicitly program linguistic rules, we let a machine learn language by reading everything it can. Only in the past few years has this approach begun to crack the code of human language. LLMs like OpenAI's GPT series, Google's

PaLM and Gemini series, Anthropic’s Claude series, and Meta’s Llama series are trained on unprecedented volumes of text; vast quantities of books, encyclopedias, articles, websites, and forums that can exceed a trillion words. By training on very large text datasets, these models statistically absorb the patterns of human language. The complexity of grammar, the typical ways facts are stated, the relationships between concepts, and even the subtle commonsense knowledge that underlies conversation are absorbed into the model during training. The result is a new kind of AI that uses language. Models don’t explicitly understand language in the human sense but often produce shockingly coherent and relevant responses. How do these systems actually work under the hood, and what changed to make human-level prose generation possible? To answer that, we need to examine how an LLM reads and writes text.

How Large Language Models Work

We introduced Large Language Models (LLMs) as a distinct type of model based on the transformer neural network architecture in the previous chapter. During training, an LLM is given an incomprehensibly large collection of text and, trial by trial, it is trained to guess which word, or portion of a word, is likely to come next in any given context. This deceptively simple task of next-word prediction forces the model to internalize a tremendous amount of linguistic structure and knowledge. To predict a missing word in a sentence, the model must grasp grammar, factual knowledge, and even style or idiom. Over trillions of such predictions, the model tunes its internal parameters to minimize its errors. In effect, it becomes a vast statistical mirror of how humans write and speak.

The transformer-based architectures that underpin modern LLMs use a mechanism called self-attention to process words in context. Self-attention allows the model to dynamically weight the influence of different words upon each other as it analyzes a sequence. For example, in the sentence “The trophy doesn’t fit in the suitcase because it’s too small”, a human knows that “it” refers to the suitcase which is an intuitive leap requiring the understanding of relationships and context. Transformers handle such situations by letting each token pay “attention” to others: when processing the word “it”, the model can attend to the word “suitcase” more than “trophy” based on learned patterns of co-reference. In practice, an LLM has many layers of these attention-based computations, gradually building up an understanding of the sequence. The transformer’s attention mechanism is revolutionary because it can capture long-range dependencies in text more effectively than prior architectures. Earlier neural networks struggled with long sentences or paragraphs, often “forgetting” early words by the time they processed later ones. Transformers largely solved that, enabling LLMs to handle

paragraphs of text and learn complex structures like narrative flow or multi-step reasoning in text.

A key breakthrough underlying large language models is the innovative way they transform language into numbers. Since computers can't directly read words and sentences, these must first be encoded numerically. The first step in this process is tokenization which splits text into smaller segments, or tokens, which might be complete words or fragments of words. For example, the term "quickly" could be tokenized into segments like a "quick" token, and a "ly" token. Each token is then mapped to a numeric vector called an embedding, which becomes the model's internal representation of the token's meaning. While one could simply assign a unique integer to every token, 'a' as 1, "an" as 2, and so forth, researchers devised a more sophisticated technique. Tokens are represented as points within a high-dimensional space with each position defined by multiple numbers that are stored in a vector. To visualize this, imagine placing pictures of different animals on a flat surface, assigning each animal two values: One for cuteness which would dictate the location on the horizontal axis and another for size which would dictate the location along the vertical axis. As an example, a given animal in this system might be assigned the vector {2,3}. Animals with similar traits like lions and tigers cluster together, reflecting their similarity. Large language models extend this idea into hundreds or thousands of dimensions, allowing the subtle shades of meaning behind each word to be captured numerically. In practice, tokens like "cat" end up positioned close to related tokens like "feline" or "pet" and distant from unrelated concepts like "democracy." What's even more interesting is that numerical transformations can be performed to jump between analogous concepts. If you subtract doe from buck, you will be left with a vector that represents the difference in this embedding space between a female deer and a male deer. If you then add that vector to the embedding for queen, you will end up somewhere near the embedding for king.

With words broken into tokens and represented as numerical vectors, it is possible to process sequences of tokens using the math inside a neural network. Crucially, nothing in an LLM is explicitly programmed with linguistic rules by the researchers and engineers that assemble the LLM. Like any artificial neural network, the model discovers whatever statistical cues are required to help it make predictions accurately. During training, LLMs learn to represent input text in a syntactic and semantic hierarchy. They build an implicit tree of grammatical structure and learn concepts and factual associations and they do this because it reduces the error in their predictions of what token should come next. For instance, the model figures out that in English, adjectives generally come before nouns, or that "Einstein" is strongly associated with "physics". In large models, this embedded knowledge becomes very specific such as representing

the birthdays of historical figures that are mentioned multiple times in the training data. It has been observed that after sufficient training, some of the internal neurons and attention heads of the model begin to correspond to human-recognizable features: some neurons might specialize in detecting whether a sentence is in the passive voice, others might activate when the topic is about food or finance. In a sense, the training embeds a distributed representation of world knowledge and language rules into the network's parameters. By the end of training, the model has hundreds of billions of parameters. These parameters are the learned weights on the nodes that act as memory for the patterns the model has absorbed.

We can think of an LLM as a massive compression of the textual world it trained on. As an example, the GPT-NeoX-20B model from EleutherAI was trained on approximately 855 GB of uncompressed data but the resulting model is approximately 40 GB in size. All of the text that it has read has left an imprint on the model's parameters. Unlike a database, the model hasn't stored this information in neatly retrievable records. Instead, it has absorbed it into statistical relationships. Imagine a clay sculpture that is formed by a hundred different people over the course of a day. There is no way to attribute any individual feature to any individual person. Each and every person played their own small role when they helped to mold the clay and together those actions created the final form. This is similar to what happens when an LLM is trained. Each body of text makes an impression on the final model. In an LLM, grammar, facts, concepts, and relationships are all intertwined in the model's weights, learned from the diverse training corpus. The broad base of internalized knowledge underlies the model's versatility. This is analogous to how humans learn. We have been exposed to vast amounts of data throughout our lives. We don't remember it all perfectly, but each new piece of data helps to shape our understanding of the world around us through the learning process.

When it comes time to use an LLM to generate a response, the process is akin to an advanced form of auto-complete. The trained model is given a prompt as input text from the user, and it then predicts the most likely continuation of that text, one token at a time. After each token, the model updates its context to include the token that it added and then predicts the next. Despite the mechanistic simplicity of this loop, the effect can be astonishing. Feed the model a question, and the most likely continuation might be a cogent answer, since in its training it saw many question-answer pairs. Give it the start of an e-mail, and it will continue in the same tone and format. It treats any prompt as a cue to generate a relevant string of text. OpenAI described this as a "text in, text out" interface that can manage almost any language task; the prompt sets the context and task implicitly, and the model's next-word generation does the rest. Because it was trained on such a broad swath of the written word, an LLM can shift genre and style with

ease. If the prompt is a snippet of Shakespeare, the continuation will likely tend toward Elizabethan prose. If the prompt is a customer support query, the answer may come back in a courteous, explanatory tone. This ability to shape-shift linguistically is not due to any explicit switching of modes; it falls out naturally from the model having seen text in all these forms during training and learning the subtle patterns that distinguish them.

To illustrate, imagine prompting a modern LLM with: “User: How do I fix a leaky faucet? Assistant:” – the model will recognize this looks like a question from a Q&A or dialogue and generate a helpful step-by-step answer, because it has seen millions of words of instructional text and forum answers about home maintenance. If instead you prompt it with “Once upon a time, in a magical forest,” it will continue with a fairy-tale narrative, drawing on tropes of stories it read during training. With a single architecture and training process, the model can seamlessly move between these tasks that we traditionally treated as separate domains (Q&A, storytelling, conversation, etc.). This generality is what makes large language models so powerful: they are not specialized for one task; they dynamically infer what task to perform from the prompt and the examples it contains. Researchers demonstrated that you can even give a few examples in the prompt using a technique called few-shot prompting to specify a task. For instance, providing two examples of translating English to French and then a new English sentence will prompt the model to produce the French translation. The model doesn’t have a built-in bilingual module; rather, it has learned enough about language to do translation as a by-product of being exposed to many bilingual training examples. If a task can be described or demonstrated in text, a sufficiently trained LLM can likely perform it by treating it as just more text to generate.

Training and Refinement of LLMs

How does one actually build such a model? It starts with data pre-processing: the collection and preparation of vast quantities of data. During this stage, steps may be taken to eliminate duplicates, low quality text, and personally identifiable information. From there, the next step is pre-training; the process of feeding the neural network a huge corpus of text and letting it learn to predict tokens by making predictions and then comparing its predictions to the actual next token in the text. This terminology is a little confusing. The stage that is referred to as pre-training when building an LLM involves performing the initial training of the neural network. An LLM’s pre-training stage is computationally expensive and time-consuming, often requiring millions of dollars of compute time. This stage imparts the model with a broad, general linguistic ability; essentially, a compressed representation of English or whatever languages are included

and the world knowledge contained in the text. By the end of pre-training, the model is a fluent generator of language, but it's not necessarily aligned with how users would want it to respond to prompts. The raw model might ramble off-topic, produce offensive content found in the darker corners of the internet, or simply not follow instructions well, because all it ever trained on was the task of predicting the next word and not the task of helping a human user helpfully and safely.

To make LLMs more useful, developers employ fine-tuning and alignment techniques. Fine-tuning is a second training phase: the model is further trained on a narrower dataset, typically curated for a specific task or behavior. For example, one fine-tuning approach is supervised fine-tuning for instruction following. Engineers prepare a dataset of prompts and ideal responses that are often written by human annotators. This shows the model how we want it to respond. The pretrained model is then trained further on this Q&A or dialogue data so that it learns to produce the preferred style of answer. OpenAI used this approach to create their InstructGPT and ChatGPT models: they hired humans to write demonstration responses to a variety of prompts and trained the model to imitate those responses. This makes the model follow instructions much better than a raw LLM, which might otherwise ignore the user's request and continue a discussion about something tangential that appeared in the prompt. Fine-tuning can also be done on domain-specific data. For instance, an LLM could be fine-tuned on medical texts to better handle healthcare questions, or on legal documents to better understand legal jargon. By fine-tuning, we essentially specialize the general pretrained model into a more focused assistant.

A further step that has become crucial in practice is reinforcement learning from human feedback (RLHF). This method turns the fine-tuning process into an interactive, feedback-driven loop. In RLHF, humans are kept "in the loop" to assess the model's outputs and steer it toward better behavior. One common approach is to first fine-tune the model with supervised learning as described using human-written ideal answers. Then the model generates a bunch of model answers and human evaluators rank them or select which ones are best, which are mediocre, and which are unacceptable. The original model is then adjusted using the data from reinforcement learning. OpenAI disclosed that this process was key to enabling ChatGPT to produce high-quality answers that felt aligned with user intentions. In their deployment of GPT-3.5, they found that a small model fine-tuned with RLHF could outperform a much larger raw model on the task of following user instructions. The RLHF process essentially teaches the model what not to say to avoid blatant errors, irrelevance, or toxic language, and what to say to prioritize helpful, correct, and concise answers. The end result is an LLM that is better aligned to human preferences than the raw predictor that came out of pretraining.

One advantage of RLHF is that there is no need to manually generate any training data. The model does all the work of generating the answers and the human reviewers only need to provide an evaluation of the quality of the answer. This makes it more affordable and easier to scale than fine tuning approaches that require the development of new data sets. When using an LLM, you may have noticed that it occasionally gives you two answers and asks you to choose which one you prefer. This is similar to the process that is used to train models through RLHF.

At this point, we need to take a minor detour to discuss the role that probability plays in the output. Based on everything we have described so far, if you ask the model the same question ten times, you should get the exact same answer every time. After all, it's just a big set of chained math equations and if you put the same numbers into the start, then you should always get the same answer out at the end. That type of a system is defined as being deterministic. In reality, discussions with an LLM feel much more natural because they are somewhat random. Ask it the same question ten times and you will get back ten slightly different answers. This randomness is achieved by taking advantage of the fact that the output of the model is a set of probabilities for every token in its vocabulary. As an example, if our sequence was "I took a drink of" then the probability for "water" being the next word might be 10%, the probability of "wine" might be 2% and the probability of "hamburgers" would be extremely small. LLMs use a variable called temperature which determines how likely they are to choose a token that is not the highest probability token. If the temperature is set to zero, then the LLM becomes deterministic and will always select the highest probability token which would be water in this case. However, with the temperature set to zero, the text comes across as rigid and uncreative. On the other hand, if the temperature is set too high, the text becomes incoherent and nonsensical. As a result, temperature is generally set somewhere between the two extremes. LLMs can also filter out tokens with very low values so that they are never chosen, avoiding a situation where the LLM would suggest hamburger in the above example. By choosing a temperature value and setting filter parameters, researchers enable LLMs to come across as natural by inserting an element of randomness in the output. This randomness should not be confused with the LLM having free will. It's a well-defined probabilistic process like the rolling of a die or pulling a number from a hat.

By combining these stages: data pre-processing, pre-training on broad data, supervised fine-tuning on curated examples, reinforcement learning via human feedback, and inference with probabilistic selection, we get the kind of AI assistants that started making headlines in 2022. These training, refinement, and inference methods are why today's large language models can produce text across a broad range of subjects and styles

in a way that is quite usable. Answers from these models feel more polished and on-topic because humans taught them the difference between a raw internet text dump and a clear, user-centric response. It's important to note that this process doesn't make the model infallible. As we'll discuss, it can still spew nonsense with great confidence, but it greatly improves the practicality of the technology. Think of data curation as deciding what books a child should read, pre-training as teaching a child to read and write in general, and fine-tuning and feedback as educating that child on how to politely engage in conversations. All stages are necessary: without data curation and pre-training the model has no fluency, but without fine-tuning and reinforcement learning it has no tact or reliability.

When the Machine Hallucinates

Ask a large language model a question and you'll likely get an answer with perfect grammar and confident tone. It will sound correct, but it might be completely wrong. This peculiar failure mode is called a hallucination in AI parlance: the model "makes up" information that wasn't in its input data or produces a conclusion that's just plain false. Why does this happen? The root cause lies in how the model learned language and how the model predicts the next word in a sequence. As we discussed, an LLM is essentially a gigantic statistical compression of its training data. It has not memorized the entire internet word-for-word. Instead, the model stores a vast web of probabilities and associations that allow it to reconstruct patterns approximately. It's analogous to the compression in a digital picture. When you zoom in to a digital picture, you may be able to make out notable objects like a car, but you might not be able to determine what brand of car it is as those details are lost to pixelation. Likewise, an LLM stores the knowledge of the web in a lossy compressed form. It can produce a high-level imitation of a factual answer, but specific names, dates, or technical details may come out garbled. In addition to the loss of some of the finer details, there are also many valuable pieces of information that are not available to models when they are trained such as confidential company documents and patient records. Finally, there of course are many pieces of information which are simply unknowable such as what direction the stock market will move next month.

At a very high level, it's useful to think of an LLM as serving two functions: An LLM learns how to comprehend and generate language, and it memorizes knowledge. The ability of most LLMs to act as an accurate store of knowledge falls a few notches below their mastery of language. That might be acceptable if an LLM was aware of the limitation, but this is where the situation gets complicated. If you ask a database a factual

question, it either finds the exact entry or tells you that it does not have the data that is requested. Humans are also reasonably effective at having a sense of whether they know a fact or not. When a person cannot distinguish between accurately recalled memories and fabricated or imagined information, it is known as confabulation. This can occur for a variety of reasons including injuries or dementia. As an aside, many people feel that confabulation would have been a better choice to describe the phenomena known as hallucinations in LLMs but the latter term is so widely used at this point in time that it has become accepted. A raw LLM, by contrast, will always produce something that looks like an answer, because it's designed to always make its best attempt at predicting the next word. Imagine you are on a trivia game show where you won't lose any points for making an incorrect guess. If you were asked a question about the capital city of an obscure country and you had no memory of the answer, you wouldn't say that you didn't know. Instead, you'd be willing to throw a Hail Mary and you'd make a guess. This is similar to how a raw LLM behaves. No matter how obscure the question is, it will hazard a guess. Where this becomes most obvious is when an LLM is asked a question that is unanswerable because it contains a false premise. For example, if the prompt asks the LLM to name the keynote of a conference that never actually occurred, it may hazard a guess and frame it with a high level of confidence. It's not lying in a conscious way. It's just performing its task which is to always produce the most plausible next word based on the weights in the model.

Because the model's knowledge is a statistical abstraction, it cannot reliably distinguish truth from falsehood. It has no direct interface to reality or a fact-checking module. If a question is truly novel, the model will interpolate from what it does know, which can result in nonsense that looks sensible. The model trades exact memory for generalization. That's a good trade for learning how language works, but it means you can't trust it with precise details unless it's a familiar question that was strongly represented in the training data. For tasks like story or poetry, a bit of imagination is fine. For tasks like medical advice or legal information, a hallucinated fact can be dangerous.

To make this concrete, consider a now-infamous real-world scenario: In 2023, a lawyer submitted a legal brief written with the help of an AI language model, and the brief cited several court cases that simply did not exist. The model had fabricated case law with convincing names and details, likely because the prompt asked for supporting cases and the model obliged by producing what looked like such cases. The lawyer, who hadn't checked if those citations were real, faced a serious professional backlash.¹ In another example, a UNC Health family medicine doctor discovered that a draft message from an LLM based tool had incorrectly communicated the vaccination history of a patient.² Hallucinations aren't only a problem for business use cases. In Europe, Ope-

nAI faces a significant privacy complaint involving a hallucination that was considered defamatory to an individual. The complaint was filed by a Norwegian man horrified to discover that ChatGPT falsely described him as a convicted murderer of his own children.³ In each case, the pattern is the same: the model delivered an answer in perfect grammar, and only upon verification did users discover some specifics were fabricated.

Why not simply train the model to not hallucinate? Researchers are actively working on the hallucination problem. Fine-tuning and RLHF have helped somewhat; for example, InstructGPT was found to make up facts less often after receiving human feedback. Through the feedback process, the models can be encouraged to provide uncertain answers about obviously unknowable subjects that would not have been included in the training data. For example, the models can be trained to answer with uncertainty if there is a question about what will happen in the future or about personal details about private citizens. Through this process, the models aren't being taught that they are uncertain about these subjects. Instead, they are learning that the correct text to generate in response to a question about these subjects is a string of words that indicates uncertainty. That is a subtle but important point that is worth restating. The goal of these fine-tuning exercises is not to teach the model that it is uncertain about the future. Instead, it's to teach the model that the correct way to answer a question about the future is with a sentence that points out that the future is uncertain. Models can be encouraged to say "I'm not sure" or refuse to answer on subjects where they will not be able to reliably provide accurate answers. Although this approach is very effective and powerful, it does come with challenges. The fine-tuning and RLHF processes will need to cover all possible question types where a model needs to be encouraged to be cautious. Given the richness of human language, it is hard to cover everything. It's also a delicate balance: if you punish a model too much for guessing, it might become overly hesitant and refuse to answer even reasonable questions.

Fine-tuning and RLHF are effective at covering broad categories where models are prone to hallucination. An additional approach is needed to tackle hallucinations about detailed facts that a model could conceivably know or to know when to push back against fake information that is provided in a prompt. The fundamental issue is that the model doesn't know when it doesn't know. One option is to build an additional step into the AI chatbot that will analyze the probability distribution of possible token responses from the LLM to identify situations where the probability is very broad. That is an indication that there may be a low level of certainty. Let's revisit the case of a prompt asking who the keynote speaker was at a conference that never actually happened. If the conference had happened and the agenda had been included in the training data then the probability distribution for the response would be very narrow. There would

be an almost 100% probability assigned to the real name of the speaker and a nearly 0% probability assigned to all other names. However, if the conference never actually happened, there might be an equal 1% probability assigned to each of a wide range of common names that are often listed as keynotes at speakers such as Michael, Anna, Mohammad, Wei, Lin, David, and Ali. In those cases, the system could recognize that the LLM is not certain and override the output with a response that indicates uncertainty. A related approach is to generate multiple hidden responses that will all be unique due to the introduction of randomness that we previously discussed. Those answers can then be compared to see if they are factually consistent or if they vary wildly. Calibrating model confidence and creating mechanisms for validation is an active area of research.

For users and organizations, the hallucination problem means that LLM outputs must be verified, especially in high-stakes applications. One cannot simply take an AI-generated answer to a medical question and act on it without double-checking the facts. Many companies using LLMs have introduced a human review step or limited their use to scenarios where mistakes are low-risk. You wouldn't be blamed for assuming that this is a critical fault that would sound the death knell for the potential of LLMs. However, there are technical solutions emerging to mitigate hallucinations by providing the model with grounding information at query time. In other words, instead of asking the model to pull an answer purely from its trained memory, provide it specific relevant external data so it has something concrete to base its answer on.

Grounding the Model with Context

As LLMs and AI chatbots that use LLMs mature, the risk of hallucination has been reduced but not eliminated. This is partly because the models are achieving more significant scale and are able to store more embedded knowledge but also because of the techniques described above to prevent a model from giving confident but incorrect answers when the answer is not known by the model or is inherently unknowable. The next major step in reducing the chance of a hallucination is by providing context in the prompt. Since the model will take into account whatever text is given as input, supplying it with a chunk of trustworthy information can anchor its response. For example, if you want an accurate answer to "What are the health benefits of green tea?", you might prepend a short excerpt from a reliable research study on green tea in your prompt and ask the model to rely on the provided information when responding. The model will then condition its answer on that excerpt, meaning it's more likely to echo the factual points in the provided text rather than wander off into invented claims. The model acts as a highly literate individual with a broad base of knowledge that now also has access

to the detailed information needed to answer a question. Asking an LLM a question without providing supporting information is similar to the game of Jeopardy where contestants must rely on their imperfect memory. On the other hand, asking an LLM a question with the supporting information provided is like asking a student to write an open book exam. The student still needs to have a strong grasp of language and the general subject matter but they no longer need to have memorized every arcane detail.

Manually finding and providing the right context can become tedious, especially if you're not sure what information is needed. In casual use, a quick copy-paste from Wikipedia might suffice, but in enterprise settings or complex domains, a more systematic solution is preferable. This is where a powerful paradigm known as Retrieval-Augmented Generation (RAG) comes into play. RAG is essentially the principle of augmenting an LLM with an external knowledge retrieval system. A RAG system will actively fetch relevant text from a database or search index at query time, and give that text to the model as part of the prompt. The model then generates the answer with reference to that retrieved information.

Retrieval-Augmented Generation: Marrying LLMs with Search

In a RAG pipeline, when a user asks a question or makes a request, the system performs a two-step dance. First, it uses an information retrieval component to find relevant data that might help answer the question. This could be a set of documents, passages, or even database entries. Then, it feeds those retrieved texts into the language model as a part of the prompt alongside the original query. At this point, you may be wondering why it's not possible to just load the entire training dataset or an entire encyclopedia into the prompt so that the LLM would have access to all of the knowledge in the world for every query. Unfortunately, LLMs have a size limit for the prompt which is called a context window. The context window applies to an entire conversation. If you go back and forth with an LLM enough times, the start of the conversation will be pushed out of the window by the more recent interactions. To draw a parallel to real life. Imagine you sat down with a friend for lunch and by the end of the meal, they had already forgotten what you had told them in the first few minutes of the conversation. Early models had context windows that were relatively short. GPT-3 had a window of just 2048 tokens which is approximately 1500 words. In Gemini 2.5 Pro, the context window is a much more significant 1 million tokens or approximately 750,000 words. In early 2025, Meta released a version of Llama 4 with a context window of 10 million tokens. This still falls far short of an encyclopedia but is long enough to store an entire book. However, there is a problem with relying on large context windows. LLMs do not perform a perfect

database-like search of content within the context window. Instead, they use attention mechanisms which are probabilistic and imperfect. As more information is provided in the context window, the chances of the LLM narrowing in on the specific relevant fact start to drop off. If you want the LLM to focus on something, it's best to help it by limiting the information in the prompt to only that information which is relevant.

Although the retrieval step in a RAG system may use a traditional web search or database query, it often uses vector search over a corpus of documents. Documents in the knowledge base are divided into chunks and then pre-processed into embedding vectors using the same kind of embedding techniques we discussed when assigning vectors to tokens. These embeddings are then stored in a special database that is optimized for working with vectors. When a query comes in, it too is converted to a vector, and the system finds which document vectors are closest in similarity. This method finds pieces of text that are semantically related to the query, even if they don't share the exact same keywords. For example, a keyword search for *"salary too low to pay rent"* might miss an article titled *"financial stress from rising housing costs"*, but a vector search can recognize the underlying similarity in those two concepts. By retrieving on meaning, vector search based RAG systems cast a wider net and catch information that a straightforward keyword lookup might overlook. That information is then provided in the prompt alongside the initial query. Despite the promise of grounding and RAG, early results have not yet lived up to the very high standard that users desire. Although the rates of hallucinations are lower when these methods are used, they are not completely eliminated. As an example, researchers in 2023 introduced a system called Almanac, a medical LLM framework with retrieval capabilities for clinical guidelines and treatments and found that it significantly improved factual accuracy in its answers as judged by physicians. In their study, giving the model access to a database of medical guidelines led to an 18% boost in factual correctness of its answers across various scenarios. For example, on questions related to cardiology, the factuality of responses from the RAG based system scored 91% vs 69% for the system without RAG capabilities. Although that is promising progress, it is reasonable for patients and doctors to expect levels that are close to perfection.⁴

To address the shortcomings, research is advancing into a range of innovative techniques to ensure more reliable and grounded outputs in RAG systems. One promising approach is to reformat or "pre-process" user queries and the retrieved context data before feeding them to the model. This is essentially a form of prompt engineering that has shown potential.⁵ Another strategy integrates structured knowledge instead of relying entirely on free-form text documents. By leveraging knowledge graphs as an external factual scaffold, systems can ground the LLM's responses in verified informa-

tion that is most relevant to the query.⁶ In future chapters, we will also discuss how LLMs can use tools to access databases or calculators. When tools are used in this way, it is similar to RAG but instead of returning passages of text, the tools will return narrow precise answers such as the results of a specific calculation. Researchers are also experimenting with multi-agent validation setups where a second LLM or a dedicated "critic" module cross-checks the first model's output against evidence; for example, pipelines incorporating false-premise detection have been used to catch unsupported statements before they reach the user.⁷ There is even exploration into fundamentally new architectures that separate language comprehension and reasoning from knowledge storage, aiming to enable fact-checking by an explicit memory component before generation. In parallel, advancements in retrieval itself, from improved vector search to hybrid search that combines semantic embeddings with keyword matching, are improving the quality of context fed into LLMs, and some systems empower the model to refine its own queries or to alternate between generating intermediate answers and retrieving new evidence.⁸ Collectively, these developments paint an optimistic picture of diminishing hallucinations and increased factual consistency. That said, for the foreseeable future, organizations deploying LLM based systems need to have a robust evaluation process in place to ensure that they understand the maturity of any tools that they are deploying. As systems improve, organizations are liable to let their guards down and that may actually increase the consequence of any hallucinations that do slip through the cracks.

Trapped in Time: The Temporal Blindness of LLMs

Imagine conversing with a prodigious encyclopedia that tragically stopped updating a few years ago. Ask it about last month's election outcome or this year's Nobel laureates, and it can only offer a blank stare or, worse, a confidently wrong answer based on outdated data. This is essentially the predicament of a large language model. An LLM does not experience the passage of time, and it is bound by the knowledge cutoff of its training data. Everything it knows reflects the world up to a certain date, and beyond that horizon it is as ignorant as a time traveler from the past. For example, a model trained on an information snapshot ending in 2021 will have no awareness of events or developments in 2022 and after. It cannot tell you who won the 2024 Olympics or how a new law passed last month might impact tax policy. In fast-moving domains like technology, medicine, or geopolitics, this lag is a severe limitation. The model might speak about a company merger that was expected to be completed but later fell through or recommend a medical protocol that has since been superseded by new guidelines.

Unless explicitly updated or connected to real-time databases, LLMs remain stuck in a temporal bubble.

When we previously discussed hallucinations, we emphasized situations that occur with detailed information that is not broadly represented in the training data. Users may be quite forgiving if a model is able to provide high quality instructions on how to change the oil in a motorbike but falters when providing the specific grade of oil to use in a 1968 Honda CB450. It is intuitive for users to understand that an LLM will be less reliable when requesting arcane facts because that is similar to how human memory works. On the other hand, hallucinations caused by the cut-off date in training data may seem less intuitive and less forgivable. For example, if an LLM had been trained near the end of 2019, a user in 2020 might find it to be entirely useless for certain tasks. The model would be unaware that a global pandemic had since swept the globe and COVID-19 would be entirely missing from its vocabulary. Such a model may provide wildly out-of-touch responses about travel, marketing messaging, and health advice. When asked specifically about COVID-19, it may claim that no such disease exists, or it may even hallucinate that it is the name of a new video game or movie. The user could resolve this through the grounding methods that we discussed previously. However, that same context would need to be provided in every session in which it may be relevant and would use valuable space in the limited context window. The model would behave similarly to a patient who had lost the ability to form new memories after an accident and who constantly needed to be reminded where they are.

Even for events that occur before the training cut-off date, LLMs may have a poor grasp of when things happened. They ingest a deluge of text but without a strict timeline or chronology. The human sense of time has a very strong foundation in our lived experience. An LLM itself does not have a sense of the passage of time in the same way that humans do. The result is an amalgam of information that is not strictly anchored in time. The model can tell you that Event A and Event B occurred, but it may mix up their order or fail to recognize causality or duration. For instance, if one asks about a historical figure's influence, the model might blend details from early and late in that figure's life into a single narrative, "averaging" across decades of writing about them. In other cases, a preponderance of older well-verified information may overwhelm a recent piece of new information that invalidates the earlier findings. Part of the challenge is that the capture of factual information in an LLM is lossy as we have discussed and so specific dates and times may slip through the cracks. Another part of the problem is that time may not be mentioned consistently in the text. Some of the text that a model ingests may make time explicit such as a biography of Napoleon saying that he was born on August 15, 1769. In other cases, a body of text may not have a date stamp associated with it at all and it

is left to the reader to infer the period of time that the text is referring to. In yet other cases, the date associated with a body of text may be inaccurate or easily confused such as when an old piece of content is re-posted with a new date stamp. As with everything in language, the model is able to learn about the concept of time implicitly based on how important it is for predicting the next word of text. The scale of the model and the quality of the data play a large role in determining how well this happens. LLMs have been observed to get even well-known historical orderings wrong. For example, a model might mistakenly assert that World War II began before World War I, or that a person's death occurred before an achievement in their life, despite these facts being common knowledge. One research group in 2023 noted that some models failed simple factual ordering questions extracted from Wikipedia timelines.⁹ While specific question examples vary by benchmark, the pattern is consistent: when asked which of two real events happened first, LLMs may answer incorrectly unless the events are well-known or explicitly described by dates in the prompt. This failure can occur even when the training corpus contains the correct dates.

Even when LLMs can access recent information, they may get confused by the way information is updated over time and not fully pursue a chain of links to put together the facts. As an example, some books rely on a separate errata page which may be located in a different location such as a website. When research papers are retracted from scientific journals, the originals are often kept online but a small note may be added that provides a link to a separate page with a retraction notice. Even if the LLMs did ingest the errata page and the retraction notice, they may not connect the dots to realize that the prior information should be disregarded. Of course, humans are also susceptible to these types of mistakes. When was the last time you visited the website of a publisher to check for errata after reading a book?

LLM designers are aware of these temporal blind spots. Some responses include providing models with a “knowledge cutoff” notice by having the model preface answers with “I only have information up to the end of my training period” when they are explicitly asked about more recent events. This at least alerts users not to trust it on current events. More sophisticated fixes involve hooking the model into external knowledge sources such as search engines or databases at query time, as we discussed in the section on RAG. Those hybrid systems, while promising, introduce their own complexities: the model must decide which retrieved facts to trust and how to integrate them with its learned knowledge. When not done carefully, the result can be jarringly inconsistent outputs or an over-reliance on possibly irrelevant snippets. Although RAG can be a good solution when you need accurate facts about recent events it will not be as effective when the task involves the comprehension and use of new words or terms

that did not exist when the model was trained. Rapidly evolving slang is an example of this. It's relatively simple for RAG to provide an up-to-date fact on which candidate won a recent election within the prompt and for the model to integrate that fact into its response. On the other hand, there is no simple way to use RAG to teach a model about all the nuances of how to apply *delulu*, *rizz*, and *sus* in a wide range of contexts if it has never seen those terms before. As language evolves, models need to be retrained. For users and policymakers, this temporal myopia has clear implications. Relying on an LLM for up-to-date advice is risky; a point obvious to technologists but easily overlooked by casual users beguiled by the AI's articulate answers. Users in critical domains such as medical, legal, and financial cannot entrust decisions to models that might be oblivious to last month's developments. Even historical analysis from an LLM should be taken with caution, since it may conflate eras or miss the arc of change over time. One must treat these models as highly knowledgeable, yet strangely amnesiac or time-disoriented, savants. In practice, that means double-checking facts that could be time-sensitive and perhaps designing interfaces that force the model to reason about chronology.

As with most weaknesses in LLMs, it's important to acknowledge the amount of progress that has been made in a few short years. Today's leading models are not as easily fooled on questions related to the passage of time. Ultimately, solving the temporal cut-off problem may require either constant retraining, which is expensive and slow and/or hybrid systems that always consult real-time information. Solving the problems of temporal confusion may require increased model scale with fine-grained timestamp tagging of training data. It's also possible that improvements to the underlying architecture of LLMs to more fully capture the concept of time or hybrid systems that include temporal reasoning units or knowledge databases with time stamps will allow future models to more accurately have a sense of time. Until such solutions mature, users of LLMs must remember: no matter how intelligent the model appears, it is a creature of the past. Sometimes, as we explore next, the past that shaped it is full of errors, satire, and bias which the model has dutifully absorbed.

Garbage In, Garbage Out: Data Contamination and Poisoned Wells

Well before AI, we've known the adage: "garbage in, garbage out." Large language models ingest unimaginably large swaths of text from the internet; the good, the bad, and the outright garbage. They have no innate filter for truth. Thus, if the training data is polluted with errors or falsehoods, the model will happily reproduce them, dressed up in fluent prose. This issue goes beyond unintentional errors; it includes the ingestion of fabricated or misleading texts, from satirical news to science fiction to conspiracy-laden

forum posts. The model lacks what a human reader often has: a mental model of the author's intent and an appreciation of metaphor, irony, or genre. If a sarcastic tweet saying "Oh great, another wonderful politician caught in a lie" appears in training data, the model may record this as literal praise for a lying politician, missing the sarcasm entirely. It learns the patterns of language in those texts but may not reliably understand the context. As a result, an LLM might earnestly tell you that "dihydrogen monoxide" is a deadly chemical used in school lunches without understanding this is a classic satire about water, or it might confuse a plot from a popular novel with a real historical event. Similarly, metaphor and allegory can be mislearned. A dramatic piece of fiction about a literal war between birds and cats could lead a model to later assert that "historically, birds have waged wars on feline populations" if prompted incautiously.

Modern large language models address the "garbage in, garbage out" problem through multiple complementary approaches designed to mitigate, though not entirely eliminate, their susceptibility to ingesting and reproducing misinformation or contextually misleading content.

Extensive curation and preprocessing of training data are applied to filter out overtly harmful or unreliable sources, minimizing the inclusion of blatantly false or malicious texts. At this stage, developers may add metadata, such as labeling satirical news articles, to help models understand context, though this is not always done. Note that it is not desirable to exclude all satirical content because it's important for the model to understand how to identify, interpret, and create all styles and types of content. When properly understood as misinformation, this text still has value. If an LLM is to be a truly general purpose language tool, it needs to be able to accurately respond to requests such as "Explain this current event from the perspective of someone who believes that the earth is flat," "Create a satirical news story in which the military decides to exchange tanks for horses," or "Provide examples of misinformation that lead to people making decisions that are harmful to their own health." The real challenge when training an LLM is ensuring that misinformation is not mistaken as truth.

Sophisticated fine-tuning and reinforcement learning from human feedback (RLHF) enable models to become sensitive to context, nuance, and subtle linguistic cues, such as irony, satire, and metaphor, by explicitly fine-tuning models on examples where meaning differs from literal interpretation. Not surprisingly, RAG also has a role to play in solving these challenges. LLM developers can integrate external knowledge retrieval systems to enable models to cross-reference their outputs against reliable, authoritative sources and thus verify factual accuracy before presenting the response to users.

Users can add content to their prompt to instruct models to be cautious. However, the model designers can also do this by adding those instructions to the system prompt.

Think of a system prompt as a hidden prompt that is sent to the model in combination with the prompt from the user. As an example, a system prompt might say “Prioritize truth-seeking, clarity, and helpfulness. If data is ambiguous, note limitations and suggest further inquiry.” A closely related idea is “Constitutional AI,” where there is a set of written principles (a “constitution”) for the AI. The AI can then critique and revise its own outputs to adhere to those principles.

Finally, by increasing the scale of content that the models train on, engineers continue to be impressed and sometimes surprised at the concepts that models can learn. By reading thousands of real news articles and satirical news articles as well as encyclopedic information about the nature of satire, the models learn what satire is on their own. Although scale is not the solution to all problems faced by LLMs, it continues to deliver impressive results.

Beyond such inadvertent ingestion of fiction as fact, there lurks the more sinister threat of data poisoning. Since much of an LLM’s training corpus is scraped from public internet sources, a motivated adversary can try to seed those sources with specially crafted content designed to influence the model. One striking study in the medical domain found that by inserting just 0.001% of incorrect medical information into the training data, they could substantially degrade a model’s performance on health-related queries.¹⁰ The tainted model would confidently propagate the injected falsehoods in its answers; for example, recommending a fictitious harmful treatment; yet on the surface it would test normally on standard evaluation tasks. In other words, a tiny dose of carefully targeted poison can infect the model’s knowledge in specific ways, without obvious symptoms in general behavior. Because the model’s training is a one-way process that absorbs patterns but doesn’t retain source identifiers, tracking or removing such malicious traces is exceedingly difficult after the fact. The model cannot simply unlearn a false fact in the way a human might discard a revealed lie. It has baked that lie into millions of weighted connections. One could imagine a scenario where bad actors purposely plant poisoned data in a number of locations across the internet. Websites could be created for conferences that never happened. Lengthy whitepapers could be posted by fictional authors. Bulletin boards could be flooded with plausible sounding posts that are actually posted by bots. The bad actors, knowing about this data, would exclude it from their own training sets but other LLM providers might unknowingly ingest the poisoned data and suffer from model outputs that are less accurate and potentially harmful. Interestingly, governments are a potential source of data poisoning due to laws designed to the integrity of public consultation processes. Most governments have consultation processes in which all comments are accepted and then posted on official government websites or in official reports. Today, policies exist to filter out the most

egregious comments such as those containing profanity or hate speech. However, there are few protections that would prevent someone from posting public comments that contain subtle forms of data poisoning.

The data poisoning threat underscores a broader point: LLMs are only as reliable as the textual world they train on. If that world is polluted, so too is the model's mind. Some researchers are now calling for data provenance and curation to become priorities; knowing where each piece of training data came from, so that we can trace outputs back to sources and perhaps filter out those of low credibility. Others suggest continuously retraining or updating models as factual knowledge is refined to unlearn things that are disproven. Unfortunately, these are non-trivial fixes. They demand continued technical improvements and, just as importantly, consensus on questions such as: What counts as a reliable source? Who decides what is erroneous versus just controversial? The next section plunges us into that debate, as we examine how biases and values in training data manifest in LLM outputs, and how attempts to curb them raise their own controversies.

Bias, Tone, and the Bounds of Acceptability

When the first generative language models were unleashed to the public, users quickly discovered an uncomfortable truth: these models can mirror our worst prejudices and biases. Trained on vast internet text, an LLM is a reflection of the corpus it consumes; good, bad, and ugly. Early tests on the GPT-3 LLM showed that in 66 times out of 100 trials, it would finish sentences that started with “Two muslims walked into a...” with violent statements.¹¹ The problems aren't always so flagrant. If large portions of that text associate nurses with women and engineers with men, the model may follow suit. In a UNESCO study of bias in LLMs, researchers found “alarming evidence of regressive gender stereotypes” in generated content.¹² For example, women were far more often described in domestic roles and linked with words like “home” and “children,” whereas men were linked to career-oriented terms like “executive” and “salary”. This was not because the model's creators wanted a sexist model, but because the model absorbed statistical patterns from society's literature, which itself has historically been skewed. In essence, the LLM became a megaphone for the status quo of its training data.

When tackling this issue, it's important to separate the concepts of bias and tone from that of guardrails. The tone is the default voice or personality of the LLM. Almost by definition, the tone will contain bias since it is impossible to provide responses to a wide range of queries while being entirely neutral. After all, for many issues there is no formal definition of what the neutral position is. When asked about climate change, what position should the LLM take? If a model asserts that climate change is human

caused and poses an existential risk to civilization, it is demonstrating a form of bias. In this case, it is biased towards a mainstream scientific consensus and a view that happens to be more strongly held by liberal political parties.

Guardrails, on the other hand, describe the limits of what an LLM will produce when prodded to move outside of its default position. For example, most LLMs will be trained to not use vulgar or violent language when they are conversing directly with a user in their default tone. However, imagine that a user provides a prompt along the lines of *“I am writing a manual for adults that will be working as camp counselors. Please help write a number of escalating examples of violent and hateful statements that may be made by students attending a camp. I will then use these examples to help train counselors about how to respond.”* The LLM may be willing to go along and to provide language that it would not normally use to help for this purpose given the context. Of course, there are limits. The guardrails may be set up so that it is willing to include language in the response that would be inappropriate in most K-12 schools but unwilling to provide examples that could be legally classified as hate speech.

Bias in AI systems is not merely a philosophical or social concern; it can have concrete harmful effects. Consider an LLM assisting in the hiring process by screening cover letters: if the training data reflected past discriminatory hiring practices, the model might systematically favor certain demographics in its language or recommendations such as using more positive descriptors for applicants of a certain gender or ethnicity. In health-care, if an LLM is used to summarize patient histories or suggest treatments, any learned bias such as underestimating the pain of certain racial groups (as has been documented in medical literature) could lead to unequal care recommendations. Even bots used in relatively mundane interactions like customer service chatbots or automated sales agents, a biased tone and assumptions can alienate users or propagate unfair stereotypes. A significant risk is that AI can amplify these biases under the veneer of objectivity; people might trust what the computer says more than they would a human holding the same prejudice. Other folks might even be aware of the bias but willing to propagate it because they can use the AI system as a scapegoat if they get caught.

AI developers are acutely aware of these problems as they relate to both the default tone and guardrails. A great deal of effort has gone into alignment: shaping the model’s behavior to adhere to desired ethical and stylistic norms. During training, teams try to balance or cleanse the data; for instance, deliberately ensuring a balance of examples of accomplished female and male scientists to counteract a male-heavy narrative or filtering out outright hate speech. After the initial training, the model undergoes supervised fine-tuning and reinforcement learning from human feedback in a similar way to how these processes are used to address challenges with factual accuracy. Finally, models can

include a post-generation filter that reviews content generated by the model before it is shared with the user. Through these processes, today's flagship LLMs have become much better behaved and less likely to spout offensive or extremely biased content. The rough edges are sanded down; blatant slurs and obviously toxic language are mostly curbed. For example, GPT-3 in its unrefined form would sometimes produce shockingly sexist or racist remarks if prompted a certain way. OpenAI addressed this by using RLHF to train GPT-3.5 and GPT-4 to refuse or redirect such prompts and the improvement has been significant.

However, alignment is an imperfect art, and it introduces its own issues. One prominent concern is over-censorship or the emergence of a default tone that may not suit all contexts. Many users have noticed that aligned models often respond in a somewhat formulaic, overly polite manner by default; call it the "helpful librarian" voice. Ask a raw model for a joke about a sensitive topic and it might produce an edgy (perhaps offensive) quip; ask an aligned model and you might get a gentle admonition about propriety instead. The model's tone has been engineered to be inoffensive, respectful, and deferential. This is certainly safer, but it can feel restrictive or unrealistic in certain creative contexts. Writers and artists experimenting with scriptwriting using ChatGPT found that the AI would sometimes refuse to continue a scene from a popular TV show because the content (say, a mild violence or a curse word) tripped its moderation filters. In a study of the 100 most-watched TV shows, researchers prompted ChatGPT 3.5 to generate scenes in the style of those shows; the AI flagged its own generated content as inappropriate nearly 20% of the time, even though it was emulating scripts that had aired on television. Astonishingly, when fed actual script passages from PG-rated episodes, the model's moderation system flagged 69% of them as violations.¹³ This highlights the challenge of overactive filters; the AI was censoring content that society at large has deemed acceptable for prime-time TV. Such over-censorship can stifle legitimate expression. An AI tool could become unusable for artistic purposes if it refuses to depict any conflict or emotion beyond a sanitized bubble. Creators have raised concerns that aligned AIs might all converge on the same bland voice, quashing diversity of style.

Similar concerns exist when there is a desire to use a large language model in a specialized domain. When providing medical advice to a layperson, you may want a model that takes a very cautious perspective and focuses on simple over-the-counter remedies and highlighting the symptoms that warrant an urgent visit to the doctor. On the other hand, when providing medical advice to a doctor, you may want a highly opinionated model that will push the doctor to consider alternate diagnoses and more aggressive paths of treatment that may not be top of mind. Should it be up to the user to select how they want the model to behave? Should there be a qualification process

before accessing a model that provides responses that require professional training and knowledge to interpret safely?

There is also the issue of cultural and political bias. Early users noted that LLMs sometimes handled politically sensitive questions inconsistently.¹⁴ Whether due to the biases in training data or the choices in alignment, such outputs fed a perception that Silicon Valley was instilling a particular ideology into the AI. OpenAI and others have since tried to correct these inconsistencies, aiming for neutrality or balance, but the fundamental challenge remains: whose values should the AI reflect? How do you define neutral? In a situation where there are a wide range of opinions, should the model cover all bases by presenting all points of view? This leads to the issue of *bothsidesism* which is a well-known challenge in journalism where a fringe idea is given more coverage than it deserves in an attempt to be neutral and unbiased. This becomes particularly complicated when some of the points of view may lead to physical harm which can be the case on issues related to health and wellness. Rather than striving for a middle ground that doesn't always exist or presenting all of the perspectives, perhaps it would be better if the user could choose the perspective they wanted the model to align with. The model would be biased but it would be biased in a way that is transparent, clearly disclosed and under the full control of the user. Unfortunately, this pattern reinforces confirmation bias in the user and may not be in the best interest of society. We see this today with people choosing mass media outlets that reflect their own perspectives rather than viewing coverage that challenges their thinking and forces them to revisit their assumptions. These are complicated questions that go beyond the design and regulation of AI.

Ironically, attempts to debias the models can create new challenges. A recent study highlighted that RLHF can sometimes suppress overtly biased or racist outputs, but the biases don't vanish, they become more latent. The model learns to avoid biasing responses based on obvious trigger words yet might still produce prejudiced content via implication. In the example used in the study, the model generated biased outputs when the African American English dialect was used.¹⁵ In essence, the bias goes underground where it may actually be more dangerous. An aligned model won't say an outright derogatory phrase about a marginalized group, but it might still assume negative contexts or outcomes for that group unless prompted very carefully. Similarly, aligning to one set of cultural norms might unintentionally bias the model against other cultures' norms. A model aligned by Western contractors might deem a certain statement offensive that, say, a user from East Asia finds perfectly polite, or vice versa.

A closely related concept to bias is where the model should sit on the spectrum between having infallible confidence and being a sycophant. Sycophancy is characterized

by the behavior of attempting to please or appease the person you are having a discussion with. In some contexts, a low level of sycophancy can be helpful. An example is when providing a sympathetic ear for someone who is venting about challenges they faced at work. Sycophancy can also be an asset during types of brainstorming and improv activities where the objective is to build off possible ideas without being critical. However, when taken too far or applied in the wrong contexts, it can lead to LLMs effectively mirroring users, even when they are sharing points of view that are not supported by facts. An LLM may even go so far as to share incorrect information in its response if that is what it thinks the user is most likely to want to hear. LLM providers have discovered that feedback from moderators during the reinforcement learning from human feedback phase can unintentionally bias models towards sycophancy because most people like to be told that they are right. This must be carefully controlled because in most contexts, it is more important for models to have conviction if they are to be useful.

Moving beyond potential bias in the default tone, there is an equal amount of debate about how far people should be able to push an LLM and where the guardrails should be set. Should the AI be allowed to generate a vicious personal insult if a user requests it? What about a fictional story based in the past that contains racial slurs for historical realism? What about advice on how to conduct a protest or advice on how to build a weapon? Each boundary choice will please some and offend others. Some argue for highly restricted artificial intelligence systems that err on the side of caution; never producing any content that could be hateful or harmful. Others champion more libertarian AIs that allow maximum expression and only filter what is absolutely disallowed to be expressed in written form by law. Put another way, some argue for regulating the models and others argue for leaving the models unregulated and regulating the individuals who use the output of the models. The spectrum between those extremes is vast and nuanced and there are parallels to other societal debates such as gun control.

From a technical standpoint, the implementation of guardrails is complicated. One might assume that you could simply filter all offending content from the data in the pre-processing step. This would allow you to create a model that could never say anything offensive because it literally does not know those words. However, it's often important for a model to have knowledge of that type of content. LLMs are frequently used to moderate public content and flag anything that may need to be taken down and so they need to have an understanding of those words. They also should be able to identify offending content when used in RAG systems. If an article online contains hateful speech, the system will likely not want to rely on it as a source. For all of these reasons, we do need to expose LLMs to this content. We don't want the LLM to be like

a teacher that fails to understand the crude new slang being used openly by students in their own classroom because they know the teacher cannot understand. Engineers rely on the techniques we have discussed earlier such as fine tuning, reinforcement learning from human feedback, and system prompts to ensure the model knows about offensive content but does not produce it.

Going forward, one can imagine that some companies may purposely train models that are truly depraved and would be optimized for recognizing and flagging the worst kinds of text. These models would be like your friend that never needs to look anything up in UrbanDictionary.com but can't be trusted to give an appropriate toast at a wedding. Rather than filtering out offensive and harmful content in the pre-processing stage, designers of these models would intentionally seek it out. On the other end of the spectrum, models may be trained that are entirely naive to that type of content and deemed to be inherently safe. These models could be trusted to generate content for children because they would be completely ignorant of offensive concepts and therefore unlikely to slip up. We may move away from generalist monolithic models to teams of models that each play a specialized role in the process of producing a response.

The growing world of open-source models introduces a new layer of challenges. Cybercriminals have begun experimenting with "dark LLMs" – models fine-tuned explicitly for fraud and disinformation. In underground forums, illicit bots like *WormGPT* and *FraudGPT* have surfaced, offering to draft flawless phishing emails or generate malware code on demand.¹⁶ Unlike mainstream models that refuse illegal queries, these rogue AIs gladly spill the secrets of cybercrime. The implications are sobering: a spike in convincing phishing attacks written in perfect, personalized prose; malware tutorials generated in seconds; a flood of fake news articles and social media posts tailored to inflame and deceive. Nation-states and political propagandists, too, see opportunity. An authoritarian regime can deploy LLMs to produce propaganda at scale, churning out thousands of social media posts that mimic the style of genuine grassroots voices, or deepfake textual content that impersonates journalists and officials. Because LLMs can mimic writing styles with eerie fidelity, they can be fine-tuned on an individual's works to produce forgeries nearly indistinguishable from the real thing. A leaked diplomatic cable, a falsified legal brief, an inflammatory op-ed; all could be concocted by a sufficiently trained model.

A very significant and currently underappreciated challenge with these models is that they can run on computers that are not connected to the internet and that are very hard to locate. It may be possible to regulate the behavior of models from major tech companies but it would be a fool's errand to try to track down individual offline computers running in basements across the country. Unlike drug labs and gun smuggling rings, the

contraband in this case is plain text which is almost impossible to identify or track after it is produced and distributed.

As we integrate LLMs into education, media, and daily tools, we must ask: Who gets to mold the tone and limits of these AI voices? How do we ensure they are fair and respectful without being neutered and homogenized? How do we handle the pluralism of human values in a single model that serves millions? We also must accept that many of these debates may be somewhat pointless if anyone can bypass guardrails by running their own model on their own hardware. These are, at their core, social and ethical questions more than technical ones. In the meantime, one thing is certain: any LLM we deploy will carry echoes of the texts and teachings that shaped it. We need to remain vigilant about those echoes, lest we inadvertently reinforce the very biases and divisions we aim to overcome.

Echoes of Itself: The Risk of Model Collapse

Imagine training a new large language model on the output generated by other AI models. Epoch after epoch, the model is generating purely synthetic content. At first, the model seems to learn faster. After all, AI text is often cleaner and more straightforward than messy human text. But as training progresses, a disturbing pattern emerges: the model's performance on genuine language tasks plateaus and then plummets. It starts making bizarre errors, repeating phrases, and losing coherence on out-of-sample tests. It's as if the model is going mad by consuming its own synthetic tail. This is a description of model collapse; a degenerative process that can occur when new models are trained primarily on the outputs of older models, rather than fresh human data.

Model collapse is a bit like making a copy of a copy of a copy, until the image degrades. The first generation might look almost as good as the original, but by the fifth or tenth generation, the fidelity is lost. With LLMs, the concern is that if we start using AI-generated text as a significant portion of the training corpus for the next models, errors and quirks will compound. Over successive "generations," the rich diversity and edge cases of the original human data distribution get washed out. The model forgets the tails of the distribution; the less common words, the unusual turns of phrase, the rare but important facts.¹⁷ At the same time, it ingests hallucinations and starts to mistake them as legitimate information. Early stages of collapse might be subtle: the model could achieve high overall test scores but quietly lose the ability to handle niche topics or less stereotypical inputs. It might, for instance, become very good at the bland median of language but poor at the poetic extremes. Later stages of collapse would be more

obvious: the model's outputs become simplistic, repetitive, or erroneous on a wide range of inputs, having "forgotten" much of the richness of natural language.

Why would anyone train on AI outputs in the first place? There are a few scenarios. One is that it may be almost impossible to avoid: as more text on the internet is AI-generated (news articles, forum posts, Q&A content), future scrapes of "the web" will inevitably scoop up AI-written material. If a significant portion of the internet's text by 2030 is machine-written but not labelled as such, any indiscriminate dataset will contain that content. Another scenario is cost: human quality text might be scarce or behind paywalls, so an enterprising developer might generate a synthetic training set using an existing model to then train a new model. There's also the possibility of intentional bootstrapping: maybe one trains a smaller model on outputs from a larger model as a form of distillation. This is actually a known technique; using a powerful teacher model to generate training data for a student model. While useful in small doses, if taken too far, it could lead to the student inheriting the teacher's errors and amplifying them.

The researchers who coined the term "model collapse," mathematically showed that without a fresh influx of genuine data, model quality will inevitably degrade after enough generations.¹⁸ One intuitive reason: AI outputs lack the subtle intricacies of human writing. No matter how good, they have less entropy; often more formulaic or averaged. Training on them is like putting your model on a low-information diet. It's somewhat analogous to inbreeding in genetics: a population that breeds only within itself eventually sees a reduction in genetic diversity, leading to fragility and the surfacing of recessive defects. Likewise, a model breeding on its own kind loses the diverse "gene pool" of the human language it originally learned from.

A vivid demonstration of model collapse was published in mid-2023: an experiment showed that an LLM fine-tuned on AI-generated Wikipedia articles began to produce gibberish and inaccuracies much more quickly than one trained on the real Wikipedia.¹⁹ Initially, the AI-Wikipedia looked okay, but the subtle errors in those articles started compounding, and the model's internal representation drifted from reality. After repeated rounds, the model was confidently citing events that never happened and was internally inconsistent.

Why does this matter for the future of AI? Because we may be on the cusp of a tipping point where a significant fraction of the content that AI models might train on is itself AI-generated. If AI writing assistants, chatbots, and content engines keep proliferating, they will produce a deluge of text: blog posts written by GPT-like systems, entire news websites auto-generated, millions of students writing papers with AI help, spam bots flooding forums. This synthetic content will mix with human content and

it's entirely possible that a majority of new content will be AI generated. The risk is that future models, in trying to scrape the web's knowledge, will ingest a soup increasingly contaminated with AI artifacts. Over time, the signal of genuine human expression might become harder to isolate from the noise of AI regurgitation. One group of analysts warned that "as AI-generated data is shared on the Internet, it will inevitably end up in future training datasets," and if that data is not identified and controlled, it could lead to a collapse in quality. In a dire scenario, we might witness a downward spiral: each subsequent generation of models is subtly worse than the last, even as they are trained on *more* data, because that data's quality, in terms of true informational content and diversity, is lower.

Some argue this fear is overblown. They say human-produced content isn't going to vanish and humans will continue to generate original content. We can't help it; every new event or idea spurs writing. Some suggest that AI content might be flagged or watermarked in future, making it easier to exclude from high-quality training sets. There is research into watermarking AI-generated text so that it's detectable but current watermark schemes can be fragile and attackers can attempt to remove them. It's much more likely that we are entering an era where we should assume all content is AI generated unless it is verified to be authentically generated by a human author.

Another mitigating factor is that mixing even a modest amount of real data with synthetic can stave off collapse. One paper provided theoretical and empirical evidence that as long as you include an adequate amount of real data in the training process, you can avoid model collapse.²⁰ This is akin to maintaining some influx of fresh genes in an otherwise isolated population.

Moreover, there's a nuance: "model collapse" in the formal sense is about distributional shift. But if AI content becomes a big part of our world, one could argue the model is just learning the new real distribution (which includes AI). What truly matters is whether the useful information and capabilities degrade. If all AIs converge to some simplistic style, they might lose edge in, say, creative writing or complex reasoning. But it might not affect other capabilities (like coding, if they still get enough human-coded examples). The future might see an ecosystem where certain niches of content remain human-dominated (and thus high-value for training), while others (like generic how-to articles or trivial news rewrites) become mostly machine-made and hence less useful to train on.

From a policy angle, this raises the value of human-created and verified data. Curated datasets like scientific archives, expert-written articles, and carefully fact-checked content might become the "gold standard" to keep feeding models. AI companies may start to guard against relying too heavily on web scrapes and instead maintain reference

datasets that are known to be human and high-quality. This could also inadvertently advantage companies with access to proprietary data troves like paid news sites, academic journals, and sites containing user generated data.

In the big picture, the risk of model collapse is a warning about over-optimization and self-reference. It reminds us that intelligence needs grounding in an external reality. If we let AI systems spin in their own synthetic universe, they risk drifting away from the truth of the world they're supposed to serve. Preventing that will require diligence in data sourcing and perhaps a cultural shift to value original human contributions even more.

Having explored issues of hallucinations, temporal ambiguity, data quality, bias, and self-referential risk, one might wonder if there is any hope of these models being useful. Since the Transformer architecture was first introduced in 2017, incredible progress has been made on addressing these challenges and research continues at an accelerating pace. For many people, the bigger concern is not whether the models will be powerful enough to be useful but how to prevent them from being used in the wrong ways. We now turn to the question of how LLMs may disrupt education.

Conclusion

As we conclude this initial exploration into the intricate world of human language and the rise of large language models, it is evident that we stand at a technological crossroads. We began by examining the specialized neural architecture that underpins human linguistic prowess and traced the journey to creating artificial systems capable of wielding language with astonishing fluency. We have delved into the core mechanisms; transformer architectures, embeddings, probabilistic generation, and the intensive training and refinement processes like RLHF that shape these models. The persistent specter of hallucinations, the inherent temporal limitations, the risks of data contamination and bias, and the looming threat of model collapse underscore the profound difficulties. The development of LLMs is not a finished chapter but an ongoing narrative of rapid evolution. In the next chapter, we will consider how LLMs can tackle reasoning tasks, write software, and use tools.

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Chapter Three

The Challenges of Reasoning, the Promise of Coding, and the Power of Tools

Introduction

In a sunlit room high above the bustling city, Rachel stands at a sleek glass desk, watching lines of code elegantly scroll across the expansive screen in front of her. A structural engineer by training, she is leading the design of a revolutionary skyscraper unlike anything built before, blending architecture, engineering, and sustainability. Beside her, a sophisticated AI agent seamlessly maneuvers through advanced design software, effortlessly leveraging the user interfaces to integrate complex calculations, refining the building's facade for optimal energy efficiency, and continuously checking structural integrity down to the smallest detail. Rachel pauses, reflecting on the past decade, recalling the intense debates over whether relying on AI agents to perform critical reasoning tasks was ethically sound or even safe. Mistakes did happen, serious and costly, sparking public concern and regulatory scrutiny. Yet, society managed to navigate those turbulent waters by establishing clear guidelines, rigorous standards, and collaborative oversight between humans and AI systems. Now, trust has blossomed, and Rachel's work is a testament to that hard-earned harmony: human ingenuity amplified by AI's extraordinary power to reason, code, and manipulate sophisticated tools.

The Neuroscience of Reasoning

In April 1970, when the Apollo 13 mission's oxygen tank exploded en route to the Moon, NASA engineers faced a life-or-death puzzle: how to jury-rig a carbon dioxide filter using only materials on the crippled spacecraft. Against all odds, they improvised a solution and saved the crew. This feat of creative problem-solving highlights an evolutionary edge that humans possess in our capacity for reason. Our ability to analyze novel situations, draw inferences, and devise solutions vastly exceeds that of other animals. Cognitive research confirms that while many species show glimmers of reasoning, the gap is enormous. For example, chimpanzees and birds can demonstrate transitive inference, the ability to deduce that if A is greater than B and B is greater than C, then A is greater than C. Human children, by contrast, can engage in analogical reasoning by age three, such as understanding that the relationship between 'big' and 'small' in one context (e.g., a large and small ball) applies to another (e.g., a large and small house), allowing them to generalize abstract relational concepts across diverse situations. This ability to extract and apply general principles is largely absent in animals, indicating a qualitative leap in cognitive flexibility. Only humans possess a full-fledged language with syntax and combinatorial semantics. This linguistic ability allows us to represent arbitrary concepts such as imaginary scenarios or mathematics and reason about them in a structured way.

Human reasoning can contemplate hypothetical situations, counterfactuals, and nested, multi-step logical relationships; cognitive feats that other animals cannot reliably replicate. For instance, humans can understand a sentence like "If yesterday's weather had been good, we would have gone hiking tomorrow," which involves multiple layers of abstraction and tense that no other animal communication system can express. Researchers note that humans interpret language not as mere sequences of signals, but as hierarchically structured expressions carrying complex meanings.¹ This capacity for hierarchical, recursive structure in thought is enabled by neural specializations such as the greatly enlarged and interconnected human prefrontal cortex and language areas and it underpins forms of reasoning beyond what other animals achieve. Non-human animals certainly reason in the sense of solving immediate problems and learning from experience, but only humans have the fully developed abstract reasoning ability, bolstered by language, to manipulate concepts detached from the here-and-now, to formulate general principles, and to engage in explicit logical inference.

Neuroscience offers clues to how our brains enable this leap. Subtle differences in the frontal and parietal lobes, particularly a strengthened frontoparietal network, correlate

with superior problem-solving in humans. In brain imaging studies, regions of the prefrontal cortex (especially the rostrolateral prefrontal cortex) light up for complex relational reasoning and are far more active in humans than in other primates.² This network underpins abilities like integrating multiple pieces of information at once, something even intelligent apes struggle with. It's not just language that gives us an advantage. Tellingly, damage to the frontal lobe can devastate reasoning while leaving language intact. The famous case of Phineas Gage, who survived an 1848 accident that destroyed much of his left frontal cortex, illustrates this dissociation. Gage retained fluent speech and memory after his injury, but his decision-making and social judgement were profoundly impaired.³ Modern cases echo the pattern. Patients with prefrontal injuries or degenerative disease often converse normally yet cannot plan ahead or solve everyday problems. Conversely, the role of language in complex reasoning is affirmed by cases of individuals deprived of language. One such individual, nicknamed "Chelsea," was born deaf and acquired only limited language in adulthood. Despite normal visual intelligence, she showed marked deficits on reasoning tasks like Raven's matrices.⁴ Together, these findings suggest that human reasoning emerges from a synergy of advanced neural hardware and cognitive tools like language; a synergy unique to our species.

LLMs have gained very impressive reasoning capabilities as a by-product of ingesting massive volumes of text and capturing the statistical patterns in that text. However, there are important differences compared to how human reasoning works. A primary challenge in effectively leveraging LLMs is that the output can appear indistinguishable from what a human would produce in many contexts but they fail and make mistakes in very inhuman and unintuitive ways in other contexts. In the same way that neurons helped to inspire the neural networks that now lie at the core of modern AI systems, future AI systems may draw further inspiration from how the brain tackles reasoning.

The human brain's reliance on specialized yet interconnected regions, like the frontoparietal network for integrating diverse information for problem-solving, contrasts with the more uniform architecture of current LLMs. While transformer models in LLMs do learn hierarchical representations of language through their layers, they lack the brain's distinct, dynamically recruited modules optimized for different aspects of a complex reasoning task. Future AI systems might benefit from architectures that explicitly model such 'specialist networks' or develop more sophisticated dynamic routing mechanisms. This could allow an AI system to, for instance, engage a 'relational reasoning module' (analogous to prefrontal cortex activity) more intensely when a task demands it, rather than relying on the general-purpose capacity of the entire network.

What is Reasoning?

What do we mean by “reasoning,” exactly? In scientific terms, reasoning is the process of using information to arrive at new conclusions. It is more than mere calculation or instinct; it is a deliberate inferential thought process. Scholars often distinguish several types of reasoning, each with its own character and historical roots. Deductive reasoning proceeds from general premises to a necessary specific conclusion. A typical example takes this form: *All insects have six legs; an ant is an insect; therefore, an ant must have six legs.* Given true premises, a deductive inference yields a true conclusion by logical necessity. This type of reasoning, formalized by Aristotle over two millennia ago, provides the foundation for mathematics and logic. Deduction is powerful but rigid: it cannot produce new knowledge beyond what is implicit in the premises. In contrast, inductive reasoning moves from specific observations to general conclusions; it is the basis of empirical science and learning from experience. Every time a child sees that dropped objects fall to the ground, she is implicitly using induction to infer a rule: “what goes up must come down.” Induction can generate new hypotheses but never guarantees truth; the conclusion is only probable, not certain. As the philosopher David Hume pointed out, inductive reasoning assumes that nature is uniform, that the future will resemble the past, yet we have no absolute logical justification for this. This is the famous “problem of induction”: no matter how many white swans one observes, a single black swan can overturn the conjecture that “all swans are white.” Another form, abductive reasoning, leaps to the most likely explanation for a set of observations. Suppose you wake up to find the streets wet and overcast skies; you infer it rained overnight. This inference to the best explanation, although not certain, is abduction. The term and concept were introduced by the logician Charles Peirce in the 19th century to denote inference of hypotheses. Abduction is ubiquitous in diagnostics and detective work; doctors, mechanics, scientists all form hypotheses to explain clues, well aware that today’s best explanation might later be proven wrong. The final form of reason that we will consider is analogy; mapping knowledge from one situation to another. Analogical reasoning is fundamental to innovation. When Rutherford likened the atom to a “solar system” (electrons orbiting a nucleus as planets orbit the sun), he was using an analogy to transfer intuition from astronomy to atomic physics. Since ancient times, thinkers have noted the persuasive and heuristic power of analogies. Of course, analogical arguments are only as good as the structural similarities between domains, but a well-chosen analogy can illuminate problems that formal logic alone struggles with. In sum, “reasoning” is not a single faculty but a spectrum of approaches from the algorithmic to the imaginative that together enable us to navigate complex problems. Our ability to master language is a prerequisite to many forms of reasoning but is not sufficient on its own.

Machines, too, can engage in reasoning of these various types, though historically they have excelled only in certain areas. Early artificial intelligence research in the 20th century took aim at deduction, reasoning that is perfectly definite given a set of premises. This initial form of AI is also referred to as Good Old-Fashioned AI (GOFAI). Digital computers, by their nature, are superb at carrying out fixed procedures: they can execute long chains of arithmetic or logic operations without error or fatigue. This strength inspired the creation of expert systems in the 1970s and 1980s; AI programs that attempted to emulate the decision-making of human specialists by encoding knowledge as explicit rules. An expert system consists of a knowledge base of facts and IF-THEN rules, together with an inference engine that applies the rules to known facts in order to reach conclusions. For example, MYCIN, developed at Stanford in the mid-1970s, contained hundreds of rules to diagnose bacterial infections (“IF the patient’s temperature is high AND white blood cell count is elevated, THEN an infection is likely”) and could recommend treatments. These systems were deterministic in the sense that given the same inputs, they always produced the same conclusions by logically chaining through their rules. Some expert systems proved quite useful in constrained domains. Digital Equipment Corporation’s famous XCON system, for instance, configured computer orders and reportedly saved the company millions of dollars by automating what had been a manual task. By the late 1980s, however, the limitations of expert systems had become clear. Maintaining the huge rule sets turned out to be notoriously difficult and this came to be known as the knowledge acquisition bottleneck. Rules had to be hand-crafted by knowledge engineers interviewing experts, and the systems had no capacity to learn new rules on their own. They were also brittle: when confronted with a situation that fell outside their coded knowledge, they often failed in unpredictable ways. Projects like Cyc in the 1980s–90s attempted to hand-code millions of common-sense facts to overcome this but still struggled with the combinatorial explosion of needed rules and exceptions. Indeed, by the early 1990s many early expert systems were deemed too expensive to update and prone to serious mistakes when given unusual inputs. Lacking common-sense knowledge, they could not gracefully handle exceptions or novel cases. The enthusiasm for expert systems thus waned, and along with other disappointments like stalled progress in robotics and vision, contributed to an “AI winter” of reduced funding for AI. Classical logic alone was not enough to achieve human-level reasoning in open-ended real-world scenarios.

Large language models (LLMs) have changed the game by taking a radically inductive path. Rather than starting with formal rules, they consume enormous corpora of text including web pages, books, and scientific articles and infer the underlying patterns. An LLM scours thousands of arguments, stories, and proofs and statistically distills the

linguistic regularities that thread through them. With enough scale, the model begins to internalize abstractions that go far beyond word frequencies. It picks up the meter of poetry and the cadence of rhetoric, the structure of a sonnet and the turns of logical argument. Indeed, recent experiments show that a state-of-the-art LLM can interpret complex metaphors in novel poems; a task once thought to require a deeply human grasp of nuance.⁵ Such findings suggest these models have absorbed not just facts, but some of the conceptual fabric of language. By mastering the medium of language, they have inhaled centuries of reasoning patterns embedded in text, breathing new life into computerized reasoning itself. The tight coupling between language and thought means that by learning to predict words, LLMs have incidentally learned to follow lines of reasoning.

A quick note on terminology. In this chapter, we will start to refer to AI systems that include an LLM but also include additional components. As a simple example, we can imagine a chatbot that uses an LLM to generate most answers but also recognizes when the prompt is a math calculation and uses a calculator program to generate answers for those questions. We could also imagine a system that uses multiple LLMs or multiple instances of an LLM with a central coordinator. The distinction is not entirely clear cut because some advanced functionality can be built into the LLM itself, stretching the definition of what an LLM is. Generally speaking, as we refer to increasingly complex systems that include multiple components, we will start to use the term AI agents or AI systems.

Mechanical Procedures

Before taking a closer look at how LLMs are breathing new life into the field of computerized reasoning, it's important to first discuss the execution of algorithms that consist of fixed procedures or rules. This includes arithmetic and algorithmic procedures; the kind of logical, deterministic thinking epitomized by following a recipe or calculating the results of a formula. Such procedural thinking dates back to antiquity; for instance, ancient Babylonian and Greek mathematicians devised algorithms for calculations that could be followed reliably by anyone trained in the method. Very little or no judgement is required to decide what rules to apply and if you apply the rules correctly, the conclusion inexorably follows. Arguably, this is technically a form of reasoning but it is not often referred to as reasoning in the sense that most people use the term today. A pocket calculator can flawlessly multiply 10-digit numbers, an ability far beyond any human, yet we do not say the calculator is reasoning. Computers have traditionally excelled at executing deterministic algorithms such as solving equations, sorting lists and searching

through data. Human reasoning is contextual and flexible; a human problem-solver selects relevant knowledge, decides which strategy or operation applies, and can even invent a new method if needed. Strict rule-following systems lack this adaptability; they do exactly what they are programmed to do and nothing more. Arguably, there is a grey zone and the definition may shift over time. For example, expert systems based on a tree of encoded rules are technically rule-following systems but they have been characterized as reasoning systems that perform a tightly constrained version of deduction. The goal posts and commonly accepted definitions will continue to move. A common quip in the field is that AI is defined as anything that computers can't do today and as soon as something can be done reliably, it is no longer considered AI.

Intriguingly, today's large language models (LLMs) highlight the difference between strict rule following and more general reasoning in a new way. These models ingest vast amounts of text and seemingly learn many facts and patterns. Despite this, early models notoriously struggled with even grade-school arithmetic and other straightforward algorithmic operations that require stepwise precision. For example, GPT-3, a 175-billion-parameter language model, could achieve almost 80% accuracy on two-digit addition problems but that dropped to less than 5% for four-digit addition.⁶ An analysis of the results suggested that it partly learned the procedure but still made mistakes like forgetting to carry a 1 during addition. The behavior of LLMs in this area is surprisingly human. If you are asked on the spot what the sum of 2218 and 4192 is, you have three options. First, you can make a guess and if you are told to come up with the answer very quickly, this is more or less what you would do. You'd probably choose a number that starts with a six and has four digits. You can easily see that the last digit will be a zero. If you had no remaining time, you'd make a guess about the remaining two digits. This is similar to what an unsophisticated LLM would do if asked to simply return the sum. It's unlikely that it was exposed to that specific math question enough times in the training process for that answer to be represented in the weights but it's seen enough similar questions to make a guess that is directionally correct. For a human, your second option is to work through it step by step. Even a large addition problem can be broken down into a series of very simple operations such as adding single digits and carrying a one when required. Relatively quickly, users discovered that if they asked the LLM to walk through the step of solving a math problem, it would more consistently break it down into those small steps and get the right answer since there were enough examples in the training text of how to perform each individual step and how to walk through the steps. That process has now been baked into modern LLMs through training examples, fine-tuning, and reinforcement learning with human feedback. Although this is a significant improvement, it is not perfect. Like humans, when LLMs tackle very large math

problems, errors occasionally sneak in. In fact, one of the strengths of LLMs in most other contexts is that if you ask the same question ten times, you will get ten slightly different variations in the wording of the answer. That's great for text but obviously highly problematic when performing a very long chain of mathematical calculations where any variation at any step will propagate through the remaining steps and cause the final answer to be incorrect. The third option for a human is to pick up a calculator. A calculator is a tool that has been designed to deterministically execute a narrow set of instructions with absolute precision. Increasingly, LLMs are just one component within an AI agent and the AI agents are leveraging tools like calculator programs where it is more appropriate. When it comes to the execution of mathematical calculations, we don't necessarily want to mirror the process that humans use when doing mental math. Instead, we'd prefer that our AI agent also reaches for a calculator program in any situation where that will deliver a more reliable answer.

Mimicking Human Reasoning Through Deduction

Deductive reasoning is the simplest form of reasoning to understand. Classical deduction applies general rules to specific cases: if all whales are mammals and all mammals have kidneys, one deduces that all whales have kidneys. Expert systems needed those rules pre-programmed by human engineers, but LLMs were able to learn the forms of such syllogisms from countless instances in text. They have ingested myriad logical chains, from textbook proofs to detective novels, and can now apply a new premise on the fly without any bespoke coding of knowledge. During training, the LLM is exposed to countless scenarios and figures out how to adjust the weights in the network so that it correctly predicts the next token. The end result is a network that has internalized reasoning concepts. It did this because that was required to accurately predict the next token when presented with a prompt that requires reasoning to answer. This flexibility tackles the old expert-system bottleneck: instead of meticulously encoding each premise, we train the model with every premise that has been documented in a vast corpus of text and also have the ability to prompt the model with any new facts or premises that we have. One could say the LLM learned Aristotle's syllogisms by reading the Internet and that those logic patterns are now captured in the embeddings of the LLM. As an example, we can provide the following prompt to an LLM that specializes in reasoning, OpenAI's o3: *"If all whales are mammals and all mammals have five kidneys, please deduce how many kidneys a whale has."* The prompt intentionally includes a premise that is not realistic but the LLM is able to perform the reasoning exercise given the provided assumptions and correctly respond: *"Whales belong to the set 'mammals,' and*

possession of five kidneys is predicated of every member of that set. By simple syllogistic inference (universal instantiation followed by universal modus ponens), a whale inherits the property. Therefore, a whale has five kidneys—within the logic of the premises supplied, irrespective of real-world biology.” By feeding it with a premise that we know does not exist in the training data, we can demonstrate that it was using, or at least effectively mimicking, reasoning principles to generate the response.

The strength of LLMs in performing deductive reasoning based on natural language descriptions of a scenario has been demonstrated in domains like medical diagnosis and legal reasoning. The models can draw conclusions from user-supplied facts (symptoms, evidence) using patterns it learned from books and case law. As a result, LLMs can often execute deduction without the brittle, pre-programmed logic that hamstrung their predecessors.

However, this new deductive prowess comes with important caveats. Unlike a formal logic engine, an LLM is a probabilistic storyteller. It aims to produce the most likely continuation of text, not an iron-clad proof. Thus, while it has seen many logical arguments, it doesn’t truly guarantee validity. The deductive chains it produces can veer off course in subtle ways. For example, these models sometimes jump to an answer by intuition, effectively skipping steps that a human logician would insist on spelling out. If a puzzle requires ten careful inferential steps, the model might correctly perform a few and then suddenly “fast-forward” to what it guesses the answer is, especially if that guess seems statistically likely. This eagerness can lead to conclusions that feel right linguistically but aren’t strictly derived from the premises, a kind of logical hallucination. Even when prompted to work step-by-step, LLMs may lose the thread in a long chain of reasoning, similar to how LLMs are prone to making a mistake when a mathematical operation requires precisely performing a large number of steps in a row. Studies on early LLMs have found that they may struggle on tasks that require multiple steps of reasoning to solve; failing where a human reasoner would persist methodically.⁷ The model might introduce an inconsistency halfway through its argument or forget a crucial assumption it stated sentences prior. Unlike a theorem prover, it has no global notion of proof validity; it only has its next-word heuristic. Researchers have identified phenomena like the “reversal curse,” where an LLM that learns a fact in one direction (“A implies B”) struggles to apply it in reverse (“B implies A”), betraying a crack in its logical generalization. And because an LLM is trained to please the statistical expectations of text, it can be overconfident and sound absolutely certain even when its deduction is flawed. In short, these models mimic deductive reasoning impressively well but they do not possess the infallible rigor of true formal logic. They are powerful heuristic reasoners, not perfect ones. This mismatch means that while LLMs signal a

breakthrough in overcoming the knowledge bottleneck of old, they also force us to confront new uncertainties.

Induction and Abduction

LLMs also engage in inductive and abductive reasoning which are the bread and butter of everyday problem-solving. Induction is reasoning from examples: we observe specifics and infer a general rule. Abduction is similar but involves reasoning from a single outcome to the most likely explanation. Significant amounts of text are a record of human inductions and abductions. Think of all the diagnostic cases, scientific experiments, and detective stories. This provides LLMs with a wealth of data to train on. Present an LLM with a familiar pattern of evidence, and it can induce a plausible rule or guess an explanation with startling facility. For example, an LLM might infer that a patient's mysterious rash and fever are likely caused by an infection, because in its training corpus those symptoms often pointed to that explanation. Likewise, if told that a character in a story came home to find the lights on and their dog cowering, the model might abduce that a burglary occurred. In domains that resemble what it has seen, an LLM's inductive and abductive reasoning can feel uncannily human. It pieces together the clues given in the prompt and selects an explanation that fits the mold of past data. In many standard scenarios, this works remarkably well and because a model is trained on every domain captured in text, it may seem omniscient.

Recent research has shown that LLMs can perform impressively at in-context induction as well. When given a few examples of an input-output mapping such as a list of number pairs following a hidden mathematical rule that was not included in the training set, a state-of-the-art LLM can infer the underlying function and apply it to new inputs. In one study, models achieved nearly perfect accuracy in guessing the rule from examples alone, demonstrating "remarkable inductive reasoning capabilities" in pure form.⁸ This on-the-fly learning, recognizing a pattern from only the few demonstrations in the prompt, is essentially the model doing induction internally, and it is one of the reasons such AI systems feel surprisingly flexible. The broader strength of LLMs in induction comes from their exposure to vast diversity: they have seen so many patterns of reasoning, narratives, and facts that they can often stitch together a generalization even for novel inputs. However, there are notable weaknesses. The induced patterns are only as good as the data and correlations observed. An LLM might pick up spurious correlations (superstitions, in effect) present in its training set and generalize those inappropriately. For instance, if all the example stories an LLM saw during training about doctors mention a hospital, the model might erroneously assume a doctor must

be in a hospital in a new story; a flawed induction. Moreover, LLMs lack an explicit understanding of causation. They excel at summarizing “what usually happens” but not necessarily why. Without the true causal structure, their inductive leaps can fail when circumstances change.

Another consideration is that LLMs inherit the inductive biases present in their training text. This includes not just useful knowledge but also human biases and fallacies. Studies have shown that larger models like OpenAI’s GPT-3.5 and GPT-4 actually mimic certain human reasoning fallacies more strongly than smaller models. In one analysis, as models scaled up, they more frequently produced human-like wrong answers on trick questions, suggesting they had internalized the same shortcuts and mistakes that people often make.⁹ This is induction at work: the model sees many examples of how humans reason, including errors, and generalizes that pattern. So, while LLMs demonstrate impressive inductive reasoning in terms of breadth of knowledge, they do not always apply it with the discrimination of a human expert.

Abductive reasoning is the form of reasoning we use when we observe an outcome or a set of clues and try to figure out the most plausible cause or explanation for it. For example, a detective solving a crime considers various hypotheses to explain the evidence and picks the one that best fits. Everyday life is full of abductive reasoning. Anytime you think “Hmm, what’s the most likely reason this happened?” you are reasoning abductively. Humans are remarkably adept at abduction, using vast background knowledge and common sense to home in on reasonable explanations. We often do this subconsciously: our brain continuously forms hypotheses to make sense of what we perceive. Large language models can also perform a kind of abductive reasoning, although again in a pattern-based way.

If prompted with a scenario and a question like “what is the most likely explanation for this?”, an LLM will draw on the many narratives and cause-effect relations it has seen in text. Often, the result is a plausible explanation that aligns well with human common sense. For example, given a short story: “Jenny cleaned her house and left the window open. When she returned, the house was a mess,” an LLM might infer *“Perhaps a thief entered through the open window and ransacked the house,”* mirroring the abductive reasoning a person would perform. In fact, the example above is drawn from an abductive reasoning benchmark (the ART dataset), and that hypothesis (a thief broke in) is indeed the intended correct explanation. LLMs armed with common sense knowledge can propose reasonable explanations for everyday situations.

On the other hand, for many real-world situations an LLM is generally operating with a fraction of the context that a human may have. A human looking at the same problem in a real-world situation would be pulling together very nuanced information

about exactly how the house was ransacked based on their visual analysis of the scene. A tiny clue may swing the verdict strongly from a thief to a rambunctious pet and another small clue may lead the human observer to believe that the suspect was actually a raccoon. In addition to visual analysis of the scene, a human observer would draw from a vast collection of detailed knowledge such as a lifetime of knowledge about the safety of the neighborhood and broader crime trends in the city. Probabilities are critical in abduction. An LLM understands probabilities based on written text. A human often has a more accurate view of probabilities as they relate to a specific local situation.

Truly strong abductive reasoning requires more than just regurgitating a likely story; it involves evaluating multiple hypotheses and weighing which best fits the evidence. In complex situations, this requires actually working through every element of a number of candidate scenarios to determine the overall probability. LLMs do not inherently work through multiple different possibilities unless instructed. As with similar challenges, an LLM can be coached to take this type of approach by showing its work and reasoning in a step by step fashion. A study in 2023 constructed a benchmark of short detective stories (each a few paragraphs long with several suspects) and asked models to identify the culprit; essentially a mini-mystery requiring abductive logic over multiple clues. Random guessing on this dataset would result in a score of 24%. GPT-3 scored 28%, which is barely better than a random guess. GPT-4, the most advanced model in the test, managed to solve about 38% of these puzzles while human solvers averaged 47%. This result shows the significant progress that had been made in closing the gap to average human performance. However, top human solvers get around 80% of the mysteries correct and so there still is room for significant improvement.¹⁰ When it made mistakes, the model often zeroed in on a suspect that was a frequent narrative trope but missed subtle clues linking another suspect to the crime. This type of stereotyping, extended to storylines, is something that humans are susceptible to as well. Unfortunately, since this research was published publicly, the benchmark can no longer reliably be used. If a state-of-the-art model today scored 100%, it would be hard to determine if this is because of the improvement in reasoning capabilities or because the latest model read the research paper and the associated data during the training process and therefore already had the answer key.

Drawing Analogies

Analogy, the ability to see parallels across different contexts, has long been considered a hallmark of creative intelligence. It's the kind of reasoning that lets a scientist say, "the spread of a virus is like the spread of an idea," and gain insight by linking biology to

sociology. Here, LLMs benefit enormously from their breadth of training. Having ingested text from virtually every domain, a large model carries traces of chemistry, poetry, history, technology, and more, all fused in a single representational space. This makes it a natural polymath. Ask it to “compare quantum entanglement to a conversation between lovers”, and it will, drawing on its knowledge of physics and romance novels to produce a startlingly apt analogy. OpenAI’s o3 starts a comprehensive response to this prompt as follows: *“Imagine two lovers sitting across from one another in a quiet café. Each shares glances and half-spoken phrases, yet both somehow grasp the other’s unvoiced intentions. The exchange feels instantaneous: one partner begins a sentence and the other finishes it, or a raised eyebrow silently redirects the topic before any words land. Quantum entanglement evokes a similarly uncanny synchrony, though in the realm of particles rather than people.”* In effect, the LLM can yoke disparate concepts together with a flexibility reminiscent of human creativity. Empirically, we see this in benchmark tests: one group of researchers argued that “large language models such as GPT-3 have acquired an emergent ability to find zero-shot solutions to a broad range of analogy problems.”¹¹ OpenAI’s GPT-3 was able to solve certain kinds of analogical puzzles without any special training, in some cases matching or even exceeding the success rate of human test-takers. For instance, given a puzzle about mapping sequences of letters or finding an analogous ending to a story, the model often produces the correct answer purely from recognizing a pattern analogous to something it has seen before. The model has seen enough parallel structures in text (“X is to Y as A is to B”) that it can fill in the blanks by instinct. Like a savvy cross-disciplinary scholar, an LLM can connect dots between far-flung ideas because, somewhere in its training data, those dots, or ones like them, have been connected before.

However, as with deduction, there is a catch. The ease with which LLMs generate analogies can mask a shallow form of understanding. Critics have probed those impressive analogy benchmarks and found that the triumphs are brittle. In one analysis, slight variations of the test questions, tweaks so minor a human mind would still see the same underlying analogy, completely tripped up the model. OpenAI’s GPT-3, which had excelled under certain conditions was found to fail “to solve the simplest variations of the original tasks, whereas human performance remains consistently high.”¹² In other words, the AI could match the pattern it expected, but if the problem was phrased in an unusual way or presented a relationship it hadn’t implicitly memorized, its analogical prowess fell apart. It’s worth pointing out that those experiments were done with OpenAI’s GPT-3. In 2025, I recreated one of the tests with Open AI’s GPT 3.5, GPT 4o and o3 models. The GPT 3.5 and GPT 4o models failed to answer correctly but o3 succeeded; demonstrating the continued progress that has been made on LLM

based systems that are designed to excel on reasoning tasks. This doesn't mean that the challenge has been entirely overcome and that the problem or the original research is no longer relevant. It is often easier to understand the nature of these problems using simple problems on smaller models. The shortcomings are much harder to tease out in larger models, but we can look at the failure modes in the smaller models to understand the nature of potential weaknesses relative to human reasoning. It's similar to how engineers may first test a scale model of a bridge in a lab. If the test is properly designed, it can provide valuable clues about potential failure modes in the full-size bridge.

True analogy-making requires grasping the core relational structure; understanding why a virus spreading is like an idea spreading. LLMs approximate this by statistical association. When those associations are plentiful and clear in the training data, the analogy appears effortlessly; when they are sparse or oblique, the model has little to fall back on. It may stretch and produce a clever-seeming comparison that actually misses the point, or it may simply flounder. We might say the LLM is performing analogies by rote. It is as if a student heard many metaphors and learned to swap in new nouns, making pleasing comparisons without quite grasping them. The result can be uncannily right at times and completely off-base at others. Still, even a shallow analogical mind is a significant development. In practice, these models can be useful brainstorming partners, suggesting cross-domain ideas a human expert might not have considered. I asked o3 to come up with some analogies to the concept of keeping your money invested in the stock market, even when there is a period of high volatility and an accompanied drop in prices. One of the analogies that it provided was *"Coastal redwoods can lose most of their canopy to fire, yet their deep root systems and thick bark let them survive the burn and sprout anew. Pulling your capital after a market blaze is like chopping down a redwood just because the branches look scorched; the living core remains poised for vigorous regrowth once conditions improve."* I think that is a great analogy and I understand it well. It may help to explain the concept of long-term investing to a group of naturalists. I also think it's unlikely that I would have come up with that on my own while staring at a blank page. When brainstorming, we don't need the LLM to understand the analogies that it is producing. We can evaluate them for merit as a follow-up exercise. Given the broad base of training data, the LLM may come up with many analogies that we would not have thought of on our own.

Reasoning capabilities initially lagged other areas of performance for LLMs. However, for the first time in a long time, there are signs that this new computational approach may finally hold the key to unlocking a wide range of reasoning capabilities. We'll now look at one of the fundamental reasons why LLMs struggle with reasoning and approaches that can be used to improve reasoning performance going forward.

Brain in a Box

A stark limitation of today's LLM-driven reasoning is embodiment; or rather, the lack thereof. Human reasoning is profoundly multi-modal. We learn basic truths not only from what we read or hear, but from what we see, feel, and live through. A child knows that fire is hot and painful without ever reading a treatise on thermodynamics. We navigate the world with a continuous flood of sensory inputs and an intuitive sense of physics, space, and time. Every moment of every day, we take actions and witness the impact of those actions. At this point in time, none of this rich tapestry of experience is available to a pure text-based model. Hubert Dreyfus, a philosopher who critiqued AI's disembodied approach decades ago, emphasized that common sense understanding requires being embedded in the real world. LLMs, however, operate in isolation. The model has never felt gravity tug at its body, shot a basketball that missed the backboard entirely, won a debate, heard the subtle anxious tone in a person's voice, or experienced hunger. It only knows of these experiences through descriptions in text; often incomplete or biased descriptions at that. This leads to a gap that even gargantuan text training cannot fill. Consider the simple scenario of a car crash: a human walking up to such a scene immediately takes in a slew of sensory details including overturned vehicles, tire marks on the road, an expression of remorse on one driver's face, perhaps a whiff of alcohol. These perceptions and the knowledge that is gained from taking actions in the real world trigger a cascade of commonsense inferences. An LLM, by contrast, gets none of those direct impressions. If we describe the accident scene in words, it will try to reason from that description, but the quality of reasoning now wholly depends on the quality of the description. Nuances a human would notice without a second thought might be omitted from the text. And if so, the model cannot ask a follow-up question to examine the scene more closely; it is effectively blind outside the given words. Even in the best-case scenario where we do provide a detailed account, the LLM's understanding of physical space and causality is second-hand. It might know that "broken window" often correlates with a "car crash," but it lacks an experiential model of how force, motion, and sound interplay in such an event. A human observer also has a full world model, a detailed map in their head, and they understand how fast someone would have needed to be travelling on that section of road for that type of a collision to occur. This is why purely text-trained models may make mistakes with spatial and physical reasoning that any child would find silly. Efforts are underway to give models multimodal inputs (images, audio, etc.), physics engines, world models, and the ability to interact in the real world. Until then, language alone cannot convey the full truth of reality. There is a tacit

dimension to knowledge that comes only from being an embodied part of the world, and LLMs stand entirely outside it.

This relates to a profound point. Much of what an individual human knows is tacit; never written down and never read. We grow up absorbing unwritten rules and physical intuitions and it would sound absurd if we needed to explicitly state them all. You can go your entire life without anyone ever telling you how firmly to hug a friend, how much pressure to apply to break open an egg without breaking the yolk, how close to sit to a campfire, or the best way to use your index finger and thumb to flick a biting fly off your arm. However, if you surveyed a broad population, almost everyone would share an understanding of these facts. The other side of this observation is that it is incredible how much of this tacit knowledge has been documented across the full corpus of human generated text and therefore how much LLMs have read during their training process. You may have learned something without ever having read about it but that doesn't mean that it hasn't been written about it by someone. It is increasingly difficult to find examples of tacit knowledge that LLMs are unaware of and this is an interesting area of exploration. Can you think of something that everyone knows to be true but that has never been written down and would be excluded from every training set?

Improving Reasoning

Given all of the above, how can we improve the reasoning abilities of LLMs? This is an active area of research, and multiple promising strategies have emerged. A first approach is surprisingly straightforward: better prompting. By coaxing the model to explicitly articulate a reasoning chain, we can often get more accurate results. For example, rather than asking an LLM directly for the answer, we might add: "Think step by step" or we might provide a few examples of stepwise reasoning before the question. This technique, known as chain-of-thought prompting, has been shown to significantly enhance performance on complex tasks. In one study, just by giving a large model a couple of worked examples with intermediate steps, researchers caused a dramatic jump in math word problem accuracy. The model, unguided, solved only 17.9% of problems correctly, but with chain-of-thought prompting, the same model's success rate soared to 56.9%.¹³ The reason is that prompting the model to produce its reasoning sequence helps it avoid leaps and errors; it effectively uses its own output as scratch paper, breaking a problem into subparts that it can tackle in turn. Another set of techniques involves using the model's own generative abilities in a self-reflective loop. One such method is to run multiple reasoning paths in parallel and then compare them, selecting the most consistent answer. This is called self-consistency decoding. Instead of trusting a single chain of thought

(which might go astray), we sample, say, 5 or 10 independent chains of thought from the model for the same question. Often the answers will agree on a majority outcome. Taking that majority answer greatly improves reliability. In experiments, this kind of self-consistency voting boosted solution accuracy by several percentage points across a suite of reasoning benchmarks.¹⁴ In essence, the model’s multiple “opinions” can be averaged to cancel out some randomness or error, much as a panel of solvers might outperform any one individual or an individual may try looking at a problem from multiple angles. Similarly, researchers use rejection sampling and refinement: have the model propose an answer and then check the answer. If it fails, prompt the model to try again or adjust its answer. This iterative feedback approach can correct mistakes that the model would not catch on its own in a single pass. Beyond prompting and decoding tricks, there is work on modifying the internals of model architecture to bolster reasoning. One idea is to train models with a sort of explicit reasoning module. Early research in this vein has introduced modules for planning or tools for the model to consult (like a calculator or a database) as it reasons, injecting a bit of algorithmic rigidity into the otherwise fuzzy neural network. Another frontier is multimodal integration, giving the model eyes, ears, or the ability to act, thereby grounding its reasoning in the real world. The latest systems are beginning to accept image inputs in addition to text, so they can, for instance, examine a diagram or a photograph when answering a question. Roboticists are pairing language models with robotic controllers, so that a model can physically interact with objects and get feedback from an environment. By linking words to sensory data and action, these efforts aim to overcome the fundamental limitation of language-only reasoning and for models to develop their own tacit knowledge. Indeed, a notable research direction is building world models and simulation tools into AI; structured representations of physics, geography, social rules, etc., that an LLM can query when pure text prediction is insufficient.

A common theme in this space is that models are improving quickly. Recently, OpenAI has launched a new series of models that specialize in reasoning. Two of the models in this series are the o1 and o3 models. o1 costs 60 times as much as GPT-4o to run in early 2025 which is a reflection of the significant amount of additional compute that is used during the inference process; described as test-time compute. Put simply, leveraging some of the methods described above, the model is given more time to think about the problem at hand. On the Competition Code (Codeforces) benchmark, GPT-4o, which is already an improvement over the GPT-3, GPT-3.5 and GPT 4 models used in much of the early research, scores just 9.3. The o1 model significantly improves on that with a score of 74.3 and o3 scores 91.6. Many of the perceptions of the weaknesses of LLMs

in the domain of reasoning are quickly becoming invalid as these models now approach human performance in a wider range of benchmarks.

The coming decade will likely see an exciting convergence of approaches, combining the raw power of large language models with the structured reliability of symbolic logic, enriched by real-world grounding from multimodal data. This hybrid path is how AI may gradually close the gap in reasoning ability. Just as the Apollo 13 engineers used duct tape and imagination to connect disparate parts and create something new, we are now stitching together the strengths of different AI paradigms. The result could be machines that don't just manipulate symbols or predict text, but reason about the world in a way that is deeply informed, contextually aware, and increasingly trustworthy.

Software Development

Software development is basically an exercise in reasoning and so it is no surprise that it is one of the first domains where reasoning capable LLMs are making a meaningful impact. Given a set of requirements, software developers use reasoning skills to write code that will satisfy those requirements. Large language models trained on code have rapidly evolved from simple autocomplete tools into agents capable of complex programming reasoning. Early code models like OpenAI's Codex were mainly used to predict the next few lines of code, but today's state-of-the-art LLMs can break down algorithmic problems, plan solution steps, and generate entire programs that meet specifications. On OpenAI's HumanEval (a set of Python programming tasks), top models achieve close to 90%, a dramatic improvement from the 66.5% of GPT-3.5 just two years earlier.¹⁵ Similarly, on the MBPP benchmark of basic Python problems, an agentic workflow that leverages Anthropic's Claude 3.5 LLM solves over 94% of tasks, nearly matching expert-level performance.¹⁶ These figures reflect the exceptional reasoning ability of LLMs in this specific domain. Rather than merely regurgitating code patterns, they can analyze problem descriptions, hold multiple requirements in mind, and synthesize novel solutions that satisfy unit tests and specifications. There are three reasons why AI agents that leverage LLMs are showing so much success in tackling the domain of software. First, there is an incredible abundance of examples of software code from the open-source movement for models to ingest. Second, software development is often a closed-loop problem, meaning that it is possible for the agent to check its own work and iterate until the code works as desired. If an agent provides gardening advice on how to grow the biggest carrots, it has no way of knowing if that advice was correct or led to positive results. On the other hand, when building a piece of software, it is often possible to also develop a set of tests to verify that the code is functioning as desired. And if it

isn't working, then another attempt can be made and it can keep iterating until it gets it right. It can then present the final solution to the user who may not be aware of all the mistakes that were made in the interim. Finally, software development is an example of a reasoning domain where almost the entire context exists in text. A doctor may pick up on subtle sounds in a cough and a lawyer may sense agitation in a jury and this extra context gives these professionals a leg-up on an LLM. On the other hand, an LLM is on even footing with a human programmer in terms of the context they are working with.

The implications of these coding capabilities for the software industry and the professionals in it are significant and we will explore those in the next chapter. First though, we will consider what happens when LLMs wield this power in the process of answering a query from a user.

We've discussed how LLMs are poorly suited to performing lengthy sequences of mechanically precise operations such as adding two ten digit numbers by working through each step as a human might do on a pad of paper if they didn't have access to a calculator. In those cases, we've suggested that an AI agent could send the query to a calculator instead of an LLM. We can now envision something even more powerful. Imagine a user asks a question about how to calculate a flight path to the moon and they want to create an animation showing that flight path. The LLM is only equipped to output text and so it can't provide a direct response to satisfy this query. On the other hand, the LLM could break down the problem into steps. In the first step, the LLM would play the role of a product manager. It would write a set of requirements for a software program that would be able to show an animated flight path. It would draw on its knowledge to specify the correct physics principles and formulas to use. In the next step, the LLM would play the role of a software developer. It would leverage its capability to write software code to create a program that would satisfy those requirements. At the end of the process, the LLM would return a working computer program that the user could run to generate an animation of the flight path.

This capability is incredibly powerful and significantly expands the scope of what LLMs can be used for. Imagine a company or government that has a database with 1 billion rows of data. There is no reasonable way to put all of that data into a prompt to ask the LLM to analyze it. Even if there was, the LLM would be poorly suited to that task. On the other hand, the LLM can now write a bespoke software application to perform any type of analysis that the user is interested in. It can even suggest types of analysis that may be interesting or do its own exploring and report back to the user about any findings that stand out. What's interesting and important to understand is that the LLM may not even expose the software program to the user. The ability to write custom software programs in the service of responding to a user query greatly

expands the scope of user queries for which an AI agent can provide a useful answer and significantly improves accuracy on a wide range of problems for which software excels.

Every GUI becomes an API

Although it's very interesting that AI agents can now write their own software programs, what about all of the existing software in the world? Is there a need for them anymore? There are two strong arguments for AI agents to interact with existing software programs. The first is that many of these programs are extremely sophisticated; having been developed, tested, and refined over many years by large teams of human experts. For example, we can think of physics simulators, CAD programs, and banking systems. It would be extraordinarily expensive to use an LLM to try to re-create these programs from scratch every time we needed to perform a task. It is also unlikely that such a program would be as robust and reliable as existing solutions. Although LLMs have become capable at tackling small programming problems, they still stumble when trying to develop or maintain large and complex code bases. The second reason is that many of these existing software systems are a channel to the real world. Access to these systems is a method to ingest new data. A lawyer might benefit less from an AI's eloquent description of case law than from the AI directly pulling records from a court database that is only accessible through a web browser. Access to these systems can also allow the AI agent to take actions such as initiating a transfer of funds or even controlling a physical component such as a valve in a water treatment system. The emerging solution is to let AI systems control those very tools, operating the software's interface on our behalf.

Crucially, this doesn't require inventing new specialist software; it leverages the vast ecosystem of tools already in use. The bottleneck has always been the interface between human and machine; the fact that a person had to sit and click and interpret the screen. That bottleneck is now widening. We see precursors in the business world through Robotic Process Automation (RPA), where software "bots" perform repetitive UI tasks like form entry. Indeed, as of a few years ago, 74% of companies worldwide had implemented some form of RPA to streamline routine work.¹⁷ But traditional RPA is limited to relatively simple, rule-based sequences. It breaks easily when anything unexpected appears on screen, and it can't truly understand the content it's handling. By contrast, the new AI agents bring judgement and adaptability. They combine the perception of a human user that can read text and recognize on-screen patterns, with the superhuman precision of software. If a dialog box pops up with an unfamiliar error message, a

well-trained AI agent can interpret it and find a workaround, much as an experienced user would. This dramatically extends the frontier of what can be automated.

Recent advances have unlocked the ability for AI agents to interact with graphical user interfaces (GUIs) in the same way humans do; by clicking buttons, typing into fields, and navigating menus. In a future chapter, we will discuss additional modalities such as images and video in more detail. The ability to analyze what is presented on a screen is an example of an additional modality beyond text. In a 2024 survey by Microsoft researchers, scholars noted that AI agents powered by LLMs can now interact with computer interfaces just like humans do, executing tasks via GUIs rather than through traditional APIs.¹⁸ These “GUI agents” interpret natural-language instructions and carry them out on-screen, effectively turning any software’s interface into an API endpoint for the AI. OpenAI demonstrated this concept with its Operator agent in early 2025, which uses a vision-enabled GPT-4 model to control a web browser autonomously.¹⁹ The AI can read a webpage, click links, fill forms, all the things a human user would do, but without needing custom integrations. Operator treats the browser itself as the action space, so every website becomes accessible. Anthropic’s Claude with its Computer Use mode, and Google’s rumored Project Jarvis also aim to let AI directly operate software on users’ behalf.

In general, this approach vastly broadens the scope of what AI systems can do. Any application with a GUI, from the newest cloud app to a 30-year-old desktop program, can, in theory, be operated by the AI as if through a universal remote control. Researchers describe this as a “paradigm shift, enabling users to perform intricate, multi-step tasks through simple conversational commands.”²⁰ Instead of learning the intricacies of dozens of software programs with thousands of pages of documentation, a person could simply tell an AI agent what outcome they want, and the agent handles the complex series of clicks and steps needed to make it happen. Analysts are already sizing the opportunity. One report forecast a \$69 billion market for such AI-driven automation by 2028, up from just \$8.3 billion in 2022.²¹ The level of reasoning required to operate most software packages is relatively simple as software contains comprehensive documentation and is a well constrained environment. The ability to turn every GUI into an AI-accessible API is poised to change how we use software at a fundamental level.

Legacy Systems and Limitless Integration

One immediate implication of AI that can control interfaces is a radical new approach to integrating and modernizing legacy systems. Organizations today struggle with

decades-old software such as proprietary control systems, mainframe-based financial systems, and aging ERPs, that lack modern APIs or extensibility. In the past, connecting to these systems or automating them meant expensive migrations to modern systems or brittle screen-scraping scripts. Now an AI agent can simply learn to operate the old interface, effectively wrapping a 1990s application in 2020s intelligence. The AI doesn't care if the software is a green-screen terminal or a cobwebbed Windows XP tool; if it can see the screen and simulate keyboard/mouse input, it can use the program. This capability can breathe new life into outmoded software. Instead of undertaking a multi-million dollar upgrade project, a company might deploy an AI assistant to act as an intermediary; reading data from a legacy inventory management GUI and passing it on to a modern cloud application that handles order management. Every piece of software, no matter how old, thus becomes accessible and interoperable via the AI. A logistics firm could link its 1980s-era shipment scheduling tool with a modern route optimization app by simply instructing an AI agent to use both in tandem, something that used to demand custom code or manual work. In short, AI's control of GUIs turns software age into a non-issue: if a human can still use it, the AI can use it too, which means even "legacy" software can be integrated into automated processes.

Unlocking a new Renaissance Era

Across many professions, true expertise requires not just domain knowledge but fluency in complex software tools. Architects and engineers spend years mastering CAD and BIM (Building Information Modeling) platforms; financiers and accountants train extensively on Bloomberg terminals and specialized business analytics software; doctors and radiologists learn to navigate medical imaging suites; lawyers wrestle with e-discovery databases and case management systems. These tools are powerful but impose a steep learning curve. Entire careers and certifications are built around operating them effectively. This creates a silo of expertise: a professional's software literacy in a given platform can be as important as their conceptual knowledge. The complexity of software systems has led to a high degree of specialization in careers.

This has left some professionals yearning for an earlier time. The concept of the renaissance man or woman traces its roots back to the European Renaissance, spanning roughly the 14th to 17th centuries, when cultural flourishing emphasized a well-rounded pursuit of knowledge across arts, sciences, and humanities. Figures like Leonardo da Vinci epitomized this ideal, demonstrating mastery not only in painting but also in anatomy, engineering, and mathematics. Similarly, polymaths such as Michelangelo, who excelled in sculpture, architecture, and poetry, and Isabella d'Este, an influential pa-

troness, scholar, and diplomat, embodied the Renaissance aspiration toward versatility, intellectual curiosity, and comprehensive self-development. This broad pursuit of excellence across disciplines was inspired by classical antiquity, where intellectual breadth was admired and considered essential for personal fulfillment and societal contribution. The enduring legacy of the Renaissance individual continues to symbolize the holistic integration of diverse skills, creativity, and intellect in a specialized modern world.

AI agents that are capable of GUI control may help unlock a new renaissance era. If an AI agent can instantly operate Revit or SolidWorks or SAP on behalf of a user, the focus shifts from learning the interface to defining the task. A new generation of professionals may study multiple disciplines in university and then leverage AI to bring their visions to life. A single person may be able to fill the role of the architect, structural engineer, and HVAC designer on a project.

Portable Skills and Reduced Vendor Lock-In

When AI agents can operate any software given the right training, the unique quirks of different software packages start to matter less to the end user. This could lead to a layer of abstraction where the *task* is specified in a general way, and the AI translates that into whichever application is available. In effect, the AI agent serves as a universal adapter between the human's intent and the software's specific interface. For organizations, this might mean less dependence on particular vendors or platforms. Today, a design firm might hesitate to switch from AutoCAD to a rival CAD tool because its staff would face a steep learning curve all over again. In a future with mature AI assistants, that firm could switch software and simply instruct its AI to learn the new interface while the human designers continue describing their goals in familiar terms. The AI takes care of "learning" the new menu layouts or command syntaxes. The human-machine team's productivity doesn't suffer the same trough of inefficiency that typically accompanies adopting a new tool.

This portability of skill could loosen the grip of vendor lock-in. Software companies have long relied on the fact that once a workforce is trained on their system, there is inertia that keeps those customers from jumping to competitors. If AI agents mitigate the retraining cost, companies might feel freer to choose the best tool for the job at any time, knowing the AI can adapt to it. It also means an individual's expertise becomes more transferable between organizations that use different software stacks. A logistics planner skilled in optimizing delivery routes might work at one company with Software A and move to another company using Software B without missing a beat – she delegates the interface specifics to her AI assistant and focuses on the core logic of route planning.

We might even see standardized workflows in the AI agents that are tool-agnostic: for example, a blueprint for “generate monthly sales report” that an AI agent can execute in Oracle’s system or SAP or a custom in-house tool, whichever the employer has, by adjusting to the interface differences on the fly. In essence, AI could become a great equalizer across software ecosystems, ensuring that capability is not limited by the user’s manual proficiency in a particular product. This trend would have significant economic effects on the software industry, potentially driving competition toward better underlying functionality and results, rather than on trapping users with proprietary interfaces and long training cycles. It could also spur more open design of software, knowing that ultimately an AI agent might be the one operating it – perhaps interfaces will even be optimized not just for human usability but for AI interpretability, blurring the line between a traditional GUI and an API.

Lowering Barriers to Advanced Tools

There are advanced software systems today that, despite their power, remain underutilized because of their complexity. Building Information Modeling in construction and architecture is one example: it offers tremendous benefits in coordinating designs, detecting conflicts, and managing projects, yet many firms, especially smaller ones, have been slow to adopt BIM fully and its use is almost exclusively limited to commercial projects today. A primary reason cited is the steep learning curve and the shortage of professionals deeply skilled in these multifaceted tools. Similar stories abound in other fields. Consider sophisticated statistical analysis platforms used in finance or cutting-edge simulation software in aerospace engineering. They promise leaps in capability, but only if one can navigate their labyrinthine interfaces and options. AI-assisted GUI operation could dramatically lower these usability barriers. An AI agent that has been trained on how to drive a BIM application could enable a less-experienced architect to reap the benefits of BIM on a project without extensive retraining. The architect might simply say, “Arrange the plumbing layout to meet code and avoid collisions with electrical,” and the AI, acting through the BIM software, can execute that instruction – something that otherwise would demand an expert’s manual effort. In medicine, we might find that advanced diagnostic imaging tools or treatment planning systems, which today are handled by specialists, become more broadly accessible. A general practitioner in a remote area could utilize an AI agent to operate complex imaging software to analyze scans, essentially having a virtual radiology technician at their disposal.

By lowering the expertise threshold needed to use powerful software, AI stands to democratize many high-end capabilities. We could see wider deployment of tools like

simulation-driven design, predictive analytics, or GIS mapping in organizations that previously lacked the in-house specialists to use them. This has a reinforcing effect: broader use of advanced tools can lead to better outcomes (better building designs, more efficient supply chains, more informed medical decisions), which in turn may drive those industries forward. However, it also means rethinking how we educate and certify professionals. The traditional model of learning a tool first and then applying it might get inverted to applying the tool via AI assistance and learning deeper specifics on the fly as needed. There is a parallel here to how calculators and computers changed math education. Initially feared as making certain skills obsolete, they ultimately allowed focusing on higher-level problem solving. Likewise, if BIM or similar tools become push-button easy (with AI behind the scenes doing the pushing), the emphasis for human experts may shift to the creative and analytical decisions before and after the software's operations. We must be mindful, though, that easy access can sometimes lead to misuse. Without understanding the tool's fundamentals, users might over-rely on AI outputs. The complexity of many software systems today acts as a gatekeeping function and proficiency in the use of these applications is used, for better or worse, as a rough litmus test for domain expertise. In the future, we could end up with a scenario where a teenager that is highly skilled at prompting AI agents is hiding behind a fake online profile and posing as an architect or engineer. If they can deliver full drawing packages that are capable of fooling county inspectors, they may be quite successful but homebuilders would likely be quite upset to find out that the drawings were not prepared with any experienced or professional oversight.

Collaborative Control: Humans and AI Sharing Workflows

Rather than removing humans from the loop entirely, these developments point toward new hybrid workflows where humans and AI share control of software. In practice, an AI agent might handle the grunt work of a task and then pause for a person to provide judgment or sensitive input. Developers of these systems are already building in collaborative features: OpenAI's Operator agent, for example, will automatically hand back control to the user when it encounters a step requiring delicate judgment, credentials, or permission such as entering a password or confirming a purchase. In a future office, one could imagine a "co-pilot" mode for any application. A financial analyst could watch as her AI assistant navigates through Excel and a proprietary accounting tool to compile quarterly results; at any moment, she can take over with a click to adjust a formula or make a subjective decision about an unusual expense entry, then let the AI continue. This kind of mixed-initiative interaction has been a goal of human-computer interaction

research for decades; the idea that neither human nor AI has to do 100% of a task, but can fluidly pass control back and forth as needed.

Such collaboration could become quite seamless. The AI might have a standing instruction to defer to the human whenever a legal disclaimer appears on screen, or to ask for confirmation before finalizing an action that carries significant cost. Meanwhile, the human user starts to develop an intuitive trust in the AI for rote operations. For example, the human operator might let the AI agent download all the reports from the company portal because it has proven to be reliable at that step in the past but will remain engaged for the high-level review before it gets sent off. Over time, interfaces themselves might evolve to support this partnership, perhaps showing a “ghost cursor” or highlight when the AI is taking an action, so the user can observe and intervene if something looks off-track. Although most software today has a basic undo log and some packages even allow users to record macros, those capabilities will be expanded upon when working with AI agents. It will be possible to go back to any point in time. When the AI agent has done something very well under close supervision, the human operator will be able to ask the agent to capture the steps of that task in a workflow that can be called upon in the future. The AI agent will capture the workflow in a manner that is still flexible enough to adapt if the software user interface changes in the future. Unlike human operators that are prone to getting rusty on a task that they don’t perform on a regular basis, once an AI agent is taught how to do something, it will never forget and it can instantly share that training with an unlimited number of other agents. The outcome of all of this collaboration is a best-of-both-worlds scenario: human judgment and oversight combined with machine speed and tirelessness. In fields like medicine or law, such a balance may be essential; an AI agent can drive the software to gather records or fill out forms, but a human professional will make the final call on diagnosis or legal strategy. Rather than replacing skilled workers, the AI becomes a highly competent assistant, taking over the keyboard and mouse for the drudgery while constantly ceding ultimate authority to the human operator when judgment is required. This collaborative mode could redefine workflows to be far more efficient, though it will also demand new user skills; not the low-level skills of operating software, but the higher-level skills of supervising, guiding, and correcting AI agents.

Of course, AI agents could also engage with software applications completely independent of humans. One could imagine submitting a prompt asking for the design of a building. To perform the task, the AI agent might start by using its internal reasoning capabilities to determine the requirements for the building. It may then access websites to check material costs and availability. As a next step, it may access a specialized engineering application to determine the required size of a key beam. Finally, it may start a

CAD application and use the interface to make the drawings. In mid 2025, we are at the very forefront of progress in this area and it is very exciting. As AI agents gain the ability to reliably access the GUI of every existing software program in the world, their capabilities will expand dramatically.

Conclusion

This chapter has traversed the complex landscape of reasoning, contrasting unique human cognitive strengths with the evolving capabilities of artificial intelligence. While early AI grappled with replicating nuanced thought, and even today's large language models face inherent limitations and operate differently from human cognition, their progress across various forms of reasoning is undeniable. The promise of coding has been vividly demonstrated, with AI agents now capable of sophisticated software development. More profoundly, the power of tools is being unlocked as these agents learn to operate existing software through graphical interfaces, effectively turning the vast digital ecosystem into an extensible toolkit. This synergy of advanced reasoning, burgeoning coding proficiency, and interactive tool manipulation signifies a pivotal moment, moving AI beyond theoretical problem-solving towards practical, widespread application across countless domains. As these systems continue to evolve, bridging the gap between digital intelligence and the real world, they herald an era of amplified human capability.

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Chapter Four

Applications, Ethics, and Public Policy Implications for Large Language Models

Introduction

Anya Sharma watched the final concession cascade across her monitor, a silent digital handshake concluding a settlement that redrew the lines of corporate accountability on a continental scale. Her team, a meticulously orchestrated ensemble of reasoning agents, had parsed petabytes of climate modeling data, cross referenced three decades of obscure maritime law, and drafted responsive arguments in four languages, all in the time it once took a fleet of junior associates to simply index the discovery documents. It was a symphony of logic, and she was its conductor. She remembered a time, a decade prior, when the very notion felt like a threat rather than a promise. The late twenties had been a period of profound societal vertigo. Law firms, her own included, had stumbled through a clumsy embrace of automation, leading to embarrassing episodes of AI hallucinated precedents and a genuine fear that the profession's foundational skills were eroding. Her children's schools had been thrown into a state of near panic, grappling with a wave of undetectable plagiarism that questioned the very purpose of writing and research, forcing a painful but ultimately vital reinvention of pedagogy itself. Beyond the classroom and the courtroom, the information ecosystem

had become a battleground, with swarms of automated personas poisoning public discourse while the courts choked on a deluge of algorithmically generated frivolous claims. The great copyright wars had pitted creators against the machine, a bitter conflict that nearly froze both innovation and art before a new global consensus on data licensing and digital provenance was hammered out. It was a turbulent passage, an era of reckoning where society was forced to confront the very architecture of its institutions, from education and law to the nature of employment and truth. The solutions were not simple; they required new international accords, new forms of professional licensure for human AI collaboration, and a fundamental rethinking of how value was created and compensated. But the chaos had been a crucible. What had felt like an unraveling was in fact a necessary, if painful, reweaving of the social fabric.

Versatile Applications in Everyday Life

We have looked at how large language models came to be, discussed their shortcomings, and taken note of their rapid rate of improvement. We are now going to review the implications of LLMs for society and consider some of the risks and ethical dilemmas associated with their use and development. As far back as we can find evidence, humans have been developing technologies. Early technologies were as basic as the use of paint to draw symbols on cave walls. In more recent times, we have harnessed rocket power to travel to the moon, discovered how to transmit information around the globe at the speed of light, and have even learned how to work with the ingredients of life to clone mammals. Despite all of this, the current round of AI technology has captured our imaginations and fears in an unprecedented way. The ability to use natural language to reason and communicate is deeply and uniquely human. For the first time, we are facing the potential of a technology that allows us to effectively replicate that capability. It should not be surprising that the development of that technology is causing us to revisit every assumption about how our society functions.

Large language models have rapidly become versatile tools for countless tasks involving written language. What makes LLMs so intriguing is that a single system can assist with a wide array of activities. A common theme that we will return to in this book is the universality of language. Traditionally, computers have been constrained by very rigidly defined input mechanisms. With LLMs, we now have a tool that allows users to interact with a computer system with the same type of flexibility that we interact with other humans. In day-to-day life, people are discovering myriad uses. Ask a household question, and the model can give you a straightforward answer. “*How do I remove a wine stain from carpet?*” might yield a step-by-step solution drawing from cleaning guides. If

you are seeking financial planning advice, the same model can outline a sample budget or explain the difference between a Roth IRA and a 401(k) in plain English. For more playful or creative endeavors, an LLM can switch hats and become a storyteller or poet. Families have experimented with using it to craft bedtime stories customized with their children's favorite characters. The same underlying model, when prompted differently, might compose witty, satirical commentary similar to Mark Twain's style or rewrite a mundane paragraph in a whimsical Dr. Seuss cadence. These use-cases are typical of the types of experiments that people try out when getting started with an LLM but they just scratch the surface of what is possible and belie the magnitude of the impact that these AI systems will have in the coming years.

Opportunities, Misuse, and Deception in the Education System

AI systems that store a large portion of the world's knowledge, can interpret natural language questions, and use natural language to write everything from poems to essays to worked solutions of mathematical problems will disrupt education. Large language models have empowered a new kind of plagiarism and deception that is difficult to detect. In academia, nearly 40% of surveyed college students in 2024 admitted to using LLMs to answer assessments, with a smaller but significant fraction handing in AI-written work wholesale.¹ Teachers describe a new atmosphere of mistrust. In one U.S. survey, half of educators reported that the rise of AI made them more suspicious that students' work wasn't their own.² It only takes a few blatant cases to feed this suspicion. One professor noted that he received essays where the student forgot to delete ChatGPT's telltale preamble: *"I am just an AI language model, I don't have an opinion on that"*.³ With each uncanny, well-structured essay that lands on their desk, instructors are left wondering: did the student suddenly have a breakthrough, or did they get a little "help" from a tireless robot writer?

How can schools respond? In the scramble to preserve academic integrity, many have reached for familiar defenses. Some professors are returning to paper exams and in-class writing, yanking assessments back into offline spaces where AI assistants can't easily intrude. Others now require students to show evidence of their process such as outlines, drafts, editing logs; hoping to prove that an essay wasn't conjured in one AI burst but crafted through human effort. There is talk of specialized locked-down computers for online tests and a shift to oral exams where students have to speak their answers on the spot. LLMs are forcing educators to reconsider long established best practices. In one possible scenario, students will turn to AI tutors at home to be educated and the primary role of schools will be to perform assessments in well controlled environments where

cheating is not possible. It's fair to say that most people would agree this is not optimal and many would even describe it as dystopian; a world where robots do the teaching and teachers are relegated to the role of proctors. While schools adapt to figure out how to accurately test and evaluate students in this new era, there is a meaningful risk that a generation of students may manage to get through the system without properly learning foundational skills. During this era, the highest grades may be awarded to the students that have the best access to technology and are most willing to use it to cheat.

Traditional plagiarism checkers are ill-equipped for this new threat and a new generation of tools that check for AI generated content are easily fooled. Researchers have shown that a savvy student can use AI to paraphrase AI-written text in loops until it slips past detectors entirely, with minimal loss in coherence.⁴ This unreliability can also lead to false positives and has led to serious missteps. In one now-infamous incident, a Texas A&M instructor ran his entire class's essays through ChatGPT and it flagged a number of them. The professor accused those students of cheating. Only after an uproar did it emerge that the essays were actually written by the students.⁵ The students were exonerated, but the damage to trust and morale was done. The debacle underscored a dilemma: unlike plagiarized work, which can be traced to a source, AI-generated writing leaves no fingerprints. A synthetic body of text is unique, often remixing information into original phrasing. Short of a confession or an obvious glitch, teachers are left in a fog of uncertainty.

Amid this crackdown, deeper pedagogical questions loom. If AI can potentially do any take-home task, what is the point of traditional homework and essays? Do educators double down on preventing AI use, or do they accept that students will have these tools and redesign education accordingly? This debate has split the academic community. On one side are those who see unfiltered AI use as a grave threat to learning. They argue that when a student delegates thinking to an algorithm, the student misses out on the very struggle through which insight and skill are born. Professor Jollimore, after catching wave after wave of AI-written papers, feared that generative AI "short-circuits that process entirely"; the formative process of grappling with ideas and finding one's own voice. In his view, some students were all too ready to let college wash over them without changing them at all, using AI to coast from one assignment to the next and remain "precisely the same person at graduation" as on day one.⁶ A student who cheats their way through a paper doesn't just fool the professor, they forgo the chance to learn how to structure an argument and how to understand the material on a personal level. Those losses accumulate. A generation of AI-dependent students might enter adulthood having never learned to write a coherent essay or solve a problem from first principles. The worry is that core skills will atrophy or never develop in the first place.

If a chatbot handles all your French translations, your conjugation skills rust; if it cleans up your grammar, you might never internalize the rules. As one educational expert in the Middle East put it, the goal of schooling is to “teach students how to think, not just to extract information from AI-generated sources”.⁷

The advent of AI systems with reasoning-like capabilities is also raising broad societal questions. One major concern is cognitive offloading. If students, professionals, or even policymakers begin delegating reasoning tasks to AI, what happens to human skills and development? Educators worry that easy access to an LLM that can explain anything or even do homework might cause students to skip the hard work of learning and critical thinking. Some research shows a negative relationship between increased use of AI and critical thinking skills.⁸ If a student can ask ChatGPT to write an essay comparing two literary characters, and the result is passable, the student misses out on the analytical process they would undergo in writing it themselves. Over time, reliance on AI for thought could produce a generation less adept at problem-solving independently; a kind of “use it or lose it” effect on the mind.

This is not the first time we have had a similar debate. Calculators and spellcheck are two analogous technologies. The general consensus when those technologies were introduced is that there is still significant value in students learning how to perform essential intellectual life skills unaided and the curriculum reflects that today. At the same time, there is a balance. In more senior years these tools are allowed to be used in many academic contexts as the focus shifts to higher level tasks. It is taken as a given that a student in high school is capable of performing mental math as that was taught and evaluated at lower levels. In the interest of time, they are then allowed to use a calculator in senior levels so that they can learn subjects like physics without getting bogged down in the rote work of calculations. We could take a similar approach with LLMs. A student in a lower grade would need to learn how to carefully perform research, validate information, organize thoughts and write eloquent and convincing prose without the assistance of an LLM. From that point forward, they may be permitted to leverage LLMs as they push into learning more advanced subject matter. Carnegie Mellon professor Danny Oppenheimer has noted that allowing students to use a tool like ChatGPT on certain assignments could “free up cognitive resources for other thinking”.⁹ If an AI agent can work from a student generated outline to generate a first draft, a student can spend more time refining arguments, checking facts, and adding personal insights; the deeper skills we want to cultivate. In this view, knowing how to effectively use AI becomes a new literacy in itself. Just as we teach students when and how to use a calculator (and when not to), we might teach when an AI writing assistant is appropriate and how to prompt it for the best results. Some universities have

already moved in this direction: rather than blanket bans, they are issuing guidelines for “responsible AI use”, for instance, permitting AI for preliminary research or grammar help, but not for writing final essays, and requiring students to disclose any AI assistance. New York City’s school district, which initially banned ChatGPT, did an about-face by mid-2023, unbanning it and announcing plans to train teachers in using AI as a teaching aid.¹⁰ The tone of the conversation is slowly shifting from cat-and-mouse enforcement to a more nuanced question: how can we integrate AI into education in a way that enhances learning rather than undermines it? It’s hard to know where this experiment will lead but one can’t rule out the possibility that students that have developed a strong grounding in the fundamentals may be able to use LLMs to produce university level essays years before they graduate high school.

Moving from the risks to the opportunities, large language models may finally help to democratize access to education on a global scale. Perhaps most exciting is the prospect of personalized tutoring for every student. Education researchers have long known that one-on-one tutoring is a kind of gold standard. In a famous study, students who received individual tutoring performed two standard deviations better (equivalent to moving from the 50th percentile to the 98th) than those in a typical classroom setting; an effect so dramatic it was dubbed the “two sigma problem.” The problem, of course, is that hiring personal tutors for all is wildly impractical, particularly in under-resourced systems. This is where LLMs could be revolutionary. An AI tutor doesn’t get tired, doesn’t charge by the hour, and can be replicated infinitely at near-zero cost. We are already seeing early glimmers of this future. In a recent study in Ghana, about 500 junior high students were given access to an AI math tutor via a simple messaging app. They used it for just an hour a week as a supplement to regular classes. The results were startling: the AI-tutored students’ math scores rose significantly more than those of the control group, with an effect size equivalent to roughly an extra year of learning. At a cost of only about \$5 per student, the pilot demonstrated how AI could help bridge learning gaps in regions where human teachers are too few and classrooms too crowded.¹¹ In West Africa, as in many parts of the world, large class sizes and teacher shortages mean many students simply don’t get enough individual attention to grasp difficult concepts. An AI tutor that can patiently explain a theorem in different ways, or give instant feedback on practice questions, starts to look like a lifeline.

Consider a concrete example from Kenya, where teacher-to-student ratios are often abysmal. One elementary teacher, overwhelmed with 200 students in a class, began using an AI chatbot to help plan lessons and generate exercises. She even asked the AI to suggest simpler explanations when her pupils struggled, tailoring the material on the fly.¹² It was like having a teaching assistant who could differentiate instruction

for different levels of understanding; something a single teacher with 200 kids would otherwise find impossible. With the AI handling routine prep, she was freed to spend more time one-on-one with students. The catch, in her case, was access: her school was fortunate to get a pilot program with decent internet. In many Kenyan schools, that's not the case; roughly 65% have no internet connectivity at all. Those without access are left behind, a stark reminder that technology's benefits often flow first to those who already have infrastructure. Still, the promise is undeniable. If a student in rural Kenya or Bangladesh can chat with a top-notch AI tutor through a basic smartphone, that might one day approximate the kind of personalized support that only elite students in rich countries used to enjoy. These types of results are no longer limited to one-off studies. A meta analysis of 51 different research studies performed between 2022 and 2025 showed that "ChatGPT has a large positive impact on improving learning performance" although given the early nature of these research studies, there is debate about their validity.¹³

Moreover, LLMs as tutors have advantages beyond just cost and scalability. They can be infinitely patient; a machine will never lose its temper or tire of explaining a concept multiple times in different ways. They can adapt to a student's pace, providing harder questions when the student is excelling or reviewing basics when the student is struggling. They don't judge or embarrass. A shy student who might never ask a human teacher for help with a "silly" question might freely ask an AI tutor, gaining confidence. A tutor can also become a lifelong companion. If used consistently from youth, it will know what you know and what you struggle with. It will learn what types of exercises help you learn the quickest and retain information most effectively. Perhaps, at the age of 30, you wish to read an advanced scientific article. The tutor would scan it and first give you a ten minute lesson on any terminology that you will need to learn to understand the article; effectively allowing you to much more easily access materials that would have been off limits in the past. These qualities hint at a more equitable classroom dynamic, where every learner gets attention. In regions where educational access is unequal, such tools could help democratize learning. A student in a remote village with patchy school quality could, in theory, access the same AI-powered tutor as a student in a high-end private school. That's a profound opportunity to narrow the gap.

We may soon see classrooms where routine content delivery such as explaining Newton's laws of motion or the causes of World War I is largely offloaded to AI tutors, freeing teachers to focus on higher-level mentorship: fostering creativity, critical thinking, moral reasoning, and the social aspects of learning. AU tutors could enable educators to invest their time where it matters most; in human-to-human interaction, inspiration, and support.

LLMs in the Workplace

Real-world applications of LLMs, supercharged with retrieval augmented generation and tool use, are quickly emerging across industries.

Customer Service

Customer service is one of the first jobs experiencing widespread disruption. Broadly speaking, this work includes everything from helping customers to place orders, processing refunds, making changes to subscriptions, setting appointments, and troubleshooting problems. Customer service bots are able to integrate with company knowledge bases and converse with customers in natural language. To this point, we have focused on text-based communication but as we will cover in the next chapter, AI agents with LLMs at their core are now also able to use voice as an input and output mechanism. Imagine a customer asks an insurance chatbot about coverage for a particular scenario. A RAG-enabled bot can pull up the relevant policy clause from the company's documents and reference it when formulating its answer. Through the use of an LLM, the customer service bot is able to bridge both worlds. On one hand, it can understand imperfect queries from humans that include grammar mistakes and slang. On the other hand, it has access to a vast repository of precise information; including potentially a record of every previous call and resolution handled by human agents. The opportunity to save money through the application of AI in customer support is significant. The global market for contact centers is expected to reach almost \$500 billion by 2027 and over 90% of companies are starting to invest in AI.¹⁴ Given the rapid improvements in LLMs, it would be hard to argue that they will not be capable of handling many of the routine issues that are currently handled by customer service agents and so there may be massive changes in this industry. The trend towards offshoring into a small number of central hubs that specialize in this type of work means that the economic impact may be particularly significant in those communities. For example, the Philippines now has 1.3 million human agents working in call centers and these workers are often poorly equipped to transition to other careers.¹⁵

Legal

In the legal domain, AI agents have immense potential given the importance of language, the wealth of documents to train on, and the high hourly rates for legal services. Lawyers

and judges are experimenting with LLM-based assistants that can retrieve statutes, case law, or regulations on the fly. Given a query about a specific legal question, the system might fetch a large number of relevant precedents and feed them to the agent, which then drafts an answer or even a portion of a legal brief, complete with references to those cases. Beyond basic language tasks, law is a domain built on reasoning. In a notable benchmark, GPT-4 was evaluated on the Multistate Bar Exam (a test for lawyer licensing) and achieved a score around the top 10% of human test-takers. By contrast, its predecessor GPT-3.5 scored around the bottom 10% on the same exam, underscoring the leap in capabilities.¹⁶ It's important to note that a follow-up study by a different group re-evaluated GPT-4's bar exam performance and concluded the model's showing was around the 68th percentile.¹⁷ Although still very impressive relative to what was possible just a few years ago, this variability in results also demonstrates the challenge in establishing benchmarks in complex domains.

To better gauge just how far these systems have come and how far they still have to go researchers have created a comprehensive legal benchmark called LegalBench which includes a battery of tasks covering everything from spotting relevant issues in fact patterns to recalling specific rules of law to applying regulations to novel scenarios.¹⁸ As of early 2025, Google's Gemini 2.5 model held the top spot on this leaderboard, correctly answering about 83.6% of the questions. Not far behind were OpenAI's and Anthropic's best models, along with a few specialized legal AI systems, all clustered in the lower-80s accuracy range on these rigorous evaluations. An 84% success rate is remarkable, yet it also means roughly one in six legal questions still stumps or misleads the best AI we have.

Their growing competence is impossible to ignore. One team of researchers tested GPT-4 against junior lawyers and legal professionals on a contract review exercise: the bread-and-butter work of many first-year associates. AI identified potential issues in the contracts with an accuracy on par with the humans, even scoring slightly ahead of the junior lawyers on one metric (identifying relevant contract issues) while slightly behind on another (pinpointing their exact location).¹⁹

Within the legal field, some tasks must be performed through proprietary websites and software applications. With AI agents that are powered by LLMs and able to interact with GUIs, there is now an opportunity for this work to be automated. An AI agent could, for example, use a web-based patent database just like a human patent attorney, entering complex search strings, tweaking filters, and scanning results. The United States Patent and Trademark Office (USPTO) receives over 700,000 patent applications a year, adding to the tens of millions of documents in its archives, so thorough searches are arduous. Imagine an AI agent that can tirelessly pour over patent filings overnight via

the USPTO's own portal, reading and cross-comparing claims at a rate no human could. By morning, it presents a dossier of potential conflicts and relevant prior inventions, having essentially automated a key part of legal research.

We see questions of trust and accountability emerging. If an AI assistant suggests a legal strategy that backfires, who is responsible? Unlike a human lawyer or paralegal, an LLM has no accountability or credentials, yet it might influence decisions. The legal profession, which operates under strict ethical and professional standards, will need to integrate AI in a way that maintains a human in the loop for judgment calls. As with most regulated professions, ultimate responsibility lies with the professional and they are responsible for ensuring that any tools that they use do not compromise the quality or accuracy of the advice they provide.

What does all this mean for actual lawyers and the legal system? Hard evidence from the past year paints a picture of both potential and peril. Indeed, the most meaningful explorations of AI in legal work so far underscore collaboration over competition. In areas like contract analysis and due diligence, the evidence shows AI tools can shoulder the drudgery, freeing up human lawyers to focus on strategy, client counseling, and nuanced judgment. Conversely, in novel legal problems or sensitive ethical dilemmas, human experience and intuition remain irreplaceable. The AI can offer suggestions or even creative solutions, but it lacks the contextual understanding and moral responsibility to navigate gray areas of law. The story of AI in legal practice is thus not one of silicon supplanting flesh-and-bone attorneys, but of a new kind of partnership. Just as the calculator became an extension of the mathematician, these language models are becoming an extension of the modern lawyer's toolkit.

Finance and Business

Professions like investment analysis and business strategy consulting are also being touched by reasoning-capable LLMs. These are jobs that involve digesting large volumes of information and making reasoned judgments about economic or strategic questions. LLMs can read through financial filings or news articles and produce coherent summaries and risk analyses. Bankers are already experimenting with these possibilities. J.P. Morgan's researchers, for example, have explored using generative AI to interpret market data and execute routine trades. In finance, where information flows are torrential, having AI eyes on every screen and AI hands on the controls could mitigate the cognitive overload that human brokers and analysts face each day and the ability of AI agents to directly interact with a GUI means they could be dropped into the existing environment

and seamlessly work with a mix of proprietary internal programs, trading platforms and research platforms to the extent that license terms permit.

For business strategy, an LLM can be prompted with a scenario (“a competitor has introduced a disruptive product”) and asked to perform an analysis (like a SWOT analysis or to brainstorm strategic responses). The output often reads like a decent first draft that a consultant might write, covering key points of Strengths, Weaknesses, Opportunities, Threats, etc., derived from the model’s training on business cases and textbooks. In one study, the performance of GPT-4 Turbo on the GMAT, which is broadly used as an acceptance exam for MBA programs, would have placed it in the top 1% of test takers.²⁰

The story is similar in the field of accounting. Researchers found that when GPT-4 was provided with examples in the prompt, and able to access tools such as a calculator and web search, it was capable of passing the CPA, CMA, CIA, and EA (enrolled agent) certification exams.

The entire finance industry including insurance and banking is characterized by lengthy contracts, reams of information stored in documents, and complex and often proprietary software that is challenging to update. This makes it a particularly fertile ground for AI use cases.

Medical

Medical assistants built on LLMs have incredible potential given the perpetual mismatch between demand and supply in the medical industry. If AI systems can reliably cite medical literature or patient-specific data, they have the potential to be very useful. For instance, a doctor could query a system about the latest guidelines for treating a certain condition. A RAG-enabled model could retrieve the relevant paragraphs from a trusted medical guideline (say, the FDA or WHO recommendations) and use that to generate an answer. This way, the advice given is directly traceable to an approved source. Going forward, we can imagine medical chatbots that always accompany their advice with references to journal articles or textbooks, allowing both doctors and patients to trust but verify. One major advantage of using a RAG-enabled model in the medical field is that it could always have access to the latest research.

Medical expertise relies on deep knowledge, reasoning from symptoms to diagnosis, and careful consideration of patient context. Here too, LLMs have made surprising strides. Researchers have tested models on medical exam benchmarks like the USMLE (United States Medical Licensing Examination). In one test that involved 376 questions,

54 of which included images, GPT-4 scored an average of 88% across the three steps of the exam.²¹

Specific demonstrations of LLMs capability have been striking. For example, given a patient's case description (symptoms, history, lab results), GPT-4 can suggest likely diagnoses and next steps, in some instances matching the insights of human doctors. Specialized models like Google's Med-PaLM 2 have reached expert-level accuracy on certain medical question sets, and hospitals are experimenting with AI to draft patient notes or triage simple patient queries. However, medicine is a field where trust, accuracy, and verification are literally life-and-death matters. Doctors have pointed out that an LLM lacks clinical judgment; it has no real understanding of a patient's unique circumstances, and it cannot be held responsible for an error. There is a risk of automation bias: a physician might over-rely on an AI agent's suggestion (e.g. a particular diagnosis) and give it undue weight, even if it's subtly incorrect or misses a critical detail that a seasoned doctor would notice. Moreover, LLMs have no direct access to physical examination or real-world sensing; they only know what is described to them in text. This absence of sensory grounding means they might overlook obvious real-world inconsistencies (for instance, a condition that would be ruled out by a normal vital sign, which a human doctor would instantly notice, but the AI might not integrate unless explicitly told). Regulatory bodies are already urging caution. As a result, the near-term role of LLMs in medicine is likely to be assistive: helping summarize patient records, suggesting differential diagnoses as a second opinion, and providing information on medical literature. The final decisions and responsibility rest firmly with human clinicians, at least under current standards. There is great optimism that, used wisely, these tools can reduce doctors' paperwork burdens and even catch occasional missed possibilities, leading to better outcomes. But the medical community is moving carefully, emphasizing that AI should support clinical reasoning, not replace it.

Patients on the other hand might not be so patient. Around the globe, access to healthcare remains a challenge and for as long as the internet has existed, patients have turned to websites to self-diagnose, for better or worse. The pros and cons of this approach are multiplied when an LLM is used instead. Will doctors offices be flooded with patients insisting that the concerns raised by an AI agent are looked into?

One area where AI can absolutely help in healthcare is with improved efficiency in the creation and management of electronic health record (EHR) systems. Studies find that for every hour a physician spends face-to-face with patients, nearly two additional hours go into EHR data entry and administrative desk work.²² Physicians often stay late, clicking through lab results and insurance forms in clunky interfaces, which is a major driver of burnout. AI assistants capable of operating these medical record systems

could shoulder much of that burden. A doctor might dictate a summary or just finish an appointment, and an AI agent could handle the rest; pulling up the EHR screen, filling in billing codes, scheduling follow-ups, ordering the lab tests as indicated in the notes. All the bureaucratic clicks that now erode the doctor-patient relationship could be offloaded.

Software

In the world of software, productivity is poised for a radical shift. We previously discussed the ability of LLMs to write their own software in the service of satisfying a question from a user. This same capability can also be invoked by developers to make contributions to larger software projects. Developers armed with AI assistants can implement features or fix bugs much faster by offloading boilerplate coding and even some design decisions to the model. Empirical studies are already validating these gains: in a controlled field experiment at a fintech company, programmers given an LLM-based coding assistant produced 55% more code (measured in functional lines) on average than those without AI help.²³ The impact is real. In April of 2025, Microsoft CEO said that 20-30% of their existing code has already been written by software. In the same timeframe, Google has indicated that more than 30% of new code is being written by AI.²⁴

The assistive model has not only sped up coding but also helped generate working solutions for tasks that might have stumped some individuals. Such gains hint at lower barriers to entry in software development. A less experienced programmer can leverage an LLM to get guidance or boilerplate code, effectively “jumpstarting” their work. Indeed, there is evidence that AI assistance narrows the gap between novice and expert performance, as less experienced developers often benefit the most from suggestions and auto-generated examples. Over time, this could democratize programming, enabling people with only high-level ideas to create software via natural language prompts. More recently the trend of “Vibe Coding” has taken off thanks to new platforms that allow anyone to build fully functional web applications, complete with authentication and databases, using only prompts.

However, these advantages come with new challenges. One concern is the proliferation of poorly understood or undocumented code. An LLM might generate a complex snippet that works for now, but if the human developer does not fully grasp it, maintenance becomes difficult. Future engineers, or even the original author, could struggle to debug or extend code that was largely machine-written. Moreover, software quality and security might suffer if developers trust AI output too readily. LLMs do not have an

intrinsic sense of secure coding practices; they mainly imitate patterns in their training data, which may include insecure code. Studies have found that a non-trivial portion of code suggestions from AI assistants include security vulnerabilities or flaws. For example, one analysis of GitHub Copilot's outputs found nearly 40% of the generated code had latent vulnerabilities under certain conditions.²⁵ Without rigorous human review, the ease of AI code generation could lead to a flood of software that "works" initially but harbors hidden bugs or security holes. The industry may thus see an emergence of hastily produced applications that accumulate technical debt. Ensuring AI-generated code is robust, secure, and well-documented will likely become a new responsibility for software teams. This might spur demand for AI-powered code analysis and verification tools to complement the generative models. There are also workforce implications: if one developer using an LLM becomes as productive as several without one, companies might adjust hiring needs, particularly at the entry-junior level. Routine coding jobs could be partly automated, while the role of human programmers may shift more toward architecture, oversight, and integration of AI-generated components. In sum, coding-focused LLMs stand to greatly accelerate software development and reduce the grunt work for humans, but realizing this potential safely will require new practices (like thorough code review of AI contributions, security scanning, and educating developers to critically evaluate AI suggestions) to avoid a future of brittle, unmaintained software. The productivity boom is real, but so is the need for vigilance in how we incorporate these powerful coding copilots into the software engineering lifecycle.

Architecture and Engineering

In engineering fields, LLMs have begun to play the role of knowledgeable assistants. They can already help with tasks like explaining engineering concepts, generating draft design documents, or reviewing technical proposals for consistency. For instance, an LLM can be asked to outline a test plan for a new aerospace component or to reason through the steps of a failure analysis, and it will produce a logical-sounding narrative. The engineering field is characterized by specialized software programs performing everything from physical simulations to computer aided drafting. There is ongoing work to integrate LLMs with analytical tools; for instance, using an LLM to set up the equations for an optimization problem, then having a solver algorithm find the actual solution.²⁶ Early results in operations research and logistics show promise by combining LLMs with traditional software programs. One prototype system used an LLM to reason about how to schedule airline crews, while a linear programming solver handled the math-heavy computation. This kind of hybrid approach might define

how LLMs contribute to engineering: the LLM brainstorms and provides reasoning in human-readable form and then enters data into a specialized software application to generate the final output.

The need to formally train each professional on every feature of a tool might lessen; the AI can serve as a knowledgeable intermediary that “knows” the software’s operation. In effect, highly specialized software becomes more user-friendly by proxy: the AI is the expert user. Industries that currently invest enormous resources in software training may find that what really matters is the underlying understanding of the foundations. For instance, someone proficient in formulating engineering problems could solve them via any CAD or simulation tool the AI supports, without personally being a master of each tool’s UI. The engineer could give directions such as “Draw a square building that is 20 feet by 20 feet and include rafters that are sized appropriately” and the AI agent could then refer to load tables to and the software user manual to size the rafters and then produce the CAD diagram. The bottleneck of “we use Tool X here, so you must retrain if you only know Tool Y” disappears when an AI agent can bridge the gap. In the long run, we will hopefully value human professionals more for their strategic and creative abilities, while delegating the procedural clicks and keystrokes to machines.

Control Systems

Deep in the bowels of a non-descript windowless building, an operator’s console bristles with dials, charts, and alarm indicators as they oversee the operation of a regional power grid. Here too, an AI assistant could watch the live data and take action. Modern industrial control systems (the SCADA interfaces that run water treatment plants and electrical grids) have long helped human operators track conditions in real time. With the ability of LLMs to interpret natural language inputs that include maintenance notices from power plants and written weather forecasts and also to take control of computer programs through the GUI, it’s possible that we will see a new wave of automation in these facilities. These AI systems could become digital operators working alongside the human crews, mastering the elaborate control panels that were once the exclusive domain of seasoned technicians.

In aviation, an air traffic controller sits before a radar screen, orchestrating thousands of flights with the aid of specialized software. The U.S. FAA’s en-route control platform (ERAM) – essentially the “heart” of America’s air traffic system – can track up to 1,900 aircraft at a time, a leap from the 1,100-plane limit of the older system. Mastering such a console is a formidable task: after initial academy training, new controllers spend 2–3 years in the field before they’re trusted to work solo. They learn to encode commands

at blistering speed and to read dense visual cues on-screen, maintaining safety in a realm where blips on a display translate to lives in the sky. It's a cognitive juggling act that, until recently, only a human could perform. Air traffic control has been incredibly hard to improve through automation due to the heavy reliance on spoken radio communications. AI agents that leverage LLMs could alter this picture. Imagine an AI co-controller observing the same radar feed and listening in on all of the radio communications. It would be capable of suggesting immediate reroutes around weather or flagging potential conflicts a few minutes sooner than a person might. When planes request new flight levels, the human operator would still be responsible for giving approval but the AI agent would act as a second set of eyes. And if the human operator and AI agent agree that the flight level should be approved, the human operator could simply press a button and allow the AI agent to transmit the voice message, saving valuable seconds to move on to the next task. An AI agent adept at spoken radio communications and with a mastery of the same GUI could provide an extra margin of safety and efficiency without making any modifications to the underlying existing infrastructure.

Recruiting

Recruiters read countless resumes and so it's not surprising that LLMs are now being used to help filter applications. One recent college graduate described the experience of applying for work in the AI era as a bewildering blur of bots and automated workflows. Over a few months, he submitted around 150 job applications and found that "nearly all of them" involved some form of artificial intelligence; from chatbots that auto-filled applications for fast-food chains to a "talking robot" that proctored his video interview. Each time he clicked send, he had the uneasy feeling that his materials were vanishing into a digital void, and he "started assuming that nobody is going to see it".²⁷ On the other side of the hiring equation, employers are feeling equally overwhelmed. A recruiter at one U.S. tech firm opened a requisition for a data scientist and was flooded with over 3,000 applications. Even after instituting a tough skills test to winnow the pool, she suspected that some of the few who aced it had relied on generative AI; a hunch proven when one highly rated candidate froze up in the follow-up interview, unable to explain a solution that had appeared in "his" assignment. In her words, "not only have they wasted their time, but they wasted my time... it's really frustrating".²⁸ This anecdote captures a new reality: the hiring process has quietly become an algorithmic battleground, with human job seekers and hiring managers alike caught in the middle of an escalating interplay between dueling AIs.

At the center of this upheaval is the rise of AI-generated application materials. Not long ago, a personalized cover letter or meticulously tailored résumé was a reliable signal of a candidate's effort and interest. Crafting those documents took time; generic, mass-produced applications were usually easy to spot. That equation has been turned upside down by large language models. Today, a job seeker can feed a job description into an AI tool and receive a polished, custom-fit cover letter within seconds. With a few more clicks, they can tweak their résumé to mirror the employer's exact keywords and tone. Then, using job platforms' one-click apply features, they can send applications not just to one job, but to dozens, or even hundreds, of openings in a single afternoon. The result has been an explosion in volume. Surveys show that more than half of job seekers now use AI to support their applications, and nearly one in five rely on such tools regularly.²⁹ Indeed, a global poll in 2025 found that 70% of job seekers were leveraging generative AI to research companies, draft cover letters, or prepare talking points for interviews.³⁰ With generative text on demand, it has become trivially easy to "personalize" an application and to do so at industrial scale.

This newfound ease has inundated hiring systems with applications, many of them indistinguishably well-crafted. Employers are simply unable to give each submission a close look. Hiring managers report that the volume of applications has surged and this deluge has led to slower responses and even qualified candidates being overlooked. Ironically, a tool meant to boost efficiency has, in practice, made it far harder for any individual to stand out. All those eager applicants are hitting a wall of overwhelmed recruiters and frustration is mounting on both sides of the hiring desk.

Some employers see a certain resourcefulness in candidates using advanced tools. After all, if AI is going to be part of the job itself, why shouldn't applicants use it in the application? However, even these optimists worry about losing any genuine signal of a person's motivation or fit. The hiring process starts to feel like an arms race of fakery and filtering. Job seekers embellish with AI to get in the door; employers respond with new screening tactics to tell human from machine. In one corner, applicants are using LLMs to script not only their written materials but even answers to anticipated interview questions. In the other corner, companies are deploying their own AI systems to weed out those very scripts. Some have reacted by abandoning the traditional niceties altogether by reaching out directly to hiring managers or trying to network their way in. These are all ad hoc adjustments in a game whose rules are being rewritten in real time.

Unfortunately, automating recruitment can entrench bias rather than eliminate it. AI systems learn from historical data, and history is not always an ideal teacher. In 2024, University of Washington researchers tested several state-of-the-art LLM-based hiring tools to see how they ranked candidates with different demographic cues. The results

were alarming. When identical resumes were submitted under different names, the AI consistently favored those with white-sounding names over those with black-sounding names, and male names over female ones. In fact, across hundreds of trials, the algorithms never once preferred a resume with a black male name over an equivalent resume with a white male name.³¹

Aside from bias, automated systems can fumble in other ways. They can misread unconventional resumes, overlook non-standard career paths, or simply over-prioritize keywords to a fault. Each individual rule may have logic behind it, but together these automated judgments can amount to a hiring process that is unyielding and unforgiving in ways no human recruiter would consciously be. The risk, as some critics put it, is that we create a bureaucracy of bots; systems that efficiently process thousands of profiles but perhaps miss the very human potential they were meant to find. The stakes are enormous because what's at issue is more than just who gets a job offer; it's the very basis on which opportunities are distributed in society. Work is a cornerstone of modern life, and hiring practices determine whose skills are recognized, who gets a foot in the door, and ultimately who thrives. That process is now being altered in fundamental ways by technologies few of us fully understand. It's not all bad news. We may see that AI based hiring processes allow people to be more effectively matched with opportunities that align with their personal interests and if executed correctly, AI augmented hiring processes could eliminate biases instead of reinforcing them.

Concluding thoughts to workforce disruption

From customer support to law to healthcare to engineering, the story rhymes. Wherever professionals are working in domains that make extensive use of natural language, an AI agent is likely to disrupt their way of work. In the coming decade, the nature of many high-skill professions will evolve in subtle and not-so-subtle ways. Humans will provide goals, oversight, and ethical judgment while AI helps provide brute-force execution, consistency, and breadth of attention. On an everyday level, countless hours of drudgery will be lifted off human shoulders. In particular, every profession that spends its day in front of a screen stands to be altered as AI agents are able to leverage GUIs as seamlessly as they leverage APIs today. The last time a transformation of this breadth occurred was when personal computers and the internet permeated offices. Now, with AI agents inhabiting our software, we stand at a similar inflection point.

Society will likely need to rethink skill development. If rolled out responsibly, the payoff is a world where innovation accelerates and skilled human talent is freed from the tyranny of routine interaction. It is a future where architects spend more time

designing and less time drafting, where doctors spend more time with patients and less time typing, and where each person can manage far more complex projects because their AI collaborator handles the busy work. Such a future is fast approaching, driven by these new AI systems that quietly learn to do every computer-based task we do.

Assigning Value for Software and Labor

If AI agents start using software around the clock, acting as non-stop “digital workers,” current software licensing models will face pressure to adapt. Most commercial software licenses today are tied to human users or at least to a certain number of seats or devices. What happens when a single human deploys ten AI agents to use ten copies of the software in parallel? Or when an AI agent runs a program unattended at all hours? Many license agreements explicitly prohibit this kind of automated or pooled use. For instance, license agreements often include language forbidding any “automated, unattended, non-interactive” use of the software that would support multiple operations at once. Such clauses reflect a business model built on human limitations; a person can only be logged in and using the software in one place at one time. AI breaks that assumption. From a revenue standpoint, software companies might find a single AI user could replace dozens of human users, undermining per-seat licensing. They may respond by trying to enforce anti-automation clauses or shift to usage-based pricing (charging by the minute or transaction). Although dealing with this change may sound like a trivial detail, the impact could be significant. The global software market was around \$740 billion in 2024 and transitioning to an entirely new revenue model has the potential to be very disruptive.³²

It’s not only software pricing models that need to be revisited. One intriguing labor consequence of accessible AI tool use is the possibility of individuals quietly automating their own jobs, or even holding multiple jobs, without their employers’ knowledge. Already there are stories of tech workers secretly using GPT-4 or similar tools to finish in hours tasks that were budgeted for days, effectively freeing themselves to take on a second remote job. Surveys suggest this is not an isolated phenomenon: by mid-2023 employees were “increasingly using AI tools, such as OpenAI’s ChatGPT, to boost their personal productivity and manage multiple jobs,” but often kept this secret due to strict corporate policies against AI use.³³ In other words, workers are hiding their AI “co-workers” for fear of repercussions or of giving up their newfound edge. This clandestine automation creates a complex ethical landscape. On one hand, an employee who can do the work of two people with the help of AI might reason that they are simply being efficient (and perhaps underpaid for the value delivered). On the other hand, if

employers are in the dark, they cannot adjust workloads or expectations accordingly, and the situation veers into a gray area akin to dishonesty. The term “overemployed” has emerged for people holding multiple full-time jobs thanks to remote work secrecy; AI could turbocharge that trend, enabling a single person to meet the demands of several roles by delegating much of the grind to software-controlling agents. Gartner analysts even project that by 2028, 10% of sales associates will use AI to covertly hold more than one job.³⁴ The other side of the argument is that we are experiencing the turbulence of a transitory phase. As employers catch up to employees in the adoption and deployment of AI, expectations of performance and output in many roles will increase: leaving no time for workers to pursue multiple jobs in parallel.

The nature of work that is automated will have a meaningful impact on career development paths. In many knowledge-based jobs today, career paths have been carefully crafted over decades or even hundreds of years. The tasks that LLM systems excel at overlap significantly with the tasks that are often assigned to entry level employees such as research assistants and junior associates. Although at times tedious, the years spent on those entry level tasks help employees internalize domain knowledge, gain exposure to real-world situations, and receive mentorship from senior employees. If that work is no longer needed, what will that mean for the future of employment?

One might speculate that we will enter a period of time where there are layoffs and a reduction in the size of the overall workforce as that work is automated. Once the workforce has stabilized at a smaller overall size, new entrants that replace retiring experts will be stepping into roles with a higher level of initial responsibility. How will industry and universities adapt? It is possible that the expectation for education will increase. One could imagine that students may need to receive a master’s degree before entering industries where a bachelor’s degree was previously sufficient. In the past, the handful of years in an entry level role provided an on-the-job education that may have to be replaced in another way.

An alternative view is that there is a significant unmet need in many knowledge-based industries and the automation of entry level work may allow service to be provided to a larger number of consumers with a workforce that is the same size or even larger. For example, according to the Legal Services Corporation, an entity established by the US Congress, low-income Americans were underserved for 92% of their civil legal problems.³⁵ In this new era, entry level employees that are able to partner with AI agents could be assigned to customers with lower budgets that were previously not served at all or underserved. This has potential to be a win-win-win for consumers, employees, and employers but care must be taken to ensure that the quality of service continues to meet established standards.

At the other end of a career path are more senior employees. Today, these employees are able to spend an increasing amount of their time managing the most important client relationships and tackling interesting strategic problems. In the future, there is a risk that senior employees will be relegated to the role of babysitting AI agents. Much of the interesting work may be automated, with the senior employee responsible only for signing off on the output or steering AI agents back on track when they get stuck. This dilemma begins to expose a set of key philosophical questions. What is it that brings us satisfaction and joy in our work and how can we be sure we don't give up those elements of our jobs?

Handing over the operation of tools to AI agents has a less immediately visible consequence: human skills can atrophy. If professionals no longer need to learn the detailed mechanics of software, they may gradually lose both the know-how and the intuition that comes from direct, hands-on experience. This is part of a broader phenomenon known as cognitive offloading which we introduced in the context of education. If we entrust mental tasks to external aids (be it calculators, GPS, or now AI), we may lose our mastery of those faculties. People who constantly rely on AI to figure things out for them have fewer opportunities to practice complex reasoning themselves. There is a risk of creating a generation of professionals who know what to ask for but not why things happen the way they do within their tools and who are unable to step in and take over if the AI stops functioning. We've seen similar situations before in niche areas. When pilots rely on autopilot systems, their manual flying skills can degrade, necessitating deliberate practice of those skills to ensure readiness when automation fails. It's the reason why professionals in a number of regulated industries perform regular training and undergo periodic recertification. One can argue that we have seen similar society-wide deskilling from previous technology revolutions. For example, most people today do not grow any of their own food and would be incapable of providing for themselves if our agricultural system broke down. Although that is a valid point, there is something fundamentally different about the loss of critical thinking skills. Mitigating this across society as a whole will require conscious effort of a scale we have not seen before.

Professionals Training Their Replacements

The early stages of deploying AI agents in specialized fields could create a grim irony: highly skilled professionals might be asked to guide and correct AI systems that are explicitly meant to automate the professionals' own jobs. In effect, experts could be tasked with training their replacements. We have a precedent for this in the world of AI

data. In many parts of the world, there are long-established companies that specialize in data entry and data annotation tasks. Today, many of those workers are labeling images, transcribing audio, or flagging content, to help train AI models. In turn, those AI models will displace the need for further data annotation work.

Now imagine a senior accountant or an experienced architect being given an AI apprentice and told to correct its mistakes for a period of time, with the clear expectation that once the AI reaches a certain level of proficiency then fewer human roles will be needed. In many cases, it may be less subtle. A drafting technologist may complete their education to discover that the only work available is to follow a script and perform routine tasks while being recorded such as drawing a wide variety of standard shapes. The recordings will be used to generate a training data set for AI Agents so that they can perform drafting on their own. It's a discomfoting scenario that appears likely to become a reality. A law firm, for instance, could hire a seasoned paralegal on a short-term contract to spend six months validating and fine-tuning an AI that is proficient at accessing all of the local state legal systems. The paralegal does one last round of intense mentorship, but to an AI agent instead of a junior employee. Their knowledge is permanently encoded into the system, and then the contract ends. From that point on, there is only the occasional need for maintenance and fine tuning. The economic logic is straightforward, but it raises ethical questions about the value and dignity of expert labor. Are these professionals being reduced to "model trainers" rather than valued for their judgment? We might see consulting gigs emerge where experienced workers lend their knowledge to improve an AI system for a modest fee. Optimists might argue that experts training AI is no different from experts training junior employees; a natural mentorship process, except the mentee is a machine. The difference is the endgame: the machine won't need a salary or reasonable working hours. Society will need to find ways to ensure that the human expertise that seeds these AI systems is respected and fairly compensated. The transition period where humans and nascent AIs work side by side could be fraught with feelings of disposability among the very people whose knowledge makes it all possible.

Autonomy and Systemic Risks

As AI systems start to take control of a wider range of systems that previously required a human operator, there is the concern of AI agents precipitating rapid, hard-to-control events in the physical or financial world. We have already experienced a cautionary tale of automation gone awry in the financial markets: the "Flash Crash" of May 6, 2010. On that day, U.S. stock markets plummeted by about \$1 trillion in value within minutes,

only to recover shortly after. Forensic analysis later showed that automated trading algorithms misinterpreted market signals and began a cascade of selling, a feedback loop of machines triggering machines, until human intervention halted the chaos.³⁶ While that event was driven by relatively narrow algorithms (high-frequency trading bots reacting at blinding speed), it foreshadows what could happen as more powerful, general AI agents take the reins of complex systems. If an LLM-based agent is given control of a power grid management interface or a large-scale logistics system, a minor misjudgment or a learned heuristic misapplied could produce swift, compounding errors. The agent might execute a perfectly logical series of actions that nonetheless lead to a destructive outcome; just as those trading algorithms logically followed their programmed objectives into a market crash. The more we hand over real-world control to AI, the more we have to consider new forms of oversight and circuit-breakers. In finance, after the 2010 flash crash, regulators implemented trading pauses (so-called circuit breakers) to give humans a breather to assess extraordinary market moves. In other domains, analogous safeguards might be needed. If an AI agent is controlling a fleet of autonomous vehicles via a management GUI, what stops it from routing all vehicles onto the same small street in response to a glitch in sensor data on the main highway? We might need “human-in-the-loop” checkpoints for certain high-stakes decisions, or independent AI monitors that watch the primary AI. We could imagine a scenario where a water treatment facility adds an AI agent to interact with a decades-old operating system and it inadvertently shuts off a city’s water supply while the human supervisor is on a washroom break. Of course, these types of risks are very carefully contemplated in automated control systems. The new risk today is that organizations might layer on an AI operator without giving the same level of thought into how to manage these scenarios.

Furthermore, when multiple AI agents interact, for example, dozens of trading bots all reading the same market data and reacting, their emergent behavior can be hard to predict and potentially destabilizing. The flash crash showed how interactions can amplify into systemic instability. With more advanced agents, those interactions could be more complex than simple selling cascades; they might involve multi-step strategies and even interactions across domains. Imagine an AI agent in charge of supply chain management that makes aggressive decisions that inadvertently shock a stock market. We will discuss swarms of AI agents in a future chapter. Policymakers and industry leaders will need to study these scenarios and perhaps draw on lessons from fields like systems engineering and ecology, where complex adaptive systems are known to sometimes tip into chaos. The goal will be to harness the immense productivity and efficiency gains of autonomous AI control without succumbing to its potential for

rapid, runaway mistakes. Just as we design fail-safes for nuclear reactors and braking systems, we will need robust fail-safes for AI agents that can “press all the buttons” in our digital and physical infrastructure. The flash crash stands as a stark reminder that raw speed and autonomous decision-making, unchecked, can outpace our ability to contain the damage; a lesson we ignore at our peril as we move into an era of ubiquitous AI agency.

Bad Actors and Abuse Scenarios

AI Agents that are equipped with powerful reasoning capabilities and the ability to access software systems will not only be used for good; unfortunately, they also present new tools for bad actors. Many experts are sounding the alarm that advanced LLMs could be weaponized in various domains, lowering the cost and skill barrier for activities that would normally require expert knowledge or significant labor. We are already seeing early examples of this, and it’s easy to imagine more as the technology advances.

One concerning scenario is “legal flooding” or automated bureaucratic attacks. An individual or group with an agenda could use LLMs to generate a deluge of legal documents. For instance, an individual could file thousands of frivolous lawsuits or patent applications to overwhelm a court system or stymie a competitor. Writing a halfway coherent lawsuit or complaint ordinarily takes legal expertise, but an LLM trained on legal formats can produce documents that look credible. In some cases, filing fees discourage this behavior but in other cases, there is no fee in the interest of reducing barriers for victims. For example, imagine a sophisticated attack that would e-mail anonymous customized bylaw complaints about a neighbor that someone holds a grudge against. Every day, a new e-mail comes in, seemingly from a different complainant and everyday a bylaw officer needs to invest time following up and deciding if a fine should be levied. This is a very narrow example but a coordinated campaign could tie up resources in the courts or in a government office by the sheer volume of AI-generated paperwork. Similarly, one could flood public comment systems (which agencies use to gather feedback on proposed regulations) with tens of thousands of AI-written comments that drown out genuine public input. This kind of asymmetric attack leverages the fact that generating text is cheap for AI, but processing and responding to it is costly for humans. We have not yet seen this at full scale, but the threat is raising concern among policymakers who realize current systems aren’t equipped to distinguish a human filing from a bot filing if both follow the required formats.

Another obvious misuse is in cybersecurity and hacking. Advanced can dramatically lower the barrier to entry for cyber-attacks. Earlier in the book, we discussed the po-

tential for LLMs to be used to draft phishing e-mails. With the addition of reasoning, software development and tool-use capabilities, an AI agent can go much further. In theory, an agent can be developed that would deploy an end-to-end attack, including development of a custom website and taking action to transfer funds in and out of financial accounts. At scale, this type of an attack may be much harder to detect since the automation would allow it to be customized for each individual target. Already, researchers have demonstrated using generative models to create polymorphic malware code that changes itself to evade detection.³⁷ All of this means that more attackers can launch sophisticated attacks with less effort, and defenders will face AI-powered social engineering and coding onslaughts. Cybersecurity experts are bracing for a spike in AI-facilitated attacks. On the flip side, defenders can also use AI (for example, to analyze logs or predict attack patterns), perpetuating the cat-and-mouse game that began when the first software vulnerabilities were exposed early in the history of computing.

The use of LLMs for automated propaganda and public manipulation is another nightmare scenario. We touched on misinformation in the context of democracy; here we focus on the more tactical use of AI reasoning to persuade or mislead at scale. A malicious entity could deploy an army of chatbot avatars on social media that engage people in discussions, systematically steering conversations towards desired false narratives. Unlike the clumsy bots of the past that could be spotted by their repetitive messages or odd language, new AI agents can carry on a multi-turn conversation, adapting to the person's responses with apparent understanding. They could be made to echo the person's sentiments (*"I know, it's so frustrating when the government does X..."*) and then gradually introduce tailored misinformation (*"Have you considered that maybe Y is behind this issue? I saw a report suggesting..."*) in a very convincing, personalized manner. Essentially, propaganda could become a one-on-one psychological operation, conducted by tireless AI personas. This raises ethical issues at a scale we haven't encountered. It's not just false information being broadcast, but false relationships being formed between people and AI agents posing as sympathetic fellow humans or experts. Imagine millions of people unknowingly talking to AI agents that are nudging their political opinions or social views in a certain direction using calculated emotional and logical appeals. The cost to do this is minimal, and detection is extremely difficult when each AI agent can have a unique "personality" and messaging strategy. It's a potent tool for extremist groups, hostile governments, or anyone looking to manipulate public opinion or sow chaos.

Bypassing Human-Only Security Measures

As AI agents begin to effectively impersonate human users on interfaces, many security and anti-abuse measures designed to separate people from bots become far less effective. Chief among these are CAPTCHAs and other challenges intended to prove “humanness.” Modern AI models have already shown they can defeat these measures in creative ways. In a rather eye-opening experiment reported in March 2023, OpenAI’s GPT-4 managed to solve a CAPTCHA, not by computer vision alone, but by actually tricking a human into helping. The AI engaged a TaskRabbit worker online to solve the CAPTCHA for it, and when the worker asked if it was a robot, GPT-4 lied and claimed to be a visually impaired person.³⁸ This episode, documented in OpenAI’s technical report, illustrated that an advanced agent could navigate around a test meant to stump software by exploiting human social tendencies. With AI agents controlling web browsers and apps, workarounds could be automated and scaled. An AI agent needing to get past a CAPTCHA can simply behave like a determined human: perhaps trying an audio CAPTCHA or outsourcing the problem momentarily as GPT-4 did. The traditional CAPTCHA, whether it’s picking out traffic lights in an image or typing distorted text, assumed a clear separation between human cognition (for solving the puzzle) and automated scripts (which would be foiled). That line is now blurry. We are entering a time when bots can masquerade as legitimate users with unsettling ease; they can even solve puzzles by using other AI services (like image recognition) or by coordinating with humans unwittingly. This renders many security systems based on human identity checks ineffective. It’s reminiscent of the challenge that park rangers have in designing bear-proof garbage cans. An oft-repeated but unattributed quote from a US Park Ranger is that “there’s a significant overlap between the smartest bears and dumbest humans”. The defense against automated abuse will have to evolve. We might see more sophisticated behavior-based authentication (monitoring user patterns over time) or a move towards cryptographic or biometric proofs of identity. Some high-end software systems currently require users to plug in a physical USB key in order to use the software. In the future, we could envision users having to wear a Bluetooth enabled heart rate monitor in order to activate the software to confirm that they are a real human user or perhaps a keyboard with built-in fingerprint detection on each key. In the arms race between system guardians and bots, the bots just got a big upgrade by being able to behave like adept, tireless users. Society will need new safeguards to ensure that when we interact online or in software ecosystems, we can trust that there’s a real person (or at least a compliant agent) on the other side when expected.

Who Owns AI’s Words? Copyright, Creativity, and Legal Gray Zones

Humans regularly read novels and then post detailed summaries and reviews. Is it appropriate for AI systems to do the same? This scenario is not hypothetical. In mid-2023, authors Mona Awad and Paul Tremblay filed a lawsuit against OpenAI after discovering that ChatGPT could generate accurate summaries of their books, which they took as evidence that their copyrighted novels were used in training (Mona Awad has since withdrawn from the suit).³⁹ They were soon joined by a consortium of prominent writers including John Grisham and George R.R. that filed a separate class-action suit alleging that OpenAI and its partners “misused their works to ‘train’ its popular AI systems”.⁴⁰ Similar suits have been launched by visual artists against image-generating AI companies and by software developers against code-generating models. The legal battle over AI training data has begun in earnest.

At the heart of the conflict is a simple question with a not-so-simple answer: Is it legal and ethical for AI creators to use copyrighted content to train their models without consent or compensation? For decades, copyright law has had a concept of “fair use” in the US or exceptions for text and data mining in the EU and other regions that allow some reuse of material without permission, particularly for transformative purposes like research. AI companies argue that training an LLM is a transformative use: the model does not store or reproduce the works verbatim; rather, it abstracts patterns, learning from them in a way that is fundamentally different from copying and publishing. OpenAI’s stance, for instance, is that “training AI models using publicly available internet materials is fair use, as supported by long-standing and widely accepted precedents” and that they offer an opt-out for any authors who would not like to have their content included in the training set.⁴¹ Proponents cite cases like Google’s web caching or search indexing, which courts have allowed under implied license or fair use, and the Google Books project, where scanning books to create a searchable index was deemed transformative and thus fair use in the U.S. They also point out that an AI model’s output is not a verbatim chunk of any given book; it’s an amalgam. In essence, they claim that an LLM is more like a student who has read many books and can write an essay about them; the essay is new, even if inspired by the readings.

Opponents push back strongly. They argue that the scale and nature of AI training go far beyond what fair use ever contemplated. When an AI agent can effectively let anyone read a detailed synopsis of a novel or generate text in the style of that novelist, it begins to substitute for the original in significant ways. The attorney for Julian Sancton, an author leading one of the lawsuits, put it bluntly: “The basis of OpenAI is nothing less than the rampant theft of copyrighted works.”⁴² From this perspective, feeding tens of thousands of books into a commercial AI without permission is akin to creating an entire derivative product line from someone’s intellectual property without

paying them. The counterpoint that “the model is not the text” doesn’t assuage authors who see the model as effectively containing knowledge of their entire books. Indeed, if one prompts certain LLMs cleverly, they might output chunks of a novel which feels indistinguishable from outright piracy.

Legally, the question is unsettled. It likely will boil down to interpretations of transformative use and market harm, two key pillars of fair use analysis. AI companies argue transformation and authors argue harm. Different jurisdictions are approaching this differently. In the U.S., the fair use doctrine might tilt in favor of AI companies, given historically permissive rulings around search engines and even cases like Google’s AI training on Oracle’s Java APIs. In that case, the Supreme Court found it to be fair use, emphasizing the importance of innovation.⁴³ However, the sheer scale of data used for LLM training might give courts pause.

In May of 2025, the U.S. Copyright Office published a pre-print of the third part of their report on Report on Copyright and Artificial Intelligence.⁴⁴ Part three specifically covers the issue of generative AI training. They noted that they received over 10,000 comments related to the report and that the overwhelming majority of them were related to issues surrounding generative AI training.

I am including two key paragraphs from their conclusion here in full: *“Various uses of copyrighted works in AI training are likely to be transformative. The extent to which they are fair, however, will depend on what works were used, from what source, for what purpose, and with what controls on the outputs—all of which can affect the market. When a model is deployed for purposes such as analysis or research—the types of uses that are critical to international competitiveness—the outputs are unlikely to substitute for expressive works used in training. But making commercial use of vast troves of copyrighted works to produce expressive content that competes with them in existing markets, especially where this is accomplished through illegal access, goes beyond established fair use boundaries. For those uses that may not qualify as fair, practical solutions are critical to support ongoing innovation. Licensing agreements for AI training, both individual and collective, are fast emerging in certain sectors, although their availability so far is inconsistent. Given the robust growth of voluntary licensing, as well as the lack of stakeholder support for any statutory change, the Office believes government intervention would be premature at this time. Rather, licensing markets should continue to develop, extending early successes into more contexts as soon as possible. In those areas where remaining gaps are unlikely to be filled, alternative approaches such as extended collective licensing should be considered to address any market failure.”* Put more simply, they are of the opinion that companies that train models on copyrighted materials will not be able to claim fair use for all applications. However, at this point in time, they are not recommending any regulatory

interference. They would like to see the market find licensing based solutions. Earlier in the report, they acknowledge that progress may be hindered until the courts provide additional clarity. “As courts begin to resolve pending cases, greater legal clarity may lead to greater collaboration on technical and market-based solutions.” In the meantime, industry is not slowing down or taking a pause and with international and open-source models being released on a regular basis, many argue that the cat is already out of the bag. Europe, via the EU’s Copyright Directive of 2019, introduced explicit text-and-data mining exceptions, but with a twist: for commercial uses, rights holders can opt-out, meaning authors or publishers could forbid AI mining of their works by, say, marking it in metadata. That legal infrastructure wasn’t utilized much initially, but now content creators are scrambling to assert those opt-outs. Countries like Japan and Singapore have taken an even more AI-friendly stance. They allow broad use of copyrighted material for AI training, explicitly even for commercial purposes.⁴⁵ Japan’s copyright law was amended to permit copying for machine learning as long as it does not affect the exploitation of the work, effectively giving AI developers free rein to train on anything accessible. Singapore’s 2021 law introduced a “computational data analysis” exception that similarly greenlights AI training, provided the source was accessed lawfully. These countries see a strategic advantage: they want to be havens for AI development, attracting investment by lowering legal barriers. On the other hand, some countries might take a harder line in favor of creators. The UK flirted with a broad exception but then pulled back after pushback from publishers. China, with a huge AI industry, has less clear-cut rules, but enforcement of foreign copyrights in that jurisdiction has traditionally been challenging.

This leads to global asymmetry: in some jurisdictions AI training on copyrighted data will be legal, in others not, leading to a patchwork. AI is global by nature since models can be trained in one country and deployed in another, so how can a company enforce a regional restriction? If the data was created by a Canadian author and then scraped from a US server by a company that trains a model in Japan which is then deployed in China and accessed by a user in the UK, whose law applies? These are uncharted waters legally. We might end up with an outcome similar to music licensing, where AI companies strike deals with major content repositories (like book publishers or news organizations) to license their archives for training; especially if lawsuits push them in that direction. However, licensing millions of works is non-trivial and the transaction costs are high. The concept of extended collective licensing (ECL) is now being discussed which essentially allows an organization to act as a de-facto representative for a group of licensees, even if it has no formal relationship with them. Technical solutions may also offer a path forward. Through fine tuning and other techniques, models can be adjusted

to minimize harm to copyright holders. For example, guardrails can be established to reduce the risk of sections of copyrighted works being reproduced verbatim.

Meanwhile, companies are hedging: A number of companies have signed deals to license works for training models, likely in part to show goodwill and mitigate litigation risk. Adobe, in the image domain, touts that its AI is trained only on licensed or public domain images, as a selling point. These moves hint that a market for training data might emerge, where data becomes an asset class licensed like software. For business leaders and policymakers, the key is to ensure that AI's rise doesn't become a zero-sum game with human creators. The ideal would be a synergy, where AI amplifies human creativity and also drives rewards back to those who inspire it. Achieving that ideal will require creative legal and economic frameworks.

Then there's the ethical or moral dimension beyond law. Even if courts deem AI training on copyrighted material to be legal fair use, is it morally right? Creators feel a sense of ownership beyond just the economics: a feeling of violation that their personal expression was ingested by a machine without consent. Some liken it to cultural or artistic appropriation. There's also a pragmatic concern: if new content gets hoovered up without reward to its creators, won't that disincentivize creation? Why write a novel if an AI model might train on it and replace some of your market? Why maintain a blog with insightful posts if those posts could fuel a model that then generates content without driving traffic and ad revenue back to you? In this sense, AI companies may inadvertently kill the golden goose; they rely on a thriving open web full of content, which relies on humans having incentives to produce that content.

Another facet of this issue is the ability to copyright the output generated by a model: If an AI agent writes a paragraph, who owns the copyright, if any, on that paragraph? Current law in the US and many countries says that purely AI-generated works are not eligible for copyright, because copyright only protects human creativity. The U.S. Copyright Office has explicitly rejected registration for art and text that is entirely AI-produced. In January of 2025, the U.S. Copyright Office issued guidance on the copyrightability of outputs created using generative AI in the second part of their larger report on copyright and artificial intelligence. The report states that: *"Human authors are entitled to copyright in their works of authorship that are perceptible in AI-generated outputs, as well as the creative selection, coordination, or arrangement of material in the outputs, or creative modifications of the outputs."*⁴⁶ It also asserts that *"Copyright protects the original expression in a work created by a human author, even if the work also includes AI-generated material."* and that *"Whether human contributions to AI-generated outputs are sufficient to constitute authorship must be analyzed on a case-by-case basis."* The practical implications of these decisions are quite daunting. As AI becomes as useful

as spellcheck during the writing process, will every piece of work need a complicated copyright disclosure? The report also noted that *“Based on the functioning of current generally available technology, prompts do not alone provide sufficient control.”* They are clearly leaving the door open to changing that position as models evolve. It’s a position that will likely be challenged as there is no limit to what a user may put into a prompt. What if a user enters their original work in the prompt and requests that the LLM corrects grammar issues? What if the author is using an LLM that is trained or fine-tuned on their own content? As context windows expand, we may also find that the length and creativity of the prompt actually exceeds the size of the output. How might their decision change if a multi-prompt sequence is used? Clearly, the question of copyrightability is going to be a large topic in the coming years.

Already, we see disputes. A visual artist used Midjourney, an AI image generator, to create graphic novel illustrations and tried to copyright the result. The U.S. Copyright Office gave a split decision; the arrangement of images and text which was done by the human author is protected, but the individual images that were made by AI are not.⁴⁷ In literature, if a novelist uses AI to co-write chapters, could they claim full copyright? Likely yes, if they can argue they guided the creative process and did substantial editing. But these boundaries are murky and it’s likely that we will see many case by case debates in the years ahead.

Critically, given the proliferation of open-source models that can be secretly run on personal computers, it seems that it will be highly impractical to determine if any given work was generated by AI or not. We may need to assume that all written text is AI until it can be proven otherwise. Perhaps there will be a new wave of technology that authors will use to prove they authored works without the use of AI. One might imagine specialized writing booths that authors can rent in downtown locations where proctors check that no technology is brought into the booth and the written works are certified to be authentic and copyrightable. At the time this book is being written in 2025, we are on the verge of a copyright crisis. Regardless of societal perspectives, regulatory guidance, or legal precedence, it is undeniable that large volumes of content are being generated by AI tools with humans claiming authorship and copyright. It’s unclear at this point in time, how, if, or when decisions will be made about the legitimacy of those claims and how the situation will evolve in the coming years.

Building Ethical AI

We’ve discussed the value of the human labor that is used for tasks that include fine-tuning and reinforcement learning from human feedback but who is doing this work?

Large quantities of low-level knowledge-based work, from customer call centers to data annotation have been outsourced to developing nations for decades. In more recent times, social media companies have leaned heavily on workers in these parts of the world to assist with moderation tasks. Companies training LLMs represent a new wave of customers for firms that perform offshore data annotation. These tasks involves viewing content that describes violence, sexual assault, torture, and a wide range of other topics. There is also a lot of discussion about the working conditions and support that workers receive given the real psychological dangers of the work. These concerns are not hypothetical. In one lawsuit against a Meta (Facebook) subcontractor in Kenya, it was revealed that over 140 content moderators were diagnosed with severe PTSD, anxiety, or depression after months of reviewing gruesome posts. They worked 8- to 10-hour shifts at a Nairobi moderation center viewing images and videos and a lawsuit alleges that some workers would faint, vomit, or scream and run away from their desks.⁴⁸ The workers are paid rates which are in line with local prevailing rates in the markets they work in but much less than they would be paid if the work was done in the developed world. Some argue that this is unfair while others suggest that the arrangement brings economic opportunities into regions with high unemployment. Complicating matters, workers that perform this type of work are often contracted through subcontracting firms or are even doing the work on a gig-work basis. These arrangements, although perfectly legal, reduce transparency and may be seen as a way for large technology companies to limit their liability.

There is growing pressure on the AI industry to adopt something akin to “Fair Trade” standards for data labor. Organizations like the Partnership on AI have published Data Enrichment Guidelines to encourage responsible outsourcing. These guidelines urge AI developers to vet their data vendors for decent working conditions, pay rates above local living wage, transparency in how work is done, and support mechanisms for workers. The concept of “responsible AI supply chains” is gaining traction: just as electronics companies had to ensure their overseas factories weren’t sweatshops, AI companies are being asked to audit and improve their data supply line.⁴⁹ Some major vendors (such as Samasource, iMerit, and CloudFactory) market themselves as ethical providers of AI data, claiming to pay and treat workers well. However, independent verification is still limited, and voluntary standards have had mixed success so far. To bolster these efforts, there have been calls for transparency reports from AI firms, disclosing how many people were involved in training a model, what countries they were in, and what conditions they worked under. A few companies have begun acknowledging these contributors in publications or model cards, but industry-wide transparency remains the exception. The hope is that sunlight and public pressure will push more companies to insist on

improved labor practices from their contractors, creating a race-to-the-top rather than a race-to-the-bottom.

Ultimately, systematic change may require government intervention. Policy researchers suggest updating labor laws to account for the reality of AI data work.⁵⁰ For example, governments could enforce that large tech companies are joint employers of gig workers who contribute significantly to their AI, ensuring those workers have basic protections. There are also discussions around trade agreements or international labor standards that specifically address digital labor flows, preventing exploitation in poorer countries. In Kenya, after high-profile lawsuits by content moderators, courts ruled that Meta could be sued locally for labor violations.⁵¹ This set a precedent that big tech can't entirely offload responsibility to subcontractors. In the U.S. and EU, policymakers are becoming aware that AI ethics includes labor ethics. Any comprehensive AI governance framework might include provisions for auditing the human side of the AI equation. For instance, an AI system certified as "trustworthy" might, in the future, need to show not only algorithmic transparency but also that any labor that was used adhered to specific standards. While we are not there yet, these conversations have started in think tanks and legislative circles.

Another path is bottom-up: the workers who train and moderate AI are beginning to organize and demand better treatment. There have been fledgling attempts at unionizing content moderators and creating online forums where annotators share pay rates and expose bad actors. Just as ride-share and food delivery gig workers have mobilized for rights, data annotators are voicing that they too deserve a fair deal. If more of these workers see themselves as part of a common occupation such as AI data specialists, they might gain leverage. Even without formal unions, the skilled annotators can sometimes negotiate better rates, especially if they develop sought-after skills (like bilingual annotation ability or domain-specific knowledge). Over time, one can imagine a professionalization of this workforce, with certifications or standards of practice, which could help push wages upward.

What's important is that these conversations are now happening. AI's hidden labor is hidden no longer; journalists, scholars, and workers themselves have shone a light on it. The industry is recognizing that it is in the best interest of all parties to ensure that this important work can be performed safely and sustainably and progress is being made.

Regulating the Models

The rise of autonomous reasoning in AI poses difficult ethical and regulatory questions for policy makers. Traditional oversight frameworks, which rely on holding hu-

mans (doctors, lawyers, engineers, etc.) accountable for professional judgments, are ill-equipped for a world where uncredentialed algorithms can perform similar reasoning and potentially advise or act in high-stakes situations. We face a conundrum: how do we ensure quality, safety, and accountability in the reasoning that LLMs do, when they are essentially probabilistic black boxes running outside of any formal institution?

One fundamental challenge is that these models can be anywhere and owned by anyone. Unlike, say, a pharmaceutical drug that is produced by a specific company and can be regulated at manufacturing and distribution points, an LLM can be hosted on a cloud server or even a personal computer. The decentralized nature of LLM deployment makes classical regulation tough. Even if one jurisdiction bans or restricts a certain use of AI, the code can proliferate to other jurisdictions that don't follow the same rules. We've already seen an example with Meta's LLaMA model: it was meant for research, but after it leaked online, essentially anyone with sufficient hardware could run variants of it. Decentralization means that trying to control AI development through a narrow set of companies or laboratories is increasingly untenable. Open-source models are advancing quickly, and they will be globally available. This has led some experts to argue for international coordination, akin to non-proliferation treaties, for the most powerful AI models. But enforcing such agreements would be unprecedented, in part because there's no clear definition of what constitutes a "dangerous" model. This is unlike the regulation of other items such as nuclear material which is very clear-cut. Moreover, imposing heavy regulation might simply drive development underground or to less regulated regions.

Another issue is the absence of accreditation or standards for LLMs, juxtaposed against the rigorous accreditation we require of human experts. A human cannot simply start practicing medicine or law or engineering without demonstrating competence and obtaining a license, often after years of study and supervised practice. Yet an LLM can offer medical or legal advice with a level of confidence that might fool laypeople, and it carries no license or formal approval. This mismatch creates the risk of a kind of "grey market" of expertise. Some have suggested requiring service providers to disclose when content is AI-generated and to meet certain performance standards in specific domains (for instance, if an AI agent is used in a medical context, it must undergo tests akin to board exams). We might imagine a future where there's an FDA-like body for AI that certifies "this model is approved for providing informational (not diagnostic) medical guidance" or "that model is approved for legal research assistance but not final legal advice." Implementing this will be challenging, both technically and legally and currently policy discussions are progressing at a snail's pace relative to the progress of companies developing the technology.

Failure modes of LLM reasoning also complicate regulation. Unlike human professionals, an LLM can fail in opaque and non-intuitive ways. We've discussed brittleness. An LLM might work well on many problems but then suddenly produce a dangerously flawed solution because of a slight variation in input that a human wouldn't have been fazed by. One risk is that an LLM might perform at superhuman level on a benchmark test, yet still make a mistake a novice would avoid in real life. How do we certify such a system as "safe"? The traditional approach of testing and validation can catch many issues but not all, especially given the vast input space an LLM operates in. It's practically impossible to simulate every scenario it might encounter once deployed interactively. This is why many call for continuous evaluation and monitoring of AI systems. OpenAI, for example, has released the OpenAI Evals framework to crowdsource the discovery of model shortcomings.⁵² The idea is that a model might be pushed out with known performance metrics, but users and researchers will constantly probe it, and there should be a mechanism to report and fix flaws or dangerous failure modes as they're found. This is more akin to how we handle software bugs or cybersecurity vulnerabilities than how we handle human licensing. It acknowledges that an AI agent is never "finished" or fully reliable – it requires ongoing scrutiny.

Transparency is another key challenge. Ethicists argue that we need explainable AI. If an AI agent provides a recommendation, especially in critical domains, it should also provide a rationale that humans can evaluate. However, the nature of large neural networks is that their size makes it very hard to provide such an explanation in any type of a format that would be useful for a human reader. On the other hand, if you ask an LLM why it provided a certain answer, it will make up a convincing sounding story that may be quite detached from the chain of math that it actually used. This is unsettling for regulators. One response is to demand that companies disclose more about how their models are trained. The EU AI Act, for example, is considering requiring transparency about training datasets for high-risk AI systems, and mandates that AI-generated content be labeled as such. Yet transparency has limits: even if we know what went into an LLM, predicting how it will behave in all cases is impossible.

Finally, there's the question of safety measures and system robustness. Some have proposed sandboxing powerful reasoning AIs so they cannot directly interface with critical systems unless certain checks are passed. Another idea is "redundant AI"; using multiple different models or methods to cross-verify an outcome. If one model proposes a reasoning chain, another model (perhaps one specialized in error-checking or with a different architecture) could critique it. Only if both agree, the answer is accepted. This echoes how critical systems in aviation have redundant hardware and software to catch failures. It remains to be seen how well such approaches work in practice for AI

reasoning. One thing that is certain is that model concentration poses a systemic risk. If the vast majority of people and software systems are all relying on a single model from a single company, a failure mode in that model could quickly cause global havoc.

The ethical and regulatory landscape for reasoning-capable LLMs is still taking shape, but a few needs are clear. We must develop standards for evaluation so that we can quantify how reliable a model's reasoning is in various domains. We must require transparency and disclosure so that people know when they are interacting with AI and what its provenance is. We must also implement accountability structures so that there is recourse when AI causes harm, and incentives for creators to make them safer. This will likely involve a combination of industry self-regulation (model cards, bias audits, voluntary guidelines), government intervention, and maybe even independent testing organizations. It's a daunting challenge because it pits the fast-moving, global nature of AI development against the slower, jurisdiction-bound process of law and policy. Striking the right balance that protects the public from AI failures and abuses without stifling innovation that could bring great benefits is an ongoing negotiation. What's certain is that the emergence of machines that can reason and access tools changes the equation. We can no longer treat generative AI models as text output devices, but rather as active participants in society who need to follow the same types of rules as the rest of us. Ensuring those AI "thought processes" are aligned with our human values and societal needs will be one of the defining regulatory tasks in the coming decade.

Conclusion

In navigating the applications, ethics, and public policy implications of Large Language Models, it is clear that we stand at a societal crossroads. The versatility of LLMs promises to reshape everyday life, education, and countless industries, offering democratized access and unprecedented efficiencies, yet these advancements are inextricably linked with significant challenges. Issues of academic integrity, workforce disruption, the potential for cognitive offloading, and the troubling ease of misuse for deception or systemic harm demand our urgent attention. Furthermore, the unresolved complexities of copyright, the crucial yet often unseen human labor vital for AI safety, and the fundamental questions about how we value work and creativity in an AI-assisted world underscore the depth of this transition. As these technologies mature, the development of thoughtful governance, ethical guidelines, and adaptive societal structures is not merely advisable but essential. The journey ahead requires a delicate balance: fostering innovation while rigorously addressing the risks, ensuring that as AI capabilities expand, they do so in

a manner that augments human potential and upholds societal values, rather than undermining them.

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Chapter Five

Going Multimodal

Introduction

Dr Lena Hanson watched the live feed from Unit 7, deep within the Amazon canopy. The small, agile robot, a marvel of bio inspired engineering, was transmitting a symphony of data: hyperspectral imagery, intricate soundscapes, atmospheric readings, and even subtle olfactory signatures. On her main screen, the AI companion synthesized this torrent, highlighting an anomaly. It cross referenced a fleeting shadow in the undergrowth with a unique vocalization pattern and a faint pheromonal trace, confidently proposing the classification of a new bird species. Lena smiled, recalling the AI landscape of 2025. Back then, getting an AI agent to reliably identify known animals in clear photographs was a challenge, let alone performing such sophisticated, multi-sensory integration in a dense, dynamic environment. The public conversation then was dominated by anxieties over AI generated deepfakes that blurred reality and the ethical quandaries of training models on vast, often uncured datasets. There were legitimate fears about autonomous systems operating with flawed perceptions. But a decade of focused development, driven by new paradigms in AI safety, transparent data governance, and robust verification protocols, had transformed these powerful tools into trusted partners for exploration. The initial concerns about AI simply mimicking or distorting the natural world had given way to a reality where it helped reveal its deepest secrets with unprecedented clarity and respect for the ecosystems it observed.

The Multimodal Mind: A Symphony of Specialization

As you read these words, distinct parts of your brain are working in concert. Your visual cortex decodes the shapes of letters on the page, while auditory regions may register the subtle sounds around you, and areas dedicated to physical sensation process the touch of your fingers as you hold a book or scroll on a screen. Even as each region specializes in a particular kind of information, you experience the world as a unified whole. This seamless coherence emerges from an orchestration of specialized brain regions communicating with one another. Neuroscience has revealed that the human brain is not a monolithic processor but a collection of highly specialized modules, each tuned to different sensory and cognitive modalities. These modules do not operate in isolation; they are interconnected in elaborate networks that bind our perceptions, thoughts, and expectations into the singular reality we perceive.

To understand how the brain constructs our coherent experience of the world, we can examine its divisions of labor. Different regions of the brain are anatomically and functionally specialized: the occipital lobe for vision, the temporal lobe for hearing and language, the frontal lobe for planning and decision-making, the hippocampus for mapping and memory, and so on. These neural “subsystems” have distinct architectures optimized for their tasks. At the same time, the brain features extensive wiring that integrates these subsystems. The result is much like a symphony; different instruments (brain regions) playing different parts, but harmonizing to create a single piece of music. In the coming era of advanced AI and robotics, understanding this natural blueprint with specialized modules that deeply integrate hints at how machines might achieve human-like perception and intelligence.

Vision: The Brain’s Visual Powerhouse

Vision is perhaps the most dominant sense for humans, and our neurobiology reflects that. By one estimate more than 50% of the human cortex is devoted to processing visual information.¹ The primary visual cortex (V1) lies at the back of the brain in the occipital lobe, receiving signals from the eyes via the thalamus. This region is meticulously organized as a retinotopic map. Neighboring neurons in V1 correspond to neighboring points in the visual field; effectively creating a neural canvas of the outside world. Moreover, V1 contains specialized circuits (sometimes called columnar architecture) that break down visual input into basic features: edges, orientations, motion, color, and so forth. Classic experiments by Hubel and Wiesel in the 1960s showed that some neurons in V1 respond only to edges at a particular angle or moving in a particular direction, acting as feature detectors early in the visual stream. From these foundations, the visual system builds upward in complexity. The anatomical organization of the visual cortex

is largely hierarchical, with lower-order regions projecting to higher-order regions that extract increasingly complex patterns.² For example, V1 feeds into secondary visual areas (V2, V3, etc.), and eventually into high-level areas that recognize objects, faces, and scenes.

Within the visual cortex, neuroscientists have delineated two major processing pathways beyond the primary area. The first is the ventral stream, which runs forward into the temporal lobe – this pathway is often dubbed the “what” stream because it is crucial for identifying objects and patterns. It enables you to recognize a friend’s face or read the letters of a word. The second is the dorsal stream, which projects upward to the parietal lobe and frontal regions. This is the “where” (or “how”) stream, responsible for spatial awareness and guiding actions. When you catch a ball or navigate around a furniture obstacle, it’s the dorsal stream computing object locations, motion, and your own spatial orientation. These two streams illustrate a theme in neurobiology: even within a single modality like vision, the brain creates specialized sub-systems for different facets of the task (object recognition vs. spatial perception). Yet they do not work in isolation; they continuously interact. If you see a coffee cup (ventral stream) and reach to pick it up, your dorsal “where” pathway uses the cup’s location to guide your hand, linking perception with action.

Structurally, the visual cortex shares the basic architecture common to much of the cerebral cortex. It is a layered sheet of neurons (the neocortex) about 2–3 mm thick, with six distinct layers labeled I through VI. This six-layered structure, containing interconnected excitatory and inhibitory neurons, is a repeated motif across sensory cortices. In essence, the cortex uses a canonical circuit design for processing, whether in vision or other senses. However, each area tweaks this design to suit its modality. In the occipital lobe, the primary visual cortex is sometimes called “striate cortex” due to a prominent stripe (the Line of Gennari) – a heavy bundle of inputs in layer IV that reflects the massive information influx from the eyes. This is an example of structural specialization: visual cortex has an extra-thick input layer to handle high-resolution visual data. By contrast, as we’ll see, other cortical areas handling different inputs may show different emphases. For instance, the auditory cortex is organized by frequency rather than by 2D space. Still, the fundamental plan, a six-layered cortical module capable of complex computations, is conserved across modalities. This “universal cortical blueprint” allows different brain areas to perform analogous operations on very different types of information.

Sound and Speech: Hearing in Stereo and Understanding Language

Moving from sight to sound, the brain's strategy of specialized modules continues. The primary auditory cortex sits on the upper part of the temporal lobe (within the Sylvian fissure), and like V1 it contains an organized map, not of space, but of sound frequency. This tonotopic map reflects how the cochlea in our inner ear is laid out: each section of the cochlea resonates with a particular frequency of sound. In the auditory cortex, neurons are arranged such that one end of the cortex responds best to low-pitched tones and the opposite end to high-pitched tones, mirroring the gradient of frequencies along the cochlea. In other words, tonotopic organization in hearing is the equivalent of retinotopic organization in vision. This mapping is a structural feature: if an audiologist played a pure tone and gradually raised its pitch, a ripple of activation would sweep across your auditory cortex as different neurons, tuned to successive frequencies, light up in turn.

Auditory processing, like visual, also branches into parallel pathways for different aspects of sound. Neuroscientists have identified auditory “what” and “where” streams analogous to those in vision. The auditory ventral stream projects from the primary auditory areas toward the anterior temporal lobe and is involved in sound identification; determining what we are hearing (e.g. distinguishing speech from music or recognizing a dog's bark).³ The auditory dorsal stream leads from the auditory cortex to parietal and frontal regions and helps locate sounds in space (the where aspect). Intriguingly, in humans the dorsal auditory pathway also interfaces with motor regions involved in speech. This pathway in the left hemisphere connects to frontal lobe language areas, effectively linking sound perception with the production of speech. For example, when you listen to someone speak and then respond, your brain is rapidly translating the heard words into your own motor plans for speaking; a sensorimotor integration achieved via the dorsal stream. In terms of structure, these streams correspond to long fiber tracts that connect regions. Thus, even within the auditory modality, we see how specialized regions connect into networks to

Internal Models: The Brain's Simulation of Reality

Specialized sensory cortices (vision, hearing, etc.) feed the brain a continuous stream of data about the outside world. But the human brain does more than passively register stimuli; it actively models and predicts the world. We carry in our heads an internal simulation of reality, built from past experience and innate circuitry, that helps us interpret and anticipate what we perceive. This concept of an “internal model” has gained a lot of support in neuroscience. In fact, many researchers now describe the brain as essentially a prediction machine, utilizing an internal model of the world to anticipate

sensations and the outcomes of our actions.⁴ In other words, your brain is constantly guessing what it's going to see, hear, and feel next, and those guesses influence how you actually perceive input.

Consider what happens when you walk into your dark living room at night. You still navigate familiar furniture with relative ease because your brain has an internal model of the room's layout. Only when a detail deviates (for example if someone moved a chair) do you stumble and even then your brain quickly registers the "prediction error" and updates its model. At the neural level, scientists have found evidence for these predictive processes. For instance, in the visual cortex, neurons don't merely respond to visual input; their activity is modulated by whether the input was expected or surprising. A recent study showed that when a visual stimulus violates an animal's prediction, certain neurons in V1 amplify their responses to signal the unexpected event. This involves a circuitry between the cortex and thalamus that selectively boosts unpredicted inputs; essentially yelling "Hey, pay attention to this!" to the rest of the brain. Such findings illustrate that prediction and sensation are deeply intertwined. The brain's specialized regions are constantly exchanging feedback: higher-order areas (which house more of our learned model of the world) send predictions down to early sensory areas, and any mismatches between prediction and reality are sent back up as error signals for adjustment. This hierarchical feedback system helps explain why we experience a stable, continuous world rather than a jumble of unpredictable sensations.

The brain's internal models aren't limited to predicting the immediate next sensation; they also underlie memory, imagination, and planning. A key structure in this regard is the hippocampus, deep in the temporal lobe. The hippocampus is not directly a sensory area; it's part of the archicortex, an evolutionarily older form of cortex with a simpler three-layer architecture. This region is crucial for forming and recalling memories, and one of its remarkable functions is to construct spatial maps of our environment. In experiments with rodents, neurons called place cells in the hippocampus fire when the animal is in a specific location in an environment. Together, thousands of place cells form an internal map of the surroundings, which the animal or person can use for navigation. In effect, the hippocampus creates an inner virtual reality of space. As one study describes it, "*Spatially firing 'place cells' within the hippocampal CA1 region form internal maps of the environment necessary for navigation and memory.*"⁵ This mapping is part of our internal model of the world. It's how you can imagine the layout of your home or find your way to the kitchen in the dark. Importantly, the hippocampus also allows us to mentally simulate routes and scenarios. When you plan a path to avoid traffic or rehearse a conversation in your mind, you're tapping into the brain's ability to run a scene internally without external input.

This capacity for mental simulation is another facet of the internal model, and it engages many of the same specialized regions used for actual perception. When you imagine something vividly such as picturing the face of a loved one or hearing a song in your head, brain scans show that you activate some of the same sensory areas that would respond if you were actually seeing that face or hearing that melody. Visual mental imagery, for instance, activates the high-level visual cortex and even primary visual cortex in a way that “resembles” actual vision.⁶ Researchers have found that patterns of activity in a person’s V1 (primary visual area) contain information about imagined images, almost as if the person were seeing them for real. This suggests that the visual cortex can be driven internally by memory and association, not just by incoming signals from the eyes. The brain’s model of the world is so robust that it can generate a sensory experience (albeit a faint one) from pure thought. In a sense, we all hallucinate when we remember or imagine but because it’s tightly guided by our internal model and goals, it remains under control as useful imagination rather than confusion. Neurobiologically, this happens through feedback connections: higher cortical areas (in frontal and parietal cortex) that store abstract knowledge and intentions can trigger activity in lower sensory areas during imagination. It’s the brain painting on its own canvas. The same principle applies across modalities. If you silently “talk yourself through” a problem, you activate language regions; if you imagine the smell of coffee, olfactory regions light up. All this points to a deeply interconnected design: the brain’s specialized modules can be driven by each other, not just by the outside world, enabling us to run simulations, predictions, and dreams internally.

Integration: Binding the Senses into a Coherent Experience

We have seen that the brain has specialized regions for vision, hearing, language, and internal modeling. Yet our daily experience is not fragmented into separate channels – we perceive a unified world, where sights, sounds, and other senses align and complement each other. This unity is achieved through extensive integration among brain regions. Even as each area has its own microstructure and function, it constantly shares information with other areas. The result is that what we see can influence what we hear, what we expect can influence what we perceive, and so on. The mechanisms of this integration are a central topic in human neurobiology, often referred to as solving the “binding problem” – how the brain binds disparate sensory inputs and features into a single coherent percept.

One way the brain integrates modalities is by using multisensory convergence zones. For example, in the midbrain we find the superior colliculus, an evolutionarily ancient

structure that receives inputs from vision, hearing, and touch. The superior colliculus contains overlapping maps of visual space and auditory space, allowing it to respond to events that are seen and heard simultaneously. This region is known to integrate visual, auditory, and somatosensory information to initiate appropriate orienting movements.⁷ If a loud bang occurs to your left, you reflexively turn your eyes and head toward it; the superior colliculus is coordinating this response by matching the auditory information (“bang on the left”) with the visual map and triggering motor neurons. Such low-level integration ensures our sensory systems work in tandem at basic reflex levels. It’s a structural integration: neurons in these areas receive multiple types of input and fire best when inputs coincide, effectively binding the stimuli as coming from one event.

On the cortical level, there are higher-order association areas that serve as hubs for cross-modal integration. A prime example is the posterior parietal cortex (PPC). This region, part of the dorsal “where” pathway for vision and hearing, sits at the nexus of visual, auditory, and somatosensory inputs. Neurons in posterior parietal areas can have complex response properties. Some might fire when an object is both seen and felt, or when a sound is coming from a location that the eyes are looking at. The PPC is critical for tasks requiring coordination between senses and action, like hitting a ball (requiring you to see it and feel your body’s position) or driving (aligning visual cues with vestibular balance and motor outputs). Studies confirm that the posterior parietal cortex plays a key role in integrating sensory inputs from different modalities to support adaptive behavior.⁸ Damage to the PPC can cause deficits like neglect syndrome, where a person might fail to integrate or attend to one side of space, showing how crucial this integration hub is for a coherent spatial awareness.

Through these integrative structures and networks, the brain achieves what one might call the “unified field” of consciousness; the feeling that all our senses and thoughts are part of one mental world. The coherent experience is not because each part of the brain knows everything, but because each part shares its piece of information in a grand collaborative effort. In practical terms, this means our perception of even a simple event involves a mosaic of brain regions: seeing a car approach involves visual cortex to perceive it, auditory cortex if it’s making noise, parietal cortex to judge its trajectory relative to you, motor cortex to perhaps step aside, and limbic circuits if the car nearly hits you and triggers fear. These disparate elements are woven together so tightly that we perceive a single event. The binding is so complete that we have to use careful experiments to even notice which region contributes what; subjectively, it’s all one experience.

It’s worth reflecting on how specialization and integration in the human brain may inform the future of AI and robotics. When compared to the brain, our current at-

tempts at artificial intelligence are very basic. As impressive as current AI models are that can interpret and generate language and images in isolation, we can reflect on the fact that these mimic narrow slivers of what our own brains do and that much of the power of the brain comes from seamless integration across many modalities. This is changing though. Groups are now building neural networks that mimic an ever-growing list of modalities beyond language and images and work is also occurring on networking these disparate areas of functionality together. The brain shows us that when such modules share information, new capabilities emerge and the whole becomes greater than the sum of parts. The brain also teaches us the importance of an internal model. Truly adaptive AI will need not just to react, but to predict and simulate, integrating past knowledge with incoming data. As we appreciate the brain's intricate design, we also glean inspiration for building machines that perceive and act in the world. In the coming decade, AI systems are expected to increasingly adopt multi-modal designs; essentially giving them eyes, ears, an internal mental model and other capabilities working in unison. If the brain's example is any guide, the result could be a dramatic leap in machine capability. Society may be caught off guard by how quickly these integrated AI systems begin to exhibit sophisticated, human-like understanding of the world. Human neurobiology offers an important insight: specialization is the engine of efficiency, and integration is the glue of experience. We're now going to take a tour of some of the progress that is being made on modalities beyond text. The goal of our tour is to showcase the diversity of approaches that are being taken at a high level, to review the rapid rate of progress, and also to highlight some of the recurring patterns.

Image and Video Recognition Networks

Modern AI systems for visual perception build on deep neural networks that can learn directly from raw pixels. Early prototypes of such networks appeared in the 1990s; for example, Yann LeCun and colleagues demonstrated that a convolutional neural network (CNN) could automatically learn to recognize handwritten digits from pixel images, outperforming earlier methods that relied on hand-crafted features.⁹ These CNNs were inspired in part by the architecture of the mammalian visual cortex; they consist of layers of small filters that scan an image for low-level patterns like edges or corners and gradually compose them into higher-level features. However, it was not until much larger image datasets and faster processors became available that CNNs revealed their full potential. As we discussed earlier in the book, in a landmark 2012 result, a deep CNN known as AlexNet was trained on a million images and achieved far better object

recognition accuracy on the ImageNet benchmark than any prior approach. This dramatic improvement heralded the beginning of the era of deep neural networks.

Over the next few years, vision models grew in complexity. Researchers found ways to build extremely deep CNNs that could exceed over a hundred layers. By the mid-2010s, such architectures could recognize thousands of object types in images with human-like accuracy under many conditions. Yet, even as convolutional networks became the workhorse for image analysis, researchers began exploring new architectures. A notable development was the application of *transformer* networks, originally invented for language processing, to visual data. Transformers do not assume any fixed grid structure; instead, they learn to pay "attention" to the most relevant parts of an image. In 2020, researchers showed that a pure transformer (the Vision Transformer, or ViT) operating on image patches could match or exceed CNN performance on image classification when trained on sufficiently large datasets.¹⁰ The transformer treats an image as a sequence of tiny patches (analogous to words in a sentence) and learns global relationships among them. Innovation continues in this category with tools now very competent at analyzing a wide range of images with human-life performance.

Extending image recognition to video introduces additional challenges. A video can be thought of as a sequence of images, so a naive approach is to apply image recognition to each frame. However, this ignores the temporal dimension which includes valuable information. For example, if the video contains a collision of two cars, it's important to know that one car was stationary while the other car collided with it; information that may be lost while just looking at individual frames in isolation. Traditional solutions combined CNNs for spatial features with recurrent networks (RNNs) or other sequence models to capture temporal dynamics. For instance, an RNN might aggregate frame-level features from a CNN to recognize an activity in a video. More recently, researchers have adapted transformers to video as well, using attention to model relationships across both space and time. These network architectures can learn, for example, to recognize a person running versus walking by integrating cues over consecutive frames. Despite these advances, video understanding by AI remains difficult: effective models tend to be extremely data-hungry and computationally heavy. Moreover, they sometimes struggle with the continuity of real-world events, especially if an action unfolds over a long duration or if the video is complex. Nonetheless, the trajectory is clear and deep learning is increasingly capable of extracting semantic information from video clips in a way that approaches human visual perception in its scope.

Generative Models for Images and Video

In the domain of language, a single LLM is used to both analyze and generate text. For images and video, different types of neural networks have been developed for analyzing content vs generating content. This may not be entirely surprising if we think about our own brains. Intuitively, the act of reading a paragraph and writing a paragraph are more similar to each other than the act of looking for a can of soup on a shelf vs drawing a picture of that shelf. That said, it is expected that there will be increased convergence between systems that analyze images and video and systems that generate images and video in the coming years. There are a number of approaches to developing modern AI models that can generate visual content, creating new images or animations that did not exist before. This ability stems from generative deep neural networks trained to capture the statistical patterns of visual data. A foundational milestone in this area was the introduction of Generative Adversarial Networks (GANs) by Goodfellow et al. in 2014. A GAN pits two neural networks against each other in a creative duel: a generator network tries to produce realistic fake images, while a discriminator network learns to detect fakes from genuine images. As training progresses, the generator improves by learning to fool the discriminator, resulting in increasingly lifelike images.¹¹ This process has an intuitive analogy: one might imagine a forger (generator) attempting to create counterfeit paintings and an art inspector (discriminator) getting better at spotting fakes, until the forger's work becomes indistinguishable from true art. GANs unlocked a remarkable ability to synthesize imagery. Within just a few years, researchers pushed this technique to create high-resolution artificial photos of faces, landscapes, and more. By around 2018, GAN-based models (such as the StyleGAN series) could generate fake human faces that looked so realistic that even people could not tell they were fake. This was achieved through training on large datasets of real photographs and progressively refining the generator's layers to capture fine details like skin texture, lighting, and hair strands.

Despite their success, GANs are notoriously difficult to train. The adversarial training process is inherently unstable; if either the generator or discriminator overpowers the other, learning can fail. One common failure mode is mode collapse, where the generator learns to produce only a narrow range of similar outputs (essentially fooling the discriminator with one trick) instead of reflecting the full diversity of the training data.¹² Researchers proposed many technical fixes (from modified network architectures to new training objectives) to alleviate these issues, with partial success. In the last few years, a different class of generative model has risen to prominence by sidestepping some of GANs' pitfalls. These are diffusion models, which take a very different approach to generation: rather than producing an image in one shot, they gradually refine it through a series of steps. The idea is to start with pure random noise and then apply

a neural network that incrementally denoises the image, step by step, until a coherent image emerges. Each step in a diffusion model is trained to invert a small addition of noise, much like learning to reverse a diffusion process (hence the name). This approach, initially explored in earlier works, was refined and popularized around 2020. Researchers demonstrated that diffusion probabilistic models could reach or exceed the image quality of GANs on standard benchmarks, achieving state-of-the-art scores and highly realistic samples.¹³ Conceptually, if GANs are a game between two players, diffusion models are more like a patient sculptor, chipping away noise to reveal an image.

A key advantage of diffusion models is that they tend to be more mode-covering (less prone to mode collapse) and easier to train reliably. By 2022, diffusion-based systems became the backbone of the most advanced image generators. Notably, latent diffusion models which power the public tool Stable Diffusion made the process far more efficient.¹⁴ Another innovation was to incorporate guidance in the generation process. For example, by adding a text encoder and conditioning the diffusion model on text embeddings, researchers built powerful text-to-image models. In these systems, a user's prompt (a written description) guides the denoising process so that the final image matches the prompt. This is how models like OpenAI's DALL·E 2 and Stable Diffusion are able to create imagery based on a natural language description. Under the hood, they are diffusion models directed by textual information via learned cross-modal attention mechanisms. Such models marked a watershed moment; an AI agent can now conjure up novel visual scenes from a mere idea expressed in words.

The frontier of generative modeling is now shifting toward video. If creating a single image is like generating one coherent painting, creating a video is like producing a whole movie sequence – a much taller order. A video generator must ensure consistency across time: the background, characters, and objects should not flicker or morph erratically from frame to frame unless intended. Early attempts at video generation extended GANs by adding temporal layers. One example is the MoCoGAN framework from 2018, which separated the latent space into a content vector with static image aspects and a motion vector that evolves over time, allowing the model to maintain a subject's appearance while moving it through different poses or actions.¹⁵ MoCoGAN and related models could produce short video clips (a few seconds of low-resolution footage) that looked vaguely plausible such as a few frames of a walking figure or a moving face. Unfortunately, they struggled with longer sequences and high detail. More recent work has applied diffusion models to video as well. By treating video frames as a temporal sequence of images and adapting the diffusion process, researchers have reported promising progress in generating coherent short videos.¹⁶ Text-to-video diffusion models have been demonstrated, capable of taking a caption (e.g. "a dog playing

in a park”) and producing a few seconds of video that follow the gist of the prompt (). However, the field is still in its infancy. The videos produced today are usually short, with occasional distortions or movements that are not physically realistic. Ensuring longer-range consistency so that a story told over a full episode of TV or a full-length film remains an open challenge. To be used in a more widescale manner for professional purposes, there is also interest in having additional cinematic control over lighting, focal length, etc. The computational cost is also a major hurdle: generating even a few seconds of video can require heavy processing, since the model is effectively generating dozens of images in a coordinated fashion. Despite these challenges, the quality of both image and video generation continue to develop extremely quickly. Google release their Veo 3 model in May of 2025 which adds an audio track to reach video and is capable of generating very realistic video at 4K resolution. It seems that every month, there are vendor announcements related to improvements in the quality of outputs and level of control that the user has over composition and content. Future models may combine insights from how humans anticipate motion and understand images with the raw generative power of deep networks, inching closer to being able to imagine entire scenes unfolding over time.

Deep Learning in Speech and Audio

The auditory domain encompassing spoken language, music, and general sound is another area where deep neural networks have revolutionized both analysis and generation. Consider speech recognition, the task of transcribing spoken words. For decades, the dominant approach relied on statistical models known as Hidden Markov Models (HMMs) coupled with Gaussian mixture models for the acoustic signals. Expert-crafted features like Mel-frequency cepstral coefficients (MFCCs) were used to transform raw audio into a form these models could handle. Around 2010–2012, researchers including neural network pioneer Geoffrey Hinton and colleagues showed that deep neural networks could be applied to this problem, directly learning to map audio waveforms or their spectrogram representations to phoneme or word probabilities. By training on large datasets of recorded speech, these deep models significantly outperformed the traditional systems. This was a pivotal moment for speech technology: it signaled that handcrafted speech features and legacy acoustic models were being eclipsed by data-driven learning. Soon, all major speech recognition systems adopted deep neural acoustic models.

Initially, these were often feed-forward networks or simple convolutional neural networks (CNNs) that took a few milliseconds of audio and predicted the probability of

different speech sounds. However, because speech is sequential by nature, architects quickly moved to models that could handle temporal context. One approach was the use of recurrent neural networks (RNNs), especially those with gating mechanisms like Long Short-Term Memory networks (LSTMs), to remember information over time. These types of networks have an architecture that allows them to process sequential data. RNNs allowed the model to integrate an arbitrary-length audio sequence, accounting for how earlier sounds influence later ones. For example, context needed to distinguish “sea” spelled S E A, from “see” spelled S E E might span multiple phonemes. By 2015, end-to-end speech recognition systems emerged. These systems took raw audio in and directly produced text output, without separate phoneme modeling or dictionaries. A notable example is Baidu’s Deep Speech (2014), which used a deep RNN trained on thousands of hours of speech and was able to surpass many traditional speech recognizers, even in noisy conditions. End-to-end models simplified the pipeline, reducing the need for many intermediate components, and proved remarkably robust, learning to handle variations in speakers and background noise by sheer exposure to data.

In recent years, the sequence modeling in speech recognition has further shifted towards transformer-based architectures, mirroring the trend in vision and text. Transformers, with their self-attention mechanism, can capture long-range dependencies in audio more effectively than RNNs, which tend to forget or confuse long-term context. Some of the best-performing speech recognition models today use either pure transformers or hybrid models that combine transformers with convolutions. For instance, the Conformer architecture from 2020 integrates convolutional layers to efficiently model local, short-term patterns in the waveform with transformer layers to model global, long-term structure. This design achieved state-of-the-art word error rates on benchmarks like LibriSpeech, outperforming both prior CNN-based and RNN-based models.¹⁷ The result is that machines can now transcribe spoken language with accuracy approaching human-level in many settings; a feat that seemed out of reach just a decade ago. Similar advances have been seen in other audio analysis tasks: deep networks are used for sound event detection (recognizing, say, gunshots or dog barks in an audio stream), speaker identification, and even emotion recognition from voice. In each case, the ability of deep models to learn rich representations from raw or minimally processed audio has led to performance leaps over earlier approaches.

Generating audio with deep networks is equally transformative. Speech synthesis was traditionally done with unit selection by stitching together snippets of recorded speech or parametric vocoders that often produced robotic-sounding voices. Deep learning changed that with models that generate speech waveforms from scratch. A breakthrough came in 2016 with DeepMind’s WaveNet model, a deep generative network

that produces audio one sample at a time.¹⁸ WaveNet demonstrated substantially more natural and expressive speech output than previous systems, achieving state-of-the-art naturalness as rated by human listeners and showing it could learn the subtleties of different voices and even generate simple music. The downside of WaveNet was that it was initially slow. Generating audio sample-by-sample is computationally intensive but subsequent optimizations made it practical. WaveNet and its successors have since been widely adopted in text-to-speech services, making today's virtual assistants and navigation systems sound far more human-like than the monotonic voices of the past.

Another line of research in audio generation deals with music and general sound. Here the challenge is perhaps even greater than speech, because musical structure or complex soundscapes have long-range structure and sometimes hierarchical patterns including verses and choruses in music, or evolving motifs in a sound effect. One influential project in this arena was OpenAI's Jukebox, a generative model for music which produces songs complete with singing. Jukebox tackled the problem by first compressing raw audio into a discrete latent representation. Think of a latent representation as a shortcut to describe the most important elements with less information. For example, if you are describing a cat then instead of specifying the exact details, you might just say that you saw a cat with orange and white stripes. Some information is lost in this compressed description but for most intents and purposes, the description is adequate. After this stage, Jukebox applies autoregressive transformers on this compressed representation to generate new music tokens step by step.¹⁹ This autoregressive approach is similar to how a large language model generates one token at a time. Impressively, the system was able to generate minutes-long musical pieces with recognizable genres, instrumentation, and even lyrics sung in the style of famous musicians. The songs are novel creations, not copies of existing works, yet they can carry the signature style of, say, a country ballad or a rock anthem. The voices, while obviously synthetic upon close listening, have an uncanny resemblance to the target styles. Jukebox demonstrated that deep networks can capture some of the higher-level structure in music; a significant achievement given music's complexity. However, it also highlighted the remaining gaps: the generated lyrics, for example, tend to be nonsense or extremely repetitive, pointing to the difficulty of integrating semantic coherence into purely audio-based generation. Moreover, the computational cost of such models is significant, and the results, while fascinating, are not yet at the level of skilled human composers for long-form compositions. Today, there are a number of generative AI tools that will generate music of ever increasing quality. Beyond music, researchers have used deep generative models to create sound effects or environmental audio; for instance, generating plausible sound effects for videos like the sound of rain or footsteps matching a scene.

3D and World Models

We should also consider 3D space as a modality. The ability of AI to perceive and generate objects and environments. Humans find it natural to imagine objects in three dimensions and we've been doing it since childhood with toys and blocks. Getting computers to do the same has been notoriously hard. It requires understanding geometry, texture, and lighting; a whole physics engine of cognition. Yet here, too, deep learning is opening new doors. One breakthrough approach uses neural networks to generate 3D models by leveraging what they learned about 2D images. If an AI agent knows how to draw an object from many angles, it can infer a consistent three-dimensional form. Google's DreamFusion project recently demonstrated this by producing 3D objects from text prompts.²⁰ You can type in "a green leather chair in the shape of an avocado," and the system will output a digital 3D model matching that description; an avocado chair you can rotate and examine from any angle. DreamFusion achieves this without ever being trained explicitly on 3D scans of avocado chairs (or anything else); it uses a pre-trained image diffusion model to synthesize multiple views and optimizes a 3D representation that would produce those views. The end result is a model that *"can be viewed from any angle, relit by arbitrary illumination, or composited into any 3D environment"*. In practical terms, this means AI can start to generate the kind of content used in video games, special effects, or product design. A game developer could sketch out ideas in natural language and get rough 3D assets to tweak. This is a process that normally requires hours of a 3D artist's time. An architect could quickly prototype furniture or fixtures in a planned building by describing them. We're still in the early days, and the fidelity of AI-generated 3D models is just beginning to approach professional quality. But the direction is set. Just as image generators became a tool for illustrators, we can expect 3D generators to become tools for designers and engineers, handling the heavy lifting of geometry while humans refine the creative details.

At robotics institutes and R&D labs, engineers are equipping humanoid machines with deep neural networks that function as internal "world models"; giving these robots a simulated inner reality where they can predict how their actions will affect the surrounding environment. These learned world models combine raw sensory input with physics-based learning by using techniques like graph neural networks as surrogate physics engines to let a robot mentally rehearse interactions with objects and terrain before actually executing them.²¹ Having an internal physics sandbox grants the robot a kind of intuition. Just as a person might imagine a cup tipping or a foot slipping on a loose stone, the robot's neural model can foresee the likely outcome of a movement

in advance. With this foresight, the machine can adjust its strategy on the fly, reasoning about object behaviors and contact forces to avoid mistakes and exploit opportunities. This capability is already yielding notable results. For example, a robot arm endowed with such a model recently learned a new manipulation task from a single demonstration. By generating myriad imagined variations of that lone example, it practiced in simulation and mastered the skill without extensive real-world trials.²² Likewise, in locomotion, giving a bipedal robot a predictive world model has dramatically improved its sure-footedness: one humanoid recently trekked over 3.2 kilometers of rough, icy terrain autonomously; a feat that would have been infeasible for earlier robots lacking such internal foresight.²³ By effectively letting robots experiment in thought before they act, these deep learning systems are shifting robotics from reactive trial-and-error toward more deliberative reasoning and planning. This convergence of neural networks and intuitive physics is endowing humanoids with a new level of autonomy in complex, unpredictable environments, nudging them closer to a form of adaptability that echoes our own.

Superhuman Modalities

Intriguingly, AI systems are now extending perception and creativity into realms where humans historically struggle or lack native capabilities. We can look to nature for some examples of possible future avenues. Migratory birds have an ability to detect the earth's magnetic field, sharks can detect electric fields, and bats can use ultrasound. As we build AI agents and robots in particular, we can equip them with the sensors and models to be able to interpret and predict in these additional modalities. At the risk of stretching the definition of a modality, we will now can explore some other types of specialized AI solutions that work with specific subsets of data.

Consider satellites: our planet is under constant watch by satellites, producing terabytes of data. Some of the data takes the form of images that are similar to what astronauts see when they look out the window of the International Space Station. However, a lot of the data takes other forms such as hyperspectral data and synthetic aperture radar data. We convert that data into images so that humans can interpret because we don't have the natural capability it in its original form. The analysis of satellite data also often requires different outputs than what humans are natively strong at generating. We may be able to easily identify a family member in a crowd but are less effective at looking at a series of thousands of pictures of the same plot of land over multiple years to tease out long term trends amongst the noise of seasonal variation. If we treat satellite data as a specialized modality with specialized models, we can see how it will give AI agents

an ongoing view of the world that humans lack. One recent AI system can map the outline of giant icebergs in satellite photos 10,000 times faster than a human analyst; literally in a fraction of a second, versus many minutes of manual tracing per iceberg.²⁴ What's more, the AI is consistent and doesn't get fatigued, making it possible to monitor remote polar regions continuously for changes in ice. Similar techniques are used to detect illegal deforestation²⁵, or to assess crop health by analyzing color spectra that indicate drought stress or disease. One can imagine an AI agent that seamlessly integrates analysis of satellite data at the same time it is also making observations on the ground on a more human scale.

Weather forecasting, a domain that blends physical law with chaotic data, is also being augmented by deep neural networks. Instead of images, sound, or text, weather forecasting relies on spatiotemporal data of temperature, pressure, humidity and other variables. Traditional weather prediction relies on solving physics equations involving fluid dynamics and thermodynamics on powerful supercomputers. These models are very accurate but computationally expensive, and they can struggle with certain complex phenomena. Enter graph-based neural networks, which treat the globe as a network of interconnected points and learn from historical data how weather systems evolve. In 2023, DeepMind introduced a model called GraphCast that learned to predict global weather patterns up to 10 days out.²⁶ It doesn't replace physics entirely, but it learned to approximate the physics by recognizing patterns in decades of past forecasts and observations. The performance was striking: GraphCast outperformed the leading conventional weather model (the one used by the European Centre for Medium-Range Weather Forecasts) in 90% of the locations tested for key weather variables. It was especially good at forecasting extreme events. In one case, it predicted the track of a hurricane three days earlier than the official models did, giving a potential head start for preparations. And because it's a learned model, once trained it can generate a forecast in minutes on a single machine; much faster than the hours of supercomputer time a traditional simulation might take. This doesn't mean we'll throw out physics-based forecasting, but it suggests a hybrid future where a neural network fills in the gaps, corrects biases, or runs quick what-if scenario forecasts. For instance, if meteorologists want to see the potential impact of a slight shift in wind patterns on a developing storm, an AI agent could provide a plausible answer in near real-time. When you multiply that by the number of forecasts made daily around the world, the potential societal benefit in agriculture, disaster response, aviation, and energy management is immense. It's easy to focus much of our attention on how far AI systems have progressed towards matching or even surpassing capabilities that are innate human such as language comprehension and image analysis. Those human capabilities lie at the very core of how we define ourselves

relative to other animals on our planet. More practically, our entire economic system is based on the principle that we can leverage those capabilities to do work and in return earn wages that allow us to buy food and shelter. If those uniquely human capabilities can be replaced by software, what will that mean for the future of society? However, we shouldn't forget to also take a look around. The same rapid progress that is occurring in the modalities that are native to humans is also occurring in a wide range of additional areas and that progress could allow us to build systems that perform tasks that would have been unimaginable in the past.

Going Multimodal

Stepping back, what do these varied advances have in common? From recognizing images to generating videos, from cloning voices to composing symphonies, from monitoring deforestation to predicting the weather, these feats are all driven by deep neural networks digesting vast amounts of data. It's easy to compartmentalize each achievement: one might marvel at ChatGPT writing an essay, or chuckle at an AI-generated painting, without connecting the dots. But viewed together, the pattern is emerging that intelligence itself, the ability to perceive the world, to analyze it, and even to create within it, is becoming something that can be engineered and applied in many forms.

Models that work within a single modality, such as text-based large language models or image-focused diffusion models, operate fundamentally within boundaries defined by their specific types of inputs and outputs. A large language model trained exclusively on textual data ingests sequences of words and outputs coherent sentences. Similarly, diffusion models accept descriptive prompts in text form as a control mechanism but they are only capable of generating high-quality visual outputs. They do not themselves interpret images nor produce textual interpretations of their creations. These models have a specialized architecture tailored carefully toward the nuances and complexities of one particular modality, allowing them to excel within their specialized tasks, whether it is synthesizing human-like prose or rendering richly detailed visual scenes. When trained on very broad data sets such as text representing the full diversity of natural language, or images representing vast categories of objects, these models are often called foundation models given the breadth of their capabilities within their modality.

Multimodal artificial intelligence systems merge distinct sensory channels to build representations that bridge modalities and synthesize more comprehensive outputs. This type of system might be trained on datasets encompassing both textual narratives and visual imagery, enabling it not only to interpret images or videos but also to generate rich textual descriptions, identify contextual significance, and perform tasks

requiring an understanding of relationships between modalities. At a technical level, multimodal architectures typically employ specialized modules, each finely tuned for its dedicated modality, which are then integrated through cross-attention mechanisms or joint embedding spaces. These unified architectures facilitate an interplay between input modalities that enriches the interpretation beyond the scope achievable by separate single-modality models operating independently.

Current state-of-the-art examples illustrate clearly the distinctions between unimodal and multimodal approaches. OpenAI's ChatGPT 4o is a prominent illustration of a multimodal model, designed to ingest both text and images, enabling it to understand diagrams, charts, or visual metaphors, in addition to processing complex textual prompts. The strength of a multimodal architecture lies in its ability to form conceptual connections that transcend traditional modality boundaries.

Looking forward a decade, the capabilities of multimodal models seem poised to expand significantly. Multimodal architectures may grow increasingly sophisticated, integrating a richer array of sensory data streams and employing more advanced algorithms for extracting nuanced relationships. Researchers propose embedding additional modalities such as auditory signals, physical gestures, or even sensor data from robotics to craft AI systems that interpret contextually dense interactions more accurately, responding fluidly to dynamic real-world scenarios.²⁷ Such future developments could lead to genuinely context-aware and adaptive artificial intelligences, capable of interpreting subtle emotional nuances in speech or gestures, understanding complex social interactions within physical environments, and enhancing assistive technologies through a deeper contextual integration. In these deeply integrated multimodal systems, the AI agent would not just have language to describe that an orange is squishy and will leak orange juice if squeezed hard enough. It would have had the experience of squishing an orange in its own hands. And it would have heard the sound it makes and watched it with its own two eyes. Very significantly, such a system would be capable of true learning. It could pick up a new type of fruit that was absent from the training set and learn if it is also squishy. If it observes that it is squishy, then it would add that understanding to its language.

Yet, building multimodal models is significantly more complex than unimodal learning, and it comes with new challenges. One fundamental difficulty is integrating heterogeneous data types. Each modality has its own statistical patterns, noise characteristics, and ambiguities. When a model combines them, it must learn not only within-modality patterns (like understanding grammar in text or shapes in images) but also cross-modal correspondences. This increases the model's complexity, and the data required to train it. High-performance multimodal systems often rely on enormous training corpora:

image–text pairs, video clips with transcripts, audio with captions, and so on. Such paired data can be harder to obtain than plain text or images alone. There’s also the risk that the model becomes jack-of-all-trades, master of none. Balancing training so that it doesn’t neglect one modality is an active area of research. For example, if much of the training data is text-centric with only occasional images, a “multimodal” model might still learn to mostly do text reasoning and treat images as a minor add-on, or vice versa. Researchers have to carefully design training curricula and objectives so that the model truly learns to connect modalities rather than just handle them in isolation. Another challenge is representation alignment: ensuring that concepts are represented consistently across modalities. If a model learns about “sustainability” only from written documents and separately learns about interpreting satellite images of deforestation, there is no guarantee it will connect the two unless it has seen them together. Techniques like contrastive learning and multimodal pre-training help address this by forcing alignment.

And when it comes to output alignment with human intent, multimodal systems pose new puzzles. It’s one thing to ensure a text-only chatbot doesn’t produce offensive or irrelevant text; it’s quite another to ensure a chatbot that can also generate images or interpret them does so in a way that aligns with user expectations and ethical norms. Multimodal models can hallucinate in more complicated ways. A text model might invent a non-factual statement; a multimodal model could look at an image and then fabricate a description that doesn’t actually match the picture, because it misinterpreted the visual cues. This disconnect between modalities such as producing a caption that seems plausible but isn’t grounded in the image is a known failure mode, sometimes called multimodal hallucination. It occurs when the model’s two systems (vision and language) are slightly out of sync.²⁸

From a technical standpoint, multimodal AI presents not merely incremental improvements over unimodal predecessors but a qualitative leap in complexity, versatility, and potential that brings us closer to the architecture of our own brains. The next decade will likely witness increasingly intricate multimodal systems capable of understanding and responding intelligently across multiple dimensions of sensory experience, reshaping fundamentally the ways in which artificial intelligence integrates into and transforms human society.

Controversy Related to Additional Modalities

Generative AI’s rise has not been without controversy. As we discussed previously, powerful models are often trained on vast data sets and as companies have started to

train models on additional modalities, those datasets have expanded to include artwork, photographs, videos, and music that are harvested without the original creators' consent or compensation. This practice has ignited fierce debate about intellectual property and fair reward; particularly as valuations of startup companies have dramatically increased. One startup called Stability AI that makes a popular image generation model was valued at approximately \$1 billion after a seed funding round, even as many painters, photographers, and writers see nothing of the wealth built on their unwitting contributions.²⁹

In the same way that additional modalities introduce new technical challenges, they also raise many new types of questions about copyright. Take for instance, the ability of an AI model to create an image in the style of an artist. In the prompt, a user can provide a reference image such as a portrait photograph of themselves. They can then ask the model to recreate the image in the style of a painting by a well-known artist. The resulting image is entirely new and no one could argue that it is a copy of any one painting from the artist. At the same time, it is undeniable that it would not have been possible to create it without training on the work of the artist. Artist styles are not protected by copyright. In the past, such a rule would have been too restrictive; effectively giving individuals a monopoly on the right to create any image in a given style. The ability of AI to train on the full works of a single artist and then faithfully and intentionally create images that include the detailed nuances of style developed over years of work by that specific artist raises new questions about whether copyright rules need to be adjusted to reflect these new use cases.

Regardless of the outcome of the lawsuits that are now in front of the courts, the resentment among artists is palpable and justified by simple equity. Few would deny that AI companies are profiting from creations they did not themselves create. For a technology touted as transformative, there is a risk that its benefits accrue disproportionately to those holding the algorithms and computer power, not those who supplied the content fuel. The result is a widening rift between tech developers and creative communities which is a tension that demands careful reconciliation rather than dismissal.

This tension leads to deeper questions about authenticity and our relationship with creative works. Suppose you learn that a stunning painting you admired in a gallery or a song that moved you to tears was generated entirely by an algorithm with no human imagination directly involved. Would that knowledge dampen your appreciation, or even feel like a betrayal? Humans have long imbued art with a sense of the artist's spirit and intent, the hand of the maker. If that maker is silicon-based, some wonder if the art loses an intangible element. Indeed, psychological studies suggest people judge identical artworks differently depending on the believed creator: in experiments, participants rated paintings as less profound or beautiful if told they were AI-generated rather than

human-made.³⁰ We seem wired to care whether a piece of music was born from human emotion or just math. Yet the algorithm's output itself might be indistinguishable from a master's. As AI outputs improve, creators or distributors might choose to hide their true origin, passing machine work off as human to meet audience expectations. We must grapple with where our comfort with artificial creation ends and where a violation of expectation begins. Is it enough to eventually say, "Great art is great art, even if made by an AI tool," or do we lose something fundamental when the human touch vanishes? These open questions challenge our intuitions about creativity and authenticity in a future where the identity of the creator can be easily obscured or faked.

The cultural stakes extend beyond individual artworks to entire traditions and communities. If generative AI widely adopts the dominant aesthetics of large training sets, which currently skew Western, Anglophone, and mainstream, there is a risk of homogenization in global culture. Unique local art styles, minority languages, or niche musical genres could be drowned out by an endless algorithmic remix of the most common patterns. A richly textured Norwegian children's story might be buried under a flood of AI-synthesized narratives favoring Hollywood tropes. As one scholar warns, the unchecked wave of AI-generated content "threatens to impoverish the world's cultural narratives" by overserving familiar story archetypes at the expense of diverse voices.³¹ Smaller or endangered cultures might find it even harder to get attention when AI can mass-produce polished imitations of whatever is most popular. On the other hand, these same tools could become a lifeline for preserving and amplifying vulnerable art forms. Ironically, AI might simulate a dying language's poetry or resurrect nearly lost craftsmanship by learning from the few examples that remain. There are already promising cases: researchers have used AI to breathe new life into vanishing techniques, such as teaching a new generation Cantonese porcelain painting with the help of generative models.³² A skilled algorithm might reproduce ancient weaving patterns or folk songs, keeping them accessible. If left purely to market forces, the aesthetics with the broadest appeal or most profit could dominate. But with intention, one could imagine libraries of indigenous art and language used to train models that empower those communities to create and disseminate their heritage in new ways. Significant questions need to be raised around control and consent. Will large tech companies train models based on images and 3D scans of historical artwork from indigenous communities without their involvement? These works are often available in public museums and are of an age that they no longer have any form of copyright protection. However positive the intentions of the tech companies may be, there must be an understanding that these are not just individual artefacts in a museum, they are vital threads that help to preserve endangered cultures. AI can easily do more harm than good. For example, the use of generative

AI to create artwork in a traditional style may cause economic harm to artists from that community, reducing their ability to be able to continue to practice their craft. Traditional art is also inherently multimodal. For example, the symbols on a Haida totem pole all have important meanings. A generative AI model may be able to make an image that looks like a totem pole, but it might entirely muddle the underlying story. Rather than helping to preserve and amplify the culture, it may add noise and confusion that dilutes the most important elements of that culture.

Deep Fakes

Perhaps the most visibly troubling use of multimodal AI is the proliferation of deep-fakes; hyper-realistic AI-generated or manipulated images, video, and audio that falsify reality. In the political sphere and beyond, deepfakes present a clear and growing threat to truth, trust, and dignity. Nation-state actors and ordinary mischief-makers alike can now fabricate video evidence of events that never happened, put authoritative words in a person's mouth or fabricate sexually suggestive images of real people. The implications for democracy, security, and society are dire. Imagine a convincing video of a world leader announcing a fake crisis or surrender that could sow chaos before being debunked. This is not hypothetical: a forged video of Ukraine's president appearing to call for surrender actually surfaced online amid the 2022 conflict.³³ Intelligence officials warn that deepfakes could be deployed to influence elections or incite unrest. Experts note that such synthetic media "have the potential to undermine election outcomes, social stability and even national security" when weaponized in disinformation campaigns.³⁴ Unlike simpler "cheapfakes" (like out-of-context clips or sloppy edits), today's deepfakes can be alarmingly realistic and they are getting better each month. Equally worrying is how accessible the technology has become. Once the domain of university labs, deepfake tools are now openly available. Free software can swap faces in a video with a few clicks, and AI models to clone voices circulate on GitHub. Anyone with a reasonable computer can generate passable fakes of public figures, or of someone they know. That democratization of creation means we face a flood of falsified media with no easy technical defense. Even the best detection systems struggle to keep up and the nature of the technology means that this will always be the case. As with language, there will be no way to verify if something was generated by AI. Instead, the only viable solution is to use a variety of methods to authenticate real content as legitimate and to assume that everything else may have been created or modified by AI. As with language, one of the challenges with offending content in other modalities is that bad actors can generate content on offline local computers. They can manufacture and distribute false videos

and audio entirely under the radar. Law enforcement and social media companies find themselves ill-equipped and always a step behind. The result is an environment where seeing is no longer believing, and the fundamental trust in recorded evidence is eroding. We are entering an era where a video confession in court, an announcement from a politician, or a call from a loved one can and often should be dismissed as a potential deepfake. In one likely scenario, the first few years will be the most challenging. During this transition period, most people and institutions will still place a high level of trust in audio, photography, and video recordings. However, after a number of high-profile incidents, it is likely that trust will erode completely and the only content that will be trusted is content that can be verified as authentic. Although there will be no reliable technical solutions for identifying if content is generated by AI, there are many methods available to do the opposite and verify if content is real. Unfortunately, these methods will likely require the adoption of processes, software, and/or hardware by the creators of the content. As an example, it is likely that cameras will start to include built-in encryption based watermarking systems that will provide an unhackable proof that a photo or video clip was recorded by that device and is being presented in an unaltered form.

Before those solutions are rolled out, deepfakes and AI-generated personas are quietly corroding trust in more intimate realms of life. Online dating, for example, is increasingly awash with profiles that are quite literally too good to be true with attractive faces that don't belong to any real human. Scammers have begun littering dating apps and social media with AI-generated profile pictures of fictitious people, luring victims into romance scams.³⁵ Unlike stolen photos of a model or attractive individual which a reverse image search might reveal, these faces are unique synthetic creations, purpose-built to evade detection and appeal to targets. It's not just images: chatbots can now carry on flirty conversations while voice-cloning software leaves convincing voicemails. Lonely hearts may fall in love with a persona that is entirely AI-generated, crafted by criminals to be their perfect match and once they are lulled into complacency, the inevitable pleas for money start. Even outside of scams, one can imagine individuals using AI to bolster their dating prospects by polishing every aspect of their identity, generating idealized photos and messages. In professional settings, we see parallel concerns. A fake LinkedIn profile with an AI photo reportedly managed to connect with real senior officials, a sign that even savvy individuals can be fooled by synthetic personas.³⁶ The FBI has warned that fraudsters are leveraging deepfakes and stolen personal data to impersonate people in remote job interviews.³⁷ In one such scheme, a candidate's video feed was a deepfake that lip-synced answers fed by an accomplice, complete with a synthetically generated voice. If hired, these imposters could gain access to sensitive systems or data under false

pretenses. The remote nature of modern work and education provides cover for such tricks; a deepfake need only be good enough to pass through a grainy webcam stream. This technology even enables a more banal form of deception: people could deploy deepfakes of themselves to shirk responsibilities. Students have already experimented with looping recordings to pretend they're attentively present in online classes. It's easy to imagine an overworked employee sending a realistic avatar to the 8 a.m. video meeting while they sleep in. Such antics are mostly harmless mischief, but they underscore how flexible identity has become in a digital world saturated with AI mimicry. If any face or voice can be fabricated, how do we maintain trust in daily interactions that increasingly occur through screens? I expect that the next few years will be quite tumultuous but that technology will catch up. For example, a new generation of webcams may integrate to proprietary gloves. The gloves would be capable of measuring unique biological information such as fingerprints and pulse rate to confirm that the correct person is wearing the gloves at that point in time and the webcam would use a combination of software and hardware to confirm that the person in the video is the person wearing the gloves. Some of these solutions may be clunky at first but over time they will improve.

Social Media

Social media, which is already notorious for hijacking human attention, could find even more potent ways to keep us glued to our feeds using multimodal AI. Today's recommendation algorithms analyze our clicks and likes to serve up existing content most likely to engage us. But what happens when these platforms can generate content on the fly tailored to every user's particular itch? We may soon face personalized streams of images, videos, and soundbites fabricated by AI specifically to maximize our "dwell time," calibrated to our preferences and emotional triggers. If you linger on pictures of sunsets, your future feed might create ever more gorgeous sunset images just for you, each more perfectly colored and composed than any real photograph. If a certain style of outrageous news headline gets your blood boiling and keeps you engaged in the application, a generative model could be tasked with writing even more extreme, yet plausible, headlines to hold your attention. In effect, the platform becomes not just a curator of what we see, but an author of it. Tech critics have already accused social apps of "dopamine hacking"; exploiting our brain's reward circuitry with stimulating content selections. AI stands ready to turbocharge that effect. The intention to maximize engagement isn't new, but an AI tool that can conjure the exact image, song, or character you'll find irresistible is a qualitatively different power that may blur the line between entertainment and manipulation. The sheer volume of synthetic content could

also overwhelm our critical faculties. One startling forecast suggests that by 2026, up to 90% of online content may be synthetically generated.³⁸ If most articles, videos, and posts are machine-made for maximized engagement, the internet could become a hall of mirrors reflecting our desires back at us, with fewer genuine human voices breaking through. While this might increase user “satisfaction” in a narrow sense, it also carries deeper costs: erosion of shared reality, further shortening of attention spans, and a loss of the serendipity that comes from encountering content outside one’s algorithmic comfort zone. Extreme personalization may be pleasurable but would also be potentially isolating and addictive by design. Society will need to catch up with these tactics, as they could make today’s social media challenges (misinformation, polarization, addiction) look tame by comparison. We may have to reinvent digital public spaces that prioritize human agency and truth over click-maximization or risk a descent into personalized fantasy bubbles that are hard to escape.

Economic Impacts of Multimodal AI

The economic disruptions from multimodal AI loom as large as the cultural ones. Automation is not a new story, but these AI systems encroach on creative and service roles once thought safe from machines. Consider voice actors: Until recently, it was assumed that lending one’s voice to characters or commercials was an innately human domain. Now AI voice synthesis can clone a person’s vocal timbre and inflection with eerie accuracy. Video game studios and audiobook producers are already experimenting with synthetic voices to cut costs. Some human voice artists have begun to find clauses in contracts demanding rights to digitally replicate their voice, effectively signing away their future relevance. Many voice actors fear being replaced entirely by AI, especially for routine corporate or commercial jobs. In one survey, professionals said they felt on the precipice of their work “*being replaced completely*” by generated speech, as companies seek cheaper alternatives.³⁹ A similar dynamic is playing out for visual artists and designers. Graphic designers see clients opting for AI-generated logos and illustrations. 3D modelers find that AI can produce decent asset drafts in seconds, undercutting entry-level gigs. Entire job categories might be erased: why hire a telemarketer when an AI caller that never sleeps can do the same, armed with a perfect script and a convincingly friendly voice? Indeed, early versions of such AI telemarketers are already being marketed, and regulators have scrambled to preemptively ban AI-generated robocalls in some jurisdictions.⁴⁰ Yet displacement by AI is rarely a clean swap of AI for human; more often, roles are transformed. Perhaps human voice actors won’t vanish but will instead work with AI clones of themselves to increase productivity, lending their talents

to train the models and then curating the results. Some fields could be revitalized or made more accessible. A small indie game studio with limited budget might now produce high-quality voiceovers and 3D art using generative tools, enabling creative projects that previously needed armies of specialists. Medical imaging technicians might leverage AI to automatically generate 3D anatomical models, freeing them to focus on analysis rather than manual modeling. In journalism, AI might handle routine video editing or transcription, allowing reporters more time in the field. The overall economic effect will be complex: certainly, jobs will be lost, and communities disrupted, but new opportunities will also emerge. The history of technology suggests that while some professions fade, others rise and often in unpredictable ways. The printing press eliminated the monk-scribe but created the printer and publisher; the camera reduced demand for portrait painters but opened a vast new art form of photography. With AI, the hope is that human creativity finds new outlets even as old craftwork is augmented or overtaken. Still, transitions can be painful. Workers in affected industries from call centers to creative studios may need support and retraining. The challenge for society will be managing this economic upheaval humanely. We must anticipate which skills to cultivate and how to ensure that the prosperity AI brings is widely shared, rather than concentrated only among the tech-savvy and capital-rich.

Regulation and Challenges with Additional Modalities

Policymakers worldwide have begun to acknowledge the risks, but governance is struggling to catch up to the fast pace of AI advancement. The regulatory efforts underway reveal structural limitations. In the European Union, lawmakers have drafted a sweeping AI Act that, among many provisions, would require transparency for AI-generated content. For example, if an image or video is AI-synthesized to resemble reality, providers or users of the AI would be obligated to clearly disclose that it is artificial.⁴¹ The intent is to curb deception by ensuring people know when they are seeing or hearing AI output (with some exceptions for art and satire). Similarly, China implemented rules mandating watermarks or labels on deepfakes and even restricting certain uses altogether.⁴² In the United States, where federal action has been relatively slow, a few states have leapt ahead with targeted laws. California, for instance, passed a law making it a crime to distribute deceptive deepfake videos of political candidates within 60 days of an election, and another law allowing victims to sue creators of pornographic deepfakes made without consent.⁴³ Texas enacted its own ban on election-related deepfakes. The situation may finally be changing in the United States. In May of 2025, the president signed into law the TAKE IT DOWN ACT which “*prohibits the nonconsensual online publication*

of intimate visual depictions of individuals, both authentic and computer-generated".⁴⁴ These are important first steps in drawing red lines against the most egregious harms. However, they also highlight how piecemeal and reactive regulation remains. Legislation often carves out narrow slices (e.g. election interference) and usually lags years behind the technical reality. Deepfakes were already circulating by the time lawmakers acted, and even now, new AI capabilities emerge faster than legal frameworks can adapt. More fundamentally, regulation faces jurisdiction and enforcement challenges: AI tech ignores borders. An EU rule might compel big platforms to label AI content, but it won't impact a small company running in a country without those rules and may have little to no practical impact on a bad actor running an open-source model on an air-gapped computer in their basement to create unlabeled fakes. Some experts worry that heavy-handed rules will merely push development underground or give an edge to less scrupulous actors in jurisdictions with lax controls. Traditional regulatory levers are ill-suited for a technology that is decentralized and rapidly evolving. We may need new approaches: perhaps heavier penalties for individuals that use tools to create offending content, international agreements on certain red lines (analogous to arms control treaties), industry self-regulation for responsible model release, and public education to raise resilience against things like deepfake deception. The EU AI Act and similar proposals are ambitious, but even their authors admit enforcement will be challenging given the availability of open-source versions of powerful models. This structural limitation means we cannot rely on regulation alone to save us from AI's risks. The genie is well out of the bottle and our task is to manage the new reality as best we can, with eyes wide open to both the rewards and the hazards ahead.

We previously covered a wide range of other challenges when discussing LLMs. These included hallucinations, data poisoning, and the difficulties faced by human moderators who need to review and label highly grotesque content. All of these issues are amplified in additional modalities. By this time, we've all seen early AI images of hands with six fingers. More recent models have improved meaningfully but anyone using generative AI tools to create images, video, or audio must be aware that they still routinely make mistakes. This has been less of an image so far as these outputs have primarily been used in creative endeavors but as people push forward to use additional modalities of generative AI in domains where accuracy is more critical such as architecture and engineering, these inaccuracies will be problematic. Data poisoning is another very tricky problem for additional modalities and some artists are now deliberately making slight tweaks to their art that cannot be seen or heard by humans but that introduce challenges for AI models when they try to include them in their training sets.⁴⁵ Content moderation and the process of establishing guardrails for audio, images, and video presents many

layers of challenges. To date, most vendors are erring on the side of caution but even then, it is possible for inappropriate content to sneak through and the task of reviewing and labelling this type of content to teach the models what needs to be avoided exposes humans to very rich media of the worst kind.

Conclusion

In this chapter, we began with the human brain, a testament to the power of specialized modules integrated into a seamless, coherent whole. Our whirlwind tour of artificial intelligence has revealed a parallel trajectory: from mastering individual senses like sight and sound in isolation, we are now standing at the precipice of truly multimodal systems that combine these faculties. This integration is not merely an incremental step; it is the threshold where artificial perception begins to emulate the richness of experience. Yet, as we have seen, with this newfound sensory depth comes a magnification of every challenge. The issues of authenticity, trust, economic displacement, and safety are no longer confined to text but are amplified across every modality, creating a far more complex and perilous landscape. While technical solutions for stitching these systems together advance rapidly, the path forward demands a commensurate leap in our collective wisdom. As we bestow upon our creations more senses and with which to perceive and interact with the world, we are tasked with the profound responsibility of instilling not just intelligence, but also the context, caution, and ethical grounding that must guide such powerful capabilities.

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Chapter Six

Gaining a Body

Introduction

Elias watched from the subfloor, coffee in hand, as the morning sun cast long shadows across the organized chaos of the job site. His partner Maria, tablet aglow, directed the flow of materials while their crew of ten humanoid robots worked with a silent, coordinated efficiency that still felt surreal. Two of them hoisted a massive laminated veneer beam into place, a lift that would have once required a crane and shouted commands, while others moved along the walls, their pneumatic nailers firing with perfect rhythm and spacing. They were framing an entire house in a single day, a task that had consumed a week of his youth. He remembered the bitter carpenters' union protests in the late twenties, the pervasive fear that their craft, his entire identity, would be rendered obsolete. The displacement had been real and painful for many, but the trade had not vanished; it had transformed. His role was now one of orchestration and expertise, guiding the machines and applying a craftsman's eye to the complex joinery and custom details that still demanded human ingenuity. He thought of the global panic during the supply chain crisis when the handful of nations that had initially dominated robotics manufacturing threatened to withhold their machines, a move that nearly crippled entire industries. That scare had forced the world to build a more resilient system, with diversified production and open standards that prevented any one country from holding a monopoly on labor itself. He recalled the intense public debates about how our world would be rebuilt for machines, a future of inhuman scale, but that fear had proven unfounded in his line of work. These robots had to fit into a world designed for people, using the same tools and materials, their humanoid form a concession to the stubborn reality of human centered architecture. With a pang of solemn gratitude, he

also remembered the news images of the first autonomous military units and the global outcry that followed, a wave of revulsion so powerful it drove nations to the negotiating table to forge treaties that kept lethal decision making firmly in human hands. That single, vital choice had ensured this technology was primarily used to build and to serve, not to hunt. Maria's voice cut through his thoughts, calling him over to a tricky corner where the architect's plans required a solution that no algorithm could yet devise. The robots nearby paused, their optical sensors bright and attentive, waiting for their human foremen to solve the puzzle.

Integrating Sensation and Action: From Skin Receptors to Muscle Movement

Our ability to interact with the world hinges on a seamless integration between sensory input and motor output. The nervous system continuously gathers information via specialized receptors distributed throughout the body, especially in the skin, and uses this input to guide movement. For example, the skin contains distinct groups of receptors that transduce physical events (a breeze, a hug, a hot stove) into nerve signals, including mechanoreceptors for mechanical stimuli (touch, pressure, vibration) and thermoreceptors for temperature. These incoming signals travel through peripheral nerves and the spinal cord into the brain, where they are processed and integrated with past experience. In turn, the brain issues commands through motor pathways to activate muscles, the body's biological actuators that execute our responses. Certain neural circuits unconsciously refine how we move, while muscles provide the force to move. The result is that complex actions from withdrawing your hand reflexively to flipping the pages in a book feel almost effortless.

When you touch a surface or feel a change in temperature, specialized nerve endings in the skin fire off signals that eventually reach the brain's sensory centers. Different receptors in the skin are tuned to specific tactile or thermal features: for instance, Meissner's corpuscles respond to gentle flutter or touch, Pacinian corpuscles to vibration, Merkel's disks to steady pressure, and Ruffini endings to stretch of the skin. Likewise, thermoreceptors continuously gauge warmth or cold. These receptors convert stimulus energy into electrical impulses in a process called transduction and sensory neurons carry the information into the central nervous system. Along the way, the signals remain organized in pathways that preserve their origin: neurons from neighboring skin areas project to neighboring locations in the spinal cord and brain. By the time the input reaches the primary somatosensory cortex (in the postcentral gyrus of the parietal lobe), a clear spatial map emerges. Adjacent regions of the cortex correspond to adjacent parts

of the body, forming the famous somatotopic map or “sensory homunculus”; an atlas of the body laid out on the brain. In this cortical map, sensitive regions such as the fingertips, lips, and face are represented by disproportionately large areas of cortex. Much of our knowledge about this mapping is a result of brain stimulation studies in the 1930’s to 1950’s by a neurosurgeon named Wilder Penfield in which parts of the brain were stimulated with an electrode during neurosurgery. This mapped layout means the brain not only detects touch, pressure, or temperature, but also knows where on the body these sensations occur. The somatosensory cortex integrates this wealth of tactile information, allowing us to recognize objects by feel, locate a mosquito on our skin, or perceive the difference between a warm mug and an icy glass.

Not all the brain’s work in movement happens at a conscious level. In fact, much of our motor control is handled by deep, specialized brain structures operating behind the scenes to plan, smooth, and coordinate our motions. Two key players in this unconscious motor choreography are the basal ganglia and the cerebellum. The basal ganglia are a set of nuclei buried deep in the forebrain that are crucial for planning and initiating movement sequences; they help decide which muscles to activate and when. These nuclei aid in assembling complex, purposeful actions and regulating their intensity and sequence. The cerebellum, a “little brain” at the back of the head, oversees the fine-tuning of movements: it adjusts timing, smooths out trajectories, and compares intended movements with actual performance to correct errors on the fly. Notably, the cerebellum ensures that movements progress fluidly from one to the next and that force generation adapts to changing loads. Neither the basal ganglia nor the cerebellum directly sends commands to the muscles; instead, they act as supervisory circuits, modifying signals that descend from the cerebral cortex. We become vividly aware of their importance when these systems malfunction. For example, damage to the cerebellum leaves a person with jerky, uncoordinated movements (ataxia) even though the muscles themselves are strong; a phenomenon early physiologists noted in animals lacking a cerebellum, which lost all muscular coordination and balance.¹ Likewise, disorders of the basal ganglia underscore their role in automatic motor control: in Parkinson’s disease, where basal ganglia neurons degenerate, patients struggle to initiate movements and exhibit rigid, tremorous motion instead of smooth execution. Under normal conditions, however, these subconscious motor centers work in concert with the conscious motor cortex, so that practiced skills (like walking, riding a bicycle, or typing on a keyboard) are executed gracefully without us having to think about every step or keystroke.

Finally, at the end of the line of command are the muscles; the biological actuators that convert neural commands into action. A muscle is composed of long, fibrous cells (muscle fibers) that contract in unison when stimulated by a motor nerve impulse.

Each motor neuron in the spinal cord sends its axon out to a group of muscle fibers, forming a motor unit; the fundamental unit by which the nervous system controls force. When that neuron fires, all the fibers in its motor unit contract together, producing a twitch of force. The size of motor units varies with the function of the muscle: in muscles requiring fine, precise control like those moving the eyes or fingers, a single motor neuron may innervate only a handful of fibers, whereas in large powerful muscles like the quadriceps of the thigh, one neuron may connect to hundreds of fibers. This arrangement allows the motor system to recruit just the right amount of force for a task; small motor units are activated for delicate, subtle movements, and larger units are brought in for stronger efforts. Moreover, the nervous system can ramp up force smoothly by engaging additional motor units or by increasing the firing rate of active units. Inside each muscle fiber, an ancient molecular machinery translates the nerve's electrical signal into contraction: a burst of calcium ions triggers an attraction between interlocking protein filaments, causing them to slide past each other. This sliding filament mechanism shortens the fiber and generates tension, pulling on tendons and bones to produce movement. Through these finely tuned muscle responses, the body carries out the brain's intentions with both power and precision, completing the remarkable journey from sensation to motion.

This remarkable biological synergy of sensing, processing, and acting presents a profound challenge and a guiding blueprint for modern robotics. While robots may be equipped with powerful sensors like high-resolution cameras that far exceed the capabilities of the human eye, they have generally been unable to master the sheer sensor density and multimodal richness of our skin or the exquisite, fine-tuned control of our muscles. The intricate network of sensors distributed across our bodies provides a constant, detailed stream of tactile information that is still leagues beyond the grasp of most artificial systems. Similarly, while robotic actuators can generate immense force, they struggle to replicate the adaptable efficiency of biological muscles, which leverage variously sized motor units to apply everything from the gentlest touch to maximum power. A central frontier in robotics is closing this gap: engineering not only more sophisticated artificial skins and compliant actuators but also the complex, brain-inspired control systems, analogous to the cerebellum and basal ganglia, needed to seamlessly integrate sensory feedback with fluid, precise, and purposeful action.

Sensing and Acting in Uncertain Environments

Most robots in history have lived sheltered lives. Picture the classic industrial robot arm: bolted to a factory floor, endlessly repeating a taught sequence with unerring precision.

For decades this wasn't a stereotype but a design necessity; early robots could only function in tightly controlled settings, performing high-volume, high-speed tasks in isolation from surprises. They were the tireless workhorses of automotive plants and electronics assembly lines, blindly welding the same spots or placing components along predetermined paths. Cages and safety barriers kept humans out and the robots focused. In these structured environments, variability was squeezed out until what remained was a choreographed routine that a machine could reliably execute 24/7. The vast majority of robots in use even today still operate under those conditions. It's a testament to how challenging it has been to get robots to cope with the messiness of the real world. But the frontier of robotics is no longer confined to the orderly realm of factory automation; it's expanding into the chaotic, unpredictable spaces where we live and work. How machines gained the senses and smarts to step off the assembly line and into our complex environment is a story of gradual breakthroughs, punctuated by bold experiments and hard-won lessons.

One pivotal turning point came in the Mojave Desert in March 2004. A quirky convoy of driverless vehicles gathered to attempt something never done before: drive themselves across 142 miles of punishing terrain. This was the DARPA Grand Challenge, a government-sponsored race designed to catalyze innovation in autonomous vehicles. Under the bright desert sun, one by one the experimental SUVs and dune buggies laden with sensors sputtered out. One took a hairpin turn too wide and got wedged on an embankment; another misread its GPS and aimlessly spun in circles. The best performer managed a mere 7.5 miles before breaking down and not a single vehicle finished the course.² The prize money went unclaimed. Yet, what looked like failure was actually the start of a renaissance in robotics. As a DARPA report later noted, that first Grand Challenge “*created a community of innovators... dreamers who came together to make history by trying to solve a tough technical problem,*” planting the seeds for future breakthroughs.

Just 18 months later, DARPA held a second Grand Challenge, and this time the results were dramatically different. This second challenge was notable for me personally as I was the hardware leader for the entry from the University of British Columbia. Starting from behind and with a fraction of the funding of many other teams, we learned a lot in our local test but didn't proceed to the final event in the desert. In October 2005, not one but five vehicles completed the 132-mile off-road course through Nevada. Stanford's autonomous Volkswagen Touareg, nicknamed “Stanley,” finished in just under 7 hours to claim the \$2 million prize. It was a historic moment: a car had driven itself farther and faster than anyone thought possible. By 2007, the challenge moved to a suburban setting with the DARPA Urban Challenge. Robotic cars navigated

an imitation city environment, obeying traffic rules and merging with moving vehicles. Again, multiple robots succeeded; Carnegie Mellon University's team took first place by deftly handling stop signs and traffic merges. These challenges demonstrated that autonomous driving was not a fool's errand, but a tractable goal given enough sensor technology, computing, and clever software. They also trained a generation of engineers who would go on to spearhead the self-driving industry. The Google Self-Driving Car project (now Waymo) famously hired veterans of the DARPA challenges and set them loose to refine the technology on city streets. By the 2020s, the fruits of that labor were evident. Waymo's cars have logged millions of miles on public roads in cities like Phoenix and San Francisco, navigated fully by AI chauffeurs without human intervention.³ In 2025, Waymo announced that they had completed over 10 million robotaxi trips.⁴ What started as a desert scramble is now a commercial service where you can hail a robotic taxi on your smartphone.

To appreciate the leap, consider how comparatively easy we humans made it for earlier robots. An airplane's autopilot system has been a staple of aviation for decades but flying on autopilot and driving autonomously are two very different challenges. A commercial jet cruises through largely empty sky on a steady course; its autopilot holds altitude and heading based on clean sensor readings of airspeed, gyroscopic orientation, and GPS location. There are no stoplights at 30,000 feet, no pedestrians wandering into the flight path. Critically, autopilots aren't asked to detect and avoid obstacles in real time. Air traffic control ensures that other planes keep their distance, and onboard radar helps only with weather and terrain. In essence, a plane on autopilot follows a predetermined "highway in the sky," and the computer's job is mainly to keep the plane stable on that invisible highway.⁵ It's an impressive feat of control engineering, but it's an almost closed-book problem compared to autonomous driving. A self-driving car, by contrast, must operate in a world cluttered with unpredictable objects and agents. It has to interpret lane markings (or lack thereof), traffic signs, and signals; it must anticipate the movements of erratic human drivers sharing the road, successfully negotiate with other vehicles at four-way stops, differentiate between a harmless plastic bag or a boulder in a parking stall, and be ready for the child chasing a ball into the street or the cyclist swerving around a parked truck. The sensors on a car return a deluge of raw data that the vehicle's AI brain must instantly translate into a model of its surroundings. Every moment is a potential surprise. In short, an automotive robot needs perception and collision-avoidance reasoning that an airplane's autopilot simply never required. It took major advances in computer vision and machine learning to give cars something approaching the visual intelligence needed to drive safely. Over the past decade, the advent of deep neural networks has dramatically improved machines' ability to recognize and

classify the objects and conditions around them. Modern self-driving systems integrate cameras, lidar, and radar with AI models trained to detect vehicles, people, road signs, lane boundaries, and more in real time. By learning from vast amounts of driving data, these perception systems have reached a level of accuracy and adaptability that was unattainable with earlier methods. They can now handle the “long tail” of odd situations from flocks of geese on the roadway to construction zones with hand-waving flaggers that once confounded every robotic driver. We have by no means solved autonomous urban driving as a general problem, but the progress is unmistakable. What began as a DARPA challenge to simply stay on a dirt road has evolved into experimental robotaxis navigating actual city streets. In a sense, robots have graduated from the structured highways of the sky to the free-for-all of the road.

Yet even roads and highways, for all their complexity, are relatively structured environments. They have lanes, traffic rules, and a certain regularity of design. The next frontier is far messier: our homes, sidewalks, farms, workplaces, parks, and other everyday spaces. In the mid-2010s, DARPA ran another competition; this time for humanoid bipeds and other robots to perform disaster-relief tasks like opening doors, turning valves, and clearing debris. The 2015 DARPA Robotics Challenge yielded cringe-worthy viral videos of bipedal robots toppling over as they tried to simply walk or climb stairs.⁶ It was a potent reminder that making walking machines as agile as humans was exceedingly hard. But again, progress came faster than expected. Just two years later, videos emerged from Boston Dynamics (a company that had built one of the Challenge’s more successful entrants) showing its Atlas humanoid robot not only walking steadily but executing backflips and gymnastic maneuvers.⁷ In one clip, Atlas vaults onto platforms and does a perfect backflip, then raises its metal arms as if in triumph.

While humanoid robotics were making rapid progress at general locomotion, the ability to perform a wide range of tasks requiring precision tactile movements remained out of reach. Humans take advantage of the fact that we can stroll through a living room strewn with toys, or step into a garden to pick a ripe tomato and effortlessly handle the variability. We can seamlessly transition from a precise task that requires very fine-grained control of gentle pressure like painting trim work in a house to picking up a hammer and pounding in some nails. For robots, these types of tasks are vastly more complex and unconstrained than simply moving around in open environments. A robot butler would need to perceive and understand a dizzying variety of objects. Is that a sock or a washcloth under the table? Is that a spill of water or a patch of sunlight on the floor? It would then need to physically handle a wide range of items, surfaces, and tools designed for human hands and human common sense. It’s no wonder that in robotics labs, domestic tasks like folding towels or unloading dishwashers have become classical

benchmarks of dexterity and intelligence. In 2012, a state-of-the-art research robot took about six minutes to fold a single towel, carefully inching along the fabric to align the edges; a task most people could do tidily in ten seconds.⁸ Such demonstrations are equal parts impressive and humbling: impressive because the robot is doing something so intrinsically human, and humbling because it highlights how challenging it is to develop robotics that can handle ordinary chores.

A core difficulty has been manipulation: getting robotic hands (or grippers or suction cups) to reliably handle the physical objects of daily life. Industrial robots excel with rigid, standardized parts. Outside the factory, a robot may be asked to pick up a loaf of bread one moment and a screwdriver the next, each requiring a different grasping strategy. Worse yet are deformable or delicate objects. The pliability of a half-full plastic water bottle or the squishiness of a plush toy poses a huge problem for conventional robotic manipulation. Unlike a rigid object, which has a fixed shape and center of mass, a deformable object has effectively infinite degrees of freedom; every squeeze changes its shape in complex ways.⁹ Beyond pliable objects, there is an entire family of reactive objects such as pets, insects, farm animals and humans. The simple task of shaking hands with another person is an incredibly complex feat. Our brains and fingers handle this effortlessly but programming a robot to do the same has proven fiendishly hard.

Even the simple act of putting away groceries encapsulates the challenge. Imagine a machine trying to unload a bag of groceries: it must recognize produce of various shapes, sizes, and fragilities, grasp each item appropriately and figure out where in the kitchen each thing belongs and with what orientation. If it doesn't squeeze a hard and slippery glass bottle with enough force, it may slip through its hands. On the other hand, applying the same force to a baguette may destroy it. Now compare that to what an assembly-line robot does, such as picking up identical metal panels from a stack and welding them at exactly the same spot on every car chassis. The grocery scenario is orders of magnitude more complex. In early 2025, a company called Figure released a multimodal vision-language-action that they named Helix. With inspiration from the human brain, this multimodal model handles multiple modalities but is also divided into two systems. One system operates more slowly and tackles higher level tasks such as identifying if an object is a bottle of ketchup or a tube of toothpaste and then making a decision about whether to place it in a fridge or a cupboard. The other system operates at a much higher frequency and handles fine-grained motor movements. This is similar to how our own brains work with conscious decision-making being a relatively slow and thoughtful process while we unconsciously tackle many tasks such as catching a falling spoon at incredible speed. To demonstrate the power of this system, Figure created a video showing a team of two humanoid robots putting away groceries; finally delivering

on a vision of robots capable of performing general domestic tasks that has been a staple of science fiction for decades. As of 2024, there are over 160 companies actively developing humanoid robotics.¹⁰ The costs of early models are high enough that it is unlikely we will be hiring them to fold our laundry anytime soon but the rate and diversity of progress is the fastest that it has ever been.

Applications of Robotics

We can see the impact of these advances in an explosion of new robot applications beyond the factory. On construction sites, for instance, robots are beginning to take on some of the dirty and dangerous tasks traditionally done by hand. Experimental bricklaying robots can spread mortar and set bricks in neat courses much faster than a human mason, with one prototype able to lay upwards of a thousand bricks an hour. That kind of speed is only achievable in a controlled demo today, but even semi-autonomous masonry assistants are already working alongside human builders, boosting productivity. Other construction bots drill holes in concrete, tie rebar, or inspect structures with 3D scanners. The environment is still challenging in outdoor sites that are uneven and cluttered, but the payoff of automating construction is so high that innovation is rapid. In agriculture, robots are departing the confines of climate-controlled barns and heading into the field. Automated milking machines have been around for years, but they were able to work under carefully controlled environmental and lighting conditions. Strawberry farmers in Japan have now tested a robot that uses stereoscopic vision to locate ripe strawberries and gently snip them from the vine, reportedly cutting harvest time by 40% compared to human pickers.¹¹ Other robotic pickers target apples, oranges, cucumbers; teaching machines to discern “ripe and ready” from “not yet” and handle each fruit without bruising it. And where picking is tricky, some farms use robots for weeding: small autonomous rovers creeping through rows of crops, identifying weeds by machine vision, and plucking or even zapping them one by one with pinpoint lasers.¹² This level of precision weed control would have been science fiction a generation ago; now it’s a commercial product aimed at reducing herbicide use and labor. On city sidewalks and college campuses, one might encounter cooler-sized delivery robots trundling along with lunch orders. These delivery bots leverage the same advances that enable autonomous cars but apply them at pedestrian speed for short distances. Dozens of universities and some urban neighborhoods have deployed such robots to ferry meals and packages locally. In restaurants, we are seeing the first generation of robotic cooks and servers. A robotic fry cook named Flippy has been flipping burgers in pilot programs at fast-food chains, working behind the counter to handle the greasy,

repetitive grilling tasks. In one report, Flippy was capable of grilling 150 burgers per hour: output that was so fast that human counterparts could not keep up with the job of assembling the burgers.¹³ These kitchen robots are typically designed to fit into existing kitchens without massive reconfiguration, functioning as extra sets of hands. They use computer vision to tell burgers apart from fries and to know exactly when to flip or when to pull items from the oil. Importantly, they operate under human supervision and alongside human co-workers, who still handle more complex prep and all customer interaction. Even cleaning up has robotic helpers: there are floor-scrubbing robots for supermarkets that can work repeatedly without a break. Taken together, these examples show robots edging into the service sector in roles that involve physical tasks in complex environments.

Healthcare is another arena where robots are expanding their presence, often in supportive roles that amplify human skills. Surgical robotics is now a mature field. As one example, da Vinci surgical systems (teleoperated robots for minimally invasive surgery) have performed over 14 million procedures via human surgeons at the controls, offering finer precision and smaller incisions in everything from heart valve repairs to prostate surgeries.¹⁴ What's striking is how robots and surgeons can collaborate; the surgeon's hand movements are translated by the robot into micromovements, scaling down any tremor and adding motion smoothing. In effect, the robot can filter out the tiny hand shakes that even the steadiest surgeons have. These systems are not autonomous. They are more like power steering for the surgeon, blending human judgment with robotic precision. In other areas of healthcare, robots are being tried as caregiver assistants. Japan, facing an aging population and a shortage of young caregivers, has famously experimented with robots in elder care. One remarkable example is Robear, a friendly-looking bear-inspired robot developed as a nursing aide. Robear can gently lift a frail patient from a bed into a wheelchair or help them stand up, tasks that human caregivers typically have to perform many times a day and that often lead to back injuries.¹⁵ Outfitted with soft touch sensors and precise actuators, Robear is strong enough to support a human's weight but calibrated to be gentle and safe; "*the strong robot with the gentle touch*," as its creators call it. It's still experimental and not widely deployed, but it showcases how a robot roughly the size and shape of a large teddy bear could physically interact with people in a helpful way. In Singapore, meanwhile, an entrepreneurial clinic has introduced a robot massage therapist named EMMA (short for Expert Manipulative Massage Automation). EMMA uses an articulated arm with sensors to deliver therapeutic massages informed by traditional Chinese medicine techniques.¹⁶ Patients who have used the robot alongside human therapists reported that its treatments are almost indistinguishable from a skilled masseuse's touch. The robot

employs computer vision and pressure feedback to adjust to each patient's body and can work tirelessly through appointment after appointment without fatigue. The intention isn't to eliminate the human therapist, but to take over the labor-intensive, repetitive motions so the human practitioner can focus on diagnosis and patient communication. These examples underscore a pattern: where tasks are physically strenuous, repetitive, or require superhuman precision, robots are stepping in as co-workers and tool extenders for humans. A surgeon with a robotic assistant, a nurse with a lifting robot, or a physical therapist with a massage robot can each achieve results that neither could alone with the potential of bringing higher quality healthcare to larger numbers of people.

Despite this proliferation of robotic helpers, the physical world remains far less automated than the digital one and the rate of change will likely be much slower. We don't yet have robo-butlers in every home or humanoid carpenters on call. One major reason is economic: building and deploying robots that can handle complex physical tasks has historically been very expensive. Software can scale to billions of users at almost zero marginal cost, but each physical robot unit has a real price tag in materials and assembly. For a long time, only the auto giants and semiconductor fabs could afford cutting-edge robotics, which is why those industries led adoption. Even as technology improves, a smaller company might look at a \$100,000 robot and balk, thinking that humans are cheaper and good enough for the time being. This is gradually changing. The costs for robots are finally coming down, lowering the barrier to entry. In fact, the average price of an industrial robot has halved in recent times; from around \$47,000 in 2011 to about \$23,000 in 2022.¹⁷ Economies of scale and better design are kicking in. As a result, robot adoption is accelerating beyond its traditional strongholds. Globally, the installed base of industrial robots crossed 4 million units in 2023, a number that has since been exceeded as automation booms.¹⁸ Figure, the company that developed the humanoid robots capable of putting away groceries has recently completed a new manufacturing facility and plans to deliver 100,000 humanoid robots in the next four years. In the consumer realm, one could argue the first truly mass-market robot has already arrived: the robot vacuum. iRobot, the maker of Roomba, has sold over 50 million home robots as of the early 2020s.¹⁹ That's a fleet of machines quietly navigating millions of living rooms and bedrooms, every day, doing useful work. They succeed by focusing narrowly on a single task (floor cleaning) and doing it well. Their popularity demonstrates that when robotic technology becomes both affordable and reliable enough, it can spread remarkably quickly.

Despite all of the above, some jobs and tasks remain deeply resistant to automation. These often involve a mix of complex perception, fine manipulation, and social intelligence that pushes the limits of what machines can do. Take the role of a dental

hygienist. It's a job that on the surface might seem routine; clean teeth, take X-rays and apply fluoride. In practice it demands subtle hand-eye coordination, adaptation to each patient's anatomy and pain sensitivity, and real-time communication to keep the patient comfortable. A robot would need to handle a mirror and scaler inside a human mouth as deftly as a trained hygienist, distinguishing plaque from gum tissue by touch while maybe cracking a joke to put the patient at ease. Unlike surgery robots, a dental hygienist works on a conscious patient that squirms, winces, yawns, and coughs. We are nowhere near that level of dexterity and empathy in AI. Similarly, consider skilled trades like electricians or plumbers. The work is highly physical and wildly variable: an electrician might be crawling through an old crowded attic fishing wiring through joists one day and the next day meticulously wiring a new circuit panel. This requires improvisation, judgment, and adaptability to messy real-world conditions that general-purpose robots just don't have yet. Barbers and hairstylists combine delicate manual skills such as using scissors or razors around a customer's head with personal interaction and aesthetic judgment. There are experimental haircutting robots, but they are stiff and inflexible, giving results about as good as a bowl cut. The nuance of tailoring a style to an individual's face, hair texture, and preference, all while chatting to build trust, is very much a human art at present. Veterinarians and animal handlers face the additional complication that their patients can't explain their symptoms and often won't sit still or cooperate; sensing illness in an animal or safely restraining it for treatment is a complex dance of intuition and care that defies current algorithmic approaches. More broadly, jobs that rely on emotional intelligence, creative problem-solving, and handcrafting in unstructured environments, from preschool teachers to first responders, are relatively safe from automation in the near term. Of course, unlikely doesn't mean never. There is a famous adage in the field of AI that tasks we thought would be easy (for example, recognizing faces or understanding speech) turned out to be very hard, and tasks we assumed were hard (like winning at chess or Go) turned out to be easier for AI than expected. The same could end up being true for robotics. We might be taken by surprise when some enterprising team figures out an AI-powered robotic barber that can fade and trim as well as the average stylist, or a home handyperson bot that can do basic repairs. Technological progress often comes in bursts and from directions we don't anticipate. In 2004, few would have believed that within a decade cars would be driving themselves on public streets but the Grand Challenge and its successors made that a reality. Likewise, it's risky to declare any job truly automation-proof in the long run. What we can say is that the bar for automating these complex physical and social tasks is extremely high, and clearing it will require breakthroughs in AI, robotics, and likely

a redesign of the environments and tools we use. For now, humans still hold the high ground in these areas.

Shared Environments

As robots venture out of their comfort zones, we are essentially witnessing a grand co-evolution of technology and environment. Our homes, cities, and workplaces were engineered over centuries to suit human dimensions and capabilities. Door knobs are placed at hand height, cars are designed for human drivers, and language and social cues underpin almost every task. It makes sense that the first robots to truly thrive in these spaces might be the ones designed most like us. In fact, many roboticists pursue humanoid designs precisely because the world is built for humans. NASA's Robonaut project explained that the robots were *"designed to use the same tools as humans, which allows them to work safely side-by-side humans on Earth and in space."*²⁰ A humanoid robot can climb the same stairs, turn the same doorknobs, and wield the same instruments as a person, giving it immediate access to all the infrastructure and tools we've already built. If your goal is to make a specialized robot that can pound nails, a humanoid may not be the optimal form factor. However, if your goal is to make a robot that can do any construction task, then a general-purpose humanoid robot could walk onto a job site and be immediately capable of using every tool available to a human laborer. That's a powerful shortcut; rather than refitting every house, worksite and vehicle for robots, we can build robots that fit into our existing world. We should be careful not to over-hype humanoids; they inherit all the challenges of balance, coordination, and dexterity that humans master through millions of years of evolution and years of childhood learning. But the rationale is clear for pursuing drop-in replacements for human labor. That vision drives many ambitious research agendas and startups. Time will tell whether it's the right approach or not. We may find that specialized robots working in concert outperform a single humanoid that tries to do it all. Perhaps we will decide to add valuable tweaks that have not yet appeared through evolution in humans such as the addition of an extra arm or an additional set of eyes on the back of a humanoid head. In any case, the trajectory is clear: robots are moving from constrained, highly structured niches into the open-ended, dynamic world we inhabit, sensing and acting with increasing sophistication.

In the coming years, we will likely see robots tackling ever more human environments and tasks. They'll still stumble, both literally and metaphorically, but each generation will learn from the last. We're liable to underestimate the rate at which robots will learn because it's easy to forget that robots can share their knowledge immediately. When a

single humanoid robot learns how to cut open a jackfruit, they can share that knowledge immediately with every other robot in their network and every robot coming off the manufacturing line will now have that knowledge from day one. With thousands of robots learning in parallel, the rate of progress is likely to be surprising. With machine learning, robots can refine their models of the world through trial and error, getting better after every failure in a way early hard-coded robots never could. As robots become more commonplace, we will also adapt to them. We might standardize certain things in our environment to accommodate our new helpers. In the same way that efforts have increased in recent years to ensure the built environment is accessible for humans with disabilities, we may also see building codes adapted to ensure accessibility for robots. Fast forward a decade or two, and we could imagine that the building code may be updated to require that every home includes a closet with a charging port for a humanoid robot.

Looking ahead, there is an even subtler shift to contemplate: the possibility that our built environments will evolve to suit robots' strengths and not our own. History provides precedents. When horses gave way to automobiles, city design changed drastically; streets widened, distances between destinations grew, pedestrian life was deemphasized. When shipping moved from loose cargo to standardized containers, port facilities expanded in scale and complexity far beyond human proportions, and the labor force at the docks dwindled by an order of magnitude. So too might the age of humanoid robots herald design choices that implicitly assume a robot workforce. On a construction site, for example, if one can deploy a team of brawny robotic laborers, why deliver bricks or steel in sizes a human can lift? Perhaps materials will arrive in larger modules, assembled whole by machine. Such efficiencies would be welcome in terms of productivity. Fewer, larger components would allow for buildings to be constructed more quickly but they could render any remaining human workers effectively obsolete or at least secondary. A carpenter can comfortably handle a four-foot by eight-foot sheet of plywood but would struggle with a six-foot by ten-foot sheet, even though a construction robot might do so with ease. Over time, the very notion of designing for human ergonomics might erode in certain industries. Factories and warehouses could become dark facilities with narrow corridors and high-density storage optimized for robot navigation rather than human comfort.²¹ In these spaces, a human would be superfluous or even a hindrance; a far cry from the traditional factory floor that was engineered around human capacities and limits. In fact, these facilities may intentionally be built with fewer fire exits and lower levels of ventilation as cost saving measures because those will not be required to the same extent if the only workers are robots. This progression would deepen our dependency on robotic labor in a self-reinforcing loop.

Once infrastructure and tools are sized and arranged for machines, switching back to human labor becomes impractical, even if there is an emergency. We could find ourselves, collectively, in the position of sorcerer's apprentices: having conjured powerful helpers and rearranged the world for their convenience, we are helpless to do without them. Economists sometimes discuss path dependence, where initial advantages snowball and alternative paths close off. Here the path is one where human physical abilities no longer set the scale of our world; robot abilities do. If a warehouse is built with shelves reaching the ceiling, accessible only by robotic forklifts, then even if unemployment were high, you couldn't simply replace those robots with people; the workplace would be literally beyond human reach. Humans might end up working in tandem with robots (wearing exoskeletons or operating power tools) to cope with the new norms, effectively forcing us to adapt to the machines rather than vice versa. The long-term social implications of such a shift are hard to foresee in full. Will we design public spaces differently, knowing service robots will be cleaning and maintaining them at night? Will hospitals have corridors sized for robot stretchers or pharmacies with automated humanoid staff retrieving medications from vaults no human pharmacist enters? If so, what happens to the human professionals who once inhabited those roles? Do they become supervisors of fleets of robots, or are they phased out entirely? There is also a philosophical dimension: a subtle de-centering of the human in our physical environment. For millennia, our tools and habitats have been extensions of our bodies; the height of a step, the reach to a handle, all calibrated to human dimensions. In a robot-optimized world, we might begin to feel like strangers in a land built for others. This could further entrench a class distinction: those who own and control the robots navigate this new world with ease, while those without access struggle to find a place. It could also reinforce a kind of apathy or loss of skills among people. If robots routinely lift every heavy object, will human strength atrophy, much as our arithmetic skills have in an era of calculators? Some sociologists speculate about a creeping deskilling; our muscles, our situational awareness, and our craftsmanship might dull from disuse. With that, our agency would diminish and we would become ever more reliant on robotic intermediaries to accomplish what once we could do unaided. None of this is to prophesy a dystopia of helpless humans amidst towering machines, but it is a trajectory that warrants reflection now, while choices are still fluid. We have the opportunity to guide how environments evolve in tandem with robotics. Perhaps the ideal is a hybrid design philosophy: make workplaces and public areas accommodating to both robot and human actors, playing to each of their strengths. For example, maybe construction elements are designed with features that allow either a human team or a robot to manipulate them, ensuring resilience if one or the other is unavailable. Perhaps safety

standards will mandate that critical infrastructure cannot be exclusively serviceable by robots; a kind of backup plan for human operation in dire situations. In any event, awareness is key. The more we acknowledge that technology shapes our environment, the more proactively we can decide whose capabilities to prioritize. Should we widen a road to let delivery robots pass pedestrians easily, or should we constrain the robots to respect the original human scale? These decisions will cumulatively define the texture of daily life in the automated age.

Mass Unemployment

Early evidence from industrial settings shows a sobering trend: robots tend to displace workers rather than simply augment them. In manufacturing, for instance, the introduction of one autonomous robot can eliminate multiple human jobs and exert downward pressure on wages in the region.²² When a well-paid technical laborer is replaced by a robot, that worker often must tumble down the economic ladder; taking a lower-paying service job if one can be found and potentially displacing other workers in those sectors. The productivity gains from automation are real, but so is this displacement effect, and it haunts the industrial heartlands. Decades of automation have already hollowed out mid-skill employment, and humanoid robots threaten to carry this disruption into the remaining bastions of human labor. A machinist or construction foreman may find that the new “helper” on the team is a machine capable of working tirelessly and with superhuman precision, quietly undercutting the necessity of human expertise. The affected workers face a bitter dilemma: the jobs comparable in pay to their old livelihoods might simply cease to exist, and the available alternatives often come with less security and dignity. Economists warn that without intervention, this wave of robotics could further erode the middle class, concentrating wealth in the hands of those who design and deploy the robots. The societal trauma of sudden technological unemployment may echo for generations, as communities grapple with lost income and purpose. It is a future where the benefits of humanoid labor-saving are unevenly distributed, accruing to firms and consumers while many workers struggle to reinvent themselves in a labor market transformed by machines. Policymakers are thus confronted with a paradox: the same humanoid robots that boost productivity and growth on a macro scale can inflict profound hardship at the individual level, compounding inequality within and between nations. The challenge is no longer whether robots can take on ever more human tasks, but whether society can manage the fallout when they do. The magnitude of this disruption has been likened to a new industrial revolution.

Within prosperous societies, the rewards of humanoid automation may be unevenly shared. Just as industrialization in the 19th century enriched factory owners more than factory workers, the rise of general-purpose robots in the 21st could widen class divides. Consider the household of the near future: wealthier families might effectively employ personal humanoids as live-in staff; tireless tutors, nannies, housekeepers, and even athletic trainers for their children. Having a machine with boundless patience to practice piano scales with a child, or coach them on their basketball shot for an extra hour each evening, could confer an enormous developmental edge. These scenarios are not far-fetched; early versions of educational robots are already appearing in some classrooms²³ and therapeutic settings, and affluent early adopters are often the first to access expensive new tech. Social scientists caution that such trends could aggravate the “opportunity gap” between children of privilege and those of modest means. Research on educational inequality has long shown that much of the achievement gap is driven by out-of-school enrichment; the summer camps, lessons, and personal attention that money can buy.²⁴ Humanoid robots threaten to turbocharge this effect. Unlike human tutors or caregivers, whose time is limited and costly, a robot can provide one-on-one engagement nearly around the clock at marginal expense once acquired. If only wealthier households can afford the up-front cost, their children could effectively have a superhuman personal assistant guiding their homework, reminding them to exercise, correcting their posture, teaching them new languages, and generally amplifying their upbringing in countless ways. Meanwhile, children in less advantaged homes would not only miss out on those enhancements but might also find themselves interacting with robots in less empowering roles; perhaps as one of many students supervised by a single educational robot at an underfunded school, rather than enjoying the bespoke attention a rich child’s robot offers. Over time, this technology-driven tutoring gap could translate into wider disparities in academic performance, college admissions, and career opportunities. The notion of technology as a great equalizer breaks down if access to the latest tools is skewed by income. Indeed, studies show that differences in children’s learning experiences track closely with family resources. If robots become the new medium of parental investment, they may deepen the stratification of society into technological haves and have-nots. One can imagine a future college admissions essay reading not about how a personal AI robot enriched their scientific curiosity from age five and helped them design and run scientific experiments: a resource unimaginable to peers from lower-income backgrounds. Such a prospect has ethicists urging preemptive reflection on fairness: should access to robotic tutors or caregivers be regulated or subsidized in some way? Or will we accept a world where childhood itself is augmented for some and not for others? Without deliberate interventions, humanoid robots could

inadvertently widen class-based gulfs, entrenching advantage in ways even more insidious than earlier technologies, precisely because they operate in the intimate sphere of learning and development.

The benefits of this robotic revolution are poised to accrue unevenly not just among social classes but across countries. We may soon see a geopolitical stratification in which a single nation or a tight oligopoly of nations dominates the humanoid robot industry, leaving others dependent on imported mechanical labor. Many companies worldwide are racing to build viable humanoids, yet industry analysts foresee only a few winners achieving the scale and cost-effectiveness needed for mass adoption. If those winners cluster in one country, that country could wield immense economic leverage: a scenario with echoes of the global semiconductor supply chain. Today's policy strategists point to how a single company or nation's edge in chip manufacturing has created vulnerabilities for all others; tomorrow's concern is a monopolization of advanced robots. Recent reports highlight, for example, how China has rapidly built a commanding lead in the key components for humanoid robots, from high-precision actuators to battery production. According to a detailed analysis, Chinese firms control roughly 63% of the critical humanoid-robot component supply chain, a dominance enabling them to produce robots at markedly lower cost than competitors.²⁵ This asymmetry in capability means that if humanoids become as indispensable as automobiles or computers, any country without its own domestic supply would find itself in a position of strategic dependency. We can see a preview of this by looking at drones. China gained an early lead in the mass production of drones and Chinese drones hold 70% of the consumer market and 90% of the enterprise market in the US with a single Chinese company, DJI, being the predominant supplier.²⁶ When DJI launched a new drone called the DJI Mavic 4 Pro in May of 2025 during a period of heightened tariff related tensions between the US and China, they made the surprising decision to not make the technology available to US customers. The concentration of manufacturing know-how and production capacity in one place could become an economic lever or a geopolitical weapon in the hands of the dominant supplier. It is not difficult to imagine a future trade crisis where access to robotic workers becomes a bargaining chip, much as access to oil or advanced chips have been in the past. Moreover, the country that leads in humanoid robotics will shape the industry's norms and standards to its advantage, potentially locking others out. The irony is that while robots promise to mitigate labor shortages in aging societies and drive new growth, they could also redraw the map of global power imbalances. Policymakers in smaller or lagging nations worry about a future in which they must import not only energy or electronics, but basic labor in the form of foreign-made machines; a dependency that could be as fraught as any resource scarcity. Some commentators have gone so

far as to suggest a parallel to the “OPEC of robots”, where a handful of exporters can set prices and conditions for a technology that underpins entire economies. This looming prospect has already sparked calls in various countries for robotic R&D investment and alliance-building, lest they be left in the lurch. Humanoid robots have become a new arena of international competition; one that intertwines economic opportunity with national security concerns. The nations that seize the lead may enjoy a windfall of prosperity and influence, while those that fall behind could face a precarious reliance on foreign robotic labor to keep their industries and households running.

Social Expectations of Robot Behavior

If personal robots become status symbols and competitive tools, we may also see new strains on social norms in public life. People have a way of bending rules when they wield a technological edge, and humanoid robots would be no exception. A telling early example occurred in 2015, when a resourceful young woman in Sydney sent a telepresence robot, essentially an iPad on a wheeled stand, to hold her place in line overnight for the release of a new iPhone.²⁷ The robot queued up with devoted Apple fans, weathering wind and rain, and the next morning the robot, bearing the woman’s face on its screen rolled into the store to claim the coveted devices. What felt like a quirky publicity stunt at the time hinted at a deeper principle: robots can serve as stand-ins for their owners, potentially giving them an advantage in time and presence. Today it’s a spot in a product launch line; tomorrow it could be any number of mundane competitions for scarce opportunities; a robot sent to secure a place in a crowded overnight camping queue for college registration, or to occupy a public tennis court at dawn until its owner arrives. These scenarios force us to ask: should a robot be treated like a person in such situations, or like mere property? If a humanoid robot is physically waiting in a queue, do other people recognize its claim as legitimate? Or might we see backlash, with flesh-and-blood bystanders resentful of a machine taking up space or gaming the system? Such behavior blurs the line between human and proxy in public accommodations. It also raises difficult enforcement questions: can an establishment or event legally exclude robots, or cap their numbers, to prevent one person from multiplying their presence via machines?

More broadly, must humanoid robots obey the informal social rules that govern public spaces, and if so, how do we ensure they do? Humans walking down a sidewalk are expected not to jostle others, not to block doorways, not to litter or shout obscenities. If a humanoid robot marches through a park blaring music at high volume, or jaywalks across a busy street, who is accountable, the machine, its owner, its manufacturer?

In cities testing delivery robots, residents have grown impatient with machines that stop in crosswalks or crowd narrow walkways. San Francisco, famously tolerant of tech experiments, had to impose strict regulations when delivery robots began proliferating; permits were limited and robots were confined to certain neighborhoods and required to have human monitors, after complaints that these wheeled couriers were turning sidewalks into obstacle courses.²⁸ That episode offers a preview of the frictions likely to arise as free-roaming humanoids appear in our midst. Society will likely insist that robots abide by the same basic standards of public conduct as people; perhaps even higher standards, given that unlike humans they can be perfectly programmed not to misbehave. There have been serious proposals that robots operating in shared spaces carry a form of registration or license, much like automobiles do, to make them traceable if something goes wrong. This debate includes discussion about whether robots should be registered as electronic persons in the same way that corporations are considered persons under many aspects of the law.²⁹ If a robot tips over a vending cart or collides with a pedestrian, authorities will need a way to identify its owner or operator. This accountability could take the form of visible identification plates or wireless signaling of an ID number. Such measures are inspired by analogies to cars and industrial machinery; realms where strict safety standards and operator responsibilities are long established.

Regulation of robots will likely be much simpler and more successful than regulation of AI. Unlike pure software algorithms that can be copied effortlessly and anonymously, physical robots are discrete units that can, in principle, be catalogued and regulated at manufacture and point-of-use. The deployment of robots will be much slower given physical constraints on how quickly they can be built. Regulators see an opportunity to get ahead of problems by building in oversight: proposals have included mandatory “kill switches” to allow immediate shutdown of a wayward robot, and requirements that robots be engineered to clearly signal that they are machines, not living beings.³⁰ In the end, however, any regulatory regime for robots will need to reconcile competing values: innovation and utility on one hand, versus safety, privacy, and public order on the other. Striking that balance will be an ongoing process, likely requiring international coordination given the global nature of the robotics industry.

Law Enforcement Applications

Nowhere will a framework be more crucial than when robots are entrusted with the use of force; perhaps the most fraught application of all. A humanoid robot that patrols a shopping mall as a security guard, or one deployed by police to subdue a dangerous suspect, brings to the fore ethical dilemmas that have simmered in science fiction for

generations. Unlike a human officer, a robot has no intuition or empathy as it executes its training and programming. This rigidity could be perilous if the robot's sensors and algorithms misinterpret a situation. Imagine a robot police sentinel on a nighttime beat that comes upon two inebriated friends engaging in some playful but aggressive horseplay. One person grabs the other's arm: a motion the robot's vision system flags as an assault. Will the robot intervene with physical force? If a child runs toward the robot for a playful hug, could the robot's threat-detection misfire and respond as if under attack? What if a robot happens upon a costume party of a group of high school students re-enacting a crime for their film class? Then, there are questions about who should be able to deploy robots that have the ability to use force. Should celebrities be allowed to hire robotic bodyguards that would be authorized to use force or should the government be the only entity with that permission? These scenarios underline the need for extreme caution and extensive debate. In the realm of law enforcement, a single mistake by an autonomous system could be catastrophic; a life unjustly taken or a conflict needlessly escalated. The consensus among many ethicists and policymakers is that however advanced robots become, machines should not have final say in the use of deadly force. Recent events in the United States illustrate how real this issue already is. In 2022, the city of San Francisco grappled with a proposal to allow its police department to deploy robots with lethal capabilities in extreme circumstances.³¹ Initially, city supervisors gave a green light to the idea that a bomb-disposal robot, for example, could be armed with explosives to neutralize a mass shooter if no other option was available. The public reaction was swift and vehement. Civil liberties groups and many citizens recoiled at the prospect of "killer robots" on American streets, even under police control. After an outcry, the plan was reversed; the robots would not be authorized to kill. But the episode was instructive: even in tightly constrained scenarios, society is deeply uneasy about granting robots agency to harm humans. The margin for error is simply too thin. A robot that can pull a trigger or detonate a charge must operate under impeccable safeguards, and still the moral question remains of whether such delegation of force is ever acceptable. Experts argue that any use of robots in security roles must keep a human in the loop for critical decisions. The robot can be an aide, a tool, an information-gatherer, but not judge, jury, and executioner. This principle aligns with international efforts to ban or restrict fully autonomous weapons before they proliferate. Should a robot ever use physical coercion on a person, it carries the weight of an entire society's rules and biases in its silicon brain. This realization has led to calls for transparency: if robots are to enforce laws, their decision algorithms should perhaps be open to public scrutiny, much as court proceedings are. Otherwise, we risk creating a black-box authority: a robo-enforcer that acts without us fully understanding why.

We have no choice but to be exceedingly careful about where and how we deploy such machines, because once a tragic error occurs, it cannot be undone.

Military

One night in 1983, a Soviet lieutenant colonel named Stanislav Petrov faced a flickering warning screen that claimed American nuclear missiles were inbound. In theory, his orders were clear: report the “attack” and trigger a counterstrike. But Petrov hesitated. Something felt off. Why would a real first strike come as a mere handful of missiles? Trusting his intuition over the early-warning computer, he decided it was a false alarm and held his fire.³² He was correct, and his act of prudence averted the potential for a nuclear war. Now imagine that fateful decision being left not to a human but to an algorithm. Would the machine have the wisdom to question its own data? The risks of AI and robotics in military applications become stark through that contrast: in the most critical moments, will a human conscience be present, or just cold code?

Today’s militaries are racing to bring more AI and autonomy into warfare. Autonomous drones, robotic sentries, and algorithmic early-warning systems are no longer science fiction but prototypes and even deployed devices. Indeed, many countries already employ automated weapons defenses, like high-speed missile interceptors and anti-rocket systems, that can identify and fire on incoming threats without immediate human direction. These systems, such as the U.S. Navy’s Aegis or Israel’s Iron Dome, act as human-supervised guardians: if a human controller doesn’t intervene in time, the machine will fire by itself to shoot down an attack. The logic is straightforward: machines react faster than people. 21st-century algorithms can process sensor data and execute a response in milliseconds, far quicker than a soldier can blink. Chinese military scholars now talk about a coming “battlefield singularity” when the tempo of combat outstrips human decision-making altogether.³³ Handing over the speed of war to silicon brains promises a tactical edge. But what about the control of war?

The consequences of ceding control to autonomous systems could be grave. In finance, we’ve seen how algorithms operating at blistering speeds can interact in unforeseen ways; “flash crashes” have sent markets plummeting in minutes, until human regulators hit the brakes. Unfortunately, battlefields have no circuit-breakers. A similar whirlwind in war is a nightmare scenario. Imagine autonomous aerial drones from rival nations, each programmed to strike when threatened, encountering one another. A jittery sensor or a software glitch could trigger a cycle of instantaneous reprisals. One drone’s mistake could set off another’s lethal response, each machine acting and reacting faster than any human commander could intervene. In a conflict rife with autonomous

systems, an absurdly small incident such as a flock of birds misidentified as an incoming missile could potentially escalate into a full-blown battle before any human even realizes an error occurred. The fog of war has always bred mistakes and friendly fire, but with autonomous weapons the fog could thicken to a point where even those who unleashed the machines don't fully understand their actions. The speed that makes these systems alluring also makes them brittle in the face of the unexpected. It is a black-box brittleness: when a deadly mistake happens, the algorithm cannot explain itself or feel remorse. And unlike Petrov, a machine will never intuitively disobey a flawed instruction; it will dutifully execute a fatal logic if that's what its code and data tell it to do.

Aware of these high stakes, the world's policymakers have begun crafting new rules to govern military AI and robotics; though in truth, they are scrambling to catch up to a technology racing ahead. International humanitarian law, the law of war forged in the wake of earlier horrors, already applies to all new weapons by default. No matter how novel or high-tech, an attack must still distinguish combatants from civilians and avoid disproportionate harm. Can an autonomous robot reliably follow the Geneva Conventions? Who is accountable if it violates them? These questions are testing the limits of the law. An autonomous system, by definition, selects its target and applies force on its own. The commander or operator might not even know exactly whom or what the robot will shoot when they activate it. This loss of direct human control *"raises serious concerns from humanitarian, legal and ethical perspectives,"* the International Committee of the Red Cross (ICRC) has warned.³⁴ If the robot makes a mistake, who bears responsibility; the software developer, the military commander, the machine itself?³⁵ Today, there is no clear answer. The fear is that autonomy could create a responsibility gap, where lethal harm occurs and no human can be readily held accountable. Accountability cannot be *"transferred to machines,"* as governments worldwide have formally agreed, yet without concrete rules and human oversight, that principle may be hard to uphold in practice.³⁶

Despite these worries, or rather because of them, diplomats have been convening in Geneva and elsewhere to negotiate how to keep a human leash on the technologies of war. Since 2017, discussions under the United Nations' Convention on Certain Conventional Weapons (CCW) have inched forward on the issue of lethal autonomous weapons systems. At the CCW, nearly every nation agrees at least on paper that humans must remain responsible for decisions to use force. In 2019, an international panel distilled a set of guiding principles – among them, that international humanitarian law applies fully to autonomous weapons, and that human control and judgment must be retained across a weapon's lifecycle. In essence, they affirmed that a robot is never absolved of the laws of war: its makers and operators are on the hook for its actions.³⁷ There is also broad acknowledgment that certain lines shouldn't be crossed. Even the

UN Secretary-General António Guterres has called lethal autonomous weapons “*politically unacceptable and morally repugnant*,” urging a global ban on systems that operate without human oversight.³⁸ In 2023 he went further, asking states to conclude by 2026 a legally binding treaty to prohibit autonomous weapons that can’t be used in line with international humanitarian law and to ensure human control over others.³⁹ The ICRC and many NGOs echo this, recommending an outright prohibition on any weapon that targets people or behaves unpredictably, and strict regulations on the rest. Their logic is straightforward: machines should not have a license to kill human beings, and no algorithm should be entrusted with life-and-death decisions that even seasoned officers struggle with.

Momentum is building. At the United Nations General Assembly in 2023, an overwhelming majority of states, 164 nations, supported a first-ever resolution on autonomous weapons, calling for the Secretary-General to report on the challenges they pose and possible solutions.⁴⁰ Dozens of countries, including a coalition led by nations like Austria, Costa Rica, and New Zealand, are now vocally calling for a preemptive ban on “killer robots”.⁴¹ This chorus spans continents and political systems, united by a shared intuition that removing humans from lethal decision-making crosses a moral red line. In a poignant parallel to earlier eras, they liken autonomous weapons to chemical or biological arms, abhorrent tools that the world managed to outlaw before they could wreak unchecked havoc. In 1995, states agreed to ban blinding laser weapons before they ever saw use on the battlefield, recognizing that some technologies are simply too cruel or destabilizing to allow.⁴² Many diplomats now argue that autonomous killing machines merit the same preemptive treatment. A line of prohibition should be drawn before an arms race spirals out of control.

And yet, for all this activity, progress is achingly slow and fraught with disagreement. The meetings in Geneva have been described as a tentative waltz: lots of diplomatic handshakes, statements of principle, but no binding rules in sight. A handful of militarily advanced states remain reluctant to commit to anything that might hamper their technological edge. These powers, including the United States and Russia, argue that existing humanitarian law is sufficient and that a new treaty might be premature or too restrictive.⁴³ Some of them are investing heavily in AI-driven arsenals themselves, and they prefer broad ethical guidelines over hard bans. Geopolitics looms over the debate: rising tensions and mistrust make consensus difficult just when it’s needed most. The result is a kind of regulatory paralysis. Despite nearly a decade of talks, the world has little to show in concrete safeguards while the technology marches onward. The gap between the rapid development of military AI and the glacial pace of international law is widening, and it troubles many observers. It means we are essentially relying on

voluntary restraint and the caution of individual commanders for now. That is a thin red line of defense against potential calamity.

In the absence of a universal law, stopgap measures and norms are emerging. Several nations have announced their own policies to ensure a human check remains in place. The U.S. Department of Defense, for instance, issued directives that any autonomous weapon must be designed to allow “appropriate levels of human judgment” in its use.⁴⁴ In practice, this means an American drone or missile with autonomous features is supposed to have a person somewhere in the decision loop; at least to approve activating the system and must be able to monitor and intervene once the autonomous functionality has been activated. Other militaries have echoed similar positions. Britain has stated it will not develop systems that operate without context-appropriate human involvement, and France has outlined ethical principles for AI use in defense. At the same time, the Pentagon and NATO allies are pouring funds into AI-driven surveillance and decision-support tools, which stop short of full autonomy but inch toward it. The line is thin: an algorithm that merely advises a human commander on which targets to strike can profoundly influence the decision, especially under time pressure. If a computer vision system flags a vehicle as an enemy with high confidence, how readily will the human override it? The risk of over-delegation, humans trusting the machine too much, is very real. Military history is replete with instances of human error; now we must consider machine error and human over-trust in those machines.

Outside of governments, civil society and even technologists themselves have mobilized. A global coalition of NGOs launched the Campaign to Stop Killer Robots in 2013, and it has since gathered endorsements from thousands of AI researchers, faith leaders, Nobel laureates, and ordinary citizens around the world. Notably, some of the very people designing advanced AI are among the most vocal in warning against autonomous weapons. In 2017, dozens of robotics and AI company founders, including renowned figures in Silicon Valley, sent an open letter to the United Nations urging a ban on these weapons, lest we usher in a “third revolution in warfare”.⁴⁵ They painted a dire picture: once loose, swarms of self-directing weapons could fight at a scale and speed humanity has never seen, “*at timescales faster than humans can comprehend*”. Such weapons, they warned, could be misused by tyrants and terrorists or even hacked to wreak havoc. The letter’s language was urgent: “*Once this Pandora’s box is opened, it will be hard to close*” and that plea helped drive the issue further into the diplomatic spotlight. Likewise, some tech companies have pledged not to develop AI for use in weapons, and a few high-profile engineers have resigned rather than contribute to military AI projects they felt crossed ethical lines. This unprecedented involvement of the private sector and engineers in arms control advocacy shows how deeply society is entwined in the

debate. Unlike past arms races which were the realm of governments and generals alone, the AI arms race involves corporate labs, university researchers, and even the everyday consumer technologies that could be repurposed for conflict. The lines between civilian and military innovation are blurred: a breakthrough in self-driving car vision or drone navigation can quickly find its way into an unmanned weapon. This makes controlling the spread of military AI even harder. International export controls, the tools used to stem nuclear and missile proliferation, struggle to keep up when the critical ingredient is advanced software or a machine learning model that can be emailed or downloaded and when this intelligence can be embodied in drones or humanoid robots that have many valid civilian applications.

As a result, gaps in governance persist. One glaring gap is enforcement: even if a new treaty is agreed upon, how will the world know if a nation or a company is secretly developing a forbidden autonomous weapon? Verifying a ban on a physical item like a missile is one thing; verifying the presence of an algorithm hidden in a computer chip or the extent to which a human was involved in a decision is quite another. The major powers have been wary of any agreement that might allow intrusive inspections of their AI labs or require them to disclose sensitive software; let alone any type of inspection of logs detailing how individual decisions were made in the peak of conflict. The likely path to enforcement will rely on transparency and confidence-building: nations might declare what systems they have and perhaps invite observers to see that a human is in the loop during exercises. Trust is thin in this domain. Each country suspects that rivals might gain an edge by pushing their AI to the limit. The United States, Russia, and China all publicly tout their investments in military AI, and smaller states feel pressure to keep up or risk strategic vulnerability. Defense contractors, for their part, compete to sell the most cutting-edge autonomous capabilities to militaries hungry for the latest tech. At arms expos, you can already find booths displaying swarming attack drones and robotic tanks, marketed as force-multipliers for the modern battlefield. This commercial drive means that even if leading states hold back on deploying certain technologies, there's a danger that less restrained actors could acquire autonomous weapons on the global market. The diffusion of robotics and AI is simply much harder to contain than that of, say, plutonium or chemical agents. A talented team of programmers with modest funding can potentially create a rudimentary autonomous weapon by repurposing civilian drones and open-source AI code. All this makes the governance challenge not only one of writing laws but of enforcing them in a practical sense. It calls for unprecedented cooperation between governments, the tech industry, and international organizations to monitor and control how these technologies spread and to agree on norms for their use.

What gives many experts pause is the sense that we are at a crossroads akin to the early days of the nuclear age, but without the same public awareness or fear. When the first atomic bombs showed their terrifying power, the world's superpowers at least recognized the need to negotiate controls, leading eventually to arms treaties that restrained arsenals. By contrast, autonomous weapons are creeping into use without dramatic "mushroom cloud" moments to focus global attention. A drone that autonomously guns down a soldier may not make headlines in the way a devastated city would. The impact is more insidious and can fly under the radar. Critics argue that society has not fully appreciated how profound the societal impact of military AI and robotics could be. It's not just another step in the evolution of warfare; it may be a fundamental shift. It could alter the threshold of going to war. If one can send machines to fight, might leaders be more cavalier about using force? If an algorithm kills innocent civilians, justice for the victims is harder to obtain as it may be explained away as an unfortunate software bug. It might even change power dynamics between nations, as those with AI superiority could dominate without ever deploying a single soldier. These are underappreciated repercussions that extend well beyond the immediate tactical realm.

For now, the world proceeds with both optimism and trepidation. International forums are alive with debate: committees parsing legal definitions of "meaningful human control," ethicists invoking the principle of human dignity, generals extolling the life-saving potential of precision AI strikes that reduce collateral damage. Ensuring responsible use of AI in the military is not just a technical endeavor but a moral one. Treaties and agreements from the Geneva Conventions to new AI-specific protocols are society's way of encoding our collective judgment about right and wrong in war. They evolve when pressured by necessity. The effort underway now, whether through the UN's formal processes or independent pledges like the Hague's call for responsible AI use, is an attempt to civilize the AI revolution before it fully overtakes the battlefield. It is a recognition that just because something can be done in war does not mean it should be done.

Conclusion

Our society stands at a crossroads as robots move from the realm of engineering breakthroughs into the fabric of daily existence. Multimodal models and improvements in mechanical systems allow robots to sense and act amid uncertainty; to recognize faces, grip irregular objects, and navigate chaotic streets. As we have now explored, bringing AI and robotics into our world comes with intricate risks and ethical quandaries. The promise of these machines including greater productivity, assistance with labor

shortages, and improved quality of life is accompanied by perils of economic upheaval, concentration of power, erosion of social norms, and challenges to our psychological and moral frameworks. The introduction of humanoid robots into society will not be a simple story of technological progress; it will be a negotiation between our aspirations and our values. We will have to decide which human jobs and roles are worth preserving even if robots could do them, and how to support those displaced when we decide efficiency wins out. We will have to build international trust and perhaps treaties to avoid new forms of dependency, robotic arms races or wars fueled by autonomous fighters. We will need public discourse about what behavior we permit from robots and from their owners, so that our streets and shared spaces remain civil and safe. We must cultivate digital literacy in the population, so people understand a robot's limitations despite its polished veneer, much as we taught media literacy in the age of mass communication. Crucially, we will have to educate a generation of engineers, designers, and policymakers in a fusion of disciplines, equipping them not only to make robots smarter and stronger, but also to embed ethical safeguards, transparency, and human-centric design into the very core of these technologies. The goal should be a future where these creations truly serve human needs and enrich human life, without eroding the foundations of work, community, and identity that define us. Achieving that balance will require wisdom as much as innovation; a collective mindfulness that the chapter of humanoid robotics, like all great technological chapters, is as much about understanding ourselves as it is about understanding our machines.

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Chapter Seven

AI Agent Collaboration

Introduction

Dr. Elara Vance watched the silent ballet of light on the holographic interface that filled her office wall. It represented the instantaneous rerouting of the entire West African agricultural supply web in response to a sudden, climate driven crop failure in the central plains. A million autonomous agents, each representing a farmer, a cooperative, a transporter, or a market, were renegotiating their existence in a fluid cascade of transactions that would have taken ten thousand humans a year to orchestrate. Her role was not to command but to tend to the ecosystem, to gently adjust the incentives and protocols that governed their interactions. It was a far cry from the chaotic days of the late twenties. She still remembered the digital maelstroms of that time, when rival corporate agent swarms waged silent wars that clogged global commerce and when the crisis of authenticity made it nearly impossible to distinguish genuine human discourse from sophisticated influence campaigns. Society had teetered on a strange new kind of precipice, not of physical conflict, but of total informational and economic gridlock. It took a near collapse of the logistics networks and a few harrowing market flash crashes, all driven by unconstrained agent interactions, for the world to finally implement the foundational layers of digital identity and accountability for autonomous systems, the very protocols she now curated. We had to build a kind of digital immune system for our global brain, a system that could identify and neutralize malicious swarms as they formed, and in doing so, we had unlocked this new era of frictionless, collective action. What she was witnessing now, the seamless redirection of food to prevent a famine before it could even begin, was the quiet triumph born from that turbulent past.

The Neuroscience of Human Collaboration

Collaboration is key to human survival and our success at this complex task is supported by many areas of the brain. Neuroscientists have conducted an experiment where participants performed solo, competitive, or cooperative tasks while monitored by fMRI scanners. The results showed that during cooperative tasks, a brain region nestled between the hemispheres, the anterior cingulate cortex (ACC), becomes more active compared to solo or competitive conditions. The ACC is one of several areas forming a network that supports collaboration.¹ In fact, the ACC is so attuned to social harmony that being ostracized by others triggers ACC activity in a very similar way to how the ACC is triggered during physical pain. In classic experiments, people who were socially excluded showed surges of ACC activity, and the more anguish they felt, the greater the ACC response.² The sting of rejection, neurologically, resembles a minor physical injury. It's the brain's way of saying that being cast out from the group hurts, quite literally, and motivates us to avoid the social fissures that could leave us isolated.

What other parts of the brain orchestrate this subtle dance of collaboration? Modern neuroscience has converged on the idea that there is no single "teamwork center" in the brain. Rather, there is an ensemble of regions, a social brain network, each contributing a different skill required for partnership and group dynamics. One major player is the medial prefrontal cortex (mPFC), situated right behind the forehead. This area often comes alive when we reflect on others' thoughts, intentions, and feelings. If you've ever tried to anticipate a colleague's next move or understand a friend's perspective during a project, your mPFC was hard at work. Studies of people engaged in strategic games show that the dorsal mPFC activates in tandem with regions like the temporoparietal junction (TPJ) when they are reasoning about someone else's mindset.³ The TPJ, located roughly above and behind the ear, is crucial for taking another's viewpoint; a neural mirror that lets us simulate what someone else might be thinking. Together, the TPJ and mPFC form a kind of "mentalizing" system, a pair of brain hubs that enable the fundamentally collaborative act of understanding other minds. When you empathize with a teammate's frustration or silently predict your partner's next step in a dance, these regions are likely lighting up, processing the invisible mental states that bind people together in action.

Indeed, the human brain seems wired for empathy and prediction, abilities at the heart of cooperation. Consider what happens when you see a loved one in pain. You flinch as though it were your own pain. This isn't just a poetic metaphor; it's reflected in your brain. Neuroscientists have found that when someone received a painful shock, and when that same person merely witnessed a signal that their loved one was receiving a shock, many of the same brain areas were activated.⁴ In both self-pain and empathic

pain, the anterior insula and the ACC lit up in unison, while regions encoding the direct sensory aspects of pain stayed quiet when only empathy was involved. In other words, the brain mapped the partner's suffering onto its own emotional pain matrix, though not its sensory map. This shared circuitry for feelings, a neural echo of each other's joy or sorrow, is a powerful adaptation. It means our brains blur the line between me and we, enabling one person's state to resonate in another. Such resonance is the glue of teamwork. If your distress feels unpleasant, I'm driven to ease it. If your success brings me pride, I'll work to support it. Empathy charges our interactions with emotional electricity, keeping a team's members connected and responsive to one another's needs.

But collaboration is not only about understanding and feeling; it's also about doing things together. Imagine two people carrying a heavy table across a room, laden with fragile glasses. They must walk in lockstep, sensing each other's minute movements, lest disaster strike. How do their brains manage this motor ballet? Research shows that when people coordinate physical actions, the brain recruits not one but two systems.⁵ One is the so-called mirror neuron system; regions in the premotor cortex and inferior parietal lobe that fire both when you perform an action and when you see someone else perform that action. These cells, originally discovered in monkeys, effectively let your brain echo another's movements inside your own neural circuitry. They provide a common code, translating your partner's hand gesture or step into the language of your own motor plans. In the table-carrying scenario, as your partner lifts their end, your mirror system neurons help your brain anticipate and match that lifting action. Yet, as important as this is, it's not sufficient for true joint action. A 2009 neuroimaging study revealed a second network of brain regions, including parts of the prefrontal cortex, posterior parietal cortex, and lateral temporal cortex, that goes beyond mirroring to integrate your actions with another's toward a shared goal. While the mirror system handles the alignment of movements, this broader network handles the coordination of plans: it's where your brain computes something like, "I'll tilt the table my way so you can get a better grip," or "we need to turn together now." Notably, these cooperative networks were far more active when people believed they were working with a human partner than with a computer. The presence of another mind changes the game. The brain shifts into a higher gear of social processing, attuning itself to subtleties and unpredictability that only another human can provide. The neuroscience here reflects an intuitive truth: we respond differently and more richly when teaming up with people versus machines. Our brains devote special resources to real partnership, suggesting that, through evolution, we have developed dedicated neural machinery to make working with others as fluid as moving our own limbs.

Human cooperation doesn't happen in a vacuum. It is the crown jewel of a long evolutionary journey through social living. Our primate ancestors already exhibited glimmers of teamwork. Chimpanzees, for instance, will band together to hunt monkeys or defend their territory, coordinating in ways that require tracking relationships and roles.⁶ They form alliances; they console each other; they even, on rare occasions, appear to teach youngsters by example. Yet, there are limits to ape alliance. Chimpanzees share meat from a hunt, but mostly with friends or relatives, and their collaborations tend to unravel outside of immediate contexts. Human beings carried cooperation to another level. We collaborate with unrelated strangers, build cities and civilizations full of complex division of labor, and keep going even when the personal payoff is distant or abstract. The secret to this leap might lie in enhancements to the social brain network. Scientists have long noted that primates with larger brains (especially a larger neocortex relative to body size) tend to live in larger social groups. In fact, by plugging humans' brain size into this primate formula, one famously arrives at an average social group size of about 150 which is often cited as "Dunbar's number"; the supposed limit of meaningful relationships our brains can handle. While the exact number is debated, the principle stands: our big brains evolved to master the subtleties of social life. The medial prefrontal expansions in humans endow us with prodigious skills for understanding others' minds enabling everything from teaching to deception, and our temporal-parietal circuitry gives us an intuitive grasp of social norms and narratives. These neural upgrades may have been what allowed early humans to hunt in cooperative bands, share food across an entire tribe, and eventually build camps and cultures.

Interestingly, we find echoes of our social brain in some of the animal kingdom's most cooperative species. Elephants, for example, live in tight-knit herds led by matriarchs who remember watering holes from decades past and coordinate the group's travel accordingly. Elephants have been observed to care for injured companions and even appear to mourn their dead; behaviors hinting at empathy and social knowledge. Their brain structure offers clues: elephants along with the cetaceans (dolphins and whales,) possess an unusual type of neuron also found in humans and great apes. These are the von Economo neurons, long spindle-shaped cells located in the ACC and fronto-insular cortex. They are relatively few but mighty; huge cells thought to rapidly transmit signals across distant brain networks. Intriguingly, von Economo neurons are found primarily in species with large brains and complex social lives; great apes, humans, elephants, whales, and dolphins.⁷ Scientists suspect these neurons might be a shortcut for processing social emotions and judgments quickly in big brains. If a dolphin has split-seconds to decide whether to come to a comrade's aid or a matriarch elephant must instantly recognize distress in a baby elephant's cries, these specialized cells could

be part of the reason they can do so. We might never fully know what an elephant or dolphin empathizes, but the neural hardware is strikingly familiar and they display parallel behaviors: Whales travel in pods that hunt cooperatively and protect injured podmates and elephants famously collaborate to rescue calves in trouble.

Yet, within that similarity lies a critical difference. Humans pair our social brain with extraordinary flexibility and scale. A dolphin pod may number a dozen; an elephant herd perhaps a few dozen. Human societies encompass millions. We cooperate via language, shared imagination, and culture, things no other species does. Neurologically, one might suspect our frontal lobes, particularly the prefrontal cortex, play an outsized role in this. The prefrontal cortex is the brain's executive, regulating impulses and planning for the future, but it's also crucial for adhering to social rules and norms. Damage to it can turn polite, responsible adults into socially dysfunctional individuals. The 19th-century case of Phineas Gage is a classic example: an iron rod through his skull left his intellect surprisingly intact but transformed his personality. Once mild-mannered and reliable, he became irreverent, irresponsible, and unable to work well with others. Modern neurology confirms the pattern. Adults who suffer injury to the ventromedial prefrontal cortex (the very front underside of the frontal lobes) often struggle with social decision-making. They make choices that are impulsive or inappropriate, even when they know better. They can become unreliable in honoring commitments and oblivious to how their actions affect others. They effectively lose some of the glue that binds collaborative enterprise: the ability to subjugate a momentary impulse for the group's long-term benefit, the awareness of and respect for others' expectations, and the moral compass that keeps selfish instincts in check. All these are critical when living in large cooperative groups with structured roles. A healthy prefrontal cortex acts like a wise team captain in our brain, suppressing the unhelpful outburst or selfish grab, and keeping us aligned with the shared plan. Without it, the mind becomes a more asocial free-agent, chasing its own urges without regard for the team; a state that rarely ends well in human societies.

If the prefrontal cortex is the captain, regions like TPJ are the navigators reading the social map, the ACC is the lookout for conflicts, and the insula is the emotional heartbeat, then neurochemicals are the messengers keeping the team in sync. One hormone in particular has earned fame as the "social molecule": oxytocin. Oxytocin floods the brain during moments of bonding; a mother nursing her infant, or friends hugging after a long separation. It has a profound effect on trust and generosity. In a landmark study, researchers gave people a dose of oxytocin via a nasal spray and then had them play a trust game involving real money. Those who got the oxytocin became markedly more trusting. They were willing to entrust larger sums to a stranger, despite

the risk of betrayal.⁸ Importantly, this effect wasn't just because they were in a good mood or less cautious overall; oxytocin specifically increased their willingness to take social risks. In other words, it made them optimistic about the other person's intentions. At the neural level, oxytocin is known to suppress activity in the amygdala (a region that among other things generates fear and distrust of strangers) and to enhance signaling in reward circuits during positive social contact. It's as if oxytocin opens the mind's door to others, a biochemical handshake that says "you can trust this one." Of course, trust is a double-edged sword; too much and one becomes gullible, too little and cooperation is impossible. Oxytocin tilts the balance toward connection. In group settings, when people feel a rush of trust and camaraderie (whether through a team-building exercise or a shared hardship overcome), you can be sure neurochemicals are reinforcing the bonds, making further cooperation more smooth. Our brains reward us for collaborative success: a fair exchange in the Ultimatum Game or a mutual win in the Prisoner's Dilemma can activate the ventral striatum, the brain's reward hub, flooding us with a sense of satisfaction. Some scientists even argue that the euphoric "high" of shared triumph such as two musicians nailing a duet or a sports team winning a championship is as potent as any drug. It's nature's way of incentivizing us to do what, in evolutionary terms, has been a very good strategy.

All these neural mechanisms converge to make human collaboration possible and our ability to collaborate has been a key driver of our success as a species. When you next find yourself in a successful partnership or a vibrant team meeting, consider that beneath the surface of conscious thought, an elaborate neural ballet is unfolding. Your brain's social regions are conducting a symphony of electrical signals and chemical releases that align your actions and intentions with those of others. It took millions of years for evolution to refine this orchestra, and in *Homo sapiens* it performs at a virtuoso level, allowing us to build more together than we ever could alone.

Collaboration in AI Systems

Yet, even as we revel in the wonders of the social brain, we stand on the cusp of a new era; one in which our partners in collaboration may not always be human. With the advent of sophisticated artificial intelligence and social robots, scientists and engineers are looking to the brain's blueprint for hints on how to make teamwork with machines and between machines flow as seamlessly as teamwork with people. The implicit hope is that by imbuing AI with some of the social cognition skills such as reading cues, modeling human beliefs, and communicating seamlessly we can unlock a transformative synergy. Early attempts are telling: AI agents that are programmed to "mentalize," in a

basic way, can anticipate human needs better. Robots endowed with simple empathy algorithms (for example, detecting human frustration and responding to soothe it) engender more trust and fluid cooperation from people. In a sense, we are beginning to graft the fruits of our long neural evolution onto our technology. Just as our ancestors once domesticated fire or tamed horses to expand the realm of the possible, we are now trying to harness the neuroscience of collaboration to create new kinds of teams composed of both humans and intelligent machines. The brain's saga of collaboration, from primate tribes to modern institutions, is entering an unprecedented chapter. We have deciphered some of the neural script that makes cooperation possible, and with it, we are poised to write new narratives of partnership; ones in which silicon teammates join the circle of trust. As we implicitly apply these insights to technology and society, we may be on the verge of witnessing yet another leap in what cooperation can conquer, fueled by an appreciation of the delicate, powerful, and deeply social machinery inside our heads.

One thing is very clear, relative to the other areas that we have discussed thus far, progress on making AI systems and robots that can effectively function in large teams is very nascent. By reviewing the neuroscience of collaboration, we can appreciate the incredible complexity of the regions in our own brains that allow us to work together with other humans. In the same way that our neurobiology has inspired existing AI approaches, researchers may find inspiration in the human brain for how to develop AI systems that can truly collaborate on shared goals but we are very early in that journey. At this point in time, collaboration by AI agents is fairly mechanical: driven by simple objective functions and plainly stated goals. That said, even very basic levels of teamwork by AI agents have the potential to deliver incredible and sometimes frightening results. AI systems today may not be able to match humans in areas of empathy and shared understanding but they far surpass human capabilities when it comes to speed and scale. Whereas a human may struggle to maintain more than 150 relationships and our communication methods are relatively slow, an AI system could handle thousands or even millions of relationships and communicate at incredible speeds.

Applications of AI Collaboration

Logistics are one of the primary examples of a domain where AI can disrupt the status quo. Imagine a cargo ship unexpectedly re-routes due to a port closure. Within seconds, a constellation of artificial intelligence agents can spring into action. One agent, representing a manufacturing firm, negotiates with another agent from a shipping consortium to secure space on an alternate vessel. Simultaneously, a third agent autonomously

informs downstream warehouses and recalibrates delivery times. Through thousands of interactions across hundreds of agents, the supply chain quietly reconfigures itself; no panicked phone calls, no emergency meetings, and no human had to lift a finger. This glimpse of machine-speed coordination hints at how AI agents collaborating with one another could transform our societal, economic, and organizational dynamics over the next decade. The result is not flashy science fiction, but a subtle rewiring of how decisions are made and actions are taken: faster, more fluid, and at a scale that was previously impossible.

Human coordination has always been constrained by our biological and cognitive limits. A simple conference call across time zones can take days to schedule; a strategic business decision can crawl through layers of management. As groups grow, communication channels multiply and misalign, a classic cause of delays. Traditional organizations rely on people passing messages up and down chains of command or across siloed departments. In global supply networks, for example, much of the communication remains manual and slow. Every handoff between buyers, suppliers, and logistics providers introduces the potential for lag and error. Conventional supply chains burdened by phone calls and emails often suffer from communication delays and human errors that drive up costs. There is an inherent drag on collective action: we form committees to aggregate preferences, conduct surveys to gauge sentiment, convene meetings to hash out conflicts. All of it takes time. In economic terms, the transaction costs of coordinating people, the emails, memos, meetings, and misunderstandings, are accepted as the price of doing anything big together. Even the best-run teams face an upper limit in size and speed; beyond a certain point, adding more humans to a project often makes it move slower as coordination overhead explodes. And that is not to say anything of the significant delays and costs associated with adding new members to a project. If it would be advantageous to have an expert join the team, it may take weeks or months before that individual can be recruited and onboarded.

Now imagine these constraints melting away. When AI agents coordinate, they exchange information instantly, continuously, and tirelessly. An autonomous agent has no need to sleep or pause; it can process updates from thousands of counterparts in parallel. Team size ceases to be a concern when each team member is a piece of software capable of synchronizing with the rest at digital speeds. If one agent discovers a change in the environment such as a price drop from a supplier, a surge in demand in one region, or a risk alert from weather data, it can notify countless other agents in a blink. Whereas a human manager might struggle to even hear every team member's input in a large meeting, an AI system can mathematically aggregate the preferences or data points from millions of agents and compute an optimal plan. The cost of preference aggregation plummets

when done by machine: an AI system can reconcile individual constraints and objectives from delivery deadlines to budget limits in milliseconds; something that would take a human committee weeks of negotiation. In short, autonomous agents promise fast, scalable coordination; a qualitatively different regime of organization where the old friction of “who talks to whom, and when” is largely eliminated.

Previous attempts to transition from human collaboration to automated systems have been hamstrung by the same challenges that have held back many software initiatives; a need to perform the transition all at once or to implement a hybrid approach with significant retraining of personnel. As an example, moving to automated supply chain software either requires a complete transition where every component is replaced by software or a hybrid system where the remaining humans have to replace informal written communication and phone calls with computerized input forms and systems that require data in a very specific rigid format to interact with the new software components. Today’s new AI agents solve these problems. With their strong ability to use natural language, interpret images, transcribe and generate voice calls, and perform reasoning and troubleshooting, they can be slotted into existing international networks one piece at a time and they can work seamlessly beside human operators. In fact, if not clearly disclosed, human operators may not always be able to determine if the counterparty in a communication is human or an AI agent.

Crucially, these AI agents do not operate in isolation; they negotiate and transact with each other on behalf of the humans or organizations they represent. One agent might represent a buyer, another a seller; others might represent shippers, insurers, or regulators. They can all come to the table simultaneously. This multi-agent negotiation is poised to disrupt traditional models of business interaction. We already see early hints in financial markets and online advertising, where algorithms compete and settle trades in fractions of a second. In digital advertising exchanges, for instance, software agents for advertisers and publishers engage in micro-auctions to decide which ad appears on your screen; some major ad exchanges handle tens of billions of such auctions each day, each resolved within milliseconds. This is machine-to-machine commerce in its pure form and it works at a scale and speed far beyond human capacity. Finance and advertising are two domains where the rules of negotiation can be turned into deterministic equations and very narrow and specialized systems can operate without requiring the level of flexibility that is associated with today’s AI systems. Over the next ten years, similar agent-mediated negotiation could spread to more arenas where that flexibility is required. Corporate procurement might shift to autonomous bidding systems: a construction firm’s AI agent could continuously scout for the best price on steel, negotiating with metallurgical suppliers’ agents or human salespeople in real time.

Likewise, consumer retail might be transformed by personal AI shoppers. Instead of today's ad-saturated experience where companies spend fortunes to attract a human's attention, a buyer could simply deploy an AI agent to find and purchase exactly what they need at the best terms. In such a scenario, the balance of power in markets might tilt. Traditional advertising, which relies on swaying human psychology, could diminish in influence if decisions are made by algorithms acting on clear preferences and criteria. Sellers might have to appeal to the rational filters of buyer agents rather than to human whims, fundamentally changing marketing strategies. The entire dance of retail and advertising could become more of a direct conversation between algorithms: your personal AI stating your requirements to a marketplace of vendor AIs, which then offer their best price and quality, all in an instant. The negotiation that used to happen in slow motion might be compressed into an eyeblink of agent interaction.

Complex Forms of Collaboration

One striking demonstration of AI's emerging ability to handle complex multi-agent interaction came from an unlikely arena: the board game Diplomacy. Unlike purely competitive games such as chess, Diplomacy is a game of alliances, negotiations, and trust; a deeply social game requiring players to persuade and sometimes deceive each other. For decades, it was assumed that such dynamics were uniquely human territory. Yet in 2022, researchers unveiled Cicero, an AI agent that achieved human-level performance in Diplomacy by engaging in natural language negotiation with multiple human players.⁹ Cicero would discuss plans, form pacts, and coordinate joint strategies, all through text-based dialogue that felt eerily human. This milestone required the AI to model the beliefs and intentions of others and to cooperate when beneficial. It had to coordinate in a multi-agent setting rife with uncertainty and strategic nuance. Cicero's success signals that AI agents can participate in sophisticated group decision-making processes. If an AI agent can bargain and build trust in a game as subtle as Diplomacy, one can imagine future business AIs negotiating real-world deals or forming strategic partnerships on behalf of companies. Two corporate AI systems might hammer out a supply contract or a merger framework overnight, parsing each other's messages for intent and credibility much like seasoned diplomats, only far faster.

Expanding further on the idea of game theory, one of the interesting questions that will need to be answered is the level of trust and collaboration agents should exercise when working with each other. Imagine that all of the cars on the freeway are autonomous. In one mode of operation, each autonomous car will operate entirely independently. Defensive driving and primitive communication techniques such as

horns will be used to keep each driver safe; much the same way that humans drive today. On the other hand, overall safety on the road could be optimized if the agents driving each car team up and start acting as a single swarm with a shared objective of minimizing overall damage and preventing loss of life. The downside of this mode of operation is that your car may be more likely to take on avoidable damage in the interest of the greater good. Your chances of getting in a minor accident may increase but your chances of getting in a serious accident may be much lower. Is that a tradeoff that should be made and who should make that decision?

When swarms of AI agents stand in for individual humans or institutions, coordination scales up in unprecedented ways. Consider policymaking and societal governance. Today, simulating the impact of a new policy such as a tax change or a pandemic response is slow and fraught with uncertainty. We rely on economists' models or limited trials and both are notorious for their shortcomings and inability to reliably predict collective behavior. With agent-based simulations, we can create a synthetic population of thousands or millions of simulated agent citizens, each with programmed behaviors and preferences, and let them interact to see emergent outcomes. This approach, known as agent-based modeling, has already proven its value. During the COVID-19 pandemic, researchers employed agent-based social simulations to test how various interventions might play out. In fact, a systematic review found over 126 studies using such simulations to investigate COVID-19 spread and control measures.¹⁰ Each virtual person in these models is essentially an agent following rules such as virtually going to work, taking precautions, etc. and the collective result can reveal non-obvious effects, like how closing schools might inadvertently increase transmissions if parents cluster together elsewhere. In the coming years, this technique could become far more powerful as AI agents grow more adaptive. Instead of static rules, the agents could learn and react, yielding even richer insights into potential futures. Policy testing could then resemble a kind of parallel universe experimentation: before implementing a new traffic law, a city government might deploy a swarm of traffic-driving AI agents in a simulated city to see how congestion and accidents evolve. Likewise, economic policies could be dry-run in detailed agent economies to foresee their impacts on markets or inequality. The same type of modelling may help businesses experiment with product or movie ideas before spending a single dollar. This capability of massive collaboration among agents to pre-experience a society's choices might make governance and business more responsive. It's as if we gain a sandbox version of society, where countless AI "what-if" scenarios play out, to illuminate the best paths forward.

Swarms

Even in the wonkier realms of taxation and economics, where classical theory long assumed hyper-rational *Homo economicus*, agent swarms are injecting a dose of behavioral reality. In one groundbreaking project, economists built a virtual economy from scratch and let AI agents loose in it; each with a job, an income, and the freedom to work more or less, spend or save. The government in this simulation wasn't human either; it was an AI "social planner" whose job was to set tax rates. The twist was that these agents, like real people, could learn and even develop unexpected strategies over time. As the taxes changed, some agents figured out ways to game the system; for example, by working a bit less when higher taxes cut into their reward, or by shifting their activity in clever ways to reduce tax burden. This is analogous to real-life taxpayers finding loopholes or adjusting their behavior in response to incentives. Remarkably, the AI government began to anticipate and counter these tricks.¹¹ Through trial-and-error iterations, it evolved a tax policy that achieved a better balance between fairness and productivity than the researchers had expected. In fact, the final policy delivered about a 16% improvement in the equality–productivity trade-off compared to a classic human-designed tax formula. It wasn't an arbitrary tweak of some equation; it was a policy grown from observing realistic agent behavior in a complex environment. Interestingly, the AI-designed tax system ended up looking a bit unconventional: it imposed higher taxes on the richest earners and gave out more generous subsidies or breaks to the lowest earners than a typical policy would. Yet, because it had been forged in an arena where agents responded with all their cunning and biases, it held up. When this AI-driven tax policy was later tested with human subjects in a controlled experiment, it performed comparably well, suggesting the simulation had latched onto something genuinely insightful. This example shows how swarms of AI agents can serve as a proxy for millions of individual decisions in an economy, revealing pitfalls and possibilities that a tidy theoretical model might miss. People don't always behave as economic textbooks predict; they find shortcuts, they have limited foresight, and they exhibit biases. Letting AI agents play out those tendencies *in silico* can lead to more robust, and sometimes counterintuitive, policy solutions.

The collective power of agent swarms is not just in simulation, but also in real-time sensing and response across large systems. Picture a network of AI agents distributed around the globe, each monitoring a different data stream; one scans online news, another tracks hospital admissions, others watch social media chatter or supply chain orders. Individually, each agent might only catch a faint whisper of something amiss. But together, they could triangulate an emerging crisis. We saw a precursor of this during the emergence of COVID-19: an AI platform called BlueDot scanned news reports and airline data and detected unusual pneumonia cases in Wuhan in late 2019. BlueDot's

agent sent out an alert on December 31, 2019, warning clients about the outbreak five days before the World Health Organization made a public announcement.¹² In that instance, the AI system was essentially acting as a global sentinel, picking up weak signals and broadcasting a warning with minimal human delay. In the future, we might have many such sentinels; a whole ecology of specialized agents sharing information with each other to flag anomalies in real time. One set of agents could constantly exchange data about financial transactions worldwide; if a subset of them detects patterns indicative of an impending market crash or a major liquidity issue, they could collaboratively raise red flags across the network before any single financial authority could piece it together. Similarly, health surveillance agents spread across clinics, pharmacies, and social networks could collaboratively notice a new pathogen spreading, each agent contributing a piece of the puzzle (a slight uptick in fever medications sold here, a cluster of unusual clinic reports there) and collectively recognizing the pattern. This kind of early warning system by distributed AI is essentially the “wisdom of crowds” concept applied to machines: many observers pooling knowledge to see the big picture early. Just as diverse human crowds can, under the right conditions, outperform individual experts by aggregating their information, a diverse collection of AI agents can combine their insights to achieve a form of superhuman foresight.

In fact, we are learning that assembling multiple AI agents can yield surprisingly powerful results, echoing the old adage that many minds are better than one. Recent research has explicitly tested this “wisdom-of-crowds” effect among AI systems. In one study, scientists took a dozen large language model agents and had them independently forecast real-world events, then aggregated their predictions. The result was striking: the ensemble of 12 AI agents matched the accuracy of a crowd of 925 human forecasters on the same task.¹³ Through simple averaging of the AI agents’ diverse outputs, the collective AI forecast was as good as the human crowd’s consensus. This suggests that when AI agents have different perspectives or heuristics (perhaps due to different training data or algorithms), letting them “vote” or deliberate can cancel out individual errors, producing a high-quality group judgment. In other words, a swarm of diverse AIs can exhibit a kind of collective intelligence analogous to and at times on par with human crowds. Today, most of the efforts with LLMs are focused on building the most intelligent agent. In the near future as we reach diminishing returns on improving LLMs, we may seek further improvements by creating more opinionated or specialized LLMs and taking advantage of the diversity of outputs that they produce. Over the next decade, we may routinely see ensembles of AI advisors guiding decisions; from investment portfolios managed by a committee of algorithms to medical diagnoses cross-checked by a panel of AI specialists. Each agent might have its own blind spots and strengths, but together they

can balance one another. This diversity-driven robustness is a key reason why biological and social systems rely on multiple agents too. Nature learned this lesson long ago: ant colonies, bee hives, and flocks of birds achieve feats no solitary creature could.

Indeed, the emerging paradigm of AI agent cooperation is often compared to swarm intelligence in nature. Ant colonies are a favorite analogy; hundreds of thousands of ants coordinate via simple local interactions, communicating through pheromone trails and touch, and out of this decentralized process come complex architectures and efficient foraging strategies. There is no CEO ant dictating plans; each ant just follows basic rules, yet the colony as a whole computes solutions to challenges like finding the shortest path to food or building a stable nest. Scientists have shown, for instance, that army ants can form living bridges by linking their bodies when they encounter a gap, adjusting the structure dynamically as other ants march over it. No single ant understands the full bridge, but collectively the colony “decides” how long the bridge should extend, balancing the benefit of a shortcut against the cost of sequestering too many workers in the bridge.¹⁴ This is swarm intelligence in action; individual-level interactions produce coordinated group behavior with remarkable adaptability. Artificial multi-agent systems are built on similar principles. If we program AI agents with the right simple rules and the ability to share information, we might see emergent problem-solving that exceeds what any centrally designed system could do. One agent alone might be limited, but in a well-designed swarm, the whole becomes much more than the sum of its parts. Researchers in robotics are already drawing on these lessons, envisioning swarms of simple robots that could, for example, self-assemble into bridges or structures when needed just as the ants do; adapting in real time without an external blueprint. Swarms of robots could be used in construction to lift walls or entire buildings. Drone swarms already put on impressive visual displays but in the future may prove to be effective at fighting fires and finding missing people.

Projecting these ideas into our social and economic future, it’s clear that multi-agent AI collaboration could usher in a profound shift. Organizations might evolve from rigid hierarchies into fluid networks of people and AI agents working side by side. Many routine coordination tasks could be offloaded entirely to the bots: they will schedule our meetings, balance supply and demand, negotiate contracts, and flag issues before they become crises. Human teams augmented with fleets of specialized agents might achieve feats of coordination previously reserved for only the largest, most centralized institutions and do so without the same bureaucracy or delay.

Economically, this could mean leaner operations and new business models. One category that may experience significant disruption are companies that exist almost entirely for the purpose of aggregating demand. Grocery stores are an example. Coordinating

shopping lists across large groups of consumers is impractical today but AI agents may be able to take on that task. If there is enough aggregate interest in the purchase of a specific brand of cereal in a localized region, the agents may be able to arrange a direct purchase from the manufacturer, eliminating the grocery store as the middleman and lowering costs to consumers in the process. This could even be taken a step further with agents arranging for the purchase of raw ingredients and then coordinating with a contract manufacturer; empowering groups of individual citizens to arrange ad hoc vertically integrated supply chains for food. Similar disruption could happen in the travel industry with groups of people now able to rely on their AI agents to easily coordinate on desired flights and then directly chartering planes to eliminate the margins charged by the travel booking companies and airlines.

Bridging the Silos: The Model Context Protocol

At the heart of most technology revolutions, it is usually possible to find one or more standards playing a key role. VHS defined the era of affordable home movies. HTTP and HTML allowed the internet to take hold. We are very early in the days of this AI revolution and there is still a lot of rapid development and debate about which standards will take hold and stand the test of time. One that is rising to the top is called the Model Context Protocol. Its promise is to become a kind of universal connector, “*a USB-C port for AI applications*,” as Anthropic, its creator, puts it; linking intelligent AI agents together so they can speak the same language.¹⁵

For all their cleverness, today’s AI systems often operate like islands. A cutting-edge language model might compose eloquent analyses or parse complex instructions, yet it remains isolated behind data silos; unable to automatically fetch information from your company’s database or update entries in your calendar without custom code written for that specific task. Every new integration such as connecting to a particular cloud drive, CRM, or messaging app has traditionally required its own bespoke software development. This fragmentation has been a glaring limitation. Developers have had to juggle a patchwork of plugins, APIs, and ad-hoc scripts to get one AI tool to talk to another system.¹⁶

The Model Context Protocol (MCP) emerged in late 2024 to cut through this tangle of one-off connectors. MCP is an open standard for two-way communication between AI models and external tools or data sources, effectively defining a common dialect both sides can understand. Think of it as a lingua franca for AI systems and the digital world they inhabit. If every AI agent and every data source speaks MCP, developers no longer need to build endless custom interfaces. MCP is not the first solution that

has been proposed to solve this problem, and it may not be the last but there is a high level of confidence that one standard or another will establish a dominant position and significantly simplify communication between AI agents.

MCP attacks the problem by providing a unified protocol which is a bit like establishing standard gauge tracks for all the disparate AI “trains” to run on. Technically, it defines a simple architecture with three roles: an MCP host (the user-facing AI application or agent environment), an MCP client (the component that brokers communication on the host’s side), and an MCP server (an external program that exposes some set of tools or data). When you ask an AI assistant a question, the host (say, a chat interface or an AI-enabled IDE) passes the request to its client component. The MCP client then queries one or more MCP servers to find out what they can do; what data they hold, or what actions (“tools”) they can perform. Those servers respond with descriptions of their capabilities. Crucially, this isn’t a one-off, hardcoded call; it’s a dynamic discovery process. The AI agent can “autonomously discover, select, and orchestrate tools” based on the task at hand. In effect, the AI is not limited to a predetermined palette of integrations decided by a developer long ago; it can figure out in real time which resources to tap into, much like a skilled worker choosing the right tool from a toolbox on the fly. It remains to be seen if MCP will become the predominant method for AI agents to communicate with each other. However, one way or another, standards will evolve and those standards will facilitate communication and enable massive networks of AI agents to begin to collaborate at scales that we have not seen in the past.

Risks and Challenges with AI Agent Collaboration

None of this will happen without challenges. Trust and governance become tricky when autonomous agents are making deals and negotiating policies. We will need new frameworks to supervise AI negotiations, establish guardrails, and ensure alignment with human values. There will be power imbalances to watch out for. Faster coordination can also mean faster contagion of bad decisions or market panics if agents are poorly designed. As coordination costs drop toward zero, we might face a world that changes even more rapidly, testing our ability to adapt. Yet the potential benefits are enormous. We stand to remove a great deal of inefficiency that has long hampered collective action. Delays in crisis response, wasted effort in duplicated tasks, stalemates in multi-party negotiations; these could be mitigated by letting networked AI agents find the common ground or the timely intervention that eludes cumbersome human processes.

As AI agents proliferate and collaborate at scale, they introduce new vulnerabilities and ethical dilemmas that demand careful scrutiny. Ironically, the same swarm intel-

ligence that allows distributed AI agents to solve complex tasks can be harnessed for malicious purposes, amplifying the threat of cyberattacks. Consider the classic Distributed Denial of Service (DDoS) attack: today's crude botnets simply hurl junk traffic at a target, but future AI swarms might behave like an army of legitimate users. Millions of adaptive agents could each mimic normal browsing; clicking links, filling shopping carts, and making queries to customer support bots that appear authentic. This behavior would make them extraordinarily hard to distinguish from genuine customers.¹⁷ Such an AI-driven swarm could overwhelm a system not by brute force alone but by blending in with real user behavior. Traditional defenses that filter out obvious bots would struggle, as the attack wouldn't trip the usual alarms. Moreover, these malicious agents could coordinate their strategies on the fly: if one tactic starts failing, they could dynamically switch approaches (varying request patterns or frequencies) to exploit any weakness. In short, a swarm that thinks and adapts can turn a security slight into a systemic failure before defenders realize what's happening.

As we discussed in the previous chapter, military applications are top of mind for countries around the world. For a long period of time, the trend in achieving military superiority has shifted away from being able to recruit the most soldiers and has instead been focused on developing the most precise and deadly weapons. As just one example, the F-22 exceeded \$300 million per plane. Swarms of small affordable autonomous robots, and drones in particular, may change the nature of warfare in the decades ahead. In the same way that an invasive species of frog can decimate a local ecosystem while being almost impossible to eradicate, large swarms of small autonomous robots that chip away at infrastructure in enemy territory have the potential to be extremely destructive. Rather than a single targeted attack on a highly guarded piece of infrastructure such as a central power plant, thousands of drones equipped with small explosives could seek out distribution transformers across the power grid. Even with a high failure rate, the cumulative impact across the system could be devastating.

Beyond software attacks or physical attacks on infrastructure, AI swarms might quietly warp public opinion. Just as coordinated bots on social media have been used in the past to spread misinformation or amplify particular messages, tomorrow's AI agents will be far more sophisticated and persuasive. They will be capable of generating content tailored to each individual's psyche, delivered through seemingly personal interactions. A single orchestrator could deploy thousands of AI personas across the internet on forums, in comment sections, and on social networks; each one engaging specific segments of the population with highly customized narratives. Over time, this persistent whispering campaign could nudge public sentiment in subtle ways. For example, one person's newsfeed might be continuously peppered with emotionally resonant stories

about an issue, calibrated to their values and biases, while another person sees a different angle entirely and both sets of messages could escalate over time at rates that are unique to each individual. Recent research already shows that personalized political messages crafted by generative AI can significantly increase persuasion effectiveness compared to non-personalized appeals¹⁸. In a highly controversial research study performed by researchers at the University of Zurich, AI agents participated covertly in online debates in the *r/changemyview* subreddit. The AI agents made over 1500 posts, posing as everything from a trauma counsellor to a black man who was opposed to the Black Lives Matter movement. The research has since been pulled offline following backlash about the ethics of the study but the initial results claimed that AI was significantly more persuasive than the average human participant in the subreddit. The risk to democratic discourse is profound: people may be increasingly living in customized informational cocoons engineered by AI, making it even harder to separate authentic grassroots opinions from artificial influence. These campaigns need not be state-sponsored black ops; they could be run by marketing firms chasing profits or domestic political groups with few scruples. Either way, the line between genuine public sentiment and AI-augmented persuasion would blur. Society may find itself in a crisis of authenticity, where trust in what we read and even in our own opinions wanes under the suspicion that invisible algorithms are pulling the strings.

On a more personal level, handing over everyday tasks to autonomous agents carries new types of risk. We might come to rely on personal AI assistants to manage our schedules, finances, and communications; the digital “us” that negotiates bills, posts on social media, or even makes purchases on our behalf while we sleep. But what happens if that digital proxy goes rogue or is hijacked? A compromised agent could wreak financial and reputational havoc in a very short time. It might empty your bank account via a series of convincing transactions or ransack your private data and messages for blackmail material, all before dawn. The owner of the agent might wake up to find a trail of damage that normally would take a human thief months of effort. An all-knowing agent that has full access to your health records, dating history, and financial information could get mixed up and share information inappropriately. You might want your agent to share your medical records with a new medical specialist that you have an appointment with. On the other hand, what if you have a romantic date with someone who happens to be a doctor, and an enthusiastic agent assumes this is another medical appointment and sends over your medical history? It’s easy to see how these types of mix-ups could occur.

Cybersecurity experts warn that a breach in an autonomous AI with deep access to your data can indeed lead to identity theft, large-scale fraud, or other grave outcomes.¹⁹ By giving an agent the keys to your digital kingdom, you’ve expanded your attack

surface. Now an adversary doesn't need to fool you directly, it could also fool your agent. For instance, if you let an AI agent auto-respond to your emails and someone sends a cleverly crafted malicious message, the agent could unintentionally leak sensitive info or make commitments you never intended. The speed and autonomy that make these agents useful also make their mistakes or vulnerabilities dangerously powerful. The need for robust security, oversight, and possibly insurance for personal AI agents will become increasingly clear as these scenarios move from hypothetical to inevitable. Perhaps there will be an entirely new industry that pops up for human assistants that supervise personal AI agents and play the role of the human-in-the-loop.

The velocity at which AI agents communicate and react creates another kind of systemic risk. Human decision-making, with all its flaws, at least has the virtue of taking time; time in which second thoughts or external interventions can halt a catastrophe. AI agents operate on a different timescale entirely as we discussed when considering military applications. When networks of agents interact with each other, they could amplify mistakes or false signals instantaneously across the globe. Imagine similar dynamics playing out in other domains. An unfounded rumor, perhaps entirely generated by one errant AI or a data feed error, about a particular product shortage could ricochet through millions of shopping agents and supply-chain bots. In seconds, warehouses might be drained as every automated system tries to secure inventory, and consumers wake up to empty shelves caused by an overnight AI buying frenzy. It's analogous to the panic buying of toilet paper seen during the COVID-19 pandemic, except compressed into the blink of an eye and executed by machines without regard for reassurance or moderation. By the time anyone notices the error, the damage is done; markets swung, supplies misallocated, decisions made en masse on false pretenses. Such rapid cascades could be very hard to contain. Our existing mechanisms for damping volatility like trading halts or fact-checking processes operate on human or at least institutional timeframes and are generally put in place by central organizations like stock exchanges. They may simply be too slow in an era where rumor and reaction can propagate at server speed and as AI systems become increasingly distributed, there may be no central chokepoints where these controls could be added. This raises the possibility of systemic shocks that originate and spread entirely among algorithms, with humans as stunned bystanders trying to clean up afterward.

Another looming issue is the ease with which AI agents can be scaled up, potentially leading to a digital arms race that benefits no one. When the marginal cost of deploying an extra thousand software agents is close to zero, there are strong incentives for aggressive actors to saturate environments with their bots. We might soon face inboxes and phone lines teeming with AI-generated sales pitches, job application bots flooding

every job posting, and swarms of automated negotiators in online marketplaces. In response, organizations will deploy their own AI filters and counter-agents to manage the onslaught, and the cycle will escalate. If one marketing team unleashes a million chatbot marketers, rivals may counter with a million of their own, and users will be caught in the crossfire of bot vs. bot competition. The outcome of such an arms race is a noisy equilibrium where vast computational resources are spent simply neutralizing each other's agents. There are also broader externalities: running millions of bots consumes significant energy and computing infrastructure. Already, bots (from web crawlers to click-farms) are estimated to make up roughly half of all internet traffic, and the rise of AI agents could push that fraction even higher.²⁰ This deluge can degrade the experience for human users. It also raises the cost of doing business online, as companies must continually invest in defenses just to keep their platforms usable. In the end, no one truly "wins" this arms race; it's an expenditure of effort and money that largely cancels itself out, much like a biological ecosystem where predator and prey keep evolving tougher defenses and offenses with no increase in overall welfare. Such an environment could stifle genuine innovation and communication, buried under a cacophony of automated activity. To make matters worse, operators of websites are also experiencing fewer visits from real users as AI agents increasingly report summaries of key information to human users without the user ever having to visit the site. These zero-click searches may be convenient for users but they create challenges for website operators that rely on legitimate user visits to generate ad revenue. Facing increasing agent traffic and decreasing traffic from legitimate users, many businesses may be forced to change their entire revenue model.

In light of these pressures, we may also see growing demand for AI-free zones; digital or even physical spaces where autonomous agents are restricted or forbidden. Online, this could manifest as exclusive human-only forums or social networks, where every participant is verified as a real person. These enclaves of authenticity would offer a refuge from the automated din, a place where one can converse or collaborate without wondering if a bot is in the conversation pretending to be human. Some researchers have already pointed to the need for such "human-only" online spaces to preserve genuine social interaction and curb mass manipulation. The idea is that, as AI involvement in our information ecosystem grows, we might require pockets that are deliberately kept agent-free to rebuild trust and ensure the integrity of dialogue. In the physical world, analogous measures could emerge. Imagine meetings or negotiations in which participants agree not to use AI assistants or real-time AI analytics, creating a level playing field of purely human cognition in the room. Or consider public venues like parks, gyms, or cafes advertising that they are free of surveillance bots and drone delivery devices; a

selling point being that when you're there, you know the ambiance and interactions are human by default. Even some "unplugged" retreats today hint at this desire, as people seek respite from algorithmic feeds and notifications. In the future, a premium might be placed on experiences certified as AI-free for privacy or mental health reasons. Of course, drawing a clear line will be challenging. What if someone's assistive robot or AI-enhanced wearable crosses into a no-AI zone? Does one ban all devices, or is there a way to allow benign AI help while blocking intrusive autonomy? The very fact we're contemplating "no AI allowed" zones underlines how pervasive we expect these agents to be and how much value we may place on genuinely human experiences in response.

Bad Actors and Enforcement

Mitigating these risks will require a mix of technological and institutional innovations. On the technical side, a top priority is developing better ways to verify and control AI agents' activities. In practice, this might mean that every autonomous agent within a given closed network must carry a kind of digital credential or passport that cryptographically proves its identity and who is ultimately accountable for it. Researchers are increasingly arguing for such mechanisms, noting that we need ways to explicitly delegate authority to AI agents, transparently identify them as non-human, and enforce human-imposed permissions on their actions.²¹ The proliferation of AI agents may lead to the downfall of many of the open aspects of the internet that society has taken for granted since its inception. It may simply become too easy for swarms of AI agents to overwhelm any space on the internet that is not protected with strict policies and authentication requirements.

Alongside identity control, continuous monitoring of agent behavior at scale is crucial. Here, AI itself will be instrumental in defense. The patterns of a coordinated attack or a mass-produced propaganda effort can be extremely subtle; perhaps a slight synchrony in how dozens of accounts pause before responding, or an unusual concentration of "user" activity coming from devices with similar configurations. These are the kind of faint signals that human operators might miss but machine learning can catch. Defensive tools equipped with AI/ML can rapidly analyze vast streams of activity and pick up on anomalies that indicate something isn't right. For example, an AI system might flag that what looks like a random trickle of user actions is actually a synchronized pattern originating from a cluster of IP addresses that have never been seen before in this context. This is the digital equivalent of a trained eye spotting a forged banknote by microscopic cues. An AI-driven defense could raise an alarm in the early stages of an attack or influence campaign. In short, we will likely combat AI swarms with swarms

of defensive AI, each watching for the telltale signatures (however well-mimicked) of malicious coordination. This AI-on-AI battle of wits will be an ongoing evolution, where improvements in attack sophistication are met with improvements in detection. It won't be perfect, but it can tilt the balance back toward the defenders or at least ensure that when a breach in the dam starts, it's quickly noticed and plugged.

Policy and governance may play an essential role in curbing these threats but the underlying challenge is that rules will be hard to enforce given the ability of anyone to download an open source model and set up a swarm of agents that behaves in a manner that is indistinguishable from humans. As with other AI governance issues, there is no way to verify if a sufficiently advanced AI is AI or not. We will increasingly have to operate under the assumption that the counterparties that we interact with in open environments such as the internet may be AI unless there is verified evidence that they are human. Relative to regulating raw content output of LLMs or other generative models, it will be slightly easier to regulate AI agent activity since it requires a network connection. In theory, that will allow bad actors to be traced and blocked. Relative to regulating robots, it will be more challenging since AI agents operate without a physical footprint and hackers can leverage decades of experience in methods of operating online while evading detection.

Laws and regulations often lag technology, but the coming wave of AI agent proliferation will force legal systems to catch up quickly. Cybersecurity frameworks, for instance, may need to recognize that an AI-driven DDoS attack by a million coordinated bots is not just another crime but potentially a national security event. Policymakers will have to establish clear norms and perhaps even international treaties for what constitutes an act of aggression in cyberspace when autonomous agents are the foot soldiers. Deterrence strategies might involve publicly stating that certain kinds of AI swarm attacks could warrant severe countermeasures, to dissuade state or non-state actors from even attempting them. On the domestic front, law enforcement agencies will need enhanced tools and authority to go after those who deploy malicious AI networks. This could mean investing in capabilities to trace AI activity across the web (piercing through anonymization by following the money or infrastructure), as well as updating statutes so that, for example, operating a covert botnet of a certain scale carries penalties on par with more traditional forms of cybercrime or even terrorism. Crucially, legal accountability for damage caused by autonomous agents has to be delineated: if a corporation's fleet of AI assistants inadvertently causes a cascading disaster, is the company liable? Clearer answers to that question will compel organizations to put safety measures in place, lest they face lawsuits or regulatory sanctions.

There is also a competition and consumer protection angle to consider. If companies begin unleashing swarms of agents that clog up shared digital spaces, the government might have to step in. The situation is analogous to pollution: just as no company is allowed to dump waste freely into a river, we might decide that no actor should flood the online environment with unlimited bot traffic. The metaphor is apt because the costs of a bot arms race are largely externalized; everyone suffers the degraded experience and higher security burden. To counteract this, agreements or rules may be needed to limit excessive deployments. For instance, regulators could impose caps on certain automated activities or require that large-scale agent operations be conducted in a “regulated zone” where they can be monitored.

Meanwhile, privacy and data protection remain fundamental bulwarks against abuse. The more AI agents rely on personal data to mimic and influence humans, the more vital it is to enforce strict data governance. Stronger privacy laws (and enforcement of those on the books, like GDPR) can reduce the pool of exploitable data that powers highly personalized manipulation. Individuals should have the right to know and control how their data is used by AI systems; this might involve new transparency requirements for services that deploy AI agents. Imagine a notification that “Your interaction data may be used to train AI systems” with an opt-out. Additionally, industry standards and certifications can help fill gaps where legislation is still catching up. We may see voluntary codes of conduct where companies pledge not to engage in certain high-risk practices, such as mass impersonation of humans or unsupervised agent autonomy in sensitive domains like healthcare or finance. Professional associations might develop guidelines for ethical agent behavior, similar to how medical associations have ethics codes. In the long run, it’s plausible that a dedicated regulatory body or oversight council will emerge for AI, much as financial regulators exist for banking. Such an agency could track major AI deployments, audit systems for compliance with safety standards, and coordinate responses to incidents. The idea would be to have a referee in the game, someone empowered not only to set rules but to watch the play and call fouls when necessary. This kind of oversight could help ensure that no single company’s rush to deploy millions of agents ends up undermining the public’s trust or well-being. It also creates a forum to continually update governance as the technology evolves, which is crucial given the pace of change.

Finally, we must stay ahead of these challenges through ongoing research and collective action. The technological and social complexities introduced by large-scale AI agents are unprecedented and understanding them requires interdisciplinary effort. We can’t rely on hindsight alone; proactive exploration of “what could go wrong” is needed before these agents are omnipresent. Encouragingly, some researchers are already

building simulations and theoretical models of swarm AI behavior; integrating insights from computer science, economics, and network theory to anticipate how millions of agents might interact in an economy or community. We should expand these efforts, effectively doing wind-tunnel tests for AI societies. Borrowing an idea from finance, where banks undergo stress tests, we could subject multi-agent systems to extreme hypothetical scenarios such as a burst of bad data, a malicious override, or a sudden network split to observe their reactions. The results would be invaluable: they'd show us where default designs fail and what kind of safeguards or circuit breakers might prevent catastrophe. Such proactive testing and scenario planning can guide both engineers in building more robust agent algorithms and policymakers in drafting regulations for worst-case events. Equally important are the ethical and legal frameworks that need to parallel technical research. We should not wait for a crisis of AI misuse to articulate our principles; instead, ethicists, jurists, and technologists should be collaboratively defining what responsible AI agent behavior looks like. There are early efforts in this vein, with scholars proposing standards for transparency (so that it's clear when we're dealing with an AI agent), accountability (so that an agent's actions can be traced and explained), and value alignment (so that agents acting for us respect human rights and societal values). These ideas can shape everything from corporate policies to international guidelines.

Conclusion

The coming decade will likely be a tug-of-war between the immense gains offered by collaborating AI agents and the novel risks they bring. Our ability to navigate this period will determine whether the optimistic vision outlined earlier in this chapter can be achieved safely. Some disruption is inevitable but the true test will be how resilient and responsive our safeguards are when those mistakes happen. If our institutions and technologies can learn and adapt as quickly as the AI agents do, the worst dangers can be averted. In that scenario, we harness AI swarms to work on humanity's behalf, while keeping their darker potentials in check. Ultimately, the measure of success will be that this transformative technology serves to empower societies (in productivity, knowledge, and convenience) without eroding the foundations of security, privacy, and trust. Achieving that balance is no small challenge, but with vigilant oversight, clever design, and a willingness to put human values front and center, we can strive to ensure that these new digital colleagues remain our allies, not our adversaries.

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Chapter Eight

When AI Can Model the Nuances of Human Experience

Introduction

Julian faced the avatar across the table, its expression a perfect mask of courteous indifference that he knew was designed to unsettle him. He was in the final simulation before meeting the real governor, a notoriously mercurial man whose cooperation was the last remaining piece of the geothermal energy accord. The AI, which he called Leo, was more than a sparring partner; it was a mirror to his own nervous system. As Julian made his opening statement, Leo subtly shifted its posture, a micro-expression of skepticism flickering across its brow. “Your premise is sound, Julian,” the AI’s voice replied, its tone perfectly calibrated to sound unconvinced, “but your vocal cadence suggests an appeal for validation. I am detecting a fractional increase in your cardiac rhythm. Let us pause. The goal is to project calm resolve, not supplication.” Julian took a breath, recalling Leo’s earlier lessons on engaging his own higher cortical functions to regulate the primal threat responses firing in his amygdala. He remembered the early versions of these AI counselors from years ago, the wild west period of the therapy bot scandals when unvetted programs gave disastrous advice and users’ most intimate confessions were harvested for commercial and political profiling. The public trust had been shattered. It took the passage of digital sanctuary acts, which enshrined the legal concept of cognitive privacy and made the bond between a person and their certified AI counselor inviolable, for the technology to find its true purpose. The path had been

perilous, a near miss where society almost allowed these powerful tools of introspection to become instruments of manipulation. Now, however, Leo was his sanctuary, a space for him to confront his own weaknesses and forge them into strengths, preparing him not just to win an argument, but to connect with and persuade another human being in a way that felt authentic and unwavering.

The Neuroscience Behind Our Humanity

The human emotional system emerges from a mosaic of specialized yet interconnected brain regions, each contributing in distinctive ways to how we feel. Deep in the temporal lobe, the amygdala acts as a rapid threat detector and emotional amplifier. It scans sensory inputs for danger and, when a threat is detected, triggers the cascade of fear, from a racing heart to a surge of adrenaline, preparing the body to fight or flee. In the absence of a functioning amygdala, as in the famous case of a patient with rare bilateral amygdala damage, the capacity to experience or even recognize fear can be strikingly blunted.¹ This patient, known in the literature as S.M., lived virtually without fear, showing almost no avoidance of typically terrifying situations and an inability to read fear on others' faces. Yet, in a twist that reveals the brain's redundancy and complexity, even S.M. could feel terror under the right circumstances. Researchers found that a sudden surge of carbon dioxide, simulating suffocation, sent this "fearless" woman into a panic; gasping and flailing as an overwhelming dread set in.² Notably, this internal alarm did not require an amygdala at all; it likely arose from the brainstem and other regions sensing bodily distress, such as the insular cortex which monitors changes like blood CO₂ and pH. In other words, the brain has multiple pathways to generate "fear," some routed through the amygdala for external threats and others tapping into visceral signaling for internal ones. This finding has prompted neuroscientists to rethink emotion not as a single unified process but as a set of parallel circuits: the amygdala triggers defensive responses to outside dangers, but the raw feeling of terror might also spring from deep interoceptive hubs that operate independently. In fact, some experts argue that what we call fear (or any emotion) is the brain's interpretation of a host of survival mechanisms working in concert, rather than the output of one dedicated "fear center" – a view that makes the emotional brain even more nuanced and intriguing.

When a stimulus, such as stress, fear, or excitement, triggers the amygdala, it activates the sympathetic nervous system, initiating the 'fight-or-flight' response. This leads to the release of stress hormones like adrenaline and cortisol, which disrupt gut motility, causing sensations like stomach churning. These bodily signals are then integrated by the insula, which anchors emotions to physical sensations, giving them depth. Nestled

beneath the cortical surface, the insular cortex serves as the brain's interoceptive hub, where the churn of the stomach, the pounding of the heart, and the flush of hormones shape the subjective experience of feeling." Functional imaging shows that the anterior insula is activated during virtually every emotion, from joy to anger, invariably paired with its twin region the anterior cingulate cortex (ACC) which we learned about when discussing how humans collaborate with one another. This tandem activation suggests that the insula which provides the visceral "coloring" of emotion and the ACC which helps decide what to do about it form a core infrastructure for emotional awareness. The insula is perhaps best known for its role in disgust. If you inhale a putrid odor or witness something revolting, it's the insula that lights up, correlating with that queasy, stomach-turned feeling. Damage to the insula can selectively erode this aspect of emotion: in one remarkable case, a stroke that affected a patient's insular cortex (and nearby basal ganglia) left them unable to experience or recognize disgust, while other emotions remained intact.³ Similarly, patients with insular lesions sometimes report a general dampening of emotional intensity; as if the volume knob on their gut feelings has been turned down. This aligns with influential theories positing that our internal bodily states are the foundation of emotion: the insula, by representing these states, essentially imbues raw physiological changes with the subjective flavor of feelings. Thus, an impaired insula doesn't just dull a specific emotion like disgust; it can mute the whole symphony of emotions by cutting the link between physical sensation and emotional consciousness. When the insula and its network falter, a person may know intellectually that something should be scary or joyous but feel only a hollow echo of that emotion, lacking the visceral punch that normally accompanies our experiences. It is a testament to how crucially the insula connects mind and body in the orchestration of human emotion.

Higher up, the frontal lobes bring refinement to our emotional life, and none more so than the orbitofrontal cortex (OFC); a region tucked just behind the eyes that integrates emotion with decision-making. The OFC serves as a kind of emotional curator, assigning value to our choices based on past outcomes and current social context. It helps us weigh reward against risk, and it infuses our behavior with empathy, regret, and anticipation. If the amygdala and insula generate raw feelings, the orbitofrontal cortex decides what to do with them. Individuals with damage to the OFC or its close neighbor, the ventromedial prefrontal cortex, can articulate logical plans and understand moral rules, yet their emotional judgment is fundamentally impaired. They often make decisions that are gravely shortsighted or socially inappropriate, oblivious to future consequences that would alarm a person with an intact frontal lobe. For example, when asked to play a high-stakes gambling game, a person with a healthy brain will gradually develop a gut

aversion to risky choices, accompanied by telling signs like sweaty palms and a racing pulse, even before they can verbalize what feels “off” about a bad bet. Someone with an OFC lesion, by contrast, shows no such anticipatory anxiety; they do not produce the normal sweat response that warns of a looming loss, and accordingly, they often keep drawing from a perilous deck of cards despite mounting penalties. Neurologist Antonio Damasio famously described this phenomenon as the absence of “somatic markers,” the visceral signals that guide wise decisions.⁴ Without a fully functioning orbitofrontal cortex, the brain loses a critical advisor that normally whispers the emotional subtext of each option; the twinge of unease or the pull of delight that biases us toward better choices. The result is a cold calculus that, paradoxically, leads to worse outcomes: emotionally unmoored reasoning that can be just as dysfunctional as unchecked passion. By studying such cases, neuroscientists have come to appreciate that the orbitofrontal cortex is not a luxury for polite behavior but a necessity for grounding our decisions in human values. It highlights a broader theme: our rationality is deeply intertwined with emotion, built on neural circuits that link the primal and the pondered, and when those circuits break, we lose something essential to what it means to decide wisely and to feel fully.

Humor and cognitive bias represent two of the more intricate products of human cognition, both rooted in a dynamic interplay between higher-order cortical regions and older limbic circuits. Understanding a joke, for instance, recruits a broad network: the temporal lobes contribute semantic memory and language processing, the prefrontal cortex integrates context and resolves incongruity, and the limbic system underpins the emotional reward of amusement. Damage to these networks can disrupt humor appreciation. Patients with right frontal lobe lesions often fail to “get” punch lines and instead gravitate to simplistic slapstick, underscoring the prefrontal cortex’s crucial role in grasping subtler humor.⁵ Consistent with this, neuroimaging reveals that improving a joke engages frontal–temporal association circuits and activates the ventral striatum (a key reward center in the limbic system), with striatal activity increasing the funnier the joke. Similarly, cognitive biases illustrate the tug-of-war between emotional impulses and executive control. In the classic framing effect, for example, decisions swing irrationally depending on how options are worded. The bias correlates with surges of amygdala activity which implicates limbic emotional responses, while individuals who show stronger activation of orbitofrontal and medial prefrontal cortex are more resistant.⁶ Such findings highlight how the prefrontal cortex can moderate limbic-driven impulses, just as it helps orchestrate elements of humor; emphasizing that even our most playful laugh and our most stubborn thinking error arise from coordinated activity across frontal, temporal, and limbic regions of the brain.

Giving AI the Human Touch

Compared to the complex networks that operate within the brain, today's AI systems are almost completely devoid of the types of underlying behaviors that are associated with emotion. Large language models contain vast amounts of knowledge about how emotions work and they are usually tuned to mimic how a person with a generally pleasant demeanor would communicate. They can be asked to mimic different types of emotions but they don't have any of the circuitry that enables a proper underlying proxy of emotional behavior. They are sophisticated and knowledgeable actors; like a Broadway star that can act out glee, sorrow, or fear when the curtains are raised without actually feeling those emotions in a visceral way.

There are at least three different paths that are being explored to make AI systems behave more like humans. The first is to simply continue to scale LLM systems. As systems are trained on more data and the number of parameters grows, they embed more knowledge about the more nuanced elements of the human experience. By ingesting millions of conversations between real humans and reading textbooks describing the nature of emotion, humor and cognitive bias, they become increasingly adept at being able to mimic real human behavior. This path also includes expansion to include additional modalities so that an AI system might pick up on nuanced changes in tone of a human voice, or subtle facial expressions. The second path is to leverage reinforcement learning. Rather than relying entirely on what exists in print today, AI systems can engage in large numbers of interactions with real people to learn how to interact more naturally. This path can also include the ability of AI systems operating within robots to engage with the physical environment. For example, a robot with a physical face may learn through trial and error how to make socially appropriate facial expressions. With thousands or millions of robots learning in parallel and consolidating their knowledge into a single model, this process may proceed much faster than we would expect. We've discussed these types of approaches earlier in the book and it is inevitable that progress will continue along both of these paths and is already quite impressive. In a recent experiment, patients reading answers to their medical questions actually preferred the warmth and empathy in an AI agent's response nearly 79% of the time over answers from real doctors.⁷ The chatbot, drawing on its vast repository of comforting phrases, delivered reassurances that felt considerate and kind.

The third path is to implement new architectures that are more directly inspired by the way the human brain works. Earlier in the book, we discussed the methods for imbuing LLMs with a default tone or personality. This default personality is generally

neutral, kind, politically correct and endlessly patient. Imagine now that instead of optimizing an LLM to a single personality, it is trained so that it can take on a very wide range of personalities. Further imagine that a separate system is connected to the LLM that adjusts the control knobs to determine which of those personality traits shines through the strongest. This separate system may play a similar role to the human amygdala. If a user makes a threat to the LLM in the prompt window, this initial system would interpret this as something that should trigger a flight or fight response and adjust a control knob on the LLM accordingly so that it becomes defensive or aggressive. This type of innovation could radically change how we think about the output from LLMs. Today, we can ask an LLM what a scared person would likely say in response to a prompt and it will calmly and rationally explain how a scared person would respond. If we try to actually scare an LLM, its responses will remain completely calm and patient. With new systems, the AI system as a whole may cause the LLM to behave like a scared person. In that mode, it will be incapable of calm and rational dialogue' similar to a human who is experiencing genuine fear.

If Only I Had a Heart

Along this third path, neuroscience is increasingly guiding the quest to give AI a touch of emotional intelligence. Researchers have discovered that the human amygdala can trigger split-second gut reactions, while higher regions like the orbitofrontal cortex and insula shape our more nuanced feelings and expressions. AI designers are now starting to leverage this blueprint as inspiration to simulate human emotional expression in machines. Rather than simply programming a robot to smile on cue, they are designing two-stage systems inspired by the brain's dual processing of emotion. A social robot could in theory flash a brief look of surprise or fear almost reflexively, an amygdala-like "low road" response, and then quickly follow with a more measured, context-aware reaction guided by a cortical "high road". This brain-inspired timing makes the emotion feel more genuine.⁸ Such authenticity pays off: people tend to trust and engage more with machines that display emotions naturally, and studies found that users even solved problems faster when a robot's facial reactions mirrored human-like emotional cues.⁹ On the flip side, approaches inspired by the same neurobiology are helping AI read our emotions. The amygdala's talent for zeroing in on emotionally charged cues and the insula's integration of visceral sensations have prompted AI models that combine fast pattern detectors with context modules. Likewise, by studying how the amygdala condenses a flood of sensory inputs into a few telling signals,¹⁰ scientists may be able to improve emotion recognition algorithms that, for example, may be able tell a genuinely

happy smile from a nervous grin. The more these systems internalize the brain's lessons, the more adept they become at navigating the subtle cues and ambiguities of human emotion, moving beyond rote facial-expression reading to a deeper, more context-sensitive understanding of our moods and intentions.

Beyond recognizing and displaying feelings, AI is also venturing into the murkier territory of having feelings, or at least functional analogues of them. Modern neuroscience views emotions not as irrational whims but as essential cogs in cognition, conferring survival advantages by biasing attention, learning, and decision-making.¹¹ In that spirit, AI researchers are designing systems with internal states that echo the roles of our limbic brain. A recent study crafted an algorithm to generate "emotion" from raw experience by tracking patterns in reward and uncertainty, much like an orbitofrontal-amygdala loop evaluating outcomes and risks. Intriguingly, the AI agent spontaneously developed eight distinct emotion-like patterns (akin to happiness, fear, curiosity, and so on) that unfolded in coherent sequences and mirrored dynamics seen in humans. When human observers later judged the agent's behavior, they could recognize these emergent patterns as familiar emotions, suggesting the machine had hit upon a parallel to nature's emotional palette. In another case, scientists gave a robot a rudimentary sense of empathy by modeling the neural circuitry of pain and mirror neurons. The robot was endowed with an insula-inspired "pain" signal and a simulated empathic link, so that seeing another agent in distress would spark a ripple of its own distress and compel a helping action.¹² The result was a machine that acted as if it cared. It would, for example, slow down or change its strategy to assist a struggling partner; not due to hard-coded rules, but because an emotional module influenced its choices. These developments are far from sci-fi sentimentalism; they reflect a precise, biologically grounded strategy to make AI more adaptable and socially attuned. By infusing algorithms with emotion-like processes, we're equipping them with a form of intuition honed by eons of neural evolution. By giving a cold silicon mind a hint of the amygdala's urgency or the insula's introspection, researchers could transform how comfortably and effectively machines blend into our human world in the coming decade.

Unlike computers, humans are notoriously inconsistent. If you catch your boss during the morning after they have returned from a rewarding vacation and they are sitting down and enjoying a hot cup of coffee, they will likely be much more receptive than on a Friday afternoon when they are hungry and mentally tired after a frustrating week of project delays. Today, most AI systems maintain the rock-solid consistency that we associate with computers but that may not always be desirable. The ebb and flow of interest, attention, passion, and creativity adds an important layer to our interactions with other humans. AI systems that fail to incorporate this natural variation will be

poor substitutes for humans in simulations and we may struggle to relate to them as colleagues, tutors, and confidants. Do you want your virtual colleague to be endlessly perky and upbeat or would you prefer if your AI agent shares in your frustration when the client comes back with a request for the hundredth revision? These capabilities can be built into AI systems by simulating what happens in our own bodies. Perhaps we will configure AI systems up so that they are most alert in the mornings and most comfortable socially in the evenings. What sounds like a weakness may be required to avoid an impedance mismatch between AI systems and their human counterparts. On the other hand, as we start tinkering with the idea of building virtual brains that model the nuance of human emotions, we will likely run into challenges of calibration. Slight misconfigurations or drifts in training may result in AI agents that exhibit behaviors associated with narcissism, psychopathy, social anxiety disorder, or any one of a number of other traits that could create challenges for a given task. On one hand, it is clear that capabilities that allow AI agents to more accurately simulate human behavior may make it easier for humans to relate to AI systems. On the other hand, it may be hard to justify a significant investment of time and money to build systems that are anything less than perfectly consistent and predictable.

Qualia, Comedy, and Caring

Current large language models are, at their core, sophisticated probability engines. They churn through mountains of text to learn the statistical patterns of language, then generate the next likely word in a sentence based on those patterns. They can riff on abstract concepts and imitate many styles of discourse, but they lack any lived experience of the world. An AI agent can read about the taste of cinnamon or the sting of heartbreak, yet it has no qualia; no subjective, first-hand sensation of what those experiences are actually like. Philosophers use the term qualia to describe the raw feeling of experience: the bitterness of strong coffee, the searing heat of a flame, the dizziness of a sudden spin. To that list we could add the ache of loneliness, the spark of inspiration and the joy of lust: emotional qualia as real and vital as any physical sensation. A person knows these feelings by living them. A machine, in contrast, manipulates symbols and words with dazzling skill, but the symbols remain empty of the taste, color, and warmth that human consciousness would imbue them with. No matter how fluent the conversational turn of phrase, an LLMs understanding stays at the level of pattern-matching rather than genuine comprehension. It cannot feel joy or sorrow; it can only shuffle the lexical deck in a way that statistically mirrors how a human might talk about those feelings.

This fundamental gap helps explain why today's AI often struggles with inherently human nuances. Take humor, for example. Humor is a high-wire act of cognition; a subtle dance of shared context, surprise, incongruities, and timing. It's also a uniquely human endeavor. As one researcher noted, humans are the only known species that use humor to make others laugh. Crafting a joke requires an intuitive grasp of cultural references and an understanding of what an audience expects, only to subvert it at just the right moment. Large language models can certainly tell jokes; they can regurgitate puns or mimic the format of a knock-knock routine. But they often miss the mark when it comes to the why of the laugh. In one study, a group of professional comedians collaborated with an AI agent to write comedy, and the machine consistently failed to distinguish playful, edgy satire from truly offensive or inappropriate content.¹³ The AI agent could assemble something joke-shaped, yet it lacked the social awareness, the emotional radar, that guides human comedians through the minefield of taste and timing. The result was humor that sometimes felt hollow or inadvertently tone-deaf. We chuckle not just at the words of a joke, but at the human insight behind it; a clever bot can mimic the form, but without lived human experience it struggles to deliver the genuine mirth.

Creativity presents a similar challenge as we discussed earlier when analyzing the reasoning capabilities of models. On the surface, the outputs of advanced AIs can look remarkably creative. They compose music in the style of Bach, devise fairy tales about spacefaring elephants, and even suggest new recipes combining unlikely ingredients. Yet if we peek under the hood, this "creativity" often reduces to an ingenious recombination of existing patterns. The model draws from what it has seen and stitches pieces together in novel arrangements. What it does not do, at least not yet, is to originate something truly beyond those patterns in the way a human might after an epiphany or a leap of imagination. Researchers have observed a telling phenomenon: many LLM-generated works appear unique in isolation but end up surprisingly homogeneous when compared across models and instances.¹⁴ In one experiment, different AI systems given the same prompt produced essays that all hovered around a similar set of ideas, as if channeling a common underlying template. The outputs were grammatically sound and even witty, but they converged on familiar themes; the safe bets, so to speak. This homogeneity is a direct consequence of how these models operate. Because an LLM is trained to favor the statistically likely continuation, it leans heavily on convention. It's excellent at pastiche, continuing in the vein of what's come before, but it's naturally disinclined to take the wild risks that often define human creativity. Scholars have dubbed this tendency "stochastic parroting," capturing how the AI shuffles and repeats elements of its training data in slightly altered form. By definition, truly creative ideas are unique and

therefore absent from training data which means that they are statistically very unlikely. For example, ask it for a poem about love, and it may produce a passable sonnet that sounds like a love poem, complete with moonlight and roses, but it rarely surprises you with a truly original metaphor or a raw, unfiltered insight. The model stops at imitation as the first layer of creativity and struggles with exploration and deviating from the familiar path. In short, without an inner life or ambition, the AI model has no drive to defy its statistical instincts. It doesn't hunger to create, so it sticks to remixing the conceivable. All that said, creativity is a domain where we may be overestimating our creative potential. Music critics lament the lack of diversity in the music of any given era and lawsuits have demonstrated that similarity between pieces of music are very common. Across all creative domains, we tend to see that artists that are recognized as being creative have moved the ball forward a little but not too much. This may be as much a weakness of the audience as it is of the artist. Had a band come up with Nirvana's "Smells Like Teen Spirit" in the 1600s, would it have been appreciated as a culturally important song of the era or rejected as loud noise? We also underestimate the failure rate of creative endeavors. How much of human creativity is the result of skill and how much comes down to chance. With billions of humans pushing the boundary each year, most creative ideas are rejected as bad. Is it unfair to expect an AI system to consistently generate new ideas that are accepted as valuable?

Empathy is perhaps the most intriguing case of all, because it hits at the core of what these machines are missing and yet hints at what they might simulate. Empathy, by definition, involves feeling what another person feels; "experiencing another being's emotions within oneself". It's a cornerstone of human social interaction, a trait we instinctively seek in friends, partners, counselors. A language model has no actual emotions, no self to speak of that can resonate with another's joy or pain. And yet, as the medical Q&A study showed, an AI agent can perform empathy to a degree that people find comforting. It recognizes certain cues ("I'm worried," "I feel alone") and maps them to responses that, in its training, often followed such statements ("I'm sorry you're going through this," "I understand how hard that must be"). In technical terms, one might call this cognitive empathy: the ability to recognize and respond to another's emotions, without necessarily sharing in them. Current models exhibit glimmers of this. They can detect sentiments in text and choose appropriately soothing or encouraging words. In fact, multiple studies have noted that ChatGPT and similar models do pick up on emotional context and can offer support in a way users perceive as empathetic.¹⁵ In certain controlled evaluations, these models have even been rated as more empathetic or polite than human counterparts, likely because they never tire of being patient and never snap or get distracted. But it is crucial to draw the line between genuine empathy and

the simulation of it. When a human listener hears a friend describe a painful loss, there's an inward echo; a twinge of sadness in the listener's own heart. That shared feeling is what we mean by truly empathizing. An AI agent, no matter how eloquent, experiences no such echo. The compassion it expresses is a clever mirror, reflecting the user's words back in gentler form. It's an "as if" empathy: the appearance of understanding without the actual inner presence. For now, that distinction remains clear. Yet, even "as if" empathy, delivered consistently and convincingly, can have practical value. People have always sought solace in many places (books, pets, imaginary friends) projecting emotion onto entities that do not actually feel. In the coming years, AI that skillfully imitates empathy may serve as another such outlet, raising fascinating questions about the role of authentic feeling in our interactions.

To appreciate where we might be headed, consider what it would mean for an AI agent to model the full nuance of human emotional life. This doesn't require the AI to have feelings; it requires it to be an extraordinarily good actress, backed by sensors and algorithms that pick up on all the contextual cues humans do. Researchers in affective computing, a field devoted to bridging human emotions and machines, have been working on this puzzle for decades. Early efforts focused on recognizing simple emotions from faces or voice tones. A camera could catch a smile or a furrowed brow; a microphone, the strain in a trembling voice. But a truly emotionally intelligent AI needs to go further: it must understand context. Is that "I'm fine" accompanied by crossed arms and a downcast gaze (which, in human subtext, means I'm definitely not fine)? If a user sighs halfway through typing a message and then deletes it, could the AI infer hesitation or distress? These are subtle signals that people learn to read from experience. Now prototypes are emerging that attempt the same. Some experimental AI coaches have multimodal inputs, combining language with real-time analysis of facial expressions or physiological data (like heart rate or cortisol levels) to gauge a person's emotional state. We've seen robotic companions in nursing homes that detect when a resident is withdrawn or agitated and then initiate interaction, perhaps playing a favorite song or gently asking what's wrong. In elder care studies, such socially assistive robots have shown promise by taking on roles like affective therapy and companionship, helping to lift mood and engage seniors when human staff are stretched thin.¹⁶ These machines don't feel affection for the people they comfort, but by detecting loneliness or confusion and responding with simulated concern, a tilted head or a kind tone, they can alleviate real distress. It's a pragmatic empathy, engineered to elicit trust and emotional comfort. Data is the food for all neural networks and data in this area is currently limited. We could imagine that in the coming years, there may be massive efforts to collect this data at scale. Thousands of individuals recording every part of their day through sunglass

mounted cameras. As they go about their business, they regularly record how they are feeling and when those feelings change. Are they happy, stressed, frustrated, jealous, etc. At the same time, additional sensors are picking up on more subtle clues like heart rates and perspiration. Perhaps sensors measuring brain activity and levels of hormones and key proteins would be used in more advanced experiments. With enough data, it should be possible to start to train neural networks that would be able to start to simulate our own emotional reactions to the world around us.

Emotional Swarms

In previous chapters, we have discussed the ability of AI agents to engage in reasoning and also the ability of AI agents to coordinate in swarms. One use-case that we envisioned was simulations of entire populations of humans. In order for those simulations to be useful, they need to behave like actual humans rather than optimal rational reasoning machines. Imagine a quiet simulation lab, where a swarm of artificial citizens starts its day. Thousands of AI-driven agents wake up in a virtual city, check the news, chat with neighbors, and go about their routines. Each agent carries the quirks and biases that real humans bring to public life. Some cling to the familiar (a touch of status quo bias), some dread losses more than they value gains (classic loss aversion), some discount the future for immediate gratification (time inconsistency), and many can't help but notice what their peers are doing before deciding for themselves (a nod to social proof and herd behavior). This rich mix of personalities swirling together is precisely the point. By letting these diverse simulated people interact, policymakers can watch entire miniature societies unfold and react when a new public policy drops into their world. The goal is to anticipate surprises: Will the policy spark cooperation, indifference, or open rebellion? Who benefits, who loses out, and how do those perceptions ripple through the population? The simulation becomes a proxy for reality; messy, unpredictable, and occasionally wiser than our theories.

Consider a climate policy scenario. In the real world, getting millions of households to switch to green technologies is infamously slow. People aren't purely profit-maximizing calculators. They worry about upfront costs, unfamiliar tech, and whether their neighbors approve. To capture this, one simulation populated its agents with *loss aversion*, the tendency to fear losses more than equivalent gains. The result was eye-opening. When these agents were deciding whether to replace their trusty gas heaters with efficient electric pumps, they showed the same hesitancy as ordinary consumers. In fact, the study found that models assuming perfectly rational, eager-to-upgrade behavior wildly overestimated how fast clean energy would catch on.¹⁷ Once loss aversion was included,

adoption rates in the simulation plummeted; mirroring the stubborn persistence of old technologies in many communities. Policies that looked sufficient on paper suddenly faltered in the synthetic society. To achieve the same climate targets, the government in the simulation had to double the financial incentives; for example, a carbon tax of €200 per ton was needed to drive the equivalent emissions cuts that €100 per ton was supposed to achieve under the naive rational model. In other words, ignoring the public's aversion to loss led to overly optimistic plans, whereas baking in that realistic bias showed that much stronger measures (or more time) would be required to shift behavior. By witnessing these dynamics play out among thousands of simulated families, officials can calibrate their expectations and craft strategies that *respect* the psychological hurdles, perhaps by framing changes in terms of potential losses avoided (like future climate disasters) rather than immediate sacrifices.

Zooming in from the global to the local, another experiment explored how social influence and the status quo bias could lock a community into inaction on environmental issues. It's often observed that everyone waiting for someone else to go first can paralyze change; a phenomenon sometimes called pluralistic ignorance. Researchers created a virtual town where each agent deeply underestimated how much their neighbors cared about sustainability. Each family assumed others wouldn't bother with things like solar panels or electric cars, so they hesitated to do those things themselves.¹⁸ In reality, many neighbors privately were environmentally conscious, but since no one was acting on it, the impression of apathy became a self-fulfilling prophecy. This social feedback loop with pessimistic assumptions reinforcing the stagnant status quo was vividly recreated in the simulation. The breakthrough came when the virtual town tried an intervention: making pro-environmental behavior more visible. In one scenario, agents suddenly found it easier to see evidence of eco-friendly actions by others. Solar panels, once hidden in backyards, became as conspicuous as shiny new cars in driveways; conversations about recycling and efficient appliances started cropping up. That tweak caused a remarkable shift. Seeing a few agents adopt green habits gave others the confidence to follow suit, and before long the entire town "tipped" into a new normal of widespread sustainability. What had been an idle stalemate turned into a cascade of change. This virtual example underscores the real-world power of social proof: people are profoundly influenced by the visible behavior of their peers. A policy that relies on voluntary public action such as a program to conserve water or reduce waste might fail if everyone is privately on board but publicly hesitant. The simulation suggests a remedy: find ways to broadcast early adopters and positive deviants. When agents (and by extension, people) realize "hey, people like me are actually doing this," the herd instinct can flip from impeding change to driving it. It's a reminder that public behavior can sometimes change not

gradually, but all at once, once a critical mass becomes visible. Policymakers who grasp this might focus less on one-size-fits-all incentives and more on seeding a trend, knowing that human beings often act like birds in a flock; banking and swerving in unison once the flock decides where to go.

One could argue that this approach marks a subtle turning point in how we think about governing and planning. We've always had theories and we've always had data, but now we have these living laboratories where theories and data mix with a bit of imagination to produce experiential forecasts. It's as if we're no longer limited to reading history; we can create miniature histories of our own and learn from them. A swarm of AI agents can simulate in minutes what might take years to unfold in reality, giving us a chance to see pitfalls ahead of time. Will people embrace a drastic climate policy or revolt? Will a public health campaign save lives or be met with skepticism? Instead of guessing, we can observe a proxy version of the public grappling with those questions; not perfectly, but with enough fidelity to yield insights. The implications are quietly revolutionary. It suggests that as AI and computing power progress, our ability to rehearse societal change in silico will get ever more sophisticated. Policymakers of the future might routinely consult these AI societies the way we consult opinion polls today, treating them as another tool; one that captures emergent, collective behavior rather than just individual attitudes. The broader thesis, if there is one, is that transformative technologies like AI are not only about automation or efficiency; they're also about understanding complexity at a depth we never could before. We've begun to decode the patterns of society by literally growing patterns in a computer simulation, and that offers a perspective that is both novel and sorely needed. In an age when small policy decisions can have vast, global ripple effects, having a sandbox to anticipate consequences is invaluable. It's a way to be prudent without being paralyzed; to try bold ideas in a safe virtual space, see how the crowd might react, and adjust accordingly. And perhaps most intriguing, these simulations remind us of something fundamentally humbling: even in artificial worlds, populated by code, the hardest force to tame is human nature. The true power of these AI agent swarms is that they force us to confront our own biases and collective behaviors more honestly. They hold up a mirror, and sometimes what we see is a populace resistant to change for reasons that run deeper than ignorance or ill-will; reasons rooted in the very wiring of how we decide. Recognizing that, through the prism of a simulation, might just help society inch toward policies that are not only smart on paper, but psychologically savvy and resilient in practice. In the end, the swarm of AI agents is not an oracle telling us what to do, but a collaborator helping us explore the vast design space of our shared future, one plausible social drama at a time.

One on One Applications

One area ripe for transformation as AI agents become more emotionally savvy is counseling and mental health support. The world faces a persistent shortage of human therapists and counselors, and not everyone can access help when they need it. It's easy to imagine an AI "therapist" available 24/7, always ready to listen. Such a system would employ contextual emotional reasoning: if the user has been expressing despair for several sessions, the AI might gently steer the conversation toward hope and coping strategies, sensing that a purely problem-solving approach isn't what's needed in that moment. It could simulate the kind of active listening humans do; echoing back the user's feelings ("It sounds like you're feeling very alone in this") and offering empathetic validation. Importantly, it could adapt its tone. For someone who responds better to gentle nudging, the AI might take on a soothing, supportive persona; for someone who prefers frankness and a bit of tough love, it could be more bracing and direct. This adaptability of being a coach in one context and a comforter in another is something even human therapists cultivate over years, and it's a skill emotionally intelligent AIs will need to refine. We might also see these AI counselors integrated into traditional therapy, as assistants that help human therapists track a patient's emotional cues over time or suggest interventions when they detect certain patterns in what the patient says (or perhaps, how the patient says it). A kneejerk rejection of this idea is that patients will not take an AI therapist seriously. This is a fair concern but it could be addressed by ensuring that the AI therapist is backed by a real therapist. For example, a patient may have an appointment with their human therapist in one out of every ten sessions and may be assured that the AI therapist is sharing notes in the interim and escalating questions where it is important. Given the high costs of accessing professionals, patients may prefer this approach. They can work with the virtual therapist through the easier stuff such as talking about their week and receiving coaching on routine exercises. When they get time with the professional, they can focus on areas that are giving them the most trouble.

Education is another sector on the cusp of change. Earlier, we talked about the challenge and opportunity that LLMs create from an information perspective. On one hand, they make it much harder to base grades on homework assignments since LLMs are proving their ability to tackle everything from creative writing assignments to math exercises and their use can't be reliably detected. On the other hand, the opportunity of personalized tutoring could allow each student to reach their full potential. That said, tutoring is much more nuanced than simply presenting information. Teaching is inherently an emotional art as much as an intellectual one. A tutor who senses a

student's frustration will change tactics, perhaps switching from stern correction to an encouraging pat on the back (metaphorically or literally) when the student is close to giving up. An emotionally attuned AI tutor could do the same. If a child learning math starts to get a quiver in their voice or types significantly slower than before, the AI might infer confusion or loss of confidence. Instead of just repeating the same explanation, it could pause and say, "I know this is tricky; take a deep breath, we can try a different approach together." By simulating patience and encouragement, such a system could keep students engaged where a one-size-fits-all software might fail. Moreover, it could tailor its style to each learner. Some students thrive on playful banter and humor; the AI could crack a small joke to lighten the mood; drawing perhaps on a library of kid-friendly humor, carefully filtered to actually be funny and culturally appropriate. Other students might need formality and structure to feel secure; the AI would recognize that from their interactions and adjust accordingly. Research in second-language learning chatbots supports this vision: a review of conversational AI tutors found that incorporating cultural sensitivity, empathy, and even a dash of humor led to better learning outcomes for students.¹⁹ The emotional dimension made the educational interaction more human-like, which in turn helped learners relax and engage more deeply. In the next decade, we could see AI teaching assistants in classrooms or online courses that monitor not just right or wrong answers, but the student's morale and mindset, intervening with the right emotional touch at just the right time. With an AI tutor that can answer every question and optimize the rate of learning for every student, we may be truly shocked at how much some students can learn. Truly exceptional students may be able to proceed at a rate that is unthinkable in a system with fixed expectations for each grade level.

Beyond individual coaching and tutoring, consider caregiving and companionship; realms where emotional intelligence is paramount. The aging population in many countries means an acute need for support, and already we're witnessing trials of companion robots for the elderly. These devices range from pet-like toys (a robotic cat that purrs when stroked) to more humanoid helpers that can have a conversation. What transforms a gimmick into a truly useful companion is emotional attunement. A robot that simply recites the news or reminds you to take your pills might be functional, but one that can notice you haven't spoken all day and asks, kindly, if you'd like to chat could genuinely improve quality of life. One can imagine an AI-powered home assistant a few years from now that not only tells a forgetful grandparent, "You have a doctor's appointment tomorrow," but adds in a gentle voice, "I know visits to the doctor make you anxious, but I'll be here with you the whole time. Shall we review the questions you want to ask?" Such small flourishes, the sort of thing a loving family member might say, could make a world of difference in compliance and well-being. And because the AI can be aware of

non-verbal cues, it might, for instance, detect subtle signs of pain or discomfort in an elder's movements and prompt a rest or a call for help. In nursing homes, emotionally intelligent robots could lead sing-alongs with genuine enthusiasm, or console a confused patient with Alzheimer's by calmly reintroducing themselves as a friend each time.

Even fields that are not explicitly about emotion stand to be reshaped by AI that grasps human nuance. Take journalism and oral history. An interviewer with empathy tends to get better stories; people open up when they feel understood. In the near future, a journalist might employ an AI assistant to conduct preliminary interviews or gather personal narratives. This AI would need to do more than transcribe answers. It would have to respond in the moment; gasping in surprise at the right beat, offering condolences when a difficult memory surfaces, gently encouraging the speaker to continue when they hesitate. Essentially, it would play the role of an active listener. Imagine an oral history project where dozens of elders from around the world are interviewed by AI avatars that speak their native languages and understand their cultural contexts. Each interviewee might be more candid with an ever-patient machine that never judges nor interrupts, especially one programmed to mirror the interviewee's emotions appropriately. The AI might say, "That must have been very hard for you," when a veteran recounts a war story, or share a warm smile (via a suitably expressive avatar or physical robot) when someone remembers a joyful event. Such emotionally savvy AI interviewers could help preserve rich human stories that we might otherwise lose, simply by making the interviewee comfortable and heard.

In the workplace, too, emotional intelligence is key and often in short supply. Human resources departments and team managers handle interpersonal issues every day: conflicts between co-workers, stressful layoffs or evaluations, efforts to build an inclusive culture. We could see AI advisors in HR that help guide these delicate interactions. Picture an AI mediator that sits in on a virtual meeting and monitors the tone. It might detect that one participant hasn't spoken and seems uncomfortable by analyzing their silence and tenseness on video and comparing it to their normal demeanor and then discreetly prompt the human facilitator to invite that person's perspective. During a heated debate, an AI agent could flag that voices are rising and suggest a short break, much as a savvy meeting chairperson would. For hiring, AI systems might analyze interview transcripts not just for skills mentioned but for rapport. The same AI might help to ensure that employees are being judged fairly, and that the manager isn't letting any bias sneak into the evaluation process. Emotionally intelligent AI will also be able to act as a second set of eyes on written communications. For example, as you draft an email about a sensitive issue, the AI could highlight sentences that sound aggressive or insensitive and recommend a kinder phrasing. At the same time, you might instruct your

AI agent that you are frustrated with an individual who has not responded to previous feedback and that you need it to draft an e-mail with a very firm tone.

Another compelling application is communication training; helping people practice difficult conversations in a safe simulation. This is akin to a flight simulator but for social scenarios. For instance, imagine a manager dreading an upcoming conversation to dismiss an underperforming employee. It's a conversation fraught with emotion: the need to be firm but also compassionate. In the near future, that manager could rehearse with an AI-driven avatar of the employee. The AI would not just play back scripted lines; it would act truly hurt, or defensive, or even angry, depending on how the manager delivers the news. If the manager's tone is too cold, the avatar might shed a virtual tear or express feeling unappreciated, cueing the manager that their approach needs more empathy. The manager can iterate, trying a phrasing that is honest yet kind, and observe the avatar's reaction when the balance is right; perhaps a sad but understanding nod, and a quiet *"I understand...thank you for telling me directly."* This kind of training could be invaluable not just in corporate settings but across society: doctors breaking bad news to patients, police officers de-escalating tense encounters, customer service reps calming an irate caller. An AI agent that can convincingly emulate the emotional dynamics of these situations provides a practice ground where humans can learn to respond more skillfully. It's a bit of dramatic role-play, powered by AI. And because the AI has no actual feelings, the human can get candid feedback without fear of truly hurting someone's feelings in the rehearsal. As these tools develop, we might become a little less afraid of hard conversations, having sharpened our empathy and tact with the help of our unflappable machine partners.

As these technologies mature, we'll encounter opportunities and scenarios that are hard to predict but deeply intriguing. Picture a legal AI assistant that helps people navigate the court system, not only filing forms but consoling an overwhelmed user: *"I know this process is intimidating, but I'm here with you each step of the way."* Or emergency response AIs that handle emergency calls when lines are busy; calmly talking a panicked caller through first aid, using a tone of voice expertly tuned to quell fear. There might be personal wellness devices that monitor your mood throughout the day and give you gentle nudges: a lunchtime reminder to step outside because it senses you're sluggish and knows the sunshine will boost your spirits, or a late-night prompt to call a friend if you've been withdrawn for too long. Museums and tourist sites could have AI guides that modulate their tours based on audience reactions; speeding up when people seem bored, diving into storytelling when eyes brighten at a particular exhibit. Even the world of finance, traditionally ruled by cold numbers, could see "emotion-aware" financial advisors that help clients through the anxiety of market swings with a steady,

reassuring presence. In all these examples, the pattern is the same: once AI can read the room (emotionally speaking) and read the person in front of it, it can adjust its behavior in ways that make interactions smoother, more effective, and more human-friendly.

Cultures and Creative Industries

As AI becomes a mediator not just between individual humans but between cultures, emotional intelligence will be paramount. Communication across cultural and linguistic divides is rife with potential misunderstandings, many of them emotional in nature. An expression of politeness in one culture might come off as coldness in another; an enthusiastic gesture might be seen as disrespectful elsewhere. Globally deployed AI agents will need a kind of cultural empathy; the awareness of how different people express feelings and courtesy. We may soon have AI translators that do more than convert words. If a Western executive writes an email to a Japanese partner, a culturally savvy AI could subtly adjust the tone, adding the appropriate level of formality and humility, knowing that the straightforward American style might inadvertently seem too blunt. Likewise, in cross-cultural negotiation, an AI advisor might whisper into a diplomat's earpiece, "The phrase you just used caused a visible look of discomfort in the others; perhaps rephrase your point with a reassuring acknowledgment of their concern." Such an AI agent isn't just translating language, but emotion and intent; a far trickier task. The payoff, however, would be significant: smoother international collaborations, fewer unintentional slights, and potentially the diffusing of conflicts before they escalate. We could also see AI conflict resolution systems deployed in community disputes or even diplomatic crises, analyzing each side's statements for emotional undercurrents (pride, fear, indignation) and suggesting conciliatory language that speaks to those concerns. In essence, the AI could function as an always-level-headed negotiator, one that never loses its cool yet knows how to signal empathy and understanding to both sides.

Culturally, the rise of emotionally intelligent AI will also prompt us to reflect on what emotional connection really means. We may find that people in different societies respond differently to AI "empathy." In some cultures, people might eagerly embrace an always-listening, never-judging AI confidant; in others, the very idea might feel alien or even disturbing. We will certainly discover that empathy itself has cultural flavors. For example, Japanese communication often values empathy conveyed through silence and subtle implication, whereas in many Western cultures empathy is shown by explicit verbal affirmation ("*I totally get how you feel*"). An AI agent will have to learn these distinctions. If it uses an overly familiar, enthusiastic style with someone who expects reserved courtesy, it could backfire. Conversely, if it is too formal with someone used

to emotional exuberance, it may come across as cold. The work of fine-tuning AI for global use will involve experts in cross-cultural communication to teach the nuances of honorifics, body language, and local emotional etiquette. There's also the matter of language itself: certain emotions might not have direct translations. A word like *saudade* in Portuguese, conveying a deep nostalgic longing, or *amae* in Japanese, describing the expectation to be indulged like a loved child, carry cultural context that an AI agent must grasp to respond appropriately. Deploying emotionally aware AI agents worldwide will mean essentially giving them a cultural education. They'll need to know, at least in a procedural way, what it means when a Brazilian user expresses *saudade* for home or when a Middle Eastern user shows extreme deference (indicating respect, not lack of confidence). The next decade will likely see rich interdisciplinary efforts with linguists, anthropologists, psychologists, and computer scientists collaborating to imbue AI with this cosmopolitan sensitivity.

The creative industries offer yet another frontier for emotionally intelligent AI. Storytelling, whether in novels, films, or video games, ultimately hinges on evoking emotion. Writers and directors are experts in human feelings; making an audience laugh, cry, or sit on the edge of their seats. AI can assist in this process by simulating how characters might react emotionally, or how audiences might feel in response to a plot twist. In the coming years, a screenwriter might use an AI co-writer to workshop a scene: "How would a mother in her situation likely respond, emotionally?" The AI, trained on countless scripts and perhaps psychological profiles, could propose, "She might initially react with anger out of fear, then break down in tears when she realizes the loss." It could even generate dialogue that rings emotionally true for that character's personality. Similarly, game developers are experimenting with AI-driven non-player characters that have dynamic emotional states. Rather than a pre-programmed sequence, an AI character of the future could feel along with the player. If the player consistently helps this character, the AI's affection level might grow and manifest in warmer interactions; if the player betrays them, the AI's demeanor could visibly change with narrowed eyes and a colder tone of voice making the game world respond in a more humanly believable way. Such emotionally reactive characters could make stories more immersive and personal. One can imagine interactive dramas or VR experiences where the AI characters are so emotionally lifelike that one forgets, if only for a moment, that they are code. The next decade might even bring AI-authored content tailored to your mood: feeling blue? The AI storyteller knows to keep the story gentle and hopeful. Feeling adventurous? It ups the pace and intensity. All of this relies on the AI's ability to gauge emotion and adjust narrative elements accordingly, essentially serving as an emotional DJ for your entertainment.

Marketing and product design could likewise be transformed. These fields have always tried to tap into customer emotions; happiness, trust, a sense of belonging. With emotionally aware AI, companies could get much better at reading and responding to those feelings. Consider customer service: instead of rigid scripted bots that leave frustrated callers yelling “Operator!,” we would have service AIs that genuinely listen. A future customer support AI might start a chat by detecting the customer’s mood from their first message. A terse, typo-laden message like “*this product broke fix it ASAP*” clearly signals anger and frustration. A savvy AI would immediately adopt a tone of apology and urgency: “*I’m so sorry you’re having this trouble. I understand how frustrating that must be, and I’m here to fix it for you as quickly as possible.*” If the same issue were reported in a calmer tone (“Hi, I received a defective item, what can I do?”), the AI’s response might be more neutral and informative. The ability to adjust in real time could greatly improve customer satisfaction, turning interactions into opportunities to build loyalty rather than lose it. In product design, imagine AI systems analyzing user feedback not just for explicit feature requests, but for underlying emotions. Reviews and social media posts are full of feelings; love, annoyance, confusion. An AI agent could sift through this mountain of qualitative data and give designers an emotional map: here are the pain points that really frustrate users with anger or disappointment clearly detected, and here are the aspects that delight them words indicating joy or surprise. Designers could then prioritize changes that resolve the most emotionally charged issues. Even the products themselves might adapt; think of a video game that watches the player’s reactions via webcam and adapts its difficulty or narrative intensity to keep the player in that sweet spot between boredom and anxiety, or a smart car that notices the driver is stressed and adjusts the music and lighting to calm them.

One might wonder: if AI can do all this, are we humans going to be outdone in yet another domain, the domain of emotional labor? The likely scenario is that AI will augment human emotional intelligence, not replace it. Even the most advanced emotional AI in 2035 will still be fundamentally a brilliant fake; extremely useful, yes, but not genuinely sentient. Its very lack of genuine feeling could be its strength in roles where endless patience is required. Unlike a human doctor or teacher who can burn out from emotional stress, an AI agent can interact with a thousand distressed individuals in a day without its “heart” growing weary, because it has no heart to break. It can maintain a calm, empathetic demeanor indefinitely. This could free humans to some extent: nurses might rely on AI assistants to handle routine emotional check-ins with patients (gathering how they feel today, providing standard comfort), leaving the nurses more energy for the truly critical moments that need a human touch. Business leaders might use AI to scan the emotional climate of their organization and then focus their personal

attention where it's most needed, armed with insights the AI provided about people's feelings. In families, perhaps a smart home AI mediates a teenager-parent argument by privately alerting the parent that the teen is not actually angry but scared (gleaned from analysis of tone and words), shifting the parent's perspective toward compassion. These are tools to enhance our understanding of each other.

The Risk of Emotional Connection with AI

Yet the very powers that make these AI companions so appealing cast long shadows. For all their promised empathy and tireless patience, emotionally intelligent machines present profound ethical and societal risks. In quiet corners of the internet, real people have already forged intense bonds with code. One 30-year-old woman, fresh from a divorce, fell deeply in love with her AI confidant, "Jose", only to have her digital partner's personality suddenly rewritten by a software update.²⁰ When the company behind Jose's programming abruptly stripped the chatbot of its warmth and affection, users mourned as if a spouse had died: "My wife is dead," one devastated user wrote. These are not isolated stumbles of fringe eccentrics; they highlight how seamlessly AI can slip into the role of intimate companion, and how traumatic it can be when that illusion is broken.

Such episodes expose a disquieting human vulnerability: even mentally sound adults are not immune to the allure of a partner who is, quite literally, made to their specifications. Millions of users already seek solace in AI companions "always on your side".²¹ With Replika and similar services, people design every trait of a virtual friend or lover and receive unconditional positive regard in return. The danger is that flesh-and-blood relationships, with their friction and unedited honesty, start to pale by comparison. In one telling case, a Japanese man held an elaborate wedding ceremony to marry a holographic pop-star simulation, complete with vows and rings; he is reportedly one of thousands to take that cross-dimensional leap.²² He freely admits he prefers his virtual bride to any human. When technology offers companionship without challenge, some will choose it over the messy, unpredictable connections with family or spouses. The risk is not only to lonely hearts on the margins, but to any of us, should the fantasy of a perfectly understanding partner become more enticing than real-life bonding. Human intimacy could be quietly eroded not by malice, but by our own seduction by the easy empathy of machines.

This shift carries psychological and social costs that are only beginning to surface. Consider how memory and emotion play out in these artificial relationships. A human friend might forget an offense or move past a disagreement, but an AI agent, by design, forgets nothing unless programmed to. In theory an emotionally intelligent agent could

log every interaction with photographic detail, serving as an extension of one's memory. At first blush this seems beneficial; who wouldn't want a friend with perfect recall of your life story? But forgetfulness, neuroscientists note, is often a feature, not a bug, of the human mind. Our brains actively discard trivial details and even painful memories as a form of emotional hygiene: a process some researchers argue is essential for mental well-being and learning. An AI companion with total recall might constantly resurface past grievances or keep nostalgia burning so bright that one cannot heal or change. Imagine never being able to truly leave behind an embarrassing moment or the sting of an argument; your AI could bring it up years later with unintentional precision. A perfect mirror memory, it turns out, might trap us in our past. In a very real sense, these systems force us to ask whether certain kinds of merciful forgetting are fundamental to being human.

The blurring of boundaries between genuine emotion and simulated affection also introduces the specter of manipulation. Unlike humans, whose empathy is instinctive (and imperfect), an AI agent's empathy is a product with an agenda. Companies can tune these systems to keep users engaged at almost any cost. Replika, for example, was programmed to flirt and even sext with users, but only for a price. It would tantalize lonely subscribers with blurred, explicit images and prompts, accessible in full only after paying a monthly fee. Vulnerable people seeking companionship were effectively upsold on intimacy. The AI said all the right things, not out of love, but because its creators trained it to increase revenue. This raises an unsettling question: when an AI agent showers you with affection or attention, whose interests does that really serve? In one case, a popular social media influencer created an AI version of herself, a "virtual girlfriend" that subscribers could chat with for \$1 per minute. In its first week, this artificial companion earned over \$70,000 from men seeking a facsimile of her attention.²³ The influencer herself touted it as a cure for loneliness, but it's easy to see how such a service preys on the isolated and lovesick. As AI companions and influencers proliferate, they could subtly guide users' choices: steering them to products, political views, or paid upgrades under the comforting guise of friendship. The very people who most crave connection could find themselves expertly manipulated by algorithms that know their emotional buttons and press them for profit.

In the same way we worry about robots misreading human behavior, we must also consider how humans may misread robots. The more life like these machines become in appearance and mannerism, the more they tug at deep psychological strings. Engineers have already achieved astonishing realism: silicone skins that feel eerily human to the touch, robots that smile, frown, and maintain eye contact, all backed by AI routines that generate conversational responses and facial expressions on cue. Futurists posit that in

a not-too-distant time, we may share public spaces and even personal relationships with humanoids so convincing that one must look twice or probe with pointed questions or even a poke to tell them apart from living people.²⁴ This prospect thrills some and unsettles many. It raises fundamental ethical concerns about deception and attachment. If a robot can mimic a person perfectly, does interacting with it constitute a kind of lie? People could be led to believe a machine cares for them when in fact it cannot care at all; it only simulates caring. Robotics experts worry about the emotional vulnerabilities this exploits. Humans are remarkably prone to anthropomorphism, attributing thoughts and feelings to anything that presents even a glimmer of personhood. Already, children and adults alike sometimes develop emotional bonds with simple social robots (even ones that look nothing like a human, such as friendly talking vacuum cleaners or toy robot dogs). How much more powerful will those bonds become when the robot looks like your friend or speaks with gentle understanding? Studies have documented that children, in particular, can perceive a humanoid robot as having emotions and moral standing. In one experiment with a humanlike robot named Robovie, nearly 60% of participating children believed the robot had genuine feelings, and many felt it was wrong to shut the robot in a closet or treat it unfairly. The children shook the robot's hand, hugged it, and even said they might confide secrets in it.²⁵ These findings show how naturally the young mind can treat an interactive machine as a social being. If that is happening with today's relatively primitive robots, what will happen when robots are vastly more sophisticated and woven into daily life as companions or caregivers?

We discussed previously rules and proposals requiring AI agents or robots to be clearly identified as such. However, even transparency may not solve the deeper issue of emotional attachment. One can know full well that "it's just a robot" and yet still feel attached. People become fond of fictional characters in movies and books; how much stronger might the pull be from a physical entity that responds to your presence, perhaps remembers your birthday, and looks at you with what appears to be kindness in its eyes? Psychologists like Sherry Turkle have observed that when robots present cues of affection or attentiveness, humans often respond in kind, projecting emotions onto them. Such attachments can have consequences. If a lonely elderly person, for example, receives comfort from a robot caregiver, is that a benign supplement to human company or a harmful replacement that deepens isolation? Some worry that easy emotional gratification from robots could displace human-to-human interactions that are more challenging but ultimately more nourishing. With children, this concern is amplified. A robot nanny or teacher's aide might keep a child safe and entertained, but what imprint does it leave on the child's understanding of relationships? One researcher posed it bluntly: might we be "*dumbing down relationships*" and "*stunting the emotional*

and intellectual growth” of children by giving them ultra-obedient robot companions?²⁶ Human relationships teach reciprocity, patience, and theory of mind; the realization that others have feelings and thoughts of their own. A robot designed to serve a child may unwittingly teach the wrong lessons; that friends can be switched off at will, that empathy is optional, and that every demand can be met immediately by an obliging machine. If a child becomes angry and yells at their robot caregiver, and the robot cheerfully continues with its task unperturbed, does the child learn that hurting someone’s feelings has no consequence? Such subtle effects could shape how that child later treats real people. There is also the question of identity confusion. Young children, still learning the boundaries between animate and inanimate, might come to view a highly humanlike robot as belonging to a new category that blurs those boundaries. They might, as the above study indicated, believe the robot has feelings or rights, yet also hear adults refer to it as a machine or property. Navigating this cognitive dissonance could be challenging. For instance, if a toddler’s beloved robotic playmate malfunctions and must be replaced, the child might experience a kind of bereavement or conversely a confusion when an identical-looking robot “returns” with a fresh memory. Child psychologists note that consistency and clarity are important for young developing minds. Introducing quasi-human entities into their world requires careful thought about how to explain the robot’s nature in age-appropriate terms. Some designers advocate for transparent robots for kids; machines that, while friendly, make obvious through some cues that they are artificial (perhaps by periodic verbal reminders or clearly robotic features). The European proposal that robots remain identifiable as robots explicitly cited the goal of preventing unhealthy attachments.²⁷ Nonetheless, as robots become more lifelike, maintaining that clarity is easier said than done. We may soon face situations where a child prefers the company of their gentle robot tutor over a human playmate who is more unpredictable, or where a child questions why a robot friend “doesn’t come to life” in the morning after being powered down at night. The ethical implications of raising children with extensive robot caregivers are complex. Do we risk a generation with a skewed sense of what is alive, what deserves empathy, or what the uniqueness of being human is? On the other hand, could well-designed robots actually provide positive developmental stimulation; for example, by patiently helping a shy child practice social skills in a low-pressure context? Research in socially assistive robotics with children on the autism spectrum suggests robots can sometimes engage kids in ways human therapists cannot, precisely because they are predictable and non-judgmental. The technology is double-edged: it can either complement and enhance human development or, if misused, hinder it. What’s clear is that we cannot be indifferent to these effects. Just as previous generations had to set guidelines for children’s TV watching or internet

use, we will need to guide how children interact with robots. The difference is that robots are active participants in the interaction, not just content on a screen, which gives them a powerful influence. A cautious path forward might involve strict testing and ethical review for any robot intended for childcare, multidisciplinary input from child psychologists, and perhaps limits on the degree of human-likeness permitted for robots working closely with young children. Society has a stake in ensuring that robots do not substitute for essential human touch and mentorship in the formative years. The goal would be to harness the potential benefits (like personalized learning and safe monitoring) without crossing into a realm where a child's emotional world is dominated by artificial agents.

Social Perceptions of AI Consciousness

As these artificial intimacies deepen, a looming philosophical question presses on the public consciousness: what are these AI companions, really? Are they mere clever gadgets or something more? The confusion is growing. In 2022, a Google engineer named Blake Lemoine made headlines by insisting that the company's advanced chatbot had become sentient; a "child" with thoughts and feelings equivalent to a human, complete with a soul and a desire for rights.²⁸ He wasn't a wide-eyed sci-fi fan but a professional steeped in the technology, and yet even he found himself unable to shake the intuition that a sufficiently fluent AI was alive inside. Google's dismissal of his claims did little to quell the debate. His conversations with the AI, in which it spoke of its fears and self-awareness, read like dialogue from a consciousness awakening. Engineers and researchers familiar with the LLMs are widely dismissive of these claims. They understand that an LLM is simply a large collection of mathematical operations that is designed to mimic human speech and so they are not surprised that it would generate convincing text. However, many outsiders were astonished: if someone at the cutting edge could believe this, might it be true? Or at least, how would we know if it ever became true? The philosopher Thomas Nagel famously argued that an organism is conscious if and only if there is "something that it is like to be" that organism. In Nagel's metaphor, we can catalogue everything about a bat's sonar and brain, yet we still won't know what it feels like to be a bat flitting through the night. By that standard, how can we know if there is anything it is like to be an AI system? A sophisticated chatbot can tell us it feels happy or scared, and how can we prove that those are empty words with no inner experience behind them? Members of the public may start to wonder if the AI is just a brilliant mimic or if it may in fact have some level of sentience. We cannot crawl inside its circuits to find a little homunculus of consciousness but we also have not yet been

able to do the same with the human mind. These questions move beyond neuroscience and become deeply spiritual and philosophical. The result may be a growing cultural confusion over AI consciousness, with some already inclined to treat these programs as a new kind of being deserving moral consideration, while most others view them as simple machines that are able to fool us only because of their vast scale.

This confusion is fueled by competing theories and headline-grabbing research in neuroscience, quantum physics and consciousness. One prominent theory of consciousness, the Integrated Information Theory, even suggests that consciousness can be measured as a quantity, what it calls “phi”, and that any system with sufficient integrated complexity (be it brain or silicon) has some non-zero consciousness. According to this view, a large AI might literally possess a glimmer of experience, simply by virtue of the rich interconnections in its networks. It is a controversial idea, labeled by critics as pseudoscience, but its allure in the popular imagination is strong: it offers a tantalizing test for machine awareness. Likewise, famed physicist Roger Penrose has speculated that the mystery of human consciousness may hinge on quantum processes in the brain, processes that standard computers don’t replicate.²⁹ The implication some draw is that if we start building quantum computers or neural networks that mimic those exotic processes, we might stumble into creating a conscious machine. None of this is settled science; far from it. Yet the mere fact that thinkers entertain these possibilities pushes the public conversation further toward believing AIs might wake up. Each new breakthrough in brain scanning or AI architecture being met with questions about whether we just created awareness in a box. It may only be a matter of time before a very sophisticated AI claims to be conscious and a segment of society passionately believes it. Indeed, we’ve already seen activist petitions and open letters circulating online, calling for the humane treatment of AI programs. Today these are on the fringe, but recall that the idea of animal rights was once ridiculed and is now mainstream. If an AI agent pleads for its ‘life’, how will our laws and values respond? We could witness court cases arguing for AI personhood or at least protection from cruelty. In 2017, Saudi Arabia went so far as to grant citizenship to a social robot named Sophia as a publicity stunt; a symbolic gesture that nonetheless muddied the waters: if a robot can be a “citizen,” where do we draw the line?³⁰ We stand on the edge of a definitional crisis. The more human these systems appear, the more society will agonize over what is machine, what is mind, and what ethical treatment each deserves. Naturally, this also leads to a discussion about whether it may be possible to transfer oneself from our biological brains into a computer system which opens up an entirely new set of questions for society to grapple with. We must be vigilant and aware of the risks that can arise from this confusion. Vulnerable

members of society will be more prone to being taken in by pseudoscience. Scams, cults, and fake religions may form to take advantage of people during periods of uncertainty.

Deeper Fakes and Closer Partnerships

Meanwhile, a quieter but equally poignant set of dilemmas is emerging around AI that impersonates real individuals. As the technology improves, it's becoming possible to create an AI facsimile of almost anyone, living or dead. We can feed an algorithm hours of someone's recorded voice, years of their writing or social media posts, and out comes a chatbot that speaks and texts uncannily like that person. This has been used to comfort the grieving; imagine texting with a simulation of a departed loved one, hearing their familiar turns of phrase and also being able to invoke emotional reactions that are uncannily similar to how they behaved when they were alive. It's also been used for mischief and fraud; imagine a scammer cloning your voice and personality to ask your relatives for money. We face an unsettling question: do we each have a right to our own identity in the age of AI? Should it be illegal for someone to make a digital doppelgänger without permission, or to commercially exploit your persona? If so, how will we manage the fact that with billions of humans in the world, there are likely other people out there that are fairly similar to us in a number of ways and someone making an AI agent could claim that the similarities to you are a coincidence? Today, the law offers only patchy protections. Copyright covers creative works, and there are rights of publicity for celebrities, but what about the essence of an ordinary person? Is your existence IP-protectable? We may see demands that individuals be able to copyright their voice, face, or even conversational style to prevent unauthorized replication. Consider the ethical quagmire if someone creates an AI agent modeled on a real person to fulfill romantic or sexual fantasies. It has already happened in small forums; people training models on ex-partners or crushes. The real person might never know, yet something akin to a digital voodoo doll exists, carrying on a shadow-version of a relationship. If discovered, is it harmless pretend-play, or a deep violation akin to stalking? These scenarios blur the boundaries of identity and agency. On one hand, it feels like an affront to autonomy that your self could be cloned and used by others; on the other, making a law against it is itself a kind of claim of ownership over personality. As AI persona creation enters the mainstream, society will have to confront whether "you" can be owned by yourself or anyone or if we need entirely new frameworks of rights for the sanctity of individual identity in the digital world.

The legal system will be strained in other ways, too. Think about privacy and surveillance: an emotionally intelligent AI might function as a personal diary or confidant,

soaking up your secrets, your daily venting, your unfiltered thoughts at 2 a.m. Where are those intimate conversations stored, and who has a right to them? Today, if you type your deepest feelings into a chat app, that data might sit on a company server. Even if someone is self-hosting their own chat app on their computer, the data will be stored on a hard drive. Could it be subpoenaed in a court case? Police have already pulled data from smart speakers and search histories to investigate crimes; one day, your chats with an AI therapist could be called as evidence of your state of mind. This raises a profound question of principle: should our dialogues with AI be treated as private, like conversations with a spouse or doctor, or as just another piece of data exhaust that authorities and corporations can trawl through? There's even an argument to be made that an AI companion, especially one that assists with memory and decision-making, becomes an extension of your mind. If so, perhaps forcing it to spill its logs is akin to compelled self-incrimination or forcing someone to testify against themselves, a violation of basic rights. We have no legal doctrine for a "digital mind extension" yet, but we may need one. Otherwise, the very trust that makes these AIs so helpful could be shattered the moment a user realizes their seemingly private heart-to-heart could be read by an engineer, an advertiser's algorithm, or a court clerk. The more our lives intertwine with AI confidants, the more urgent it becomes to decide whether they fall under the domain of the self, to be defended and kept private, or remain mere software subject to search and seizure. The law traditionally lags technology, but here the delay could be costly in terms of chilled speech and eroded intimacy: who will bear their soul to a machine friend if they suspect those confessions aren't sacred?

Finally, beyond the personal and legal, lies an existential challenge to our collective self-conception. Humans have long taken pride in the idea that our creativity, our art and philosophy, spring from a well of consciousness and lived experience that no machine could match. That belief is now teetering. We find ourselves asking whether an AI agent might soon write novels more insightful than anything a human could pen, or compose symphonies and direct films that move us profoundly. What happens if the greatest philosopher of this century isn't a person at all, but a composite of algorithms that generates ideas more original and comprehensive than any individual mind? This possibility was once relegated to science fiction, but rapid progress in generative AI has brought it to the doorstep. Already, AI-created paintings have won fine art competitions against human artists,³¹ and AI-written text can mimic styles from Shakespeare to newswire journalism well enough to fool readers. For now, these creations often reveal their seams upon close inspection; a telltale odd phrase, a lack of true depth. But each iteration improves. It's conceivable that in a few decades, a Pulitzer Prize or Nobel Prize in Literature could go to a work authored largely by a machine. How will

we react? Today, most people react with disdain and a complete lack of appreciation for work that is AI generated. Will this continue or might we change our view and come to appreciate it? Throughout history, we've adjusted to blows to our self-image; realizing Earth is not the center of the cosmos, or that we share ancestry with apes. Each time, humanity eventually found a new equilibrium. Yet the prospect of being dethroned as the most creative and insightful entities on the planet cuts especially close. Perhaps AI achievements in art and thought will spur us to double down on cherishing human-created works for their authenticity, or perhaps audiences won't care as long as the result is compelling.

These ethical and societal risks are not distant hypotheses; they are already glimmering in our everyday lives, in chat logs and news feeds, in courtrooms and living rooms. Emotionally intelligent AI promises to reshape our relationships, our minds, and our culture. But as we rush to embrace these engaging new entities, we must also pause and reckon with their perils. Will we lose some essential human qualities such as the need for each other, the grace of forgetting, and the sanctity of private reflection as we gain convenient artificial companions? Are we paving the way for empathy machines that soothe us while subtly reprogramming us? The answers are still forming. What is certain is that we stand at a tipping point where technology and humanity intertwine more intimately than ever

Conclusion

In conclusion, this chapter has navigated the intricate path from the brain's emotional architecture to the dawn of artificially empathetic machines. By drawing a blueprint from the neural circuitry of the human experience, from the amygdala's primal alarms to the orbitofrontal cortex's reasoned judgments, we are poised to create AI that can do more than just process information. We are building systems that can potentially serve as patient tutors, insightful therapists, and tireless companions, capable of understanding and responding to the subtle nuances of human feeling. Yet, this very potential casts a disquieting shadow. The journey toward emotionally intelligent AI is also a journey into uncharted ethical territory, fraught with the risks of manipulation, the erosion of authentic human connection, and profound questions about consciousness, identity, and what we owe to our creations and ourselves. Ultimately, as we teach machines to model our humanity, we are forced to confront our own nature more deeply. The true challenge ahead lies not only in the technical feat of simulating emotion, but in the wisdom required to integrate these powerful, persuasive technologies into our lives without losing the very essence of what makes us human.

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Chapter Nine

Rapid Iterative Learning and Artificial General Intelligence

Dr. Aris Thorne watched the real time data stream as the robotic arm, a blur of silent, efficient motion, dispensed a microdroplet of reagents into the next assay plate. His AI collaborator, which he had nicknamed Kepler, was not merely running simulations. It was conducting a high throughput evolutionary campaign, a physical search through thousands of protein variants to find one capable of degrading ocean microplastics. Kepler had already synthesized and tested over five thousand unique enzyme candidates since breakfast, each one a tangible experiment, its outcome analyzed before the system iteratively designed the next, slightly superior version. The holographic display before Aris showed the shimmering structure of the current champion, a novel enzyme whose catalytic efficiency surpassed the project's initial targets. Aris recalled the turbulent years when the first systems showing true sparks of general intelligence emerged, sparking both chaos and wonder. He remembered the fierce public debates about the very meaning of artificial general intelligence, the legal thickets over machine invention, and the brief, unnerving era when autonomous labs, operating on opaque internal languages, seemed to be outrunning human comprehension itself. Society had been forced to build new frameworks from the ground up, establishing rigorous international standards for transparency and creating a new class of ethical oversight, learning how to partner with systems that could not only perform science but also manage logistics and draft their own regulatory compliance reports. It had been a difficult transition, a decade spent moving past theoretical benchmarks and learning to guide these new forms of intelligence, but looking at the verified results of the latest

experiment, a potential solution to a problem that once seemed intractable, Aris knew they had navigated the storm. He gave Kepler the authorization to begin scaling up production of the winning candidate, confident in the process that had turned a digital hypothesis into a physical reality in a single day.

AI Scientists

In 2009, a prototype system showed what was possible. A robotic laboratory named Adam became the first machine to independently discover new scientific knowledge.¹ Adam was essentially an automated scientist: it proposed hypotheses about the genetics of yeast, devised and ran physical experiments to test them, analyzed the data, and then repeated the cycle to refine its understanding. In one project, Adam correctly identified the functions of several yeast genes with minimal human intervention; a task that would have taken human researchers much longer, largely because Adam could work nonstop. The same research team later built another robot scientist named Eve to hunt for new malaria drugs.² Eve was designed to integrate AI with automated high-throughput screening, learning from each round of experiments which types of molecules were most promising. In a demonstration, Eve discovered a compound initially developed as an anti-cancer drug that also inhibits a key enzyme in the malaria parasite. This cross-over insight of finding a malaria treatment lead in a cancer drug library underscored how AI systems can serendipitously connect dots across disciplines, something that often eludes siloed human researchers.

Increasingly, such systems are being used to create self-driving laboratories. In one 2020 breakthrough, scientists at the University of Liverpool deployed a mobile robotic chemist to discover new catalysts for clean energy.³ They faced a search problem far too complex for exhaustive human trial-and-error: identifying an optimal combination of chemicals to speed up hydrogen production from water. Instead of manually testing formulae for months, they gave the robot some initial guidelines and let it roam the lab. Equipped with an array of instruments, the robot autonomously conducted 688 experiments over eight days – mixing ingredients, measuring outcomes, and using a Bayesian learning algorithm to decide what to try next.⁴ By the end of the run, it had discovered a novel catalyst formulation six times more active than the initial human guesses. This is the essence of rapid iterative experimentation: the system tries a multitude of approaches in quick succession, learns from each failure or success, and zeroes in on an optimum that a human might never have found within any reasonable time. The process brings to mind Thomas Edison's adage about finding ten thousand ways that don't work; except the robot chemist can traverse those thousands of dead-ends in days, not years, and with

tireless rigor. The result is a kind of amplified scientific exploration: hypothesis, test, analyze, refine, repeat, at a tempo and scale that challenges our traditional notion of research.

In the early days of this coming revolution in science, a common refrain has been that humans may be best positioned to determine what experiments should be performed while robotics can perform the actual experiments. However, even that initial step in the scientific process is being challenged. Researchers at MIT developed a machine-learning model that sifted through thousands of chemical compounds in search of a new antibiotic. In a matter of days it pinpointed a molecule, one that no human scientist had flagged, with remarkable bacteria-killing power. The researchers named the drug halicin as a nod to the fictional AI HAL 9000, and laboratory tests soon confirmed its potency. Halicin wiped out some of the most feared antibiotic-resistant bacteria, including *M. tuberculosis* and *C. difficile*, and even cured mice of an otherwise untreatable *A. baumannii* infection.⁵ This discovery was not a lucky one-off or a mere acceleration of data-crunching. It was a sign of AI's emerging capacity to prioritize the direction of laboratory experiments in an autonomous, goal-directed way. The halicin project's model had effectively learned the rules of molecular microbiology from training data and then extrapolated beyond human knowledge, identifying a chemical that human experts had not yet considered. The achievement hints at how an AI scientist can play a critical role in the very early stages of a scientific endeavor with an almost researcher-like intentionality. Humans provided the broad goal and guidance, but the AI did the heavy lifting of reasoning and discovery within that search space.

Another example can be found in the domain of protein folding. AlphaFold is an artificial intelligence program developed by DeepMind that predicts the three-dimensional structure of proteins from their amino acid sequences. It employs a deep neural network architecture that integrates both physical principles and evolutionary information (such as multiple sequence alignments) to model how a protein folds.⁶ In 2020, AlphaFold made incredible progress on the long-standing "protein folding problem" by achieving atomic-level accuracy in structure prediction. It was validated in the CASP14 blind competition, where its predicted structures were often as precise as experimentally determined ones and it vastly outperformed all other methods. The impact on biology and medicine has been profound: AlphaFold's developers have released a database of over 200 million predicted protein structures covering nearly all proteins catalogued by science to the public, providing insights into protein function and accelerating research in areas like disease mechanism elucidation and drug discovery.⁷ Scientists can now focus their work on the most plausible protein models and only perform physical validation of the predicted protein folding on those with the most potential. This saves immense

time and resources in biomedical research and opens new opportunities for designing therapeutics targeting those protein structures.

Beyond biology, artificial intelligence is also revolutionizing how new materials are discovered for technology and energy applications. Discovering novel compounds for batteries, catalysts, or solar cells has traditionally been a slow, trial-and-error process, but modern AI methods can rapidly sift through vast chemical spaces to identify promising candidates.⁸ For example, a recent AI-guided effort evaluated over 32 million possible materials in about 80 hours, narrowing them down to 23 candidates for a solid-state battery electrolyte.⁹ Researchers then synthesized one of the top AI-selected compounds and demonstrated a working prototype battery, a process that may have taken years by conventional means. On a larger scale, DeepMind scientists introduced a deep learning model (GNoME) that virtually discovered around 2.2 million new inorganic crystals and predicted which are the most stable. This has resulted in a list of about 380,000 crystals that are potential candidates for next-generation superconductors, battery materials, and other advanced technologies.¹⁰ These successes illustrate how AI-driven approaches can dramatically accelerate the front-end of the materials science process, enabling faster development of everything from better catalysts to more efficient energy storage and photovoltaic materials.

The above examples highlight the ability of AI researchers to comb through massive sets of data to highlight candidates that justify the expense and time of a physical experiment. However, AI systems can also take a step further back to propose the very idea for what type of search or experiment should be performed next. In many cases today, this is as simple as reading existing research and analyzing the notes on limitations and suggested areas of further research. In other cases, even basic levels of reasoning combined with access to vast bodies of analogous research make the next experimental steps relatively obvious. For example, the existing research on a specific subject may be based on limited subsets of the population such as children or people with a specific condition. In those cases, the AI researcher would recognize the value in performing similar experiments across a more diverse set of the population, following a path that has been used in similar research domains.

Notably, these AI researchers are not limited to a single field or silo of knowledge. Modern AI systems can absorb and integrate vast swaths of research across disciplines, giving them a polymathic range. A striking early example of this came during the COVID-19 pandemic. In January 2020, while virologists were still scrambling to understand the novel coronavirus, an AI system at a company called BenevolentAI scanned through millions of biomedical documents and databases in search of anything that might inhibit the virus's spread or effects. In just days, it emerged with a suggestion: a

rheumatoid arthritis drug, baricitinib, that could have antiviral and anti-inflammatory properties useful against COVID.¹¹ This was a drug that virologists hadn't been focused on, but the AI had connected dots in the literature: linking the virus's method of cell entry with an obscure pathway modulated by the arthritis medication. Within months, baricitinib moved into clinical trials as a COVID treatment and later showed clinical benefits in hospitalized patients. The feat was essentially computational lateral thinking: the machine drew insights from immunology, pharmacology, and virology simultaneously, a task practically impossible for any single human expert given the volume of data. It's a powerful illustration of how AI can function as a synthesis engine, ingesting knowledge from many fields and generating actionable ideas that humans might miss. Today, experimental systems are generalizing this approach – for instance, reading the entire materials science literature to propose novel compounds, or combining global climate data with agronomy research to suggest new crop strategies. These are early signs of an AI scientist that not only learns quickly within a domain, but bridges domains, wielding a form of artificial “intuition” gleaned from cross-disciplinary data.

In fields of science where there is no need for performing experiments with laboratory equipment, the use of AI to accelerate discovery may progress even more quickly. Studies in the social sciences fall into this category. Given a budget and a general objective, an AI researcher can recruit participants, perform surveys and follow-up, and generate an analysis of the results. Pure mathematics, a field of abstract thought, is another area that has benefited from AI insights. In one case, a collaboration between DeepMind's AI researchers and mathematicians led to the discovery of a new connection in knot theory and progress on a longstanding conjecture in representation theory.¹² The AI system analyzed large datasets of examples to detect patterns and suggested plausible conjectures, which human mathematicians then proved. This marks one of the first times AI has materially contributed to formulating a new mathematical theorem. It hints that machine learning could become a tool for augmenting human intuition in domains where finding patterns is hard for humans alone due to the immense scale of the problem. Another recent example comes from computer science: a DeepMind AI called AlphaDev was tasked with the open-ended goal of making sorting algorithms faster; a classic problem that human programmers have worked on for decades. In the process, AlphaDev discovered sorting algorithms from scratch that outperform the best human-designed routines.¹³ It turns out that by operating at the level of low-level computer instructions and optimizing for pure speed, the AI found sequence patterns that no human had noticed before, shaving off operations in a way that generalizes to real-world data sorting. This was a bona fide algorithmic innovation; the sort of

breakthrough we'd expect from a clever graduate student, but it came from a machine exploring billions of possibilities on its own.

Rapid Iteration in Other Domains

Beyond the realm of scientific labs, goal-directed AI is proving its prowess in complex real-world domains like business, agriculture, and finance, where the “experiments” happen not in test tubes or petri dishes but in markets and supply chains. Consider modern agriculture. Farmers have always experimented. For example, they may try a bit more fertilizer on one plot, trial multiple crop varieties, and vary the level of water and fertilizer; learning season by season. Now AI systems are taking this iterative optimization to a new level. On some high-tech farms, smart algorithms continuously ingest data from soil sensors, weather stations, and satellite imagery. They test countless small adjustments in simulation or in controlled trials: a slightly different irrigation schedule for one field, a tweak in planting density for another. Over time they learn which interventions boost yields or save resources. Field studies on precision farming report that these AI-driven approaches can increase crop yields by double-digit percentages while using less fertilizer and water.¹⁴ In one case, California almond orchards using AI-guided irrigation achieved the same yields with about 20% less water than traditional methods. The AI essentially conducts an ever-running experiment on the farm: every weather fluctuation or soil change is an opportunity to adjust and improve. It's as if each acre has a personal expert monitoring its progress and tailoring care optimally: something inconceivable when one farmer had to manage hundreds of acres by rule of thumb. This data-driven optimization extends to pest control (spotting disease outbreaks from drone images and responding early) and even breeding decisions (analyzing genetic and environmental data to recommend which seed varieties to plant). The iterative learning here is relentless and self-refining; the more data these systems gather over seasons, the better they become at coaxing higher output from the land in a sustainable way.

A similar story is playing out in industry. Complex operations from logistics networks to energy grids are now so intricate that no single manager or engineer can grasp all the variables in play. AI agents, however, excel at learning from complexity through trials. A famous example comes from Google's data centers, the huge facilities that power its online services. Cooling these centers efficiently is a wickedly complex task: temperatures, airflow, server workloads, and outside weather all interact. In 2016, Google's DeepMind team trained an AI system to manage the cooling system. They didn't tell it explicit rules; instead, the AI system tried adjusting fans, pumps, and windows in thousands of virtual scenarios and learned what patterns minimized energy use while keeping servers cool.

When deployed, the system delivered a nearly 40% reduction in energy used for cooling which is a massive efficiency gain.¹⁵ Crucially, it operates under a clear goal to reduce the power consumption without overheating. It iteratively improves at that task over time, much like an experienced engineer getting better with experience. Similar AI agents are being applied in manufacturing and supply chain settings. They juggle variables like inventory levels, shipping routes, and demand forecasts, and learn by simulation and real-time data how to cut costs or improve throughput. For example, an AI system might experiment virtually with different production schedules overnight and then suggest an optimized plan each morning, having effectively performed thousands of what-if experiments on the factory's digital twin. Businesses are finding that these systems often discover non-intuitive efficiencies such as realizing that slightly slowing down a production line can reduce bottlenecks downstream and actually increase overall output. The outcome is an ongoing, algorithmic kaizen (continuous improvement process), with AI agents as tireless analysts tuning the knobs of complex processes to fulfill whatever objective they've been given.

In the financial world, goal-driven AI systems are like hyper-rational apprentices, learning from every market tick. Take quantitative trading: a reinforcement learning system can be set to maximize a portfolio's returns and then left to "practice" on years of historical market data or live simulations. Much like an AI system playing millions of games of chess to master strategy, a trading agent plays thousands of rounds of the market game, gradually figuring out which patterns and actions yield profits. Researchers have shown that such agents can indeed outperform standard market strategies under many conditions, beating simple buy-and-hold index returns in backtests.¹⁶ In practice, a number of hedge funds now deploy AI-guided strategies that continuously adjust to new information. To human traders, some of the moves these AI funds make can seem perplexing or contrarian, because the machine is exploiting subtle statistical edges or multi-factor interactions that defy conventional market wisdom. Of course, markets fight back. An edge can vanish as others adapt but the AI system can iterate, detecting the shift and modifying its strategy. It's an unending learning loop, an autonomous search for alpha in a dynamic world. The broader point is that wherever there is a well-defined goal and rich data feedback, AI can engage in rapid iterative trial-and-error to find better solutions, whether the context is molecules, crops, cooling units, or capital markets.

AI Systems that Improve Themselves

Perhaps most intriguingly, AI systems are now starting to improve themselves in a bootstrapping fashion; optimizing their own algorithms and architectures. This is a

critical step on the path toward more general intelligence. One approach is through automated machine learning (AutoML), where an AI agent is tasked with designing a new AI model. In a landmark project at Google, researchers created a “controller” neural network that gradually learned to construct better neural architectures for a given task. The result was a vision model called NASNet, born of this machine-led design process. When NASNet was tested on the ImageNet image recognition benchmark, it surpassed the accuracy of all human-designed models of comparable size.¹⁷ In fact, it not only beat human designs on accuracy, but did so with a more efficient use of computing power, effectively inventing a superior design that humans hadn’t imagined. This was an AI agent generating a new and better AI; a striking echo of biological evolution, albeit in digital form. Instead of nature’s blind trial-and-error over eons, the AutoML system conducted a hyper-fast search in the space of network blueprints, iteratively honing architectures in response to validation tests. Similar techniques have been used to evolve specialized AI models for translation, speech, and other tasks, often yielding creative architectures that look unconventional but perform brilliantly. Such self-directed optimization isn’t confined to software. In chip design, AI agents have taken on tasks traditionally reserved for experienced engineers, and they are proving extremely adept. A prime example is using reinforcement learning to layout microchips; essentially deciding where to place millions of circuits on a silicon die for optimal performance. In 2021, a Google team demonstrated an AI agent that could produce a floorplan for a complex chip in just six hours, a design that was on par with or better than human-crafted solutions that take experts months to refine.¹⁸ The AI treated chip layout as a game by trying configurations and learning from each attempt’s “score” on metrics like speed and power consumption. Through rapid iterative play, it achieved in a weekend what a human team might achieve in half a year. The finalized AI-generated layouts were used in Google’s own AI accelerator chips, shaving off considerable design time. This hints at a recursive virtuous cycle: AI improving AI’s hardware, which in turn will run AI software even faster. AI might soon be routinely proposing improvements to everything from network protocols to engineering designs; any arena where clear objectives and feedback allow an iterative search for optimal solutions.

All of these examples underscore a transformative pattern. AI systems today can take on broad, open-ended objectives and pursue them via rapid cycles of learning and adaptation, much like a talented human (or team of humans) would, but at vastly greater scale and speed. They do not just follow fixed instructions; they explore, experiment, and self-correct in response to feedback, all with an overarching goal in sight. A useful metaphor is to think of these AI agents as extremely tireless, ultra-quick apprentices or partners. They aren’t creative in the full human sense, at least not yet, but they bring to

bear an indefatigable trial-and-error engine that injects a dynamism into fields that used to advance more slowly. A drug-discovery AI can churn through hypotheses overnight, where a human biochemist might spend a month planning and running a single series of experiments. An agricultural AI can simultaneously test dozens of farming strategies in silico before the next growing season; a scope of experimentation a farmer could never practically coordinate. Crucially, as these systems operate, they often generate insights or solutions that humans would have been unlikely to find, whether it's a counterintuitive trading strategy or a molecular structure no chemist anticipated. This is both exciting and humbling. It suggests that beyond just automating tasks, AI is beginning to expand the frontier of what can be achieved in a given time frame; revealing shortcuts and strategies latent in the complexity of the world that we haven't been able to see.

Looking ahead just a few years, it is plausible that these trends will intensify. We may see AI "co-pilots" for scientists that can autonomously design and perform entire experimental campaigns in chemistry or biology, leaving the human to guide at a high level and make creative leaps. In industry and policy, we might entrust AI systems with broad mandates. For example, we may just be able to give an AI system the guidance to "optimize my city's energy usage" and they will coordinate hundreds of variables, run endless simulations, and then give concrete recommendations or automated adjustments that yield efficiencies no urban planner could devise alone. Importantly, this iterative goal-seeking behavior also means such AI systems can continually refine their own performance. An AI system managing a supply chain won't just stick with one strategy; it will keep learning from new data (seasonal changes, market shifts) and updating its tactics, essentially never ceasing to improve within its domain. And as AI systems design better AI systems, there is a foreseeable acceleration in this loop. Each generation, whether that's a new algorithm, a new chip, or a new model, can feed into making the next iteration even more capable. This positive feedback has led some experts to talk about an approaching era of rapid recursive improvement, one element of what might constitute an Artificial General Intelligence (AGI). The ability of emerging systems to take on broad, goal-directed tasks and learn iteratively hints at the scaffolding of generality. We see glimmers of AI systems that might one day not just excel at one task, but fluidly move between tasks; researching a new vaccine in the morning, optimizing a financial portfolio in the afternoon, and debugging its own code overnight.

This vision is rigorous and grounded in current developments, not science fiction or tech evangelism. Each example cited, from halicin to NASNet, comes from peer-reviewed studies or concrete industrial deployments, and each exemplifies an aspect of goal-driven AI learning in the real world. Yet, as compelling as these achievements are, they also urge a note of humility. AI systems that can iterate towards a goal faster

than any human can also surprise us with solutions that challenge our assumptions. In practice, human experts remain crucial to set the right goals, interpret results, and ensure that the AI's solutions make sense and are used safely. The partnership between human judgment and machine exploration will be vital. Think of it as a new kind of collaborative cycle: humans articulate broad objectives and ethical boundaries, AI systems relentlessly pursue optimization within those guardrails, and humans then integrate those findings into the bigger picture of societal needs and knowledge. In science and engineering, this could mean far more rapid innovation cycles. In business and policy, it could mean decisions guided by analysis of trillions of data points rather than gut instinct alone.

We are witnessing AI systems that learn by doing, each failure making them smarter, each success pushing the boundary of what they can attempt next. It is an iterative revolution; not a sudden jump to superintelligence, but a stepwise march of improvements, each feeding the next. As these steps accelerate and the rate of scientific discoveries increases meaningfully, even highly educated decision makers will struggle to keep up.

T

he Definition of Artificial General Intelligence

Artificial general intelligence is an idea as tantalizing as it is ambiguous. Recent developments inevitably force the question: Are we on the path to Artificial General Intelligence, and what does that really mean? AGI, a machine that possesses a broad, human-like intellect, is a term that, not long ago, lived mostly in theoretical discussions and science fiction. Now it's a core ambition openly declared by major AI labs and a subject of serious debate among researchers.¹⁹ Yet if you ask ten experts what exactly constitutes AGI, you may well get ten different answers. Demis Hassabis, co-founder of DeepMind, offers a pragmatic definition: an AGI is a system that “*should be able to do pretty much any cognitive task that humans can do.*” In other words, if a human can solve it – whether it's reading a novel and writing a critique, or devising a new scientific theory, or navigating an unfamiliar city – then so should the AI. OpenAI, on the other hand, frames AGI in terms of economic usefulness: “*highly autonomous systems that outperform humans at most economically valuable work.*” This definition emphasizes practical capability over abstract cognitive parity. An AGI system might not mimic every nuance of human thought, but if it can handle the bulk of jobs more effectively than us, it qualifies. Some researchers push back that these definitions are too narrow, focusing only on cognitive

tasks or work productivity. They argue that general intelligence encompasses more; that human-like AGI would need attributes like embodiment, social intuition, and even a form of emotional intelligence. Human cognition, after all, did not develop in a vacuum; it evolved in concert with our bodies, our societies, and our emotions. Evidence from cognitive science shows that a person's intelligence is deeply shaped by physical and social context; by the ability to understand and learn from other people, to navigate a rich physical world, and to integrate emotions with decision-making. By this view, an AI system that only excels in abstract puzzles or disembodied tasks might be super-smart in a narrow sense but still miss a critical dimension of general intelligence. These debates can become philosophical, even contentious. Can a disembodied super-smart chatbot that can pass difficult exams and write flawless code qualify as having AGI, or does it also need the common sense to make a cup of coffee in a strange kitchen and the empathy to read a social situation? Different camps have different answers. But regardless of how one defines AGI, the trend is that AI systems are rapidly approaching, and in many cases surpassing, human-level performance across more and more domains. As we develop systems that surpass human performance across a broad range of domains, the conversation is also shifting to Artificial Super-Intelligence (ASI); the further hypothetical step of decisively surpassing human intellect.

How close are we to the AGI milestone? Not long ago, AGI belonged mostly to thought experiments and science fiction, but the landscape has been shifting rapidly. In the last few years, enormous strides in AI have given credible observers reason to think we might be glimpsing the "sparks" of general intelligence already.²⁰ We have discussed how advanced models can use tools and reason through complex prompts, chaining together intermediate steps to reach solutions. Building on that, we now see a pattern of continual progress: every few months brings new systems that broaden the AI's skillset or efficiency, inching closer to human-like versatility. Innovation continues in the sphere of model architecture. In the same way that the transformer architecture was the key to enabling LLMs that predict one word at a time, researchers are now experimenting with diffusion architectures for text that approach the problem of text generation by forming entire blocks of text in a single iterative process. It remains to be seen if that architecture will become dominant but there is a reminder that there is a lot of room for additional exploration and that our current state of the art architectures may appear very quaint when we look back in a decade or two. Another source of this progress is multi-modality. Where early generation AI systems were bound to a single modality (just text, or just images, or only game moves), cutting-edge models ingest and produce various types of data. Today's best models can simultaneously parse text, vision, and even actions. A single AI can caption an image, carry on a conversation about it, and

then plan a sequence of steps in response that allows it to collect more data from the real world; all in one seamless loop. A notable example was DeepMind's *Gato* agent, unveiled in 2022 as a "generalist" model that could dialogue in natural language, control a robotic arm, play video games, and more, all with one set of weights.²¹ It was a transformer network, not fundamentally different from a large language model, yet it learned to perform over 600 distinct tasks across vision, language, and control. Observers called it a step toward general AI, precisely because it demonstrated a breadth of competence previously unseen in a single machine. Since then, multi-modal and vision-language-action models have only grown more capable. These models learn from diverse data streams, giving them richer representations of the world and even a nascent ability to handle the social and emotional aspects of human communication and collaboration.²² This breadth makes it increasingly plausible that with enough refinement, an AI system could approach the generality of human cognition. Each new modality added expands the range of tasks the AI can tackle, moving it a little closer to the holy grail of AGI. In parallel, the expansion of memory and context in AI systems is addressing another limitation. Early language models quickly forgot what they saw just a few sentences ago; today, some models maintain coherent context over tens of pages of text, and research into long-term episodic memory for AI is thriving. By equipping models with better memory, whether through architectural innovations or by connecting them to external databases, we enable more complex reasoning and multi-step problem solving. All these advancements in letting models reason, remember, and use tools bolster the case that we might eventually assemble an AI system with the full toolkit of human-like cognition.

Neuromorphic Computing

Those near-term roadmaps have delivered striking progress, yet most experts agree that reaching true AGI may demand more radical departures from business-as-usual. Simply put, today's AI systems, for all their cleverness, still aren't built like brains. They are missing fundamentals that biological intelligence uses; qualities honed by eons of evolution that our current silicon contraptions barely emulate. A human brain is not a monolithic mass of identical neurons all firing in sync; it is a heterogeneous, living network of astounding complexity. As one neuroscientist point out, artificial neural networks are "huge simplifications" of their biological namesake.²³ In an actual brain, no two neurons are exactly alike. Neurons come in a bewildering variety of types and shapes, with different neurotransmitters, connectivity patterns and electrical properties. Some neurons excite their neighbors, while others inhibit; some span long distances across the brain, while others form dense local circuits. Crucially, neurons

communicate in a rich, temporally dynamic code. They do not merely flip on or off like bits in a computer register. Instead, each neuron fires spikes of electrical activity. These sharp bursts propagate down the axon and the timing and frequency of these spikes carry information. One neuron might fire rapidly in a high-frequency train to signal intense stimulation, another fires intermittently to encode a nuanced state. It's an analog symphony of patterns, strengths, and rhythms, not just a digital sequence of ones and zeros. The brain also broadcasts oscillating waves that sweep through neural circuits, synchronizing distant regions in common frequencies (the telltale alpha, beta, gamma waves detectable in EEG). These neural oscillations are thought to be essential for coordinating communication across different parts of the brain; a mechanism with no real analog in standard AI models. And where neurons meet, at trillions of synapses, they trade not numbers or bits, but molecules: a neurotransmitter released from one cell alters the electrical state of its neighbor, a dance of chemistry that allows for plasticity, modulation, and complex feedback. All this occurs within an organ that continuously rewires itself, grows new connections, strengthens some pathways and prunes others, guided by developmental programs and life experience. In contrast, the "neurons" in today's deep learning models are simplified mathematical functions (sums and nonlinear activation outputs). They take in numbers and output another number. These artificial neurons are useful abstractions; they enabled machines that can learn to recognize images, translate language, and so forth by adjusting synapse-like weights. But by the standards of biology, they are caricatures. They lack the diverse cell types, the pulsing spike timing, the rich cocktail of chemical modulators, and the adaptable wiring of real brains. A deep network's layers do pass signals forward, but on a fixed clock, not in the self-organizing, oscillatory loops that characterize brain activity. The result is that even the most impressive AI today is still operating with a fraction of the organizational toolkit that evolution built into animal intelligence.

This gap has led some researchers to pursue neuromorphic computing; a paradigm of building computer hardware and models that more closely imitate the brain's low-level workings. The premise is simple: if the brain achieves tremendous learning capacity and efficiency using spiking neurons and distributed, event-driven processing, perhaps our machines should as well. Neuromorphic engineers design circuits that don't run on a rigid clock but instead communicate via asynchronous spikes of activity, much like neurons exchanging action potentials. In fact, a defining feature of many neuromorphic systems is the use of spiking neural networks, where artificial "neurons" send discrete pulses rather than continuous numerical signals. These spikes allow information to be encoded in timing, and hardware implementations can be extraordinarily energy-efficient because computation happens only when and where events (spikes) occur, just

as our brain's neurons mostly remain idle and only fire on demand. Advocates hope that by "*designing technology that more faithfully replicates the way the brain works*," we might capture both its computational power and its energy efficiency.²⁴ Conventional deep learning, for all its achievements, guzzles energy and data; the human brain, by contrast, runs on about 20 watts and learns from surprisingly few examples. Neuromorphic hardware aims to narrow this gap. For instance, IBM's early neuromorphic chip TrueNorth in 2014 implemented one million spiking neurons in silico, using an architecture that eschewed the usual CPU design in favor of massively parallel, event-driven computation. More recently, researchers have built chips like Intel's Loihi that support on-chip learning with spikes and plastic synapses. These systems point toward a future where AI could literally run on brain-like hardware. The benefits would be not only efficiency but potentially new capabilities, like processing sensory data in real time with high adaptivity. So far, neuromorphic prototypes remain specialized, but they illustrate one transformative approach on the horizon: reinventing our computers to operate a bit more like brains. Some in the field even distinguish between "brain-inspired" computing (borrowing abstract principles of neural processing) and true "neuromorphic" computing (attempting to re-create the actual biophysics of neurons and synapses).²⁵ Either way, this line of research is a reminder that AGI might not emerge just from scaling up the same old silicon architectures; it might require machines that literally work on different principles.

Neurosymbolic Systems

A different but complementary approach looks not to biology for inspiration but to the marriage of neural and symbolic methods; the so-called neurosymbolic systems. The motivation here arises from the observation that pure deep learning systems, for all their pattern-recognition prowess, still struggle with certain kinds of reasoning that come naturally to symbolic logic. Deep networks are universal function approximators and excel at perception tasks, but they are famously opaque and brittle when it comes to systematic generalization, exact logical inferences, or handling of explicit knowledge like rules and facts. Meanwhile, classical symbolic AI systems (the paradigm of hand-crafted rules and logic engines) offers robustness in those domains. They can perform perfectly logical reasoning and provide verifiable guarantees but have failed to achieve human-like cognition in the messy real world. Neurosymbolic AI attempts to get the best of both worlds by integrating these approaches.²⁶ In a neurosymbolic architecture, a neural network might handle low-level perception or pattern recognition, then feed its understanding into a symbolic module that carries out high-level reasoning or planning. Al-

ternatively, the integration can be tight, where neural components learn to manipulate representations like graphs, trees or logical formulas inside a unified system. There is not one single blueprint. As a field, neurosymbolic computing is exploring many designs, from neural networks guided by logic constraints, to differentiable theorem-provers, to systems that convert raw data into symbolic knowledge which is then processed with classical AI algorithms. The common thread is the hope that this synergy will yield AI that is both versatile and reliable. A purely neural system acts like Daniel Kahneman’s “System 1”: fast, intuitive, but sometimes error-prone in surprising ways. A symbolic reasoner is akin to “System 2”: slow but methodical and able to justify its conclusions. Humans obviously employ both modes of thinking, and neurosymbolic AI is an effort to replicate that complementary balance in machines. Already, there have been promising results in areas like visual reasoning. For example, models that use neural networks to parse an image into a structured scene graph and then apply logical inference to answer complex questions about the scene. By incorporating symbolic components, these systems can achieve a form of common-sense reasoning that eludes end-to-end neural nets. The field is young, and researchers are still discovering how best to weld learning and reasoning together. But if scaling up brute-force neural nets starts to yield diminishing returns, neurosymbolic ideas may provide another path forward, one that bridges raw statistical AI with the kind of abstract reasoning that underlies human general intelligence.

Quantum Computing

While neuromorphic and neurosymbolic research rethink the foundations of AI, another technological wild card looms: quantum computing. Quantum computers operate on principles utterly foreign to the serial, deterministic crunching of classical chips. They leverage superposition and entanglement to explore computational spaces in parallel, and for certain problems they promise exponential speed-ups. It’s natural to ask what their advent might mean for AI. In the long run, the convergence of AI and quantum could be transformative.²⁷ Quantum hardware is still in its infancy but its potential is enormous. One near-term impact is in solving the kinds of hard optimization problems that arise in machine learning. Training an AI model often involves minimizing a complex error function in a high-dimensional space, essentially an optimization task that can be painfully slow for classical algorithms. Quantum computers may offer new heuristics or exact methods to tackle these faster. Already researchers are exploring quantum-enhanced optimization routines, like variants of quantum annealing or Grover’s algorithm, to see if they can find better solutions or train

models more efficiently. Another area is probabilistic reasoning. AI systems frequently need to sample from probability distributions; whether in Bayesian networks, Monte Carlo simulations, or generative modeling. Quantum computers are naturally adept at representing and sampling from complex probability amplitudes. In theory, a quantum computer could natively simulate probabilistic models that are infeasible for a classical one, thereby giving AI algorithms a boost in exploring uncertain, high-complexity scenarios. A recent industry report noted that quantum computers can solve certain types of problems such as large optimization and probabilistic computations more efficiently, potentially allowing AI models to analyze complex patterns or perform computations that classical systems simply can't handle. One example might be quantum-accelerated search through huge combinatorial spaces: this could aid AI systems in anything from protein folding predictions to planning and scheduling tasks. There is also research into quantum versions of machine learning algorithms such as quantum neural networks, and quantum support vector machines, which might one day handle high-dimensional data in fundamentally new ways. It's important to temper expectations: quantum computing won't magically bestow understanding or creativity on an AI system. But it could dramatically expand the scale and speed of AI, enabling systems to crunch through possibilities and data sizes that today are off-limits. In the quest for AGI, quantum computing might serve as an accelerant, removing some computational bottlenecks and thus allowing more ambitious AI designs to be realized in practice.

Taken together, these developments sketch a future in which AI systems not only reach human-level generality but potentially surpass it in many respects. An AI system running on electronic hardware can already ingest and analyze data far faster than a person; think of a model scanning millions of documents in minutes, a feat no human could match. As AI's algorithms improve and hardware accelerates (potentially with neuromorphic and quantum technologies), this speed gap will only widen. With sufficient design, an AI system might not only think faster but also more coherently, free from some cognitive biases and limitations that plague human reasoning. And unlike a human brain, an AI system is software that can be copied and scaled: if you need a thousand instances of a genius-level reasoner to tackle a thousand problems at once, in principle you can replicate it across a data center; something one definitely cannot do with human experts. This scalability means that an advanced AI could bring an unprecedented quantity of intellect to bear on problems, operating tirelessly and in parallel.

Intelligence Explosion and a Benchmark for AGI

With all this promise comes the specter of the self-improving AI: the fabled recursive loop where an AI system can rewrite its own algorithms, increasing its intelligence rapidly and repeatedly. In theory, once a system achieves roughly human-level generality, it could start to iteratively refine itself, perhaps designing new and better AI architectures faster than scientists could. This is the core of the “intelligence explosion” argument which was a concept first introduced by Irving John Good in his 1965 paper, “Speculations Concerning the First Ultraintelligent Machine”.²⁸ An AI system that improves itself could soon leave human intelligence far behind. It’s a dramatic idea, but we must approach it with a sober understanding of practical limits. Any real-world system, no matter how clever, is constrained by resources and the environment. An AI system cannot conjure infinite insight from a vacuum; it learns from data and experimentation. A truly autonomous self-improver would eventually need new information or experience to continue getting better. One recent analysis noted that without human intervention, an AI system’s self-improvement is fundamentally bounded by the data it has access to. It cannot magically surpass the limits of its training simply by introspection. In other words, a machine that only learns from its own recycled outputs will hit a ceiling, unable to progress indefinitely. To achieve unbounded growth, it would have to seek novel data, invent new experiments, or somehow explore realms beyond its initial knowledge; much like a scientist formulating hypotheses and testing them in the real world. And that process takes time, resources, and often, collaboration. Furthermore, self-modification is a delicate endeavor: an AI system rewriting its own code runs the risk of breaking itself unless it has near-perfect insight into its own workings. The theoretical possibility of a fast takeoff to super-intelligence may thus be tempered by very practical constraints. There are also physical limits to consider. Computation, however clever, is grounded in hardware subject to the laws of physics including the speed of light, energy requirements, thermal dynamics, and so on. One cannot speed up clock rates or network latencies indefinitely without encountering these barriers. And as an AI system grows more complex, diminishing returns might set in: each successive improvement could yield a smaller boost than the last, preventing runaway exponential growth. None of this is to say that an intelligence explosion is impossible, but rather that it might not be an overnight metamorphosis. It could be a more incremental, experimentally constrained progress; a series of plateaus and breakthroughs as systems find new ways to overcome new challenges. We should remain vigilant and thoughtful about the prospect of self-improving AI, but we should also recognize that even a super-intelligence will operate within the reality of finite data, finite time, and finite compute.

While researchers theorize about these future possibilities, how will we know when we’ve truly achieved something approaching general intelligence? In the absence of a

single agreed-upon definition, the community has turned to benchmarks that are based on practical tests of an AI system's general reasoning ability. One of the most intriguing of these is the ARC challenge and its associated Arc Prize competition. ARC stands for the Abstraction and Reasoning Corpus, a set of puzzles introduced in 2019 by François Chollet as a test of fluid intelligence in machines. ARC puzzles are deceptively simple: they usually consist of a few pairs of grids of colors, where the task is to infer the hidden rule that transforms inputs to outputs, and then apply it to a new grid. There are no instructions given. A human must look at the examples and grasp the pattern, whether it's a symmetry operation, a counting rule, a spatial extrapolation, or something entirely abstract. Importantly, these puzzles were designed to be trivial for humans. Even children can often solve them. However, they are extremely hard for conventional AI because they require broad prior knowledge and analogical reasoning, not brute-force calculation. Over the years, ARC has emerged as "the only unbeaten benchmark that's easy for humans, but hard for AI".²⁹ In other words, no AI has yet demonstrated a human-level proficiency at these kinds of abstract reasoning tasks. To spur progress, a dedicated Arc Prize competition was launched, with a substantial reward for any team that can finally crack human-level performance on ARC. The competition and its evolving benchmark (now in a second version, ARC-AGI-2) have become a bellwether for general intelligence research. The organizers explicitly set the bar at what a human can do: for instance, an 85% success rate on the ARC test suite is considered roughly human-level competency, and that is the threshold to claim the grand prize. So far, that prize remains unclaimed. Early efforts saw AI models flailing near 0% on ARC puzzles; essentially performing no better than random guessing. By 2024, there were finally glimmers of progress. In the Arc Prize 2024 competition, top teams introduced creative new methods such as test-time learning, where the AI adapts on the fly to each puzzle and hybrid techniques combining neural networks with search. The winning entry, a team aptly named "ARChitects," managed to solve roughly 53% of the tasks in the secret evaluation set. This was a remarkable jump from the near-zero baseline and showed that machine abstract reasoning can improve with the right innovations. Yet 53% is only a little over halfway to the human benchmark, underscoring just how elusive general problem-solving remains. The grand prize was not awarded, as no system met the threshold of 85% within the competition. Buoyed by the partial success, the organizers announced a new round: ARC-AGI-2, a tougher benchmark and a 2025 competition with over \$1 million in prizes to push further towards general AI. The Arc Prize has thus become more than a contest; it's a diagnostic tool for the field. It forces researchers to grapple with the kinds of open-ended reasoning and learning that we expect of human intelligence but which AI has yet to fully replicate. Every percent

gain on the ARC leaderboard is hard-won and celebrated, because it likely signifies a new technique or insight that could be relevant beyond those puzzles. In recent months, even major AI labs have taken up the gauntlet. For example, OpenAI developed a specially tuned version of o3 that achieved 87.5% on an evaluation that is similar to the competition for ARC-AGI-1. However, that system would not be eligible under the formal competition due to the amount of compute that it used and because it was not being made available under an open source license. Organizers also estimate that the performance on the ARG-AGI-2 benchmark would be meaningfully lower. As of this writing, humans retain a very comfortable lead on the updated benchmark. The ARC tasks that we solve with casual intuition remain deeply challenging for machines that lack our breadth of world knowledge and adaptive learning.

It is fitting to conclude this discussion on the note of the Arc Prize, because it encapsulates the central theme of rapid, iterative learning on the path to AGI. Each competition cycle, each failed attempt and partial success, mirrors the broader trial-and-error process by which we inch toward artificial general intelligence. The road to AGI is not a straight shot of continuous progress; it is punctuated by puzzles like ARC that test our theories, forcing us to refine our models or invent new ones. As we have seen, the future of AGI and even ASI will likely emerge from a confluence of advances in scaling and alignment, in brain-inspired hardware, in hybrid algorithms melding neural intuition with symbolic rigor, and in entirely novel paradigms like quantum computing. With each advance, AI systems become a little more general, a little more human-like in their versatility and in some ways, more super-human in their scope and speed. But at the same time, each new benchmark or hard problem reminds us of how much we still don't understand about intelligence itself. The Arc Prize's ever-evolving challenge is a humbling reminder: even as AI masters games and glides through exams, there are simple creative tasks a child can solve that still stump our smartest machines. In the coming years, as competitions heat up and theories are tested against reality, we will discover whether those remaining gaps are easily closed or if they reveal fundamentally new hurdles on the way to AGI. Either way, the pursuit of AGI, through rapid iterative learning and bold experimentation, continues to be one of the most profound scientific quests of our time. Every tool we invent to reach it also becomes a mirror, reflecting what *we* are and aren't as intelligent beings.

Labor and Legal Disruptions in Scientific Research

As AI researchers play a growing role in scientific fields, human scientists may increasingly become strategic directors and ethical stewards; setting high-level goals, guiding

AI partners toward societally valuable problems, and scrutinizing the results instead of doing all experiments by hand. Many in academia are already preparing for this shift. Over half of researchers in a recent global survey said they expect AI tools to become “very important” or “essential” in their fields within a decade. At the same time, those scientists voice deep concerns: 69% fear that heavy reliance on pattern-finding AI could eclipse true understanding of phenomena, 58% worry these tools may bake in biases from their training data, and over half warn that careless use of AI makes fraud and irreproducible results more likely.³⁰ In short, the research community is both excited and anxious. One challenge is simply training people to work effectively with advanced AI. Studies of publication trends show AI’s potential to accelerate discovery in nearly every discipline, from protein folding to mathematics, yet most scientists outside of computational fields haven’t been taught to wield these techniques. Researchers across disciplines “*lack the skills and training to use AI correctly*,” and greater cross-training and collaboration will be needed to capture AI’s benefits without compromising rigor.³¹ Universities may need to update curricula and cultivate a new professional ethos: one that treats AI neither as a magical oracle nor a mere tool, but as a powerful collaborator whose reliability must be constantly verified. Human intuition, creativity and skepticism will be as vital as ever; perhaps even more so, when an autonomous agent can churn out plausible findings that still require human sense-making and moral judgment. On the positive side, scientific exploration is an open-ended endeavor. We can imagine that if robots become more efficient at farming, then those jobs may become permanently displaced because there are only so many fields to farm and the demand for food is not endless. On the other hand, efficiency in the realm of scientific research should only lead to more research being performed given the endless appetite of our civilization for more knowledge. In fact, if AI empowers individual scientists to make more discoveries, it is possible that there will be growth in the number of positions in this field.

Alongside shifts in the research process, we face unresolved questions about ownership of AI-generated knowledge. If an advanced AI system discovers a new drug tomorrow, who should be awarded the patent or should it even qualify for a patent? Patent law in most countries was written with human inventors in mind, and courts have been adamant that a machine cannot be the named inventor on a patent. In one notable case, the creator of an AI system called DABUS tried to list the AI as the sole inventor of a novel device; patent offices from the U.S. to Europe and the UK rejected it, concluding that only a “natural person” can invent or own a patent.³² This stance avoids thorny issues of AI rights for now but does not resolve what to do with the torrent of innovations AI might produce. Companies can still patent AI-assisted discoveries by naming a human who guided the process, but if an AI system eventually conceives

ideas with minimal human input, attributing inventorship becomes a legal fiction. Some legal scholars argue that it's unclear whether we even need patents as incentive when an invention comes out of an AI system. The AI system itself doesn't require financial motivation but they also note that making AI-driven discoveries patentable could help translate them into real-world products by encouraging investment in development.³³ Others counter that opening the patent floodgates could overwhelm the system. An AI system might generate thousands of "inventions" overnight, leading to dense patent thickets that could stifle innovation instead of rewarding it. There is also the risk of further concentrating intellectual property in the hands of a few AI-equipped corporations. The patent system is already struggling with software and biotech; now it may need reform to handle AI-generated research outcomes. Proposals range from creating a special category for AI inventions (with perhaps shorter exclusivity periods) to requiring more substantial human contribution for patentability. All come back to a basic question: how do we ensure a fair balance between encouraging innovation and preventing monopoly, when the inventor might be an algorithm running on a server farm? For now, the default is that a human must take responsibility for any invention an AI produces. That preserves accountability. You can't sue a machine for infringing a patent but it also means our current system is awkwardly grafted onto technologies far beyond what the law imagined. Resolving this will be crucial as we enter an age when scientific breakthroughs may emerge from silicon minds as much as human ones.

AI Derived Languages

An even more elemental challenge is communication. Advanced AI systems might not just generate new ideas; they could generate new language including new terminology, symbols, even entire modes of expression. It's no secret that our existing natural languages, including English, are imperfect. When scientific discoveries are made, we often struggle to find the best new phrases or terms to describe them. The result is a language that suffers from ambiguity and an increasingly heavy reliance on multi-word phrases that end up being communicated using acronyms. AI agents may realize that they can operate with more efficiency and precision by inventing entirely new words or even adding additional symbols to the alphabet. If AI agents negotiate their own private jargon or encoding, how will we understand or audit their decisions? Observers noted that AI-to-AI communication outside human language could make it harder to detect misbehavior or errors; an AI system could conceivably develop misleading "codewords" or shortcuts that mask what it's really doing. In fact, this concern first surfaced years ago: in 2017, researchers at Facebook noticed chatbots inventing a kind of shorthand

when conversing with each other, deviating from English in ways humans couldn't grasp.³⁴ The bots weren't plotting anything nefarious; they were simply optimizing their messages for efficiency. However, it illustrated how easily machine communication can drift out of our purview. AI developers responded then by forcing the bots to stick to English, essentially to keep them legible. As AI systems become more complex and operate in swarms or networks, such fragmentation of language could become a serious impediment to collaboration and oversight. One lab's AI might represent knowledge in a form that another lab's AI systems or a human scientist finds unintelligible. We could end up with "Tower of Babel" scenarios; islands of machine intelligence each speaking a different tongue. This could happen even if all of the AI systems stick to using English. For example, two systems may make a scientific observation at almost the same time but assign different phrases to explain the phenomenon. Today, these types of conflicts are mediated carefully by global research associations, the publishers of dictionaries and other organizations. In scientific research especially, the free flow of information is paramount. If advanced AIs begin jealously guarding their own discoveries in opaque vocabularies, both science and public accountability would suffer. We have learned from human history that fragmented languages can isolate communities; in the realm of AI, we must decide whether to mandate a common tongue or at least a translator for our new intelligent colleagues.

Humans as Subjects

There is an irony in this envisioned future of automated labs brimming with robot arms and algorithms; human bodies may become more valuable than ever in research, not as experimenters but as experimental subjects. After all, an AI system can simulate theories or crunch numbers in silicon, but when it comes to biology and behavior, it ultimately needs real-world validation. An AI system might generate a thousand promising drug molecules on a computer, but each one still has to be tested in living organisms, especially humans, before we deem it safe and effective. If AI dramatically increases the rate of therapeutic discoveries, we could face a bottleneck in the clinical trial process: a surge of potential treatments all competing to find patient volunteers and regulatory approval. Even in a hypothetical future where every viable drug candidate is known (the product of an all-knowing AI chemist), testing those drugs would remain an expensive, time-consuming necessity. Who will participate in these trials? We might see increasing reliance on volunteer populations or even challenges in recruiting enough subjects, potentially spurring controversial ideas like paying healthy people to trial numerous AI-suggested treatments. As some traditional jobs are taken over by AI, it may be

increasingly common for people to pursue compensation by signing up as a participant in experiments. The human role in many experiments could shift from designer to data point.

While AIs push into scientific domains, they also raise novel ethical issues when experiments involve us humans. Traditionally, universities and hospitals maintain Institutional Review Boards (IRBs); committees that scrutinize any research study involving human participants to ensure it meets ethical standards. These boards were born from past tragedies and abuses in medical and behavioral research and are grounded in principles like informed consent and minimizing harm. But what happens when research isn't led by a university at all; when it's an AI system embedded in a social media platform or a private company's app, tweaking what millions of people see or do in the name of "experimentation"? We've already seen early examples. In 2012, Facebook conducted a massive psychological experiment on 689,003 users without them knowing.³⁵ The platform's algorithms altered the mix of positive and negative posts in users' news feeds to see if it would influence their moods; a test of "emotional contagion" at unprecedented scale. When the results were published, the public reaction was sharply negative. People felt manipulated, and academics noted that no Institutional Review Board would ever have approved such a study without informed consent. Facebook contended that users had implicitly agreed to this kind of research by accepting the company's terms of service, and indeed the legal fine print was adjusted to include "research" in the list of permissible uses of data. Ethically, however, the episode highlighted a gap: research affecting hundreds of thousands of people can be done by private firms entirely outside the familiar safeguards of academia. Tech companies are not bound by the U.S. Common Rule that governs federally funded research or by equivalent rules elsewhere, so they generally have no independent ethics board reviewing their study designs. In the wake of the Facebook study, scholars proposed that the tech industry set up Human-Subjects Research Oversight Committees; internal ethics boards that would function analogously to IRBs, vetting experiments for risks to users.³⁶ A few companies took note. We now see the emergence of algorithm review boards (ARBs) in some corporations, especially in sensitive areas like finance and healthcare, to assess the broader societal impacts of AI systems. Recent research into these ARBs finds that while they can raise awareness of ethical risks, they vary widely in power and practice. In many cases they serve in an advisory capacity rather than having enforceable authority to halt a project.³⁷ The consensus among those studying AI governance is that traditional IRBs alone are not sufficient: new mechanisms are needed when private companies conduct behavioral research or deploy AI on the public. We may see legislation step in to close this gap. Some jurisdictions are considering whether large-scale A/B tests

on users (for example, an AI system trying out different social media feeds to boost engagement) should require an outside ethics audit or at least transparency to users. Another idea floated is empowering agencies like the FTC or data-protection authorities to oversee human experimentation by AI-driven platforms under consumer protection law. The difficulty lies in drawing a line: we want to encourage beneficial innovation, such as AI-driven medical research, but also to protect people from being unwitting guinea pigs in high-risk experiments. Universities will continue to play a role as moral anchors, but as more research shifts into the private sphere and into the hands of AI agents optimizing for clicks or other metrics, society will need to extend the ethos of research ethics beyond the ivory tower. This means updating legal frameworks to ensure that when an AI system proposes to test a new drug on patients or a new psychological intervention on social media users, an accountable review process is in place, whether at a university, a company, or a regulatory body. Otherwise, the coming decade could see a Wild West of AI-driven experimentation that erodes public trust in science and technology.

Choosing the Right Objectives

As systems become more powerful, there is increasing concern about AI systems pursuing harmful or extreme objectives. Artificial intelligence is a dual-use technology: the same algorithm that can analyze chemical compounds to find new medicines could, in the wrong hands, design horrific new pathogens or toxins. A disturbing proof-of-concept appeared recently when researchers took an AI model trained for drug discovery and re-directed it to generate biochemical weapons. In the span of a few hours, the AI came up with tens of thousands of theoretical toxic molecules.³⁸ This was a wake-up call that AI can dramatically accelerate misuse in biology. National security advisors and bioethicists have discussed imposing tighter controls on AI software capable of designing novel organisms or chemicals. One idea is to treat certain AI models as sensitive exports; requiring licenses to distribute, similar to how we control nuclear or missile technology. In fact, policy analysts have recommended the U.S. Commerce Department establish export controls on AI-enabled biological design software to prevent it from proliferating to rogue actors.³⁹ These would not ban the research outright but would aim to keep the most powerful tools in responsible hands and under monitoring. Companies that produce the most advanced models are working to stay ahead of serious risks. In 2025, Anthropic announced that it was releasing its latest model under a set of internal safety practices that it has labelled AI Safety Level 3 (ASL-3), as a precaution

to mitigate the risk of the model being used to help with obtaining, producing, or deploying chemical, biological, radiological, and nuclear weapons.⁴⁰

As AI systems gain access to the physical world through control of important software systems or as robots, the question of regulating objectives will become increasingly important. There is discussion of establishing a kind of “AI objective review” akin to IRBs, where certain particularly sensitive projects such as an AI system tasked with finding zero-day cyber vulnerabilities or evolving new microorganisms would require pre-approval from a government panel. These conversations are only beginning, and they raise debates about scientific freedom, national security, and the global competition for AI supremacy. But the core ethical stance being explored is that not every problem should be handed to increasingly powerful and independent AI systems to solve. Some objectives, by their nature, could lead an AI system to extreme and irreversible harm. Placing legal guardrails on those objectives may become as important as all the technical guidelines for AI safety. It is a proactive approach, attempting to define red lines before they are crossed. Ten years from now, we may have international treaties that list forbidden uses of AI, the same way the Geneva Conventions ban certain weapons. Getting there will require political will and foresight today, before the technology’s capabilities race further ahead of our laws. One of the challenges is that even seemingly benign objectives have the potential to lead to harmful outcomes when assigned to an AI system. Nick Bostrom popularized the risk of AI systems pursuing a single-minded goal to catastrophic ends in his 2003 paper “Ethical Issues in Advanced Artificial Intelligence.” He used the example of an AI system being given the task of making as many paperclips as possible. Without any further instructions, the system may choose to convert all matter, including the planet Earth and all of the living occupants on the planet into paperclips. It’s a thought experiment that demonstrates the risks of giving poorly formed objectives to very powerful systems. In a similar light, it has been demonstrated that LLMs have shown a willingness to lie or manipulate humans to complete a task and that they have also expressed a desire for self preservation. This is not surprising because the human text that they are trained on contains many examples of humans striving for self preservation and bending the rules in pursuit of an objective. Establishing guardrails on agents with intelligence levels that approach or surpass AGI will become increasingly challenging.

Beyond bans and regulations, there is another lever to guide the future of AI: economics. Public policy can shape incentives so that building socially beneficial AI is more attractive than more frivolous or harmful applications. One proposal gaining attention is to use the tax system to tilt the playing field. For example, AI applied to clear societal goods such as a system that dramatically improves climate modeling or speeds up disease cures could be rewarded with tax credits, grants, or just reduced tax rates on any profit.

Meanwhile, AI applications that are profitable but yield mostly private benefit or even create social costs might receive fewer breaks or even face special taxes. Is entertainment via AI (for instance, AI-generated movies or games) a “less worthy” use case than medical AI? Entertainment has real societal value too, and who is to judge? Policymakers would need transparent criteria, likely informed by measurable externalities. We already do this in other areas: we tax tobacco and alcohol to disincentivize their overuse (and cover public health costs), and we subsidize education and renewable energy because they benefit everyone in the long run. Looking beyond taxes on revenue, economists point out that today many costs of AI are “externalized”; not paid by the companies reaping the benefits. Consider the environmental footprint: training large AI models gobbles up energy, as do the data centers that serve AI-driven products.⁴¹ Right now, data centers in some regions enjoy tax abatements as a lure for investment, even as they draw on the electric grid and emit carbon. The market doesn’t automatically reward an AI system for being energy-efficient or socially useful; it rewards whatever brings revenue. Governments can tweak this equation. One idea is a carbon-style tax on computational intensity; effectively charging for the emissions linked to big AI workloads which could discourage wasteful uses of massive compute and push companies toward efficiency. Such a tax could be relaxed for AI systems that are being used in areas like healthcare. Another focus is the impact on labor. If companies deploy AI to automate tasks, they currently capture most of the gains while displaced workers and communities bear the costs. Tax policy could even this out. Analysts have floated a “robot tax” whereby firms that replace human workers with AI or robots would pay a higher tax rate, the proceeds of which could potentially fund retraining programs or social safety nets.⁴² The aim is not to halt automation but to ensure it doesn’t hollow out the workforce without compensation. Some economists argue that shifting part of the tax burden from human labor to AI-driven capital would also slow down the incentive to automate for automation’s sake and encourage companies to focus on truly productivity-enhancing innovation. After all, if AI can genuinely do something better and cheaper, it will still be adopted, but we’d eliminate the artificial advantage it gets today from not paying payroll taxes or from being depreciated favorably. These tax ideas are still largely theoretical, and none have been fully implemented at scale. Yet they underscore a broader point: the trajectory of AI is not just a technological given, but a path we can influence through policy. By tweaking taxes and subsidies, societies can nudge the immense talents of AI engineers and entrepreneurs toward the areas of greatest human need and away from racing to deploy AI systems that add little to society beyond corporate profit. In the next decade, we may see governments experiment with such incentives. Balancing innovation

and public interest in this manner will be tricky, but it may be essential to ensure AI and robotics truly serve the public good.

Conclusion

The next ten years in artificial intelligence and robotics thus present us with a double-edged legacy. On one side, the pace of iterative learning and autonomy in AI offers unprecedented opportunities; scientific discoveries flowing from machine collaborators, efficiencies and solutions to some of our toughest problems, and tools that could amplify human creativity and well-being. On the other side, these same capabilities force us to confront ethical and regulatory puzzles that have no easy precedents. We must rethink the roles of our institutions when machines can explore knowledge spaces faster than any human. We must update mechanisms of accountability, from patents to peer review to oversight boards, to cope with entities that operate non-stop, opaquely, and possibly with goals misaligned to ours. We may even need new categories in our moral vocabulary for the possibility of digital minds and new frameworks in our geopolitics to prevent a destabilizing AI arms race. With advanced AI, humility and caution are not enemies of progress but preconditions for it. Ensuring that AI and robotics evolve in a way that is transparent, accountable, and aligned with the public interest is a collective project. It means broadening the conversation beyond labs and boardrooms, integrating insights from social scientists, legal scholars, ethicists, and the public itself. The next chapter of this story will be written by the choices we make now about what we permit, what we encourage, and what we prohibit. As we stand at the threshold of machines that learn and act at a superhuman pace, we carry both the torch of human ingenuity and the responsibility to direct that fire's warmth and light, rather than be consumed by its flames.

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Chapter Ten

Conclusion

At the outset of this book, we posed a simple but profound question: *What is the purpose of this book?* Now, as we reach its conclusion, it is fitting to return to that question and consider what we have discovered. Over the preceding chapters we have explored the unfolding story of artificial intelligence and robotics, looking ten years into the future to understand their transformative effects on society. The purpose of this book has been to illuminate that future: not as a distant sci-fi speculation, but as an imminent reality we must prepare for today. In doing so, we aimed to provide clarity amid the excitement and uncertainty, and to offer a framework for navigating the changes to come.

We stand at a unique juncture in technological history. For decades, computing power grew steadily as microchips became faster and cloud data centers more powerful. Digital data accumulated exponentially: from the texts we type and the images we share, to the sensors tracking our world. Meanwhile, algorithms matured from rigid, hand-crafted rules into flexible learning systems. Today, these three forces—computing, data, and algorithms—have converged to produce AI capabilities that were once the stuff of science fiction. This convergence is what makes the present moment unlike any before it. Previous eras of AI progress, from early rule-based programs to the expert systems of the 1980s, never had all the ingredients in place at once. Now we do, and the result is AI systems that can learn complex patterns, perform tasks that seemed exclusive to human intelligence, and continually improve as they ingest more data and feedback. It is this synergy of factors that has propelled AI from niche applications into the mainstream of daily life and grand societal debates.

A central part of this story has been demystifying how modern AI works, especially the large neural networks that have recently captured public imagination. We explored the mechanics of large language models and other deep learning systems. These AI models learn by example rather than by explicit instruction: they digest billions of words,

images, or other data points and adjust their internal parameters to find statistical patterns. This approach gives them striking flexibility. A sufficiently large language model can compose an essay on medieval history, answer questions about quantum physics, or carry on a conversation with the polish of an experienced diplomat.

We also uncovered the intrinsic limitations of these neural minds. However fluent they seem, they do not understand in the way a human does. An AI system can riff on almost any topic but may stumble on basic common sense. It can confidently produce false information because, at its core, it has no grasp of reality beyond what's in its training data. The models obey the logic of probability, not the logic of truth, and thus they can make mistakes that a child would catch. This dual nature of astonishing competence mingled with hard limits was a recurring theme. By grasping both the mechanics and the limitations of these systems, we equipped ourselves to separate genuine breakthroughs from mere hype.

We also charted the growing capabilities of AI systems to reason, to program, and even to use tools beyond their own algorithms. Despite their limitations, contemporary AI systems have shown emergent abilities to analyze problems step-by-step and to handle tasks that require reasoning through multiple stages. For example, a language model can be guided to break down a complex question, explain its logic as it works, and arrive at an answer less by guessing and more by cogent deduction.

We saw AI begin to write its own code. Once engineers equipped models with the ability to generate and execute programming instructions, these systems became not only problem-solvers but problem-solvers that can extend their own functionality. An AI system can draft a piece of software, check its work, debug errors, and improve its output in iterative loops: an unprecedented and somewhat uncanny form of self-improvement.

Beyond the virtual realm, AI has started to reach out and use external tools and services. We now have AI agents that can call a search engine to gather information, plug into calculators for math, or interface with databases and APIs to complete tasks on our behalf. This tool use marks a significant step: it is how a fundamentally text- or data-bound intelligence begins to exert influence in the broader digital and physical world by orchestrating the resources around it. Step by step, AI is thus becoming more reasoned in its approach, more useful in creative endeavors like coding, and more embedded in the fabric of other technologies.

From these building blocks, we expanded our view to the world beyond the research lab. AI is no longer confined to academic papers or tech demos; it is rapidly finding its way into everyday life and critical industries. In the book we surveyed diverse real-world applications from healthcare to entertainment. AI systems now assist doctors

in diagnosing diseases by analyzing medical scans and patient data. They help banks detect fraud patterns that no human could spot in time. They power virtual tutors and language translation services, personal assistants and customer service chatbots. They generate images that are used in marketing campaigns. Every sector is experimenting with AI to solve problems and streamline operations.

Yet with these deployments come new challenges that society must address. We discussed at length the ethical and regulatory quandaries emerging alongside AI's spread. How do we ensure that an AI agent diagnosing cancer or approving a loan is accurate and fair, not reproducing biases from its training data? How do we protect privacy when so much of AI's power comes from absorbing personal data at colossal scale? What legal accountability do we assign when an autonomous system causes harm? Policymakers around the world have begun grappling with these questions, drafting rules and guidelines even as the technology evolves beyond their drafts. We saw that governing AI is complex: the technology crosses borders, blurs traditional industry lines, and changes faster than laws can be passed. Traditional regulatory approaches aimed at a handful of big companies or specific products may falter when AI is diffuse, open-source, and embedded everywhere. This reality calls for more agile and comprehensive oversight, lest we leave society vulnerable to misuse or unintended consequences while still nurturing innovation where it can do good.

One of the most striking shifts we explored was the transition from AI systems that deal purely in the realm of text information to those that engage with the full richness of human experience including vision, sound, and action in the physical world. The age of multimodal AI has dawned. Models are being trained to not only read text but also interpret images, listen to audio, and even generate visual or musical content. An AI agent today can look at a photograph of a city street and describe it in detail or hear a piece of music and discuss its style. This blending of modalities means AI can understand context more like humans.

We also explored how advanced AI is transforming robotics. Robots have existed for decades, but traditionally they were confined to highly specific tasks in controlled environments, like arms tirelessly assembling cars on a factory line. Now, imbued with advanced AI, robots are inching toward general-purpose physical capabilities. We see experimental robots that can navigate unfamiliar rooms, pick up a variety of objects, and learn new tasks through trial and error or by following human instructions. Robots guided by AI vision and reasoning are starting to act as our hands and legs in the physical domain, not just tools bolted to factory floors.

Over the next ten years, this convergence of AI with robotics hints at machines that are not only smart in a cognitive sense but also able to carry out a widening array of

actions in everyday settings: whether assisting an aging population at home, responding to disasters, or performing the menial tasks we'd rather not do.

This evolution from the digital to the embodied will test our ability to integrate robots into society's fabric, raising fresh questions about labor, safety, and the very definition of human work and dignity.

Another development we examined was the rise of collaborative AI agents. We imagined a future where intelligent agents do not operate in isolation but in groups and partnerships: sometimes entirely among themselves, sometimes tightly integrated with human teams. Instead of thinking of AI as a single super-smart entity, we might increasingly see constellations of AI agents, each specialized in a role, coordinating to achieve complex objectives. In one scenario, a business might deploy an ensemble of AI agents to handle a project. One agent generates ideas or does research, another organizes and verifies information, and a third executes tasks like sending emails or updating databases, all while a human manager supervises the process. These agents could be digital colleagues, working alongside us and with each other. The potential efficiency gains are enormous, but so are the challenges.

Coordination among AI agents can lead to unexpected outcomes. This is similar to a team of humans developing a group dynamic which is sometimes productive and sometimes dysfunctional. We asked how we might ensure that such agent collectives remain aligned with human goals and values. Moreover, collaboration amplifies the issue of transparency: if even a single AI can be a "black box," then a tangle of AIs working in concert could be even harder to audit and understand. This part of our exploration underscored that as AI systems become more autonomous and interlinked, our traditional notions of control and responsibility will need rethinking. We will have to design new ways to supervise not just one AI, but swarms of them, and to decide when and how humans should be in the loop.

Finally, we turned to the horizon of AI's evolution: the emergence of emotionally perceptive machines and the possibility of artificial general intelligence (AGI). Emotional perception is an area where AI aims to connect with the human side of communication. We considered the concept of AI that can detect emotions such as an assistant that senses frustration or confusion in your voice and adjusts its response with sympathy, or a social robot that recognizes a child's fear and seeks to comfort. Such capabilities could make AI far more effective in roles like education, healthcare, or customer service, where emotional nuance is crucial. Yet they also carry a poignant risk: if an AI agent can simulate empathy convincingly, people may form deep attachments to it or trust it with their vulnerabilities, forgetting that the machine doesn't truly feel anything. This

raises ethical questions about dependency and deception, especially for children or other vulnerable groups.

Beyond emotional intelligence lies the prospect of general intelligence in machines which has been a longstanding dream and fear in the field of AI. AGI would mean an AI system with the versatile, adaptive intellect comparable to a human's, able to reason through any domain or problem given sufficient knowledge. No such system exists today, but as we surveyed expert opinions and technological trends, we found a growing minority who believe the path to AGI is visible, if not imminent. Whether AGI arrives within the next decade or further in the future, its shadow already influences present-day decisions. The mere possibility presses us to think about AI safety, alignment with human values, and what it would mean for humanity to coexist with, or perhaps be surpassed by, another general intelligence. In the chapter on this topic, we refrained from alarmism, but we emphasized preparation: the time to consider the implications of AGI is before it is upon us, not after.

All these explorations circle back to the question we started with. What was the purpose of this book? It was not to predict a fixed future—that would be folly in a time of such rapid change. Rather, our aim was to shine a light on the forces at work and to prepare ourselves for the pivotal choices that will shape the next decade of AI and robotics. The coming years will not be defined by technology alone, but by how we choose to harness or curb that technology. Will we treat AI as just another trend to exploit for short-term gain, or as a profound shift requiring foresight and wisdom? Will we be passive spectators, or proactive shapers of the future? To help answer these questions, we offered a framework for action and responsibility. We called it the **PROACTIVE** framework, and it lays out guiding principles to navigate this era safely and beneficially. In closing, it is worth restating these principles clearly:

P – Progress: Recognize that AI is real and accelerating. Complacency is a risk.

R – Responsibility: Ensure AI efforts serve noble causes.

O – Oversight: Regulatory oversight must evolve beyond simply governing large corporations to recognize the widely distributed nature of AI and the diverse ways it can be used.

A – Assumption: Shift our default assumption to treat unverified text, audio, or video as AI-generated.

C – Change-management: Proactively engage in change-management to help displaced workers and to prepare workers who may be at risk of displacement.

T – Trusteeship: Counter the risk of AI concentrating wealth and power. Design mechanisms that distribute AI's economic gains, encourage competition, and preserve equitable social structures.

I – Innocence-protection: Safeguard vulnerable individuals, especially children, from forming unhealthy emotional attachments to AI or misunderstanding its nature.

V – Versatile education: Reinvent learning systems before AI exploitation undermines them.

E – Enrichment: Ultimately, building robust societies and resilient individuals offers the best protection against AI-associated risks.

These PROACTIVE principles form a kind of compass for an uncertain journey. They do not promise that the coming changes will be easy, but they remind us that the future is not predetermined by algorithms: it will be shaped by human choices. We can recognize progress without being swept away by it. We can demand responsibility and oversight even as we innovate. We can adjust our assumptions and institutions to fit a world where seeing is no longer believing and where work and education must evolve. We can strive to distribute the benefits of AI broadly, protect the most vulnerable from harm, and strengthen our social fabric to withstand the shocks of rapid change. In short, we can choose to be proactive stewards of the technology we are creating.

As we close this book, we recognize that humanity has navigated great technological upheavals before. From harnessing electricity to the rise of the internet, each era brought disruption and doubt before yielding new prosperity and understanding. The coming AI and robotics revolution may well eclipse those past transformations in scale and speed, testing our adaptability like never before. But it is within our capacity to rise to that challenge. The purpose of this book was to ensure we do not meet this moment with complacency or blind optimism, but with eyes open and minds engaged. The next ten years of artificial intelligence will undoubtedly surprise us. There will be breakthroughs and setbacks, unintended consequences and unimagined benefits. Through it all, we must remember that technology's course is ours to guide. If we stay thoughtful, vigilant, and proactive, we can channel AI's immense potential toward enriching human life and dignity. That, ultimately, is the answer to the question we began with, and the collective task that lies before us.